

AptitudeX 2,000-Question Bank

50 topics ? 20 worked examples + 20 practice questions per topic ? distinct model tags ? practice solutions withheld until the answer-key section.

Percentages ? 40 questions

WORKED EXAMPLES

Q1 ? Direct computation ? Beginner

A data analyst asks for 15% of a dataset containing 140 records. What is the selected count?

Options: 25 | 19 | 23 | 21

Answer: 21

Solution: $15/100 \times 140 = 21$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q2 ? Reverse value ? Easy

A ■160 project budget reserves 20% for testing. How much is reserved?

Options: 38 | 29 | 35 | 32

Answer: 32

Solution: $20/100 \times 160 = 32$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q3 ? Missing base ? Medium

An employee earns ■180 and receives a 25% performance bonus. Find the bonus.

Options: 45 | 50 | 40 | 55

Answer: 45

Solution: $25/100 \times 180 = 45$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q4 ? Successive change ? Hard

A warehouse stores 200 units; 30% must be inspected. How many units are inspected?

Options: 54 | 66 | 60 | 72

Answer: 60

Solution: $30/100 \times 200 = 60$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q5 ? Comparison trap ? Placement

In a poll of 220 voters, 35% choose option A. How many voters choose A?

Options: 93 | 85 | 77 | 69

Answer: 77

Solution: $35/100 \times 220 = 77$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q6 ? Ratio form ? Beginner

A service invoice of ■240 attracts a 40% levy. Find the levy amount.

Options: 96 | 106 | 86 | 116

Answer: 96

Solution: $40/100 \times 240 = 96$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q7 ? Weighted result ? Easy

A training batch has 260 seats and 45% are reserved for interns. How many seats?

Options: 117 | 129 | 105 | 141

Answer: 117

Solution: $45/100 \times 260 = 117$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q8 ? Replacement cycle ? Medium

A measurement is 280; an audit reports 10% of it as the affected portion. Find that portion.

Options: 31 | 34 | 25 | 28

Answer: 28

Solution: $10/100 \times 280 = 28$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q9 ? Unit conversion ? Hard

A retailer applies 15% markdown to an item priced at ■300. Find the markdown amount.

Options: 50 | 55 | 40 | 45

Answer: 45

Solution: $15/100 \times 300 = 45$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q10 ? Work-rate model ? Placement

A placement cohort contains 320 candidates and 20% clear the cutoff. How many clear?

Options: 70 | 76 | 64 | 58

Answer: 64

Solution: $20/100 \times 320 = 64$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q11 ? Speed-time model ? Beginner

A data analyst asks for 25% of a dataset containing 340 records. What is the selected count?

Options: 85 | 94 | 76 | 103

Answer: 85

Solution: $25/100 \times 340 = 85$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q12 ? Table lookup ? Easy

A ■360 project budget reserves 30% for testing. How much is reserved?

Options: 130 | 97 | 119 | 108

Answer: 108

Solution: $30/100 \times 360 = 108$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q13 ? Chart comparison ? Medium

An employee earns ■380 and receives a 35% performance bonus. Find the bonus.

Options: 133 | 146 | 120 | 159

Answer: 133

Solution: $35/100 \times 380 = 133$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q14 ? Multi-step word problem ? Hard

A warehouse stores 400 units; 40% must be inspected. How many units are inspected?

Options: 160 | 176 | 144 | 192

Answer: 160

Solution: $40/100 \times 400 = 160$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q15 ? Maximum / minimum ? Placement

In a poll of 420 voters, 45% choose option A. How many voters choose A?

Options: 227 | 170 | 208 | 189

Answer: 189

Solution: $45/100 \times 420 = 189$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q16 ? Least / greatest value ? Beginner

A service invoice of ■440 attracts a 10% levy. Find the levy amount.

Options: 52 | 40 | 48 | 44

Answer: 44

Solution: $10/100 \times 440 = 44$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q17 ? Remainder cycle ? Easy

A training batch has 460 seats and 15% are reserved for interns. How many seats?

Options: 76 | 83 | 62 | 69

Answer: 69

Solution: $15/100 \times 460 = 69$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q18 ? Divisibility test ? Medium

A measurement is 480; an audit reports 20% of it as the affected portion. Find that portion.

Options: 96 | 116 | 86 | 106

Answer: 96

Solution: $20/100 \times 480 = 96$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q19 ? Factorisation ? Hard

A retailer applies 25% markdown to an item priced at ■500. Find the markdown amount.

Options: 151 | 112 | 138 | 125

Answer: 125

Solution: $25/100 \times 500 = 125$. Shortcut: Use a fraction–percentage equivalent before multiplying.

Q20 ? Approximation ? Placement

A placement cohort contains 520 candidates and 30% clear the cutoff. How many clear?

Options: 156 | 172 | 140 | 188

Answer: 156

Solution: $30/100 \times 520 = 156$. Shortcut: Use a fraction–percentage equivalent before multiplying.

PRACTICE ? solutions hidden below

Q21 ? Options elimination ? Beginner

A data analyst asks for 35% of a dataset containing 540 records. What is the selected count?

Options: 189 | 170 | 208 | 227

Q22 ? Boundary case ? Easy

A ■560 project budget reserves 40% for testing. How much is reserved?

Options: 268 | 202 | 246 | 224

Q23 ? Case split ? Medium

An employee earns ■580 and receives a 45% performance bonus. Find the bonus.

Options: 313 | 235 | 287 | 261

Q24 ? Contradiction check ? Hard

A warehouse stores 600 units; 10% must be inspected. How many units are inspected?

Options: 72 | 66 | 60 | 54

Q25 ? Statement I test ? Placement

In a poll of 620 voters, 15% choose option A. How many voters choose A?

Options: 93 | 111 | 84 | 102

Q26 ? Statement II test ? Beginner

A service invoice of ■640 attracts a 20% levy. Find the levy amount.

Options: 154 | 115 | 141 | 128

Q27 ? Combined statements ? Easy

A training batch has 660 seats and 25% are reserved for interns. How many seats?

Options: 165 | 182 | 148 | 199

Q28 ? Counting restriction ? Medium

A measurement is 680; an audit reports 30% of it as the affected portion. Find that portion.

Options: 184 | 204 | 224 | 244

Q29 ? Circular arrangement ? Hard

A retailer applies 35% markdown to an item priced at ■700. Find the markdown amount.

Options: 295 | 220 | 270 | 245

Q30 ? Complement method ? Placement

A placement cohort contains 720 candidates and 40% clear the cutoff. How many clear?

Options: 346 | 259 | 317 | 288

Q31 ? Pattern: differences ? Beginner

A data analyst asks for 45% of a dataset containing 740 records. What is the selected count?

Options: 333 | 366 | 300 | 399

Q32 ? Pattern: ratios ? Easy

A ■760 project budget reserves 10% for testing. How much is reserved?

Options: 68 | 76 | 84 | 92

Q33 ? Pattern: alternating ? Medium

An employee earns ■780 and receives a 15% performance bonus. Find the bonus.

Options: 141 | 105 | 129 | 117

Q34 ? Pattern: squares / cubes ? Hard

A warehouse stores 800 units; 20% must be inspected. How many units are inspected?

Options: 160 | 176 | 144 | 192

Q35 ? Inference from facts ? Placement

In a poll of 820 voters, 25% choose option A. How many voters choose A?

Options: 184 | 205 | 226 | 247

Q36 ? Assumption check ? Beginner

A service invoice of ■840 attracts a 30% levy. Find the levy amount.

Options: 302 | 227 | 277 | 252

Q37 ? Cause vs correlation ? Easy

A training batch has 860 seats and 35% are reserved for interns. How many seats?

Options: 361 | 271 | 331 | 301

Q38 ? Grammar agreement ? Medium

A measurement is 880; an audit reports 40% of it as the affected portion. Find that portion.

Options: 352 | 387 | 317 | 422

Q39 ? Vocabulary in context ? Hard

A retailer applies 45% markdown to an item priced at ■900. Find the markdown amount.

Options: 364 | 487 | 446 | 405

Q40 ? Mixed placement challenge ? Placement

A placement cohort contains 920 candidates and 10% clear the cutoff. How many clear?

Options: 110 | 83 | 101 | 92

Profit & Loss ? 40 questions

WORKED EXAMPLES

Q41 ? Direct computation ? Beginner

A reseller buys inventory for ■225 and reports the placement-day selling result. Find the answer represented by the profit model. Original assessment data: An article costs ■225 and is sold at 10% profit. Find its selling price.

Options: 297.5 | 222.5 | 272.5 | 247.5

Answer: 247.5

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 247.5. Shortcut: Multiply directly by 1.10.

Q42 ? Reverse value ? Easy

A vendor compares cost and selling price in a campus-canteen case. Using the stated percentage model, find the result. Original assessment data: An article costs ■250 and is sold at 15% profit. Find its selling price.

Options: 345.5 | 258.5 | 316.5 | 287.5

Answer: 287.5

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 287.5. Shortcut: Multiply directly by 1.15.

Q43 ? Missing base ? Medium

A marketplace seller records ■275 as cost. Determine the selling outcome under the question's profit rate. Original assessment data: An article costs ■275 and is sold at 20% profit. Find its selling price.

Options: 330 | 363 | 297 | 396

Answer: 330

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 330. Shortcut: Multiply directly by 1.20.

Q44 ? Successive change ? Hard

A shopkeeper changes the marked price and discount on a festive order. Find the resulting profit/loss quantity. Original assessment data: An article costs ■300 and is sold at 25% profit. Find its selling price.

Options: 337 | 375 | 451 | 413

Answer: 375

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 375. Shortcut: Multiply directly by 1.25.

Q45 ? Comparison trap ? Placement

A finance intern audits an invoice bought for ■325. Calculate the requested profit or loss result. Original assessment data: An article costs ■325 and is sold at 30% profit. Find its selling price.

Options: 422.5 | 464.5 | 506.5 | 380.5

Answer: 422.5

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 422.5. Shortcut: Multiply directly by 1.30.

Q46 ? Ratio form ? Beginner

A trader prepares a margin report for goods costing ■350. Find the required margin answer. Original assessment data: An article costs ■350 and is sold at 5% profit. Find its selling price.

Options: 441.5 | 404.5 | 367.5 | 330.5

Answer: 367.5

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 367.5. Shortcut: Multiply directly by 1.05.

Q47 ? Weighted result ? Easy

A store sells a batch of goods and asks for its unit-economics result. Use the given profit data. Original assessment data: An article costs ■375 and is sold at 10% profit. Find its selling price.

Options: 371.5 | 412.5 | 453.5 | 494.5

Answer: 412.5

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 412.5. Shortcut: Multiply directly by 1.10.

Q48 ? Replacement cycle ? Medium

A placement assessment gives a retail transaction with base ■400. Find the requested result. Original assessment data: An article costs ■400 and is sold at 15% profit. Find its selling price.

Options: 552 | 506 | 460 | 414

Answer: 460

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 460. Shortcut: Multiply directly by 1.15.

Q49 ? Unit conversion ? Hard

A reseller buys inventory for ■425 and reports the placement-day selling result. Find the answer represented by the profit model. Original assessment data: An article costs ■425 and is sold at 20% profit. Find its selling price.

Options: 510 | 561 | 612 | 459

Answer: 510

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 510. Shortcut: Multiply directly by 1.20.

Q50 ? Work-rate model ? Placement

A vendor compares cost and selling price in a campus-canteen case. Using the stated percentage model, find the result. Original assessment data: An article costs ■450 and is sold at 25% profit. Find its selling price.

Options: 674.5 | 618.5 | 562.5 | 506.5

Answer: 562.5

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 562.5. Shortcut: Multiply directly by 1.25.

Q51 ? Speed-time model ? Beginner

A marketplace seller records ■475 as cost. Determine the selling outcome under the question's profit rate. Original assessment data: An article costs ■475 and is sold at 30% profit. Find its selling price.

Options: 555.5 | 679.5 | 617.5 | 741.5

Answer: 617.5

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 617.5. Shortcut: Multiply directly by 1.30.

Q52 ? Table lookup ? Easy

A shopkeeper changes the marked price and discount on a festive order. Find the resulting profit/loss quantity. Original assessment data: An article costs ■500 and is sold at 5% profit. Find its selling price.

Options: 525 | 578 | 472 | 631

Answer: 525

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 525. Shortcut: Multiply directly by 1.05.

Q53 ? Chart comparison ? Medium

A finance intern audits an invoice bought for ■525. Calculate the requested profit or loss result. Original assessment data: An article costs ■525 and is sold at 10% profit. Find its selling price.

Options: 635.5 | 693.5 | 519.5 | 577.5

Answer: 577.5

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 577.5. Shortcut: Multiply directly by 1.10.

Q54 ? Multi-step word problem ? Hard

A trader prepares a margin report for goods costing ■550. Find the required margin answer. Original assessment data: An article costs ■550 and is sold at 15% profit. Find its selling price.

Options: 569.5 | 632.5 | 695.5 | 758.5

Answer: 632.5

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 632.5. Shortcut: Multiply directly by 1.15.

Q55 ? Maximum / minimum ? Placement

A store sells a batch of goods and asks for its unit-economics result. Use the given profit data. Original assessment data: An article costs ■575 and is sold at 20% profit. Find its selling price.

Options: 621 | 828 | 759 | 690

Answer: 690

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 690. Shortcut: Multiply directly by 1.20.

Q56 ? Least / greatest value ? Beginner

A placement assessment gives a retail transaction with base ■600. Find the requested result. Original assessment data: An article costs ■600 and is sold at 25% profit. Find its selling price.

Options: 900 | 750 | 825 | 675

Answer: 750

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 750. Shortcut: Multiply directly by 1.25.

Q57 ? Remainder cycle ? Easy

A reseller buys inventory for ■625 and reports the placement-day selling result. Find the answer represented by the profit model. Original assessment data: An article costs ■625 and is sold at 30% profit. Find its selling price.

Options: 812.5 | 893.5 | 731.5 | 974.5

Answer: 812.5

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 812.5. Shortcut: Multiply directly by 1.30.

Q58 ? Divisibility test ? Medium

A vendor compares cost and selling price in a campus-canteen case. Using the stated percentage model, find the result. Original assessment data: An article costs ■650 and is sold at 5% profit. Find its selling price.

Options: 614.5 | 682.5 | 750.5 | 818.5

Answer: 682.5

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 682.5. Shortcut: Multiply directly by 1.05.

Q59 ? Factorisation ? Hard

A marketplace seller records ■675 as cost. Determine the selling outcome under the question's profit rate. Original assessment data: An article costs ■675 and is sold at 10% profit. Find its selling price.

Options: 890.5 | 816.5 | 742.5 | 668.5

Answer: 742.5

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 742.5. Shortcut: Multiply directly by 1.10.

Q60 ? Approximation ? Placement

A shopkeeper changes the marked price and discount on a festive order. Find the resulting profit/loss quantity. Original assessment data: An article costs ■700 and is sold at 15% profit. Find its selling price.

Options: 805 | 886 | 724 | 967

Answer: 805

Solution: Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 805. Shortcut: Multiply directly by 1.15.

PRACTICE ? solutions hidden below**Q61 ? Options elimination ? Beginner**

A finance intern audits an invoice bought for ■725. Calculate the requested profit or loss result. Original assessment data: An article costs ■725 and is sold at

20% profit. Find its selling price.
Options: 870 | 1044 | 957 | 783

Q62 ? Boundary case ? Easy

A trader prepares a margin report for goods costing ■750. Find the required margin answer. Original assessment data: An article costs ■750 and is sold at 25% profit. Find its selling price.
Options: 1125.5 | 843.5 | 1031.5 | 937.5

Q63 ? Case split ? Medium

A store sells a batch of goods and asks for its unit-economics result. Use the given profit data. Original assessment data: An article costs ■775 and is sold at 30% profit. Find its selling price.
Options: 906.5 | 1007.5 | 1108.5 | 1209.5

Q64 ? Contradiction check ? Hard

A placement assessment gives a retail transaction with base ■800. Find the requested result. Original assessment data: An article costs ■800 and is sold at 5% profit. Find its selling price.
Options: 840 | 756 | 924 | 1008

Q65 ? Statement I test ? Placement

A reseller buys inventory for ■825 and reports the placement-day selling result. Find the answer represented by the profit model. Original assessment data: An article costs ■825 and is sold at 10% profit. Find its selling price.
Options: 907.5 | 998.5 | 1089.5 | 816.5

Q66 ? Statement II test ? Beginner

A vendor compares cost and selling price in a campus-canteen case. Using the stated percentage model, find the result. Original assessment data: An article costs ■850 and is sold at 15% profit. Find its selling price.
Options: 1173.5 | 879.5 | 1075.5 | 977.5

Q67 ? Combined statements ? Easy

A marketplace seller records ■875 as cost. Determine the selling outcome under the question's profit rate. Original assessment data: An article costs ■875 and is sold at 20% profit. Find its selling price.
Options: 1050 | 1155 | 945 | 1260

Q68 ? Counting restriction ? Medium

A shopkeeper changes the marked price and discount on a festive order. Find the resulting profit/loss quantity. Original assessment data: An article costs ■900 and is sold at 25% profit. Find its selling price.
Options: 1125 | 1238 | 1012 | 1351

Q69 ? Circular arrangement ? Hard

A finance intern audits an invoice bought for ■925. Calculate the requested profit or loss result. Original assessment data: An article costs ■925 and is sold at 30% profit. Find its selling price.
Options: 1442.5 | 1082.5 | 1322.5 | 1202.5

Q70 ? Complement method ? Placement

A trader prepares a margin report for goods costing ■950. Find the required margin answer. Original assessment data: An article costs ■950 and is sold at 5% profit. Find its selling price.
Options: 997.5 | 1097.5 | 897.5 | 1197.5

Q71 ? Pattern: differences ? Beginner

A store sells a batch of goods and asks for its unit-economics result. Use the given profit data. Original assessment data: An article costs ■975 and is sold at 10% profit. Find its selling price.
Options: 1072.5 | 1179.5 | 965.5 | 1286.5

Q72 ? Pattern: ratios ? Easy

A placement assessment gives a retail transaction with base ■1000. Find the requested result. Original assessment data: An article costs ■1000 and is sold at 15% profit. Find its selling price.
Options: 1380 | 1035 | 1265 | 1150

Q73 ? Pattern: alternating ? Medium

A reseller buys inventory for ■1025 and reports the placement-day selling result. Find the answer represented by the profit model. Original assessment data: An article costs ■1025 and is sold at 20% profit. Find its selling price.
Options: 1476 | 1107 | 1353 | 1230

Q74 ? Pattern: squares / cubes ? Hard

A vendor compares cost and selling price in a campus-canteen case. Using the stated percentage model, find the result. Original assessment data: An article costs ■1050 and is sold at 25% profit. Find its selling price.
Options: 1312.5 | 1443.5 | 1181.5 | 1574.5

Q75 ? Inference from facts ? Placement

A marketplace seller records ■1075 as cost. Determine the selling outcome under the question's profit rate. Original assessment data: An article costs ■1075 and is sold at 30% profit. Find its selling price.
Options: 1397.5 | 1537.5 | 1257.5 | 1677.5

Q76 ? Assumption check ? Beginner

A shopkeeper changes the marked price and discount on a festive order. Find the resulting profit/loss quantity. Original assessment data: An article costs ■1100 and is sold at 5% profit. Find its selling price.
Options: 1387 | 1039 | 1271 | 1155

Q77 ? Cause vs correlation ? Easy

A finance intern audits an invoice bought for ■1125. Calculate the requested profit or loss result. Original assessment data: An article costs ■1125 and is sold at 10% profit. Find its selling price.

Options: 1237.5 | 1361.5 | 1113.5 | 1485.5

Q78 ? Grammar agreement ? Medium

A trader prepares a margin report for goods costing ■1150. Find the required margin answer. Original assessment data: An article costs ■1150 and is sold at 15% profit. Find its selling price.

Options: 1322.5 | 1454.5 | 1190.5 | 1586.5

Q79 ? Vocabulary in context ? Hard

A store sells a batch of goods and asks for its unit-economics result. Use the given profit data. Original assessment data: An article costs ■1175 and is sold at 20% profit. Find its selling price.

Options: 1692 | 1269 | 1551 | 1410

Q80 ? Mixed placement challenge ? Placement

A placement assessment gives a retail transaction with base ■1200. Find the requested result. Original assessment data: An article costs ■1200 and is sold at 25% profit. Find its selling price.

Options: 1800 | 1350 | 1650 | 1500

Simple Interest ? 40 questions

WORKED EXAMPLES

Q81 ? Direct computation ? Beginner

A loan repayment case uses the quantities in this question. Determine the requested result using the direct computation model: Find the simple interest on ■1250 at 5% per annum for 2 years.

Options: 151 | 112 | 138 | 125

Answer: 125

Solution: Context: loan repayment. $SI = PRT/100 = 1250 \times 5 \times 2/100 = \blacksquare 125$. Shortcut: Cancel factors with 100 before multiplying.

Q82 ? Reverse value ? Easy

A fixed deposit case uses the quantities in this question. Determine the requested result using the reverse value model: Find the simple interest on ■1500 at 6% per annum for 3 years.

Options: 324 | 243 | 297 | 270

Answer: 270

Solution: Context: fixed deposit. $SI = PRT/100 = 1500 \times 6 \times 3/100 = \blacksquare 270$. Shortcut: Cancel factors with 100 before multiplying.

Q83 ? Missing base ? Medium

A education loan case uses the quantities in this question. Determine the requested result using the missing base model: Find the simple interest on ■1750 at 7% per annum for 4 years.

Options: 490 | 539 | 441 | 588

Answer: 490

Solution: Context: education loan. $SI = PRT/100 = 1750 \times 7 \times 4/100 = \blacksquare 490$. Shortcut: Cancel factors with 100 before multiplying.

Q84 ? Successive change ? Hard

A working-capital advance case uses the quantities in this question. Determine the requested result using the successive change model: Find the simple interest on ■2000 at 8% per annum for 5 years.

Options: 720 | 880 | 800 | 960

Answer: 800

Solution: Context: working-capital advance. $SI = PRT/100 = 2000 \times 8 \times 5/100 = \blacksquare 800$. Shortcut: Cancel factors with 100 before multiplying.

Q85 ? Comparison trap ? Placement

A savings account case uses the quantities in this question. Determine the requested result using the comparison trap model: Find the simple interest on ■2250 at 9% per annum for 1 years.

Options: 242.5 | 182.5 | 222.5 | 202.5

Answer: 202.5

Solution: Context: savings account. $SI = PRT/100 = 2250 \times 9 \times 1/100 = \blacksquare 202.5$. Shortcut: Cancel factors with 100 before multiplying.

Q86 ? Ratio form ? Beginner

A equipment finance case uses the quantities in this question. Determine the requested result using the ratio form model: Find the simple interest on ■2500 at 10% per annum for 2 years.

Options: 600 | 450 | 550 | 500

Answer: 500

Solution: Context: equipment finance. $SI = PRT/100 = 2500 \times 10 \times 2/100 = \blacksquare 500$. Shortcut: Cancel factors with 100 before multiplying.

Q87 ? Weighted result ? Easy

A invoice delay case uses the quantities in this question. Determine the requested result using the weighted result model: Find the simple interest on ■2750 at 11% per annum for 3 years.

Options: 816.5 | 1089.5 | 907.5 | 998.5

Answer: 907.5

Solution: Context: invoice delay. $SI = PRT/100 = 2750 \times 11 \times 3/100 = \blacksquare 907.5$. Shortcut: Cancel factors with 100 before multiplying.

Q88 ? Replacement cycle ? Medium

A personal loan case uses the quantities in this question. Determine the requested result using the replacement cycle model: Find the simple interest on ■3000 at 12% per annum for 4 years.

Options: 1728 | 1296 | 1584 | 1440

Answer: 1440

Solution: Context: personal loan. $SI = PRT/100 = 3000 \times 12 \times 4/100 = \blacksquare 1440$. Shortcut: Cancel factors with 100 before multiplying.

Q89 ? Unit conversion ? Hard

A loan repayment case uses the quantities in this question. Determine the requested result using the unit conversion model: Find the simple interest on ■3250 at 4% per annum for 5 years.

Options: 650 | 715 | 585 | 780

Answer: 650

Solution: Context: loan repayment. $SI = PRT/100 = 3250 \times 4 \times 5/100 = \blacksquare 650$. Shortcut: Cancel factors with 100 before multiplying.

Q90 ? Work-rate model ? Placement

A fixed deposit case uses the quantities in this question. Determine the requested result using the work-rate model model: Find the simple interest on ■3500 at 5% per annum for 1 years.

Options: 175 | 193 | 157 | 211

Answer: 175

Solution: Context: fixed deposit. $SI = PRT/100 = 3500 \times 5 \times 1/100 = \blacksquare 175$. Shortcut: Cancel factors with 100 before multiplying.

Q91 ? Speed-time model ? Beginner

A education loan case uses the quantities in this question. Determine the requested result using the speed-time model model: Find the simple interest on ■3750 at 6% per annum for 2 years.

Options: 405 | 495 | 450 | 540

Answer: 450

Solution: Context: education loan. $SI = PRT/100 = 3750 \times 6 \times 2/100 = \blacksquare 450$. Shortcut: Cancel factors with 100 before multiplying.

Q92 ? Table lookup ? Easy

A working-capital advance case uses the quantities in this question. Determine the requested result using the table lookup model: Find the simple interest on ■4000 at 7% per annum for 3 years.

Options: 1008 | 756 | 924 | 840

Answer: 840

Solution: Context: working-capital advance. $SI = PRT/100 = 4000 \times 7 \times 3/100 = \blacksquare 840$. Shortcut: Cancel factors with 100 before multiplying.

Q93 ? Chart comparison ? Medium

A savings account case uses the quantities in this question. Determine the requested result using the chart comparison model: Find the simple interest on ■4250 at 8% per annum for 4 years.

Options: 1360 | 1496 | 1224 | 1632

Answer: 1360

Solution: Context: savings account. $SI = PRT/100 = 4250 \times 8 \times 4/100 = \blacksquare 1360$. Shortcut: Cancel factors with 100 before multiplying.

Q94 ? Multi-step word problem ? Hard

A equipment finance case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Find the simple interest on ■4500 at 9% per annum for 5 years.

Options: 1822 | 2025 | 2228 | 2431

Answer: 2025

Solution: Context: equipment finance. $SI = PRT/100 = 4500 \times 9 \times 5/100 = \blacksquare 2025$. Shortcut: Cancel factors with 100 before multiplying.

Q95 ? Maximum / minimum ? Placement

A invoice delay case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Find the simple interest on ■4750 at 10% per annum for 1 years.

Options: 571 | 523 | 475 | 427

Answer: 475

Solution: Context: invoice delay. $SI = PRT/100 = 4750 \times 10 \times 1/100 = \blacksquare 475$. Shortcut: Cancel factors with 100 before multiplying.

Q96 ? Least / greatest value ? Beginner

A personal loan case uses the quantities in this question. Determine the requested result using the least / greatest value model: Find the simple interest on ■5000 at 11% per annum for 2 years.

Options: 990 | 1100 | 1210 | 1320

Answer: 1100

Solution: Context: personal loan. $SI = PRT/100 = 5000 \times 11 \times 2/100 = \blacksquare 1100$. Shortcut: Cancel factors with 100 before multiplying.

Q97 ? Remainder cycle ? Easy

A loan repayment case uses the quantities in this question. Determine the requested result using the remainder cycle model: Find the simple interest on ■5250 at 12% per annum for 3 years.

Options: 1890 | 2079 | 1701 | 2268

Answer: 1890

Solution: Context: loan repayment. $SI = PRT/100 = 5250 \times 12 \times 3/100 = \blacksquare 1890$. Shortcut: Cancel factors with 100 before multiplying.

Q98 ? Divisibility test ? Medium

A fixed deposit case uses the quantities in this question. Determine the requested result using the divisibility test model: Find the simple interest on ■5500 at 4% per annum for 4 years.

Options: 880 | 968 | 792 | 1056

Answer: 880

Solution: Context: fixed deposit. $SI = PRT/100 = 5500 \times 4 \times 4/100 = \blacksquare 880$. Shortcut: Cancel factors with 100 before multiplying.

Q99 ? Factorisation ? Hard

A education loan case uses the quantities in this question. Determine the requested result using the factorisation model: Find the simple interest on ■5750 at 5% per annum for 5 years.

Options: 1725.5 | 1293.5 | 1581.5 | 1437.5

Answer: 1437.5

Solution: Context: education loan. $SI = PRT/100 = 5750 \times 5 \times 5/100 = \blacksquare 1437.5$. Shortcut: Cancel factors with 100 before multiplying.

Q100 ? Approximation ? Placement

A working-capital advance case uses the quantities in this question. Determine the requested result using the approximation model: Find the simple interest on ■6000 at 6% per annum for 1 years.

Options: 432 | 360 | 396 | 324

Answer: 360

Solution: Context: working-capital advance. $SI = PRT/100 = 6000 \times 6 \times 1/100 = \blacksquare 360$. Shortcut: Cancel factors with 100 before multiplying.

PRACTICE ? solutions hidden below**Q101 ? Options elimination ? Beginner**

A savings account case uses the quantities in this question. Determine the requested result using the options elimination model: Find the simple interest on

■6250 at 7% per annum for 2 years.
Options: 875 | 1051 | 963 | 787

Q102 ? Boundary case ? Easy

A equipment finance case uses the quantities in this question. Determine the requested result using the boundary case model: Find the simple interest on

■6500 at 8% per annum for 3 years.
Options: 1872 | 1404 | 1716 | 1560

Q103 ? Case split ? Medium

A invoice delay case uses the quantities in this question. Determine the requested result using the case split model: Find the simple interest on ■6750 at 9% per annum for 4 years.

Options: 2916 | 2187 | 2673 | 2430

Q104 ? Contradiction check ? Hard

A personal loan case uses the quantities in this question. Determine the requested result using the contradiction check model: Find the simple interest on

■7000 at 10% per annum for 5 years.
Options: 3500 | 3850 | 3150 | 4200

Q105 ? Statement I test ? Placement

A loan repayment case uses the quantities in this question. Determine the requested result using the statement i test model: Find the simple interest on ■7250 at 11% per annum for 1 years.

Options: 717.5 | 797.5 | 877.5 | 957.5

Q106 ? Statement II test ? Beginner

A fixed deposit case uses the quantities in this question. Determine the requested result using the statement ii test model: Find the simple interest on ■7500 at 12% per annum for 2 years.

Options: 2160 | 1620 | 1980 | 1800

Q107 ? Combined statements ? Easy

A education loan case uses the quantities in this question. Determine the requested result using the combined statements model: Find the simple interest on

■7750 at 4% per annum for 3 years.
Options: 930 | 1023 | 837 | 1116

Q108 ? Counting restriction ? Medium

A working-capital advance case uses the quantities in this question. Determine the requested result using the counting restriction model: Find the simple interest on ■8000 at 5% per annum for 4 years.

Options: 1600 | 1760 | 1440 | 1920

Q109 ? Circular arrangement ? Hard

A savings account case uses the quantities in this question. Determine the requested result using the circular arrangement model: Find the simple interest on

■8250 at 6% per annum for 5 years.
Options: 2971 | 2227 | 2723 | 2475

Q110 ? Complement method ? Placement

A equipment finance case uses the quantities in this question. Determine the requested result using the complement method model: Find the simple interest on ■8500 at 7% per annum for 1 years.

Options: 595 | 655 | 535 | 715

Q111 ? Pattern: differences ? Beginner

A invoice delay case uses the quantities in this question. Determine the requested result using the pattern: differences model: Find the simple interest on

■8750 at 8% per annum for 2 years.
Options: 1400 | 1540 | 1260 | 1680

Q112 ? Pattern: ratios ? Easy

A personal loan case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Find the simple interest on ■9000 at 9% per annum for 3 years.

Options: 2430 | 2673 | 2187 | 2916

Q113 ? Pattern: alternating ? Medium

A loan repayment case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Find the simple interest on

■9250 at 10% per annum for 4 years.
Options: 4070 | 4440 | 3330 | 3700

Q114 ? Pattern: squares / cubes ? Hard

A fixed deposit case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Find the simple interest on

■9500 at 11% per annum for 5 years.
Options: 6271 | 4702 | 5748 | 5225

Q115 ? Inference from facts ? Placement

A education loan case uses the quantities in this question. Determine the requested result using the inference from facts model: Find the simple interest on

■9750 at 12% per annum for 1 years.
Options: 1170 | 1287 | 1053 | 1404

Q116 ? Assumption check ? Beginner

A working-capital advance case uses the quantities in this question. Determine the requested result using the assumption check model: Find the simple interest on ■10000 at 4% per annum for 2 years.

Options: 720 | 960 | 880 | 800

Q117 ? Cause vs correlation ? Easy

A savings account case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Find the simple interest on

■10250 at 5% per annum for 3 years.

Options: 1537.5 | 1691.5 | 1383.5 | 1845.5

Q118 ? Grammar agreement ? Medium

A equipment finance case uses the quantities in this question. Determine the requested result using the grammar agreement model: Find the simple interest on ■10500 at 6% per annum for 4 years.

Options: 2520 | 2772 | 2268 | 3024

Q119 ? Vocabulary in context ? Hard

A invoice delay case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Find the simple interest on ■10750 at 7% per annum for 5 years.

Options: 3386.5 | 3762.5 | 4514.5 | 4138.5

Q120 ? Mixed placement challenge ? Placement

A personal loan case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Find the simple interest on ■11000 at 8% per annum for 1 years.

Options: 1056 | 792 | 968 | 880

Compound Interest ? 40 questions

WORKED EXAMPLES

Q121 ? Direct computation ? Beginner

A investment ladder case uses the quantities in this question. Determine the requested result using the direct computation model: Find the compound interest on ■1500 at 10% annually for 2 years.

Options: 315 | 347 | 315.0000000000003 | 283

Answer: 315.0000000000003

Solution: Context: investment ladder. $CI = 1500[(1+10/100)^2-1] = \blacksquare 315.0000000000003$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q122 ? Reverse value ? Easy

A inflation forecast case uses the quantities in this question. Determine the requested result using the reverse value model: Find the compound interest on ■2000 at 20% annually for 2 years.

Options: 968 | 880 | 792 | 879.9999999999999

Answer: 879.9999999999999

Solution: Context: inflation forecast. $CI = 2000[(1+20/100)^2-1] = \blacksquare 879.9999999999999$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q123 ? Missing base ? Medium

A loan balance case uses the quantities in this question. Determine the requested result using the missing base model: Find the compound interest on ■2500 at 5% annually for 2 years.

Options: 256.25 | 282.25 | 230.25 | 256.2500000000001

Answer: 256.2500000000001

Solution: Context: loan balance. $CI = 2500[(1+5/100)^2-1] = \blacksquare 256.2500000000001$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q124 ? Successive change ? Hard

A annual deposit case uses the quantities in this question. Determine the requested result using the successive change model: Find the compound interest on ■3000 at 10% annually for 2 years.

Options: 567 | 630.0000000000006 | 693 | 630

Answer: 630.0000000000006

Solution: Context: annual deposit. $CI = 3000[(1+10/100)^2-1] = \blacksquare 630.0000000000006$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q125 ? Comparison trap ? Placement

A fund NAV case uses the quantities in this question. Determine the requested result using the comparison trap model: Find the compound interest on ■3500 at 20% annually for 2 years.

Options: 1539.9999999999998 | 1386 | 1694 | 1540

Answer: 1539.9999999999998

Solution: Context: fund NAV. $CI = 3500[(1+20/100)^2-1] = \blacksquare 1539.9999999999998$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q126 ? Ratio form ? Beginner

A equipment lease case uses the quantities in this question. Determine the requested result using the ratio form model: Find the compound interest on ■4000 at 5% annually for 2 years.

Options: 410.0000000000001 | 451 | 410 | 369

Answer: 410.0000000000001

Solution: Context: equipment lease. $CI = 4000[(1+5/100)^2-1] = \blacksquare 410.0000000000001$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q127 ? Weighted result ? Easy

A savings target case uses the quantities in this question. Determine the requested result using the weighted result model: Find the compound interest on ■4500 at 10% annually for 2 years.

Options: 1040 | 850 | 945 | 945.0000000000008

Answer: 945.0000000000008

Solution: Context: savings target. $CI = 4500[(1+10/100)^2-1] = \blacksquare 945.0000000000008$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q128 ? Replacement cycle ? Medium

A rate comparison case uses the quantities in this question. Determine the requested result using the replacement cycle model: Find the compound interest on ■5000 at 20% annually for 2 years.

Options: 2199.9999999999995 | 1980 | 2420 | 2200

Answer: 2199.9999999999995

Solution: Context: rate comparison. $CI = 5000[(1+20/100)^2-1] = \blacksquare 2199.9999999999995$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q129 ? Unit conversion ? Hard

A investment ladder case uses the quantities in this question. Determine the requested result using the unit conversion model: Find the compound interest on ■5500 at 5% annually for 2 years.

Options: 563.75 | 619.75 | 507.75 | 563.7500000000002

Answer: 563.7500000000002

Solution: Context: investment ladder. $CI = 5500[(1+5/100)^2-1] = \blacksquare 563.7500000000002$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q130 ? Work-rate model ? Placement

A inflation forecast case uses the quantities in this question. Determine the requested result using the work-rate model model: Find the compound interest on ■6000 at 10% annually for 2 years.

Options: 1260 | 1386 | 1134 | 1260.0000000000011

Answer: 1260.0000000000011

Solution: Context: inflation forecast. $CI = 6000[(1+10/100)^2-1] = \blacksquare 1260.0000000000011$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q131 ? Speed-time model ? Beginner

A loan balance case uses the quantities in this question. Determine the requested result using the speed-time model model: Find the compound interest on ■6500 at 20% annually for 2 years.

Options: 2860 | 3146 | 2574 | 2859.9999999999995

Answer: 2859.9999999999995

Solution: Context: loan balance. $CI = 6500[(1+20/100)^2-1] = \blacksquare 2859.9999999999995$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q132 ? Table lookup ? Easy

A annual deposit case uses the quantities in this question. Determine the requested result using the table lookup model: Find the compound interest on ■7000 at 5% annually for 2 years.

Options: 717.5000000000002 | 789.5 | 717.5 | 645.5

Answer: 717.5000000000002

Solution: Context: annual deposit. $CI = 7000[(1+5/100)^2-1] = \blacksquare 717.5000000000002$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q133 ? Chart comparison ? Medium

A fund NAV case uses the quantities in this question. Determine the requested result using the chart comparison model: Find the compound interest on ■7500 at 10% annually for 2 years.

Options: 1575 | 1733 | 1417 | 1575.0000000000014

Answer: 1575.0000000000014

Solution: Context: fund NAV. $CI = 7500[(1+10/100)^2-1] = \blacksquare 1575.0000000000014$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q134 ? Multi-step word problem ? Hard

A equipment lease case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Find the compound interest on ■8000 at 20% annually for 2 years.

Options: 3520 | 3872 | 3168 | 3519.9999999999995

Answer: 3519.9999999999995

Solution: Context: equipment lease. $CI = 8000[(1+20/100)^2-1] = \blacksquare 3519.9999999999995$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q135 ? Maximum / minimum ? Placement

A savings target case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Find the compound interest on ■8500 at 5% annually for 2 years.

Options: 871.2500000000003 | 784.25 | 958.25 | 871.25

Answer: 871.2500000000003

Solution: Context: savings target. $CI = 8500[(1+5/100)^2-1] = \blacksquare 871.2500000000003$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q136 ? Least / greatest value ? Beginner

A rate comparison case uses the quantities in this question. Determine the requested result using the least / greatest value model: Find the compound interest on ■9000 at 10% annually for 2 years.

Options: 2079 | 1890 | 1701 | 1890.0000000000016

Answer: 1890.0000000000016

Solution: Context: rate comparison. $CI = 9000[(1+10/100)^2-1] = \blacksquare 1890.0000000000016$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q137 ? Remainder cycle ? Easy

A investment ladder case uses the quantities in this question. Determine the requested result using the remainder cycle model: Find the compound interest on ■9500 at 20% annually for 2 years.

Options: 4180 | 4598 | 3762 | 4179.999999999999

Answer: 4179.999999999999

Solution: Context: investment ladder. $CI = 9500[(1+20/100)^2-1] = \blacksquare 4179.999999999999$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q138 ? Divisibility test ? Medium

A inflation forecast case uses the quantities in this question. Determine the requested result using the divisibility test model: Find the compound interest on ■10000 at 5% annually for 2 years.

Options: 1025.0000000000005 | 1128 | 1025 | 922

Answer: 1025.0000000000005

Solution: Context: inflation forecast. $CI = 10000[(1+5/100)^2-1] = \blacksquare 1025.0000000000005$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q139 ? Factorisation ? Hard

A loan balance case uses the quantities in this question. Determine the requested result using the factorisation model: Find the compound interest on ■10500 at 10% annually for 2 years.

Options: 2205.0000000000002 | 1984 | 2426 | 2205

Answer: 2205.0000000000002

Solution: Context: loan balance. $CI = 10500[(1+10/100)^2-1] = \blacksquare 2205.0000000000002$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

Q140 ? Approximation ? Placement

A annual deposit case uses the quantities in this question. Determine the requested result using the approximation model: Find the compound interest on ■11000 at 20% annually for 2 years.

Options: 4839.999999999999 | 4356 | 5324 | 4840

Answer: 4839.999999999999

Solution: Context: annual deposit. $CI = 11000[(1+20/100)^2-1] = \blacksquare 4839.999999999999$. Shortcut: For two years, $CI = SI + P(R/100)^2$.

PRACTICE ? solutions hidden below**Q141 ? Options elimination ? Beginner**

A fund NAV case uses the quantities in this question. Determine the requested result using the options elimination model: Find the compound interest on

■11500 at 5% annually for 2 years.
Options: 1178.75 | 1296.75 | 1060.75 | 1178.7500000000005

Q142 ? Boundary case ? Easy

A equipment lease case uses the quantities in this question. Determine the requested result using the boundary case model: Find the compound interest on ■12000 at 10% annually for 2 years.
Options: 2520.0000000000023 | 2268 | 2772 | 2520

Q143 ? Case split ? Medium

A savings target case uses the quantities in this question. Determine the requested result using the case split model: Find the compound interest on ■12500 at 20% annually for 2 years.
Options: 5500 | 6050 | 5499.999999999999 | 4950

Q144 ? Contradiction check ? Hard

A rate comparison case uses the quantities in this question. Determine the requested result using the contradiction check model: Find the compound interest on ■13000 at 5% annually for 2 years.
Options: 1332.5 | 1465.5 | 1199.5 | 1332.5000000000005

Q145 ? Statement I test ? Placement

A investment ladder case uses the quantities in this question. Determine the requested result using the statement i test model: Find the compound interest on ■13500 at 10% annually for 2 years.
Options: 2835 | 3119 | 2551 | 2835.0000000000027

Q146 ? Statement II test ? Beginner

A inflation forecast case uses the quantities in this question. Determine the requested result using the statement ii test model: Find the compound interest on ■14000 at 20% annually for 2 years.
Options: 6160 | 6776 | 5544 | 6159.999999999999

Q147 ? Combined statements ? Easy

A loan balance case uses the quantities in this question. Determine the requested result using the combined statements model: Find the compound interest on ■14500 at 5% annually for 2 years.
Options: 1486.25 | 1635.25 | 1337.25 | 1486.2500000000005

Q148 ? Counting restriction ? Medium

A annual deposit case uses the quantities in this question. Determine the requested result using the counting restriction model: Find the compound interest on ■15000 at 10% annually for 2 years.
Options: 2835 | 3150 | 3465 | 3150.0000000000027

Q149 ? Circular arrangement ? Hard

A fund NAV case uses the quantities in this question. Determine the requested result using the circular arrangement model: Find the compound interest on ■15500 at 20% annually for 2 years.
Options: 6819.999999999999 | 6138 | 7502 | 6820

Q150 ? Complement method ? Placement

A equipment lease case uses the quantities in this question. Determine the requested result using the complement method model: Find the compound interest on ■16000 at 5% annually for 2 years.
Options: 1640 | 1804 | 1476 | 1640.0000000000005

Q151 ? Pattern: differences ? Beginner

A savings target case uses the quantities in this question. Determine the requested result using the pattern: differences model: Find the compound interest on ■16500 at 10% annually for 2 years.
Options: 3465 | 3812 | 3118 | 3465.000000000003

Q152 ? Pattern: ratios ? Easy

A rate comparison case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Find the compound interest on ■17000 at 20% annually for 2 years.
Options: 6732 | 7479.999999999999 | 7480 | 8228

Q153 ? Pattern: alternating ? Medium

A investment ladder case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Find the compound interest on ■17500 at 5% annually for 2 years.
Options: 1793.7500000000007 | 1614.75 | 1972.75 | 1793.75

Q154 ? Pattern: squares / cubes ? Hard

A inflation forecast case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Find the compound interest on ■18000 at 10% annually for 2 years.
Options: 4158 | 3780 | 3402 | 3780.000000000003

Q155 ? Inference from facts ? Placement

A loan balance case uses the quantities in this question. Determine the requested result using the inference from facts model: Find the compound interest on ■18500 at 20% annually for 2 years.
Options: 8140 | 8954 | 8139.999999999999 | 7326

Q156 ? Assumption check ? Beginner

A annual deposit case uses the quantities in this question. Determine the requested result using the assumption check model: Find the compound interest on ■19000 at 5% annually for 2 years.
Options: 1947.5000000000007 | 1752.5 | 2142.5 | 1947.5

Q157 ? Cause vs correlation ? Easy

A fund NAV case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Find the compound interest on ■19500 at 10% annually for 2 years.

Options: 4095.0000000000036 | 3685 | 4505 | 4095

Q158 ? Grammar agreement ? Medium

A equipment lease case uses the quantities in this question. Determine the requested result using the grammar agreement model: Find the compound interest on ■20000 at 20% annually for 2 years.

Options: 8800 | 7920 | 9680 | 8799.999999999998

Q159 ? Vocabulary in context ? Hard

A savings target case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Find the compound interest on ■20500 at 5% annually for 2 years.

Options: 2101.25 | 2311.25 | 1891.25 | 2101.250000000001

Q160 ? Mixed placement challenge ? Placement

A rate comparison case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Find the compound interest on ■21000 at 10% annually for 2 years.

Options: 4410 | 4851 | 4410.000000000004 | 3969

Ratio & Proportion ? 40 questions

WORKED EXAMPLES

Q161 ? Direct computation ? Beginner

A team allocation case uses the quantities in this question. Determine the requested result using the direct computation model: ■168 is divided in the ratio 3:5. Find the first share.

Options: 57 | 75 | 69 | 63

Answer: 63

Solution: Context: team allocation. Total parts = 8; first share = $168 \times 3/8 = \blacksquare 63$. Shortcut: Find one part, then multiply.

Q162 ? Reverse value ? Easy

A recipe scaling case uses the quantities in this question. Determine the requested result using the reverse value model: ■242 is divided in the ratio 4:7. Find the first share.

Options: 97 | 88 | 79 | 106

Answer: 88

Solution: Context: recipe scaling. Total parts = 11; first share = $242 \times 4/11 = \blacksquare 88$. Shortcut: Find one part, then multiply.

Q163 ? Missing base ? Medium

A project staffing case uses the quantities in this question. Determine the requested result using the missing base model: ■184 is divided in the ratio 5:3. Find the first share.

Options: 115 | 127 | 103 | 139

Answer: 115

Solution: Context: project staffing. Total parts = 8; first share = $184 \times 5/8 = \blacksquare 115$. Shortcut: Find one part, then multiply.

Q164 ? Successive change ? Hard

A screen resolution case uses the quantities in this question. Determine the requested result using the successive change model: ■264 is divided in the ratio 6:5. Find the first share.

Options: 172 | 130 | 158 | 144

Answer: 144

Solution: Context: screen resolution. Total parts = 11; first share = $264 \times 6/11 = \blacksquare 144$. Shortcut: Find one part, then multiply.

Q165 ? Comparison trap ? Placement

A map scale case uses the quantities in this question. Determine the requested result using the comparison trap model: ■225 is divided in the ratio 2:7. Find the first share.

Options: 50 | 55 | 45 | 60

Answer: 50

Solution: Context: map scale. Total parts = 9; first share = $225 \times 2/9 = \blacksquare 50$. Shortcut: Find one part, then multiply.

Q166 ? Ratio form ? Beginner

A fleet split case uses the quantities in this question. Determine the requested result using the ratio form model: ■156 is divided in the ratio 3:3. Find the first share.

Options: 78 | 86 | 70 | 94

Answer: 78

Solution: Context: fleet split. Total parts = 6; first share = $156 \times 3/6 = \blacksquare 78$. Shortcut: Find one part, then multiply.

Q167 ? Weighted result ? Easy

A budget share case uses the quantities in this question. Determine the requested result using the weighted result model: ■243 is divided in the ratio 4:5. Find the first share.

Options: 108 | 119 | 130 | 97

Answer: 108

Solution: Context: budget share. Total parts = 9; first share = $243 \times 4/9 = \blacksquare 108$. Shortcut: Find one part, then multiply.

Q168 ? Replacement cycle ? Medium

A mixture ratio case uses the quantities in this question. Determine the requested result using the replacement cycle model: ■336 is divided in the ratio 5:7. Find the first share.

Options: 168 | 126 | 154 | 140

Answer: 140

Solution: Context: mixture ratio. Total parts = 12; first share = $336 \times 5/12 = \blacksquare 140$. Shortcut: Find one part, then multiply.

Q169 ? Unit conversion ? Hard

A team allocation case uses the quantities in this question. Determine the requested result using the unit conversion model: ■261 is divided in the ratio 6:3. Find the first share.

Options: 208 | 174 | 191 | 157

Answer: 174

Solution: Context: team allocation. Total parts = 9; first share = $261 \times 6/9 = \blacksquare 174$. Shortcut: Find one part, then multiply.

Q170 ? Work-rate model ? Placement

A recipe scaling case uses the quantities in this question. Determine the requested result using the work-rate model model: ■210 is divided in the ratio 2:5. Find the first share.

Options: 72 | 66 | 60 | 54

Answer: 60

Solution: Context: recipe scaling. Total parts = 7; first share = $210 \times 2/7 = \blacksquare 60$. Shortcut: Find one part, then multiply.

Q171 ? Speed-time model ? Beginner

A project staffing case uses the quantities in this question. Determine the requested result using the speed-time model model: ■310 is divided in the ratio 3:7. Find the first share.

Options: 93 | 111 | 84 | 102

Answer: 93

Solution: Context: project staffing. Total parts = 10; first share = $310 \times 3/10 = \blacksquare 93$. Shortcut: Find one part, then multiply.

Q172 ? Table lookup ? Easy

A screen resolution case uses the quantities in this question. Determine the requested result using the table lookup model: ■224 is divided in the ratio 4:3. Find the first share.

Options: 154 | 115 | 141 | 128

Answer: 128

Solution: Context: screen resolution. Total parts = 7; first share = $224 \times 4/7 = \blacksquare 128$. Shortcut: Find one part, then multiply.

Q173 ? Chart comparison ? Medium

A map scale case uses the quantities in this question. Determine the requested result using the chart comparison model: ■330 is divided in the ratio 5:5. Find the first share.

Options: 165 | 182 | 148 | 199

Answer: 165

Solution: Context: map scale. Total parts = 10; first share = $330 \times 5/10 = \blacksquare 165$. Shortcut: Find one part, then multiply.

Q174 ? Multi-step word problem ? Hard

A fleet split case uses the quantities in this question. Determine the requested result using the multi-step word problem model: ■442 is divided in the ratio 6:7. Find the first share.

Options: 184 | 204 | 224 | 244

Answer: 204

Solution: Context: fleet split. Total parts = 13; first share = $442 \times 6/13 = \blacksquare 204$. Shortcut: Find one part, then multiply.

Q175 ? Maximum / minimum ? Placement

A budget share case uses the quantities in this question. Determine the requested result using the maximum / minimum model: ■175 is divided in the ratio 2:3. Find the first share.

Options: 63 | 84 | 77 | 70

Answer: 70

Solution: Context: budget share. Total parts = 5; first share = $175 \times 2/5 = \blacksquare 70$. Shortcut: Find one part, then multiply.

Q176 ? Least / greatest value ? Beginner

A mixture ratio case uses the quantities in this question. Determine the requested result using the least / greatest value model: ■288 is divided in the ratio 3:5. Find the first share.

Options: 97 | 108 | 119 | 130

Answer: 108

Solution: Context: mixture ratio. Total parts = 8; first share = $288 \times 3/8 = \blacksquare 108$. Shortcut: Find one part, then multiply.

Q177 ? Remainder cycle ? Easy

A team allocation case uses the quantities in this question. Determine the requested result using the remainder cycle model: ■407 is divided in the ratio 4:7. Find the first share.

Options: 148 | 163 | 133 | 178

Answer: 148

Solution: Context: team allocation. Total parts = 11; first share = $407 \times 4/11 = \blacksquare 148$. Shortcut: Find one part, then multiply.

Q178 ? Divisibility test ? Medium

A recipe scaling case uses the quantities in this question. Determine the requested result using the divisibility test model: ■304 is divided in the ratio 5:3. Find the first share.

Options: 190 | 209 | 171 | 228

Answer: 190

Solution: Context: recipe scaling. Total parts = 8; first share = $304 \times 5/8 = \blacksquare 190$. Shortcut: Find one part, then multiply.

Q179 ? Factorisation ? Hard

A project staffing case uses the quantities in this question. Determine the requested result using the factorisation model: ■429 is divided in the ratio 6:5. Find the first share.

Options: 280 | 211 | 257 | 234

Answer: 234

Solution: Context: project staffing. Total parts = 11; first share = $429 \times 6/11 = \blacksquare 234$. Shortcut: Find one part, then multiply.

Q180 ? Approximation ? Placement

A screen resolution case uses the quantities in this question. Determine the requested result using the approximation model: ■360 is divided in the ratio 2:7. Find the first share.

Options: 80 | 88 | 72 | 96

Answer: 80

Solution: Context: screen resolution. Total parts = 9; first share = $360 \times 2/9 = \blacksquare 80$. Shortcut: Find one part, then multiply.

PRACTICE ? solutions hidden below**Q181 ? Options elimination ? Beginner**

A map scale case uses the quantities in this question. Determine the requested result using the options elimination model: ■246 is divided in the ratio 3:3. Find

the first share.

Options: 123 | 135 | 111 | 147

Q182 ? Boundary case ? Easy

A fleet split case uses the quantities in this question. Determine the requested result using the boundary case model: ■378 is divided in the ratio 4:5. Find the first share.

Options: 202 | 151 | 185 | 168

Q183 ? Case split ? Medium

A budget share case uses the quantities in this question. Determine the requested result using the case split model: ■516 is divided in the ratio 5:7. Find the first share.

Options: 259 | 237 | 215 | 193

Q184 ? Contradiction check ? Hard

A mixture ratio case uses the quantities in this question. Determine the requested result using the contradiction check model: ■396 is divided in the ratio 6:3. Find the first share.

Options: 264 | 290 | 238 | 316

Q185 ? Statement I test ? Placement

A team allocation case uses the quantities in this question. Determine the requested result using the statement i test model: ■315 is divided in the ratio 2:5. Find the first share.

Options: 90 | 99 | 108 | 81

Q186 ? Statement II test ? Beginner

A recipe scaling case uses the quantities in this question. Determine the requested result using the statement ii test model: ■460 is divided in the ratio 3:7. Find the first share.

Options: 166 | 124 | 152 | 138

Q187 ? Combined statements ? Easy

A project staffing case uses the quantities in this question. Determine the requested result using the combined statements model: ■329 is divided in the ratio 4:3. Find the first share.

Options: 188 | 207 | 169 | 226

Q188 ? Counting restriction ? Medium

A screen resolution case uses the quantities in this question. Determine the requested result using the counting restriction model: ■480 is divided in the ratio 5:5. Find the first share.

Options: 240 | 264 | 216 | 288

Q189 ? Circular arrangement ? Hard

A map scale case uses the quantities in this question. Determine the requested result using the circular arrangement model: ■637 is divided in the ratio 6:7. Find the first share.

Options: 323 | 352 | 265 | 294

Q190 ? Complement method ? Placement

A fleet split case uses the quantities in this question. Determine the requested result using the complement method model: ■250 is divided in the ratio 2:3. Find the first share.

Options: 90 | 100 | 110 | 120

Q191 ? Pattern: differences ? Beginner

A budget share case uses the quantities in this question. Determine the requested result using the pattern: differences model: ■408 is divided in the ratio 3:5. Find the first share.

Options: 153 | 168 | 138 | 183

Q192 ? Pattern: ratios ? Easy

A mixture ratio case uses the quantities in this question. Determine the requested result using the pattern: ratios model: ■572 is divided in the ratio 4:7. Find the first share.

Options: 208 | 229 | 187 | 250

Q193 ? Pattern: alternating ? Medium

A team allocation case uses the quantities in this question. Determine the requested result using the pattern: alternating model: ■424 is divided in the ratio 5:3. Find the first share.

Options: 319 | 238 | 292 | 265

Q194 ? Pattern: squares / cubes ? Hard

A recipe scaling case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: ■594 is divided in the ratio 6:5. Find the first share.

Options: 324 | 356 | 292 | 388

Q195 ? Inference from facts ? Placement

A project staffing case uses the quantities in this question. Determine the requested result using the inference from facts model: ■495 is divided in the ratio 2:7. Find the first share.

Options: 110 | 121 | 132 | 99

Q196 ? Assumption check ? Beginner

A screen resolution case uses the quantities in this question. Determine the requested result using the assumption check model: ■336 is divided in the ratio 3:3. Find the first share.

Options: 202 | 151 | 185 | 168

Q197 ? Cause vs correlation ? Easy

A map scale case uses the quantities in this question. Determine the requested result using the cause vs correlation model: ■513 is divided in the ratio 4:5. Find the first share.

Options: 274 | 205 | 251 | 228

Q198 ? Grammar agreement ? Medium

A fleet split case uses the quantities in this question. Determine the requested result using the grammar agreement model: ■696 is divided in the ratio 5:7. Find the first share.

Options: 290 | 319 | 261 | 348

Q199 ? Vocabulary in context ? Hard

A budget share case uses the quantities in this question. Determine the requested result using the vocabulary in context model: ■531 is divided in the ratio 6:3. Find the first share.

Options: 319 | 389 | 354 | 424

Q200 ? Mixed placement challenge ? Placement

A mixture ratio case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: ■420 is divided in the ratio 2:5. Find the first share.

Options: 144 | 108 | 132 | 120

Averages ? 40 questions

WORKED EXAMPLES

Q201 ? Direct computation ? Beginner

A class performance case uses the quantities in this question. Determine the requested result using the direct computation model: Find the average of 11, 15, 19, 23, 27.

Options: 23 | 17 | 21 | 19

Answer: 19

Solution: Context: class performance. The sum is 95. Divide by 5 to get 19. Shortcut: The middle of an arithmetic sequence is its average.

Q202 ? Reverse value ? Easy

A weekly sales case uses the quantities in this question. Determine the requested result using the reverse value model: Find the average of 12, 16, 20, 24, 28.

Options: 24 | 22 | 20 | 18

Answer: 20

Solution: Context: weekly sales. The sum is 100. Divide by 5 to get 20. Shortcut: The middle of an arithmetic sequence is its average.

Q203 ? Missing base ? Medium

A sensor readings case uses the quantities in this question. Determine the requested result using the missing base model: Find the average of 13, 17, 21, 25, 29.

Options: 21 | 23 | 19 | 25

Answer: 21

Solution: Context: sensor readings. The sum is 105. Divide by 5 to get 21. Shortcut: The middle of an arithmetic sequence is its average.

Q204 ? Successive change ? Hard

A team output case uses the quantities in this question. Determine the requested result using the successive change model: Find the average of 14, 18, 22, 26, 30.

Options: 22 | 24 | 20 | 26

Answer: 22

Solution: Context: team output. The sum is 110. Divide by 5 to get 22. Shortcut: The middle of an arithmetic sequence is its average.

Q205 ? Comparison trap ? Placement

A match scores case uses the quantities in this question. Determine the requested result using the comparison trap model: Find the average of 15, 19, 23, 27, 31.

Options: 27 | 21 | 25 | 23

Answer: 23

Solution: Context: match scores. The sum is 115. Divide by 5 to get 23. Shortcut: The middle of an arithmetic sequence is its average.

Q206 ? Ratio form ? Beginner

A delivery time case uses the quantities in this question. Determine the requested result using the ratio form model: Find the average of 16, 20, 24, 28, 32.

Options: 24 | 26 | 22 | 28

Answer: 24

Solution: Context: delivery time. The sum is 120. Divide by 5 to get 24. Shortcut: The middle of an arithmetic sequence is its average.

Q207 ? Weighted result ? Easy

A salary bands case uses the quantities in this question. Determine the requested result using the weighted result model: Find the average of 17, 21, 25, 29, 33.

Options: 25 | 28 | 22 | 31

Answer: 25

Solution: Context: salary bands. The sum is 125. Divide by 5 to get 25. Shortcut: The middle of an arithmetic sequence is its average.

Q208 ? Replacement cycle ? Medium

A quality samples case uses the quantities in this question. Determine the requested result using the replacement cycle model: Find the average of 18, 22, 26, 30, 34.

Options: 32 | 23 | 29 | 26

Answer: 26

Solution: Context: quality samples. The sum is 130. Divide by 5 to get 26. Shortcut: The middle of an arithmetic sequence is its average.

Q209 ? Unit conversion ? Hard

A class performance case uses the quantities in this question. Determine the requested result using the unit conversion model: Find the average of 19, 23, 27, 31, 35.

Options: 33 | 24 | 30 | 27

Answer: 27

Solution: Context: class performance. The sum is 135. Divide by 5 to get 27. Shortcut: The middle of an arithmetic sequence is its average.

Q210 ? Work-rate model ? Placement

A weekly sales case uses the quantities in this question. Determine the requested result using the work-rate model model: Find the average of 20, 24, 28, 32, 36.

Options: 28 | 31 | 25 | 34

Answer: 28

Solution: Context: weekly sales. The sum is 140. Divide by 5 to get 28. Shortcut: The middle of an arithmetic sequence is its average.

Q211 ? Speed-time model ? Beginner

A sensor readings case uses the quantities in this question. Determine the requested result using the speed-time model model: Find the average of 21, 25, 29, 33, 37.

Options: 29 | 32 | 26 | 35

Answer: 29

Solution: Context: sensor readings. The sum is 145. Divide by 5 to get 29. Shortcut: The middle of an arithmetic sequence is its average.

Q212 ? Table lookup ? Easy

A team output case uses the quantities in this question. Determine the requested result using the table lookup model: Find the average of 22, 26, 30, 34, 38.

Options: 36 | 27 | 33 | 30

Answer: 30

Solution: Context: team output. The sum is 150. Divide by 5 to get 30. Shortcut: The middle of an arithmetic sequence is its average.

Q213 ? Chart comparison ? Medium

A match scores case uses the quantities in this question. Determine the requested result using the chart comparison model: Find the average of 23, 27, 31, 35, 39.

Options: 31 | 34 | 28 | 37

Answer: 31

Solution: Context: match scores. The sum is 155. Divide by 5 to get 31. Shortcut: The middle of an arithmetic sequence is its average.

Q214 ? Multi-step word problem ? Hard

A delivery time case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Find the average of 24, 28, 32, 36, 40.

Options: 29 | 32 | 35 | 38

Answer: 32

Solution: Context: delivery time. The sum is 160. Divide by 5 to get 32. Shortcut: The middle of an arithmetic sequence is its average.

Q215 ? Maximum / minimum ? Placement

A salary bands case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Find the average of 25, 29, 33, 37, 41.

Options: 33 | 36 | 30 | 39

Answer: 33

Solution: Context: salary bands. The sum is 165. Divide by 5 to get 33. Shortcut: The middle of an arithmetic sequence is its average.

Q216 ? Least / greatest value ? Beginner

A quality samples case uses the quantities in this question. Determine the requested result using the least / greatest value model: Find the average of 26, 30, 34, 38, 42.

Options: 40 | 31 | 37 | 34

Answer: 34

Solution: Context: quality samples. The sum is 170. Divide by 5 to get 34. Shortcut: The middle of an arithmetic sequence is its average.

Q217 ? Remainder cycle ? Easy

A class performance case uses the quantities in this question. Determine the requested result using the remainder cycle model: Find the average of 27, 31, 35, 39, 43.

Options: 35 | 39 | 31 | 43

Answer: 35

Solution: Context: class performance. The sum is 175. Divide by 5 to get 35. Shortcut: The middle of an arithmetic sequence is its average.

Q218 ? Divisibility test ? Medium

A weekly sales case uses the quantities in this question. Determine the requested result using the divisibility test model: Find the average of 28, 32, 36, 40, 44.

Options: 32 | 36 | 44 | 40

Answer: 36

Solution: Context: weekly sales. The sum is 180. Divide by 5 to get 36. Shortcut: The middle of an arithmetic sequence is its average.

Q219 ? Factorisation ? Hard

A sensor readings case uses the quantities in this question. Determine the requested result using the factorisation model: Find the average of 29, 33, 37, 41, 45.

Options: 41 | 45 | 33 | 37

Answer: 37

Solution: Context: sensor readings. The sum is 185. Divide by 5 to get 37. Shortcut: The middle of an arithmetic sequence is its average.

Q220 ? Approximation ? Placement

A team output case uses the quantities in this question. Determine the requested result using the approximation model: Find the average of 30, 34, 38, 42, 46.

Options: 46 | 42 | 38 | 34

Answer: 38

Solution: Context: team output. The sum is 190. Divide by 5 to get 38. Shortcut: The middle of an arithmetic sequence is its average.

PRACTICE ? solutions hidden below**Q221 ? Options elimination ? Beginner**

A match scores case uses the quantities in this question. Determine the requested result using the options elimination model: Find the average of 31, 35, 39, 43, 47.

Options: 39 | 35 | 43 | 47

Q222 ? Boundary case ? Easy

A delivery time case uses the quantities in this question. Determine the requested result using the boundary case model: Find the average of 32, 36, 40, 44, 48.

Options: 40 | 44 | 36 | 48

Q223 ? Case split ? Medium

A salary bands case uses the quantities in this question. Determine the requested result using the case split model: Find the average of 33, 37, 41, 45, 49.

Options: 49 | 37 | 45 | 41

Q224 ? Contradiction check ? Hard

A quality samples case uses the quantities in this question. Determine the requested result using the contradiction check model: Find the average of 34, 38, 42, 46, 50.

Options: 42 | 46 | 38 | 50

Q225 ? Statement I test ? Placement

A class performance case uses the quantities in this question. Determine the requested result using the statement i test model: Find the average of 35, 39, 43, 47, 51.

Options: 39 | 47 | 43 | 51

Q226 ? Statement II test ? Beginner

A weekly sales case uses the quantities in this question. Determine the requested result using the statement ii test model: Find the average of 36, 40, 44, 48, 52.

Options: 44 | 48 | 40 | 52

Q227 ? Combined statements ? Easy

A sensor readings case uses the quantities in this question. Determine the requested result using the combined statements model: Find the average of 37, 41, 45, 49, 53.

Options: 55 | 40 | 50 | 45

Q228 ? Counting restriction ? Medium

A team output case uses the quantities in this question. Determine the requested result using the counting restriction model: Find the average of 38, 42, 46, 50, 54.

Options: 46 | 41 | 51 | 56

Q229 ? Circular arrangement ? Hard

A match scores case uses the quantities in this question. Determine the requested result using the circular arrangement model: Find the average of 39, 43, 47, 51, 55.

Options: 57 | 52 | 47 | 42

Q230 ? Complement method ? Placement

A delivery time case uses the quantities in this question. Determine the requested result using the complement method model: Find the average of 40, 44, 48, 52, 56.

Options: 53 | 58 | 43 | 48

Q231 ? Pattern: differences ? Beginner

A salary bands case uses the quantities in this question. Determine the requested result using the pattern: differences model: Find the average of 41, 45, 49, 53, 57.

Options: 49 | 54 | 44 | 59

Q232 ? Pattern: ratios ? Easy

A quality samples case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Find the average of 42, 46, 50, 54, 58.

Options: 45 | 55 | 50 | 60

Q233 ? Pattern: alternating ? Medium

A class performance case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Find the average of 43, 47, 51, 55, 59.

Options: 51 | 56 | 46 | 61

Q234 ? Pattern: squares / cubes ? Hard

A weekly sales case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Find the average of 44, 48, 52, 56, 60.

Options: 62 | 47 | 57 | 52

Q235 ? Inference from facts ? Placement

A sensor readings case uses the quantities in this question. Determine the requested result using the inference from facts model: Find the average of 45, 49, 53, 57, 61.

Options: 48 | 53 | 58 | 63

Q236 ? Assumption check ? Beginner

A team output case uses the quantities in this question. Determine the requested result using the assumption check model: Find the average of 46, 50, 54, 58, 62.

Options: 64 | 59 | 54 | 49

Q237 ? Cause vs correlation ? Easy

A match scores case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Find the average of 47, 51, 55, 59, 63.

Options: 55 | 61 | 67 | 49

Q238 ? Grammar agreement ? Medium

A delivery time case uses the quantities in this question. Determine the requested result using the grammar agreement model: Find the average of 48, 52, 56,

60, 64.

Options: 68 | 62 | 56 | 50

Q239 ? Vocabulary in context ? Hard

A salary bands case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Find the average of 49, 53, 57, 61, 65.

Options: 51 | 57 | 69 | 63

Q240 ? Mixed placement challenge ? Placement

A quality samples case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Find the average of 50, 54, 58, 62, 66.

Options: 70 | 58 | 64 | 52

Time & Work ? 40 questions

WORKED EXAMPLES

Q241 ? Direct computation ? Beginner

A software sprint case uses the quantities in this question. Determine the requested result using the direct computation model: A finishes a job in 9 days and B in 14 days. How long do they take together?

Options: 6.48 | 5.48 | 4.48 | 7.48

Answer: 5.48

Solution: Context: software sprint. Combined rate = $1/9+1/14$. Time = $126/23 = 5.48$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q242 ? Reverse value ? Easy

A construction crew case uses the quantities in this question. Determine the requested result using the reverse value model: A finishes a job in 10 days and B in 16 days. How long do they take together?

Options: 6.15 | 7.15 | 5.15 | 8.15

Answer: 6.15

Solution: Context: construction crew. Combined rate = $1/10+1/16$. Time = $160/26 = 6.15$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q243 ? Missing base ? Medium

A data migration case uses the quantities in this question. Determine the requested result using the missing base model: A finishes a job in 11 days and B in 18 days. How long do they take together?

Options: 7.83 | 8.83 | 6.83 | 5.83

Answer: 6.83

Solution: Context: data migration. Combined rate = $1/11+1/18$. Time = $198/29 = 6.83$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q244 ? Successive change ? Hard

A exam paper review case uses the quantities in this question. Determine the requested result using the successive change model: A finishes a job in 12 days and B in 20 days. How long do they take together?

Options: 6.5 | 7.5 | 8.5 | 9.5

Answer: 7.5

Solution: Context: exam paper review. Combined rate = $1/12+1/20$. Time = $240/32 = 7.5$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q245 ? Comparison trap ? Placement

A factory line case uses the quantities in this question. Determine the requested result using the comparison trap model: A finishes a job in 13 days and B in 22 days. How long do they take together?

Options: 10.17 | 7.17 | 9.17 | 8.17

Answer: 8.17

Solution: Context: factory line. Combined rate = $1/13+1/22$. Time = $286/35 = 8.17$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q246 ? Ratio form ? Beginner

A event setup case uses the quantities in this question. Determine the requested result using the ratio form model: A finishes a job in 14 days and B in 13 days. How long do they take together?

Options: 6.74 | 7.74 | 8.74 | 5.74

Answer: 6.74

Solution: Context: event setup. Combined rate = $1/14+1/13$. Time = $182/27 = 6.74$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q247 ? Weighted result ? Easy

A content pipeline case uses the quantities in this question. Determine the requested result using the weighted result model: A finishes a job in 15 days and B in 15 days. How long do they take together?

Options: 8.5 | 9.5 | 6.5 | 7.5

Answer: 7.5

Solution: Context: content pipeline. Combined rate = $1/15+1/15$. Time = $225/30 = 7.5$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q248 ? Replacement cycle ? Medium

A audit team case uses the quantities in this question. Determine the requested result using the replacement cycle model: A finishes a job in 16 days and B in 17 days. How long do they take together?

Options: 10.24 | 7.24 | 9.24 | 8.24

Answer: 8.24

Solution: Context: audit team. Combined rate = $1/16+1/17$. Time = $272/33 = 8.24$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q249 ? Unit conversion ? Hard

A software sprint case uses the quantities in this question. Determine the requested result using the unit conversion model: A finishes a job in 8 days and B in 19 days. How long do they take together?

Options: 5.63 | 6.63 | 7.63 | 4.63

Answer: 5.63

Solution: Context: software sprint. Combined rate = $1/8+1/19$. Time = $152/27 = 5.63$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q250 ? Work-rate model ? Placement

A construction crew case uses the quantities in this question. Determine the requested result using the work-rate model model: A finishes a job in 9 days and B in 21 days. How long do they take together?

Options: 7.3 | 8.3 | 6.3 | 5.3

Answer: 6.3

Solution: Context: construction crew. Combined rate = $1/9+1/21$. Time = $189/30 = 6.3$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q251 ? Speed-time model ? Beginner

A data migration case uses the quantities in this question. Determine the requested result using the speed-time model model: A finishes a job in 10 days and B in 12 days. How long do they take together?

Options: 4.45 | 5.45 | 6.45 | 7.45

Answer: 5.45

Solution: Context: data migration. Combined rate = $1/10+1/12$. Time = $120/22 = 5.45$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q252 ? Table lookup ? Easy

A exam paper review case uses the quantities in this question. Determine the requested result using the table lookup model: A finishes a job in 11 days and B in 14 days. How long do they take together?

Options: 8.16 | 7.16 | 6.16 | 5.16

Answer: 6.16

Solution: Context: exam paper review. Combined rate = $1/11+1/14$. Time = $154/25 = 6.16$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q253 ? Chart comparison ? Medium

A factory line case uses the quantities in this question. Determine the requested result using the chart comparison model: A finishes a job in 12 days and B in 16 days. How long do they take together?

Options: 6.86 | 7.86 | 8.86 | 5.86

Answer: 6.86

Solution: Context: factory line. Combined rate = $1/12+1/16$. Time = $192/28 = 6.86$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q254 ? Multi-step word problem ? Hard

A event setup case uses the quantities in this question. Determine the requested result using the multi-step word problem model: A finishes a job in 13 days and B in 18 days. How long do they take together?

Options: 8.55 | 9.55 | 6.55 | 7.55

Answer: 7.55

Solution: Context: event setup. Combined rate = $1/13+1/18$. Time = $234/31 = 7.55$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q255 ? Maximum / minimum ? Placement

A content pipeline case uses the quantities in this question. Determine the requested result using the maximum / minimum model: A finishes a job in 14 days and B in 20 days. How long do they take together?

Options: 10.24 | 7.24 | 9.24 | 8.24

Answer: 8.24

Solution: Context: content pipeline. Combined rate = $1/14+1/20$. Time = $280/34 = 8.24$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q256 ? Least / greatest value ? Beginner

A audit team case uses the quantities in this question. Determine the requested result using the least / greatest value model: A finishes a job in 15 days and B in 22 days. How long do they take together?

Options: 9.92 | 8.92 | 7.92 | 10.92

Answer: 8.92

Solution: Context: audit team. Combined rate = $1/15+1/22$. Time = $330/37 = 8.92$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q257 ? Remainder cycle ? Easy

A software sprint case uses the quantities in this question. Determine the requested result using the remainder cycle model: A finishes a job in 16 days and B in 13 days. How long do they take together?

Options: 7.17 | 9.17 | 6.17 | 8.17

Answer: 7.17

Solution: Context: software sprint. Combined rate = $1/16+1/13$. Time = $208/29 = 7.17$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q258 ? Divisibility test ? Medium

A construction crew case uses the quantities in this question. Determine the requested result using the divisibility test model: A finishes a job in 8 days and B in 15 days. How long do they take together?

Options: 4.22 | 5.22 | 6.22 | 7.22

Answer: 5.22

Solution: Context: construction crew. Combined rate = $1/8+1/15$. Time = $120/23 = 5.22$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q259 ? Factorisation ? Hard

A data migration case uses the quantities in this question. Determine the requested result using the factorisation model: A finishes a job in 9 days and B in 17 days. How long do they take together?

Options: 5.88 | 6.88 | 4.88 | 7.88

Answer: 5.88

Solution: Context: data migration. Combined rate = $1/9+1/17$. Time = $153/26 = 5.88$ days. Shortcut: For two workers, use $ab/(a+b)$.

Q260 ? Approximation ? Placement

A exam paper review case uses the quantities in this question. Determine the requested result using the approximation model: A finishes a job in 10 days and B in 19 days. How long do they take together?

Options: 6.55 | 7.55 | 8.55 | 5.55

Answer: 6.55

Solution: Context: exam paper review. Combined rate = $1/10+1/19$. Time = $190/29 = 6.55$ days. Shortcut: For two workers, use $ab/(a+b)$.

PRACTICE ? solutions hidden below**Q261 ? Options elimination ? Beginner**

A factory line case uses the quantities in this question. Determine the requested result using the options elimination model: A finishes a job in 11 days and B in

21 days. How long do they take together?

Options: 8.22 | 9.22 | 6.22 | 7.22

Q262 ? Boundary case ? Easy

A event setup case uses the quantities in this question. Determine the requested result using the boundary case model: A finishes a job in 12 days and B in 12 days. How long do they take together?

Options: 5 | 8 | 7 | 6

Q263 ? Case split ? Medium

A content pipeline case uses the quantities in this question. Determine the requested result using the case split model: A finishes a job in 13 days and B in 14 days. How long do they take together?

Options: 6.74 | 7.74 | 5.74 | 8.74

Q264 ? Contradiction check ? Hard

A audit team case uses the quantities in this question. Determine the requested result using the contradiction check model: A finishes a job in 14 days and B in 16 days. How long do they take together?

Options: 7.47 | 9.47 | 6.47 | 8.47

Q265 ? Statement I test ? Placement

A software sprint case uses the quantities in this question. Determine the requested result using the statement i test model: A finishes a job in 15 days and B in 18 days. How long do they take together?

Options: 8.18 | 9.18 | 7.18 | 10.18

Q266 ? Statement II test ? Beginner

A construction crew case uses the quantities in this question. Determine the requested result using the statement ii test model: A finishes a job in 16 days and B in 20 days. How long do they take together?

Options: 10.89 | 7.89 | 9.89 | 8.89

Q267 ? Combined statements ? Easy

A data migration case uses the quantities in this question. Determine the requested result using the combined statements model: A finishes a job in 8 days and B in 22 days. How long do they take together?

Options: 6.87 | 7.87 | 4.87 | 5.87

Q268 ? Counting restriction ? Medium

A exam paper review case uses the quantities in this question. Determine the requested result using the counting restriction model: A finishes a job in 9 days and B in 13 days. How long do they take together?

Options: 4.32 | 5.32 | 6.32 | 7.32

Q269 ? Circular arrangement ? Hard

A factory line case uses the quantities in this question. Determine the requested result using the circular arrangement model: A finishes a job in 10 days and B in 15 days. How long do they take together?

Options: 5 | 8 | 7 | 6

Q270 ? Complement method ? Placement

A event setup case uses the quantities in this question. Determine the requested result using the complement method model: A finishes a job in 11 days and B in 17 days. How long do they take together?

Options: 6.68 | 7.68 | 5.68 | 8.68

Q271 ? Pattern: differences ? Beginner

A content pipeline case uses the quantities in this question. Determine the requested result using the pattern: differences model: A finishes a job in 12 days and B in 19 days. How long do they take together?

Options: 7.35 | 9.35 | 6.35 | 8.35

Q272 ? Pattern: ratios ? Easy

A audit team case uses the quantities in this question. Determine the requested result using the pattern: ratios model: A finishes a job in 13 days and B in 21 days. How long do they take together?

Options: 8.03 | 9.03 | 7.03 | 10.03

Q273 ? Pattern: alternating ? Medium

A software sprint case uses the quantities in this question. Determine the requested result using the pattern: alternating model: A finishes a job in 14 days and B in 12 days. How long do they take together?

Options: 8.46 | 7.46 | 6.46 | 5.46

Q274 ? Pattern: squares / cubes ? Hard

A construction crew case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: A finishes a job in 15 days and B in 14 days. How long do they take together?

Options: 7.24 | 8.24 | 6.24 | 9.24

Q275 ? Inference from facts ? Placement

A data migration case uses the quantities in this question. Determine the requested result using the inference from facts model: A finishes a job in 16 days and B in 16 days. How long do they take together?

Options: 8 | 10 | 9 | 7

Q276 ? Assumption check ? Beginner

A exam paper review case uses the quantities in this question. Determine the requested result using the assumption check model: A finishes a job in 8 days and B in 18 days. How long do they take together?

Options: 6.54 | 5.54 | 4.54 | 7.54

Q277 ? Cause vs correlation ? Easy

A factory line case uses the quantities in this question. Determine the requested result using the cause vs correlation model: A finishes a job in 9 days and B in 20 days. How long do they take together?

Options: 6.21 | 7.21 | 5.21 | 8.21

Q278 ? Grammar agreement ? Medium

A event setup case uses the quantities in this question. Determine the requested result using the grammar agreement model: A finishes a job in 10 days and B in 22 days. How long do they take together?

Options: 7.88 | 8.88 | 6.88 | 5.88

Q279 ? Vocabulary in context ? Hard

A content pipeline case uses the quantities in this question. Determine the requested result using the vocabulary in context model: A finishes a job in 11 days and B in 13 days. How long do they take together?

Options: 4.96 | 5.96 | 6.96 | 7.96

Q280 ? Mixed placement challenge ? Placement

A audit team case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: A finishes a job in 12 days and B in 15 days. How long do they take together?

Options: 8.67 | 7.67 | 6.67 | 5.67

Time, Speed & Distance ? 40 questions

WORKED EXAMPLES

Q281 ? Direct computation ? Beginner

A delivery route case uses the quantities in this question. Determine the requested result using the direct computation model: A vehicle travels at 35 km/h for 3 hours. What distance does it cover?

Options: 127 | 94 | 116 | 105

Answer: 105

Solution: Context: delivery route. Distance = speed $\times$ time = $35 \times 3 = 105$ km. Shortcut: Confirm units before calculation.

Q282 ? Reverse value ? Easy

A commute case uses the quantities in this question. Determine the requested result using the reverse value model: A vehicle travels at 40 km/h for 4 hours. What distance does it cover?

Options: 160 | 176 | 144 | 192

Answer: 160

Solution: Context: commute. Distance = speed $\times$ time = $40 \times 4 = 160$ km. Shortcut: Confirm units before calculation.

Q283 ? Missing base ? Medium

A train journey case uses the quantities in this question. Determine the requested result using the missing base model: A vehicle travels at 45 km/h for 5 hours. What distance does it cover?

Options: 225 | 248 | 202 | 271

Answer: 225

Solution: Context: train journey. Distance = speed $\times$ time = $45 \times 5 = 225$ km. Shortcut: Confirm units before calculation.

Q284 ? Successive change ? Hard

A drone flight case uses the quantities in this question. Determine the requested result using the successive change model: A vehicle travels at 50 km/h for 6 hours. What distance does it cover?

Options: 360 | 330 | 300 | 270

Answer: 300

Solution: Context: drone flight. Distance = speed $\times$ time = $50 \times 6 = 300$ km. Shortcut: Confirm units before calculation.

Q285 ? Comparison trap ? Placement

A cycling stage case uses the quantities in this question. Determine the requested result using the comparison trap model: A vehicle travels at 55 km/h for 2 hours. What distance does it cover?

Options: 132 | 99 | 121 | 110

Answer: 110

Solution: Context: cycling stage. Distance = speed $\times$ time = $55 \times 2 = 110$ km. Shortcut: Confirm units before calculation.

Q286 ? Ratio form ? Beginner

A race track case uses the quantities in this question. Determine the requested result using the ratio form model: A vehicle travels at 60 km/h for 3 hours. What distance does it cover?

Options: 180 | 198 | 162 | 216

Answer: 180

Solution: Context: race track. Distance = speed $\times$ time = $60 \times 3 = 180$ km. Shortcut: Confirm units before calculation.

Q287 ? Weighted result ? Easy

A courier service case uses the quantities in this question. Determine the requested result using the weighted result model: A vehicle travels at 65 km/h for 4 hours. What distance does it cover?

Options: 260 | 286 | 234 | 312

Answer: 260

Solution: Context: courier service. Distance = speed $\times$ time = $65 \times 4 = 260$ km. Shortcut: Confirm units before calculation.

Q288 ? Replacement cycle ? Medium

A network latency case uses the quantities in this question. Determine the requested result using the replacement cycle model: A vehicle travels at 30 km/h for 5 hours. What distance does it cover?

Options: 180 | 135 | 165 | 150

Answer: 150

Solution: Context: network latency. Distance = speed $\times$ time = $30 \times 5 = 150$ km. Shortcut: Confirm units before calculation.

Q289 ? Unit conversion ? Hard

A delivery route case uses the quantities in this question. Determine the requested result using the unit conversion model: A vehicle travels at 35 km/h for 6 hours. What distance does it cover?

Options: 252 | 231 | 210 | 189

Answer: 210

Solution: Context: delivery route. Distance = speed $\times$ time = $35 \times 6 = 210$ km. Shortcut: Confirm units before calculation.

Q290 ? Work-rate model ? Placement

A commute case uses the quantities in this question. Determine the requested result using the work-rate model model: A vehicle travels at 40 km/h for 2 hours. What distance does it cover?

Options: 80 | 88 | 72 | 96

Answer: 80

Solution: Context: commute. Distance = speed $\times$ time = $40 \times 2 = 80$ km. Shortcut: Confirm units before calculation.

Q291 ? Speed-time model ? Beginner

A train journey case uses the quantities in this question. Determine the requested result using the speed-time model model: A vehicle travels at 45 km/h for 3 hours. What distance does it cover?

Options: 163 | 121 | 149 | 135

Answer: 135

Solution: Context: train journey. Distance = speed $\times$ time = $45 \times 3 = 135$ km. Shortcut: Confirm units before calculation.

Q292 ? Table lookup ? Easy

A drone flight case uses the quantities in this question. Determine the requested result using the table lookup model: A vehicle travels at 50 km/h for 4 hours. What distance does it cover?

Options: 240 | 180 | 220 | 200

Answer: 200

Solution: Context: drone flight. Distance = speed $\times$ time = $50 \times 4 = 200$ km. Shortcut: Confirm units before calculation.

Q293 ? Chart comparison ? Medium

A cycling stage case uses the quantities in this question. Determine the requested result using the chart comparison model: A vehicle travels at 55 km/h for 5 hours. What distance does it cover?

Options: 275 | 303 | 247 | 331

Answer: 275

Solution: Context: cycling stage. Distance = speed $\times$ time = $55 \times 5 = 275$ km. Shortcut: Confirm units before calculation.

Q294 ? Multi-step word problem ? Hard

A race track case uses the quantities in this question. Determine the requested result using the multi-step word problem model: A vehicle travels at 60 km/h for 6 hours. What distance does it cover?

Options: 360 | 396 | 324 | 432

Answer: 360

Solution: Context: race track. Distance = speed $\times$ time = $60 \times 6 = 360$ km. Shortcut: Confirm units before calculation.

Q295 ? Maximum / minimum ? Placement

A courier service case uses the quantities in this question. Determine the requested result using the maximum / minimum model: A vehicle travels at 65 km/h for 2 hours. What distance does it cover?

Options: 156 | 117 | 143 | 130

Answer: 130

Solution: Context: courier service. Distance = speed $\times$ time = $65 \times 2 = 130$ km. Shortcut: Confirm units before calculation.

Q296 ? Least / greatest value ? Beginner

A network latency case uses the quantities in this question. Determine the requested result using the least / greatest value model: A vehicle travels at 30 km/h for 3 hours. What distance does it cover?

Options: 99 | 90 | 81 | 108

Answer: 90

Solution: Context: network latency. Distance = speed $\times$ time = $30 \times 3 = 90$ km. Shortcut: Confirm units before calculation.

Q297 ? Remainder cycle ? Easy

A delivery route case uses the quantities in this question. Determine the requested result using the remainder cycle model: A vehicle travels at 35 km/h for 4 hours. What distance does it cover?

Options: 140 | 154 | 126 | 168

Answer: 140

Solution: Context: delivery route. Distance = speed $\times$ time = $35 \times 4 = 140$ km. Shortcut: Confirm units before calculation.

Q298 ? Divisibility test ? Medium

A commute case uses the quantities in this question. Determine the requested result using the divisibility test model: A vehicle travels at 40 km/h for 5 hours. What distance does it cover?

Options: 200 | 220 | 180 | 240

Answer: 200

Solution: Context: commute. Distance = speed $\times$ time = $40 \times 5 = 200$ km. Shortcut: Confirm units before calculation.

Q299 ? Factorisation ? Hard

A train journey case uses the quantities in this question. Determine the requested result using the factorisation model: A vehicle travels at 45 km/h for 6 hours. What distance does it cover?

Options: 324 | 243 | 297 | 270

Answer: 270

Solution: Context: train journey. Distance = speed $\times$ time = $45 \times 6 = 270$ km. Shortcut: Confirm units before calculation.

Q300 ? Approximation ? Placement

A drone flight case uses the quantities in this question. Determine the requested result using the approximation model: A vehicle travels at 50 km/h for 2 hours. What distance does it cover?

Options: 100 | 110 | 90 | 120

Answer: 100

Solution: Context: drone flight. Distance = speed $\times$ time = $50 \times 2 = 100$ km. Shortcut: Confirm units before calculation.

PRACTICE ? solutions hidden below**Q301 ? Options elimination ? Beginner**

A cycling stage case uses the quantities in this question. Determine the requested result using the options elimination model: A vehicle travels at 55 km/h for 3

hours. What distance does it cover?

Options: 165 | 182 | 148 | 199

Q302 ? Boundary case ? Easy

A race track case uses the quantities in this question. Determine the requested result using the boundary case model: A vehicle travels at 60 km/h for 4 hours. What distance does it cover?

Options: 288 | 216 | 264 | 240

Q303 ? Case split ? Medium

A courier service case uses the quantities in this question. Determine the requested result using the case split model: A vehicle travels at 65 km/h for 5 hours. What distance does it cover?

Options: 391 | 292 | 358 | 325

Q304 ? Contradiction check ? Hard

A network latency case uses the quantities in this question. Determine the requested result using the contradiction check model: A vehicle travels at 30 km/h for 6 hours. What distance does it cover?

Options: 180 | 198 | 162 | 216

Q305 ? Statement I test ? Placement

A delivery route case uses the quantities in this question. Determine the requested result using the statement i test model: A vehicle travels at 35 km/h for 2 hours. What distance does it cover?

Options: 63 | 77 | 70 | 84

Q306 ? Statement II test ? Beginner

A commute case uses the quantities in this question. Determine the requested result using the statement ii test model: A vehicle travels at 40 km/h for 3 hours. What distance does it cover?

Options: 144 | 108 | 132 | 120

Q307 ? Combined statements ? Easy

A train journey case uses the quantities in this question. Determine the requested result using the combined statements model: A vehicle travels at 45 km/h for 4 hours. What distance does it cover?

Options: 180 | 198 | 162 | 216

Q308 ? Counting restriction ? Medium

A drone flight case uses the quantities in this question. Determine the requested result using the counting restriction model: A vehicle travels at 50 km/h for 5 hours. What distance does it cover?

Options: 250 | 275 | 225 | 300

Q309 ? Circular arrangement ? Hard

A cycling stage case uses the quantities in this question. Determine the requested result using the circular arrangement model: A vehicle travels at 55 km/h for 6 hours. What distance does it cover?

Options: 330 | 363 | 396 | 297

Q310 ? Complement method ? Placement

A race track case uses the quantities in this question. Determine the requested result using the complement method model: A vehicle travels at 60 km/h for 2 hours. What distance does it cover?

Options: 120 | 132 | 108 | 144

Q311 ? Pattern: differences ? Beginner

A courier service case uses the quantities in this question. Determine the requested result using the pattern: differences model: A vehicle travels at 65 km/h for 3 hours. What distance does it cover?

Options: 195 | 215 | 175 | 235

Q312 ? Pattern: ratios ? Easy

A network latency case uses the quantities in this question. Determine the requested result using the pattern: ratios model: A vehicle travels at 30 km/h for 4 hours. What distance does it cover?

Options: 144 | 108 | 132 | 120

Q313 ? Pattern: alternating ? Medium

A delivery route case uses the quantities in this question. Determine the requested result using the pattern: alternating model: A vehicle travels at 35 km/h for 5 hours. What distance does it cover?

Options: 211 | 157 | 193 | 175

Q314 ? Pattern: squares / cubes ? Hard

A commute case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: A vehicle travels at 40 km/h for 6 hours. What distance does it cover?

Options: 240 | 264 | 216 | 288

Q315 ? Inference from facts ? Placement

A train journey case uses the quantities in this question. Determine the requested result using the inference from facts model: A vehicle travels at 45 km/h for 2 hours. What distance does it cover?

Options: 81 | 108 | 90 | 99

Q316 ? Assumption check ? Beginner

A drone flight case uses the quantities in this question. Determine the requested result using the assumption check model: A vehicle travels at 50 km/h for 3 hours. What distance does it cover?

Options: 180 | 135 | 165 | 150

Q317 ? Cause vs correlation ? Easy

A cycling stage case uses the quantities in this question. Determine the requested result using the cause vs correlation model: A vehicle travels at 55 km/h for 4 hours. What distance does it cover?

Options: 264 | 242 | 220 | 198

Q318 ? Grammar agreement ? Medium

A race track case uses the quantities in this question. Determine the requested result using the grammar agreement model: A vehicle travels at 60 km/h for 5 hours. What distance does it cover?

Options: 300 | 330 | 270 | 360

Q319 ? Vocabulary in context ? Hard

A courier service case uses the quantities in this question. Determine the requested result using the vocabulary in context model: A vehicle travels at 65 km/h for 6 hours. What distance does it cover?

Options: 351 | 390 | 468 | 429

Q320 ? Mixed placement challenge ? Placement

A network latency case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: A vehicle travels at 30 km/h for 2 hours. What distance does it cover?

Options: 72 | 60 | 66 | 54

Trains ? 40 questions

WORKED EXAMPLES

Q321 ? Direct computation ? Beginner

A rail crossing case uses the quantities in this question. Determine the requested result using the direct computation model: A 110 m train crosses a pole in 9 seconds. Find its speed in km/h.

Options: 51.2 | 44 | 47.6 | 40.4

Answer: 44

Solution: Context: rail crossing. Speed = $110/9 = 12.22$ m/s = 44 km/h. Shortcut: Multiply m/s by 3.6.

Q322 ? Reverse value ? Easy

A platform clearance case uses the quantities in this question. Determine the requested result using the reverse value model: A 120 m train crosses a pole in 10 seconds. Find its speed in km/h.

Options: 50.4 | 39.6 | 46.8 | 43.2

Answer: 43.2

Solution: Context: platform clearance. Speed = $120/10 = 12$ m/s = 43.2 km/h. Shortcut: Multiply m/s by 3.6.

Q323 ? Missing base ? Medium

A two-train dispatch case uses the quantities in this question. Determine the requested result using the missing base model: A 130 m train crosses a pole in 11 seconds. Find its speed in km/h.

Options: 42.55 | 46.15 | 38.95 | 49.75

Answer: 42.55

Solution: Context: two-train dispatch. Speed = $130/11 = 11.82$ m/s = 42.55 km/h. Shortcut: Multiply m/s by 3.6.

Q324 ? Successive change ? Hard

A signal timing case uses the quantities in this question. Determine the requested result using the successive change model: A 140 m train crosses a pole in 12 seconds. Find its speed in km/h.

Options: 38.4 | 49.2 | 45.6 | 42

Answer: 42

Solution: Context: signal timing. Speed = $140/12 = 11.67$ m/s = 42 km/h. Shortcut: Multiply m/s by 3.6.

Q325 ? Comparison trap ? Placement

A freight corridor case uses the quantities in this question. Determine the requested result using the comparison trap model: A 150 m train crosses a pole in 13 seconds. Find its speed in km/h.

Options: 41.54 | 45.14 | 48.74 | 37.94

Answer: 41.54

Solution: Context: freight corridor. Speed = $150/13 = 11.54$ m/s = 41.54 km/h. Shortcut: Multiply m/s by 3.6.

Q326 ? Ratio form ? Beginner

A metro overtaking case uses the quantities in this question. Determine the requested result using the ratio form model: A 160 m train crosses a pole in 14 seconds. Find its speed in km/h.

Options: 48.34 | 44.74 | 41.14 | 37.54

Answer: 41.14

Solution: Context: metro overtaking. Speed = $160/14 = 11.43$ m/s = 41.14 km/h. Shortcut: Multiply m/s by 3.6.

Q327 ? Weighted result ? Easy

A track maintenance case uses the quantities in this question. Determine the requested result using the weighted result model: A 170 m train crosses a pole in 15 seconds. Find its speed in km/h.

Options: 37.2 | 40.8 | 44.4 | 48

Answer: 40.8

Solution: Context: track maintenance. Speed = $170/15 = 11.33$ m/s = 40.8 km/h. Shortcut: Multiply m/s by 3.6.

Q328 ? Replacement cycle ? Medium

A station approach case uses the quantities in this question. Determine the requested result using the replacement cycle model: A 180 m train crosses a pole in 8 seconds. Find its speed in km/h.

Options: 88.2 | 77.4 | 84.6 | 81

Answer: 81

Solution: Context: station approach. Speed = $180/8 = 22.5$ m/s = 81 km/h. Shortcut: Multiply m/s by 3.6.

Q329 ? Unit conversion ? Hard

A rail crossing case uses the quantities in this question. Determine the requested result using the unit conversion model: A 190 m train crosses a pole in 9 seconds. Find its speed in km/h.

Options: 76 | 79.6 | 72.4 | 83.2

Answer: 76

Solution: Context: rail crossing. Speed = $190/9 = 21.11$ m/s = 76 km/h. Shortcut: Multiply m/s by 3.6.

Q330 ? Work-rate model ? Placement

A platform clearance case uses the quantities in this question. Determine the requested result using the work-rate model model: A 200 m train crosses a pole in 10 seconds. Find its speed in km/h.

Options: 72 | 75.6 | 79.2 | 68.4

Answer: 72

Solution: Context: platform clearance. Speed = $200/10 = 20$ m/s = 72 km/h. Shortcut: Multiply m/s by 3.6.

Q331 ? Speed-time model ? Beginner

A two-train dispatch case uses the quantities in this question. Determine the requested result using the speed-time model model: A 210 m train crosses a pole in 11 seconds. Find its speed in km/h.

Options: 65.13 | 68.73 | 75.93 | 72.33

Answer: 68.73

Solution: Context: two-train dispatch. Speed = $210/11 = 19.09$ m/s = 68.73 km/h. Shortcut: Multiply m/s by 3.6.

Q332 ? Table lookup ? Easy

A signal timing case uses the quantities in this question. Determine the requested result using the table lookup model: A 220 m train crosses a pole in 12 seconds. Find its speed in km/h.

Options: 73.2 | 66 | 69.6 | 62.4

Answer: 66

Solution: Context: signal timing. Speed = $220/12 = 18.33$ m/s = 66 km/h. Shortcut: Multiply m/s by 3.6.

Q333 ? Chart comparison ? Medium

A freight corridor case uses the quantities in this question. Determine the requested result using the chart comparison model: A 230 m train crosses a pole in 13 seconds. Find its speed in km/h.

Options: 63.69 | 67.29 | 60.09 | 70.89

Answer: 63.69

Solution: Context: freight corridor. Speed = $230/13 = 17.69$ m/s = 63.69 km/h. Shortcut: Multiply m/s by 3.6.

Q334 ? Multi-step word problem ? Hard

A metro overtaking case uses the quantities in this question. Determine the requested result using the multi-step word problem model: A 240 m train crosses a pole in 14 seconds. Find its speed in km/h.

Options: 58.11 | 61.71 | 65.31 | 68.91

Answer: 61.71

Solution: Context: metro overtaking. Speed = $240/14 = 17.14$ m/s = 61.71 km/h. Shortcut: Multiply m/s by 3.6.

Q335 ? Maximum / minimum ? Placement

A track maintenance case uses the quantities in this question. Determine the requested result using the maximum / minimum model: A 250 m train crosses a pole in 15 seconds. Find its speed in km/h.

Options: 67.2 | 56.4 | 63.6 | 60

Answer: 60

Solution: Context: track maintenance. Speed = $250/15 = 16.67$ m/s = 60 km/h. Shortcut: Multiply m/s by 3.6.

Q336 ? Least / greatest value ? Beginner

A station approach case uses the quantities in this question. Determine the requested result using the least / greatest value model: A 260 m train crosses a pole in 8 seconds. Find its speed in km/h.

Options: 117 | 120.6 | 113.4 | 124.2

Answer: 117

Solution: Context: station approach. Speed = $260/8 = 32.5$ m/s = 117 km/h. Shortcut: Multiply m/s by 3.6.

Q337 ? Remainder cycle ? Easy

A rail crossing case uses the quantities in this question. Determine the requested result using the remainder cycle model: A 270 m train crosses a pole in 9 seconds. Find its speed in km/h.

Options: 108 | 111.6 | 104.4 | 115.2

Answer: 108

Solution: Context: rail crossing. Speed = $270/9 = 30$ m/s = 108 km/h. Shortcut: Multiply m/s by 3.6.

Q338 ? Divisibility test ? Medium

A platform clearance case uses the quantities in this question. Determine the requested result using the divisibility test model: A 280 m train crosses a pole in 10 seconds. Find its speed in km/h.

Options: 108 | 104.4 | 100.8 | 97.2

Answer: 100.8

Solution: Context: platform clearance. Speed = $280/10 = 28$ m/s = 100.8 km/h. Shortcut: Multiply m/s by 3.6.

Q339 ? Factorisation ? Hard

A two-train dispatch case uses the quantities in this question. Determine the requested result using the factorisation model: A 290 m train crosses a pole in 11 seconds. Find its speed in km/h.

Options: 102.11 | 91.31 | 98.51 | 94.91

Answer: 94.91

Solution: Context: two-train dispatch. Speed = $290/11 = 26.36$ m/s = 94.91 km/h. Shortcut: Multiply m/s by 3.6.

Q340 ? Approximation ? Placement

A signal timing case uses the quantities in this question. Determine the requested result using the approximation model: A 300 m train crosses a pole in 12 seconds. Find its speed in km/h.

Options: 90 | 93.6 | 86.4 | 97.2

Answer: 90

Solution: Context: signal timing. Speed = $300/12 = 25$ m/s = 90 km/h. Shortcut: Multiply m/s by 3.6.

PRACTICE ? solutions hidden below**Q341 ? Options elimination ? Beginner**

A freight corridor case uses the quantities in this question. Determine the requested result using the options elimination model: A 310 m train crosses a pole in

13 seconds. Find its speed in km/h.
Options: 85.85 | 89.45 | 82.25 | 93.05

Q342 ? Boundary case ? Easy

A metro overtaking case uses the quantities in this question. Determine the requested result using the boundary case model: A 320 m train crosses a pole in 14 seconds. Find its speed in km/h.
Options: 89.49 | 78.69 | 85.89 | 82.29

Q343 ? Case split ? Medium

A track maintenance case uses the quantities in this question. Determine the requested result using the case split model: A 330 m train crosses a pole in 15 seconds. Find its speed in km/h.
Options: 79.2 | 82.8 | 75.6 | 86.4

Q344 ? Contradiction check ? Hard

A station approach case uses the quantities in this question. Determine the requested result using the contradiction check model: A 340 m train crosses a pole in 8 seconds. Find its speed in km/h.
Options: 153 | 156.6 | 149.4 | 160.2

Q345 ? Statement I test ? Placement

A rail crossing case uses the quantities in this question. Determine the requested result using the statement i test model: A 350 m train crosses a pole in 9 seconds. Find its speed in km/h.
Options: 147.2 | 136.4 | 143.6 | 140

Q346 ? Statement II test ? Beginner

A platform clearance case uses the quantities in this question. Determine the requested result using the statement ii test model: A 360 m train crosses a pole in 10 seconds. Find its speed in km/h.
Options: 136.8 | 126 | 133.2 | 129.6

Q347 ? Combined statements ? Easy

A two-train dispatch case uses the quantities in this question. Determine the requested result using the combined statements model: A 370 m train crosses a pole in 11 seconds. Find its speed in km/h.
Options: 121.09 | 124.69 | 117.49 | 128.29

Q348 ? Counting restriction ? Medium

A signal timing case uses the quantities in this question. Determine the requested result using the counting restriction model: A 380 m train crosses a pole in 12 seconds. Find its speed in km/h.
Options: 114 | 117.6 | 110.4 | 121.2

Q349 ? Circular arrangement ? Hard

A freight corridor case uses the quantities in this question. Determine the requested result using the circular arrangement model: A 390 m train crosses a pole in 13 seconds. Find its speed in km/h.
Options: 115.2 | 104.4 | 111.6 | 108

Q350 ? Complement method ? Placement

A metro overtaking case uses the quantities in this question. Determine the requested result using the complement method model: A 400 m train crosses a pole in 14 seconds. Find its speed in km/h.
Options: 99.26 | 102.86 | 106.46 | 110.06

Q351 ? Pattern: differences ? Beginner

A track maintenance case uses the quantities in this question. Determine the requested result using the pattern: differences model: A 410 m train crosses a pole in 15 seconds. Find its speed in km/h.
Options: 98.4 | 102 | 94.8 | 105.6

Q352 ? Pattern: ratios ? Easy

A station approach case uses the quantities in this question. Determine the requested result using the pattern: ratios model: A 420 m train crosses a pole in 8 seconds. Find its speed in km/h.
Options: 196.2 | 185.4 | 192.6 | 189

Q353 ? Pattern: alternating ? Medium

A rail crossing case uses the quantities in this question. Determine the requested result using the pattern: alternating model: A 430 m train crosses a pole in 9 seconds. Find its speed in km/h.
Options: 179.2 | 168.4 | 175.6 | 172

Q354 ? Pattern: squares / cubes ? Hard

A platform clearance case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: A 440 m train crosses a pole in 10 seconds. Find its speed in km/h.
Options: 158.4 | 162 | 154.8 | 165.6

Q355 ? Inference from facts ? Placement

A two-train dispatch case uses the quantities in this question. Determine the requested result using the inference from facts model: A 450 m train crosses a pole in 11 seconds. Find its speed in km/h.
Options: 147.27 | 150.87 | 143.67 | 154.47

Q356 ? Assumption check ? Beginner

A signal timing case uses the quantities in this question. Determine the requested result using the assumption check model: A 460 m train crosses a pole in 12 seconds. Find its speed in km/h.
Options: 145.2 | 134.4 | 141.6 | 138

Q357 ? Cause vs correlation ? Easy

A freight corridor case uses the quantities in this question. Determine the requested result using the cause vs correlation model: A 470 m train crosses a pole in 13 seconds. Find its speed in km/h.

Options: 130.15 | 133.75 | 126.55 | 137.35

Q358 ? Grammar agreement ? Medium

A metro overtaking case uses the quantities in this question. Determine the requested result using the grammar agreement model: A 480 m train crosses a pole in 14 seconds. Find its speed in km/h.

Options: 123.43 | 127.03 | 119.83 | 130.63

Q359 ? Vocabulary in context ? Hard

A track maintenance case uses the quantities in this question. Determine the requested result using the vocabulary in context model: A 490 m train crosses a pole in 15 seconds. Find its speed in km/h.

Options: 124.8 | 114 | 121.2 | 117.6

Q360 ? Mixed placement challenge ? Placement

A station approach case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: A 500 m train crosses a pole in 8 seconds. Find its speed in km/h.

Options: 232.2 | 221.4 | 228.6 | 225

Problems on Ages ? 40 questions

WORKED EXAMPLES

Q361 ? Direct computation ? Beginner

A family tree case uses the quantities in this question. Determine the requested result using the direct computation model: A father is 3 times his son's age. If the son is 9, find the father's age.

Options: 30 | 33 | 24 | 27

Answer: 27

Solution: Context: family tree. Father's age = $3 \times 9 = 27$ years. Shortcut: Check whether ages are present or shifted.

Q362 ? Reverse value ? Easy

A career timeline case uses the quantities in this question. Determine the requested result using the reverse value model: A father is 4 times his son's age. If the son is 10, find the father's age.

Options: 48 | 36 | 44 | 40

Answer: 40

Solution: Context: career timeline. Father's age = $4 \times 10 = 40$ years. Shortcut: Check whether ages are present or shifted.

Q363 ? Missing base ? Medium

A parent-child interview case uses the quantities in this question. Determine the requested result using the missing base model: A father is 5 times his son's age. If the son is 11, find the father's age.

Options: 67 | 49 | 61 | 55

Answer: 55

Solution: Context: parent-child interview. Father's age = $5 \times 11 = 55$ years. Shortcut: Check whether ages are present or shifted.

Q364 ? Successive change ? Hard

A birth-year audit case uses the quantities in this question. Determine the requested result using the successive change model: A father is 2 times his son's age. If the son is 12, find the father's age.

Options: 24 | 26 | 22 | 28

Answer: 24

Solution: Context: birth-year audit. Father's age = $2 \times 12 = 24$ years. Shortcut: Check whether ages are present or shifted.

Q365 ? Comparison trap ? Placement

A generation puzzle case uses the quantities in this question. Determine the requested result using the comparison trap model: A father is 3 times his son's age. If the son is 13, find the father's age.

Options: 43 | 47 | 35 | 39

Answer: 39

Solution: Context: generation puzzle. Father's age = $3 \times 13 = 39$ years. Shortcut: Check whether ages are present or shifted.

Q366 ? Ratio form ? Beginner

A team seniority case uses the quantities in this question. Determine the requested result using the ratio form model: A father is 4 times his son's age. If the son is 14, find the father's age.

Options: 62 | 68 | 50 | 56

Answer: 56

Solution: Context: team seniority. Father's age = $4 \times 14 = 56$ years. Shortcut: Check whether ages are present or shifted.

Q367 ? Weighted result ? Easy

A historic record case uses the quantities in this question. Determine the requested result using the weighted result model: A father is 5 times his son's age. If the son is 15, find the father's age.

Options: 83 | 91 | 67 | 75

Answer: 75

Solution: Context: historic record. Father's age = $5 \times 15 = 75$ years. Shortcut: Check whether ages are present or shifted.

Q368 ? Replacement cycle ? Medium

An age-ratio case uses the quantities in this question. Determine the requested result using the replacement cycle model: A father is 2 times his son's age. If the son is 16, find the father's age.

Options: 32 | 35 | 38 | 29

Answer: 32

Solution: Context: age-ratio case. Father's age = $2 \times 16 = 32$ years. Shortcut: Check whether ages are present or shifted.

Q369 ? Unit conversion ? Hard

A family tree case uses the quantities in this question. Determine the requested result using the unit conversion model: A father is 3 times his son's age. If the son is 17, find the father's age.

Options: 51 | 56 | 61 | 46

Answer: 51

Solution: Context: family tree. Father's age = $3 \times 17 = 51$ years. Shortcut: Check whether ages are present or shifted.

Q370 ? Work-rate model ? Placement

A career timeline case uses the quantities in this question. Determine the requested result using the work-rate model model: A father is 4 times his son's age. If the son is 18, find the father's age.

Options: 72 | 79 | 86 | 65

Answer: 72

Solution: Context: career timeline. Father's age = $4 \times 18 = 72$ years. Shortcut: Check whether ages are present or shifted.

Q371 ? Speed-time model ? Beginner

A parent-child interview case uses the quantities in this question. Determine the requested result using the speed-time model model: A father is 5 times his son's age. If the son is 19, find the father's age.

Options: 95 | 115 | 85 | 105

Answer: 95

Solution: Context: parent-child interview. Father's age = $5 \times 19 = 95$ years. Shortcut: Check whether ages are present or shifted.

Q372 ? Table lookup ? Easy

A birth-year audit case uses the quantities in this question. Determine the requested result using the table lookup model: A father is 2 times his son's age. If the son is 20, find the father's age.

Options: 40 | 44 | 48 | 36

Answer: 40

Solution: Context: birth-year audit. Father's age = $2 \times 20 = 40$ years. Shortcut: Check whether ages are present or shifted.

Q373 ? Chart comparison ? Medium

A generation puzzle case uses the quantities in this question. Determine the requested result using the chart comparison model: A father is 3 times his son's age. If the son is 21, find the father's age.

Options: 63 | 69 | 75 | 57

Answer: 63

Solution: Context: generation puzzle. Father's age = $3 \times 21 = 63$ years. Shortcut: Check whether ages are present or shifted.

Q374 ? Multi-step word problem ? Hard

A team seniority case uses the quantities in this question. Determine the requested result using the multi-step word problem model: A father is 4 times his son's age. If the son is 22, find the father's age.

Options: 88 | 106 | 97 | 79

Answer: 88

Solution: Context: team seniority. Father's age = $4 \times 22 = 88$ years. Shortcut: Check whether ages are present or shifted.

Q375 ? Maximum / minimum ? Placement

A historic record case uses the quantities in this question. Determine the requested result using the maximum / minimum model: A father is 5 times his son's age. If the son is 23, find the father's age.

Options: 139 | 103 | 127 | 115

Answer: 115

Solution: Context: historic record. Father's age = $5 \times 23 = 115$ years. Shortcut: Check whether ages are present or shifted.

Q376 ? Least / greatest value ? Beginner

A age-ratio case case uses the quantities in this question. Determine the requested result using the least / greatest value model: A father is 2 times his son's age. If the son is 24, find the father's age.

Options: 53 | 58 | 43 | 48

Answer: 48

Solution: Context: age-ratio case. Father's age = $2 \times 24 = 48$ years. Shortcut: Check whether ages are present or shifted.

Q377 ? Remainder cycle ? Easy

A family tree case uses the quantities in this question. Determine the requested result using the remainder cycle model: A father is 3 times his son's age. If the son is 25, find the father's age.

Options: 75 | 91 | 67 | 83

Answer: 75

Solution: Context: family tree. Father's age = $3 \times 25 = 75$ years. Shortcut: Check whether ages are present or shifted.

Q378 ? Divisibility test ? Medium

A career timeline case uses the quantities in this question. Determine the requested result using the divisibility test model: A father is 4 times his son's age. If the son is 26, find the father's age.

Options: 124 | 114 | 104 | 94

Answer: 104

Solution: Context: career timeline. Father's age = $4 \times 26 = 104$ years. Shortcut: Check whether ages are present or shifted.

Q379 ? Factorisation ? Hard

A parent-child interview case uses the quantities in this question. Determine the requested result using the factorisation model: A father is 5 times his son's age. If the son is 27, find the father's age.

Options: 163 | 121 | 149 | 135

Answer: 135

Solution: Context: parent-child interview. Father's age = $5 \times 27 = 135$ years. Shortcut: Check whether ages are present or shifted.

Q380 ? Approximation ? Placement

A birth-year audit case uses the quantities in this question. Determine the requested result using the approximation model: A father is 2 times his son's age. If the son is 28, find the father's age.

Options: 62 | 68 | 50 | 56

Answer: 56

Solution: Context: birth-year audit. Father's age = $2 \times 28 = 56$ years. Shortcut: Check whether ages are present or shifted.

PRACTICE ? solutions hidden below**Q381 ? Options elimination ? Beginner**

A generation puzzle case uses the quantities in this question. Determine the requested result using the options elimination model: A father is 3 times his son's

age. If the son is 29, find the father's age.

Options: 87 | 105 | 96 | 78

Q382 ? Boundary case ? Easy

A team seniority case uses the quantities in this question. Determine the requested result using the boundary case model: A father is 4 times his son's age. If the son is 30, find the father's age.

Options: 144 | 108 | 132 | 120

Q383 ? Case split ? Medium

A historic record case uses the quantities in this question. Determine the requested result using the case split model: A father is 5 times his son's age. If the son is 31, find the father's age.

Options: 155 | 171 | 139 | 187

Q384 ? Contradiction check ? Hard

A age-ratio case case uses the quantities in this question. Determine the requested result using the contradiction check model: A father is 2 times his son's age. If the son is 32, find the father's age.

Options: 70 | 76 | 64 | 58

Q385 ? Statement I test ? Placement

A family tree case uses the quantities in this question. Determine the requested result using the statement i test model: A father is 3 times his son's age. If the son is 33, find the father's age.

Options: 99 | 119 | 89 | 109

Q386 ? Statement II test ? Beginner

A career timeline case uses the quantities in this question. Determine the requested result using the statement ii test model: A father is 4 times his son's age. If the son is 34, find the father's age.

Options: 164 | 122 | 150 | 136

Q387 ? Combined statements ? Easy

A parent-child interview case uses the quantities in this question. Determine the requested result using the combined statements model: A father is 5 times his son's age. If the son is 35, find the father's age.

Options: 175 | 193 | 157 | 211

Q388 ? Counting restriction ? Medium

A birth-year audit case uses the quantities in this question. Determine the requested result using the counting restriction model: A father is 2 times his son's age. If the son is 36, find the father's age.

Options: 86 | 65 | 72 | 79

Q389 ? Circular arrangement ? Hard

A generation puzzle case uses the quantities in this question. Determine the requested result using the circular arrangement model: A father is 3 times his son's age. If the son is 37, find the father's age.

Options: 133 | 100 | 122 | 111

Q390 ? Complement method ? Placement

A team seniority case uses the quantities in this question. Determine the requested result using the complement method model: A father is 4 times his son's age. If the son is 38, find the father's age.

Options: 152 | 167 | 137 | 182

Q391 ? Pattern: differences ? Beginner

A historic record case uses the quantities in this question. Determine the requested result using the pattern: differences model: A father is 5 times his son's age. If the son is 39, find the father's age.

Options: 195 | 215 | 175 | 235

Q392 ? Pattern: ratios ? Easy

A age-ratio case case uses the quantities in this question. Determine the requested result using the pattern: ratios model: A father is 2 times his son's age. If the son is 40, find the father's age.

Options: 72 | 88 | 80 | 96

Q393 ? Pattern: alternating ? Medium

A family tree case uses the quantities in this question. Determine the requested result using the pattern: alternating model: A father is 3 times his son's age. If the son is 41, find the father's age.

Options: 147 | 111 | 135 | 123

Q394 ? Pattern: squares / cubes ? Hard

A career timeline case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: A father is 4 times his son's age. If the son is 42, find the father's age.

Options: 168 | 185 | 151 | 202

Q395 ? Inference from facts ? Placement

A parent-child interview case uses the quantities in this question. Determine the requested result using the inference from facts model: A father is 5 times his son's age. If the son is 43, find the father's age.

Options: 193 | 215 | 237 | 259

Q396 ? Assumption check ? Beginner

A birth-year audit case uses the quantities in this question. Determine the requested result using the assumption check model: A father is 2 times his son's age. If the son is 44, find the father's age.

Options: 106 | 79 | 97 | 88

Q397 ? Cause vs correlation ? Easy

A generation puzzle case uses the quantities in this question. Determine the requested result using the cause vs correlation model: A father is 3 times his son's age. If the son is 45, find the father's age.

Options: 135 | 149 | 121 | 163

Q398 ? Grammar agreement ? Medium

A team seniority case uses the quantities in this question. Determine the requested result using the grammar agreement model: A father is 4 times his son's age. If the son is 46, find the father's age.

Options: 184 | 202 | 166 | 220

Q399 ? Vocabulary in context ? Hard

A historic record case uses the quantities in this question. Determine the requested result using the vocabulary in context model: A father is 5 times his son's age. If the son is 47, find the father's age.

Options: 235 | 259 | 211 | 283

Q400 ? Mixed placement challenge ? Placement

A age-ratio case case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: A father is 2 times his son's age. If the son is 48, find the father's age.

Options: 116 | 86 | 106 | 96

Probability ? 40 questions

WORKED EXAMPLES

Q401 ? Direct computation ? Beginner

A quality inspection case uses the quantities in this question. Determine the requested result using the direct computation model: A bag has 3 red and 5 blue balls. What is the probability of drawing red?

Options: $\frac{3}{8}$ | $\frac{5}{8}$ | $\frac{3}{5}$ | $\frac{1}{8}$

Answer: $\frac{3}{8}$

Solution: Context: quality inspection. Favourable = 3; total = 8. $P = \frac{3}{8} = \frac{3}{8}$. Shortcut: Simplify the final fraction.

Q402 ? Reverse value ? Easy

A card draw case uses the quantities in this question. Determine the requested result using the reverse value model: A bag has 4 red and 7 blue balls. What is the probability of drawing red?

Options: $\frac{7}{11}$ | $\frac{4}{7}$ | $\frac{4}{11}$ | $\frac{1}{11}$

Answer: $\frac{4}{11}$

Solution: Context: card draw. Favourable = 4; total = 11. $P = \frac{4}{11} = \frac{4}{11}$. Shortcut: Simplify the final fraction.

Q403 ? Missing base ? Medium

A lottery selection case uses the quantities in this question. Determine the requested result using the missing base model: A bag has 5 red and 9 blue balls. What is the probability of drawing red?

Options: $\frac{9}{14}$ | $\frac{1}{14}$ | $\frac{5}{14}$ | $\frac{5}{9}$

Answer: $\frac{5}{14}$

Solution: Context: lottery selection. Favourable = 5; total = 14. $P = \frac{5}{14} = \frac{5}{14}$. Shortcut: Simplify the final fraction.

Q404 ? Successive change ? Hard

A server failure case uses the quantities in this question. Determine the requested result using the successive change model: A bag has 6 red and 4 blue balls. What is the probability of drawing red?

Options: $\frac{3}{5}$ | $\frac{4}{10}$ | $\frac{6}{4}$ | $\frac{1}{10}$

Answer: $\frac{3}{5}$

Solution: Context: server failure. Favourable = 6; total = 10. $P = \frac{6}{10} = \frac{3}{5}$. Shortcut: Simplify the final fraction.

Q405 ? Comparison trap ? Placement

A candidate choice case uses the quantities in this question. Determine the requested result using the comparison trap model: A bag has 7 red and 6 blue balls. What is the probability of drawing red?

Options: $\frac{1}{13}$ | $\frac{7}{6}$ | $\frac{7}{13}$ | $\frac{6}{13}$

Answer: $\frac{7}{13}$

Solution: Context: candidate choice. Favourable = 7; total = 13. $P = \frac{7}{13} = \frac{7}{13}$. Shortcut: Simplify the final fraction.

Q406 ? Ratio form ? Beginner

A medical screening case uses the quantities in this question. Determine the requested result using the ratio form model: A bag has 2 red and 8 blue balls. What is the probability of drawing red?

Options: $\frac{1}{5}$ | $\frac{8}{10}$ | $\frac{2}{8}$ | $\frac{1}{10}$

Answer: $\frac{1}{5}$

Solution: Context: medical screening. Favourable = 2; total = 10. $P = \frac{2}{10} = \frac{1}{5}$. Shortcut: Simplify the final fraction.

Q407 ? Weighted result ? Easy

A urn experiment case uses the quantities in this question. Determine the requested result using the weighted result model: A bag has 3 red and 3 blue balls. What is the probability of drawing red?

Options: $\frac{1}{2}$ | $\frac{1}{6}$ | $\frac{3}{6}$ | $\frac{3}{3}$

Answer: $\frac{1}{2}$

Solution: Context: urn experiment. Favourable = 3; total = 6. $P = \frac{3}{6} = \frac{1}{2}$. Shortcut: Simplify the final fraction.

Q408 ? Replacement cycle ? Medium

A risk estimate case uses the quantities in this question. Determine the requested result using the replacement cycle model: A bag has 4 red and 5 blue balls. What is the probability of drawing red?

Options: $\frac{5}{9}$ | $\frac{4}{9}$ | $\frac{4}{5}$ | $\frac{1}{9}$

Answer: $\frac{4}{9}$

Solution: Context: risk estimate. Favourable = 4; total = 9. $P = \frac{4}{9} = \frac{4}{9}$. Shortcut: Simplify the final fraction.

Q409 ? Unit conversion ? Hard

A quality inspection case uses the quantities in this question. Determine the requested result using the unit conversion model: A bag has 5 red and 7 blue balls. What is the probability of drawing red?

Options: $\frac{5}{12}$ | $\frac{7}{12}$ | $\frac{5}{7}$ | $\frac{1}{12}$

Answer: $\frac{5}{12}$

Solution: Context: quality inspection. Favourable = 5; total = 12. $P = \frac{5}{12} = \frac{5}{12}$. Shortcut: Simplify the final fraction.

Q410 ? Work-rate model ? Placement

A card draw case uses the quantities in this question. Determine the requested result using the work-rate model model: A bag has 6 red and 9 blue balls. What is the probability of drawing red?

Options: $\frac{6}{9}$ | $\frac{2}{5}$ | $\frac{9}{15}$ | $\frac{1}{15}$

Answer: $\frac{2}{5}$

Solution: Context: card draw. Favourable = 6; total = 15. $P = \frac{6}{15} = \frac{2}{5}$. Shortcut: Simplify the final fraction.

Q411 ? Speed-time model ? Beginner

A lottery selection case uses the quantities in this question. Determine the requested result using the speed-time model model: A bag has 7 red and 4 blue balls. What is the probability of drawing red?

Options: 4/11 | 7/4 | 7/11 | 1/11

Answer: 7/11

Solution: Context: lottery selection. Favourable = 7; total = 11. $P = 7/11 = 7/11$. Shortcut: Simplify the final fraction.

Q412 ? Table lookup ? Easy

A server failure case uses the quantities in this question. Determine the requested result using the table lookup model: A bag has 2 red and 6 blue balls. What is the probability of drawing red?

Options: 1/8 | 1/4 | 6/8 | 2/6

Answer: 1/4

Solution: Context: server failure. Favourable = 2; total = 8. $P = 2/8 = 1/4$. Shortcut: Simplify the final fraction.

Q413 ? Chart comparison ? Medium

A candidate choice case uses the quantities in this question. Determine the requested result using the chart comparison model: A bag has 3 red and 8 blue balls. What is the probability of drawing red?

Options: 3/11 | 8/11 | 3/8 | 1/11

Answer: 3/11

Solution: Context: candidate choice. Favourable = 3; total = 11. $P = 3/11 = 3/11$. Shortcut: Simplify the final fraction.

Q414 ? Multi-step word problem ? Hard

A medical screening case uses the quantities in this question. Determine the requested result using the multi-step word problem model: A bag has 4 red and 3 blue balls. What is the probability of drawing red?

Options: 1/7 | 3/7 | 4/7 | 4/3

Answer: 4/7

Solution: Context: medical screening. Favourable = 4; total = 7. $P = 4/7 = 4/7$. Shortcut: Simplify the final fraction.

Q415 ? Maximum / minimum ? Placement

A urn experiment case uses the quantities in this question. Determine the requested result using the maximum / minimum model: A bag has 5 red and 5 blue balls. What is the probability of drawing red?

Options: 5/5 | 5/10 | 1/10 | 1/2

Answer: 1/2

Solution: Context: urn experiment. Favourable = 5; total = 10. $P = 5/10 = 1/2$. Shortcut: Simplify the final fraction.

Q416 ? Least / greatest value ? Beginner

A risk estimate case uses the quantities in this question. Determine the requested result using the least / greatest value model: A bag has 6 red and 7 blue balls. What is the probability of drawing red?

Options: 6/13 | 7/13 | 6/7 | 1/13

Answer: 6/13

Solution: Context: risk estimate. Favourable = 6; total = 13. $P = 6/13 = 6/13$. Shortcut: Simplify the final fraction.

Q417 ? Remainder cycle ? Easy

A quality inspection case uses the quantities in this question. Determine the requested result using the remainder cycle model: A bag has 7 red and 9 blue balls. What is the probability of drawing red?

Options: 7/16 | 7/9 | 9/16 | 1/16

Answer: 7/16

Solution: Context: quality inspection. Favourable = 7; total = 16. $P = 7/16 = 7/16$. Shortcut: Simplify the final fraction.

Q418 ? Divisibility test ? Medium

A card draw case uses the quantities in this question. Determine the requested result using the divisibility test model: A bag has 2 red and 4 blue balls. What is the probability of drawing red?

Options: 4/6 | 1/6 | 1/3 | 2/4

Answer: 1/3

Solution: Context: card draw. Favourable = 2; total = 6. $P = 2/6 = 1/3$. Shortcut: Simplify the final fraction.

Q419 ? Factorisation ? Hard

A lottery selection case uses the quantities in this question. Determine the requested result using the factorisation model: A bag has 3 red and 6 blue balls. What is the probability of drawing red?

Options: 1/9 | 1/3 | 6/9 | 3/6

Answer: 1/3

Solution: Context: lottery selection. Favourable = 3; total = 9. $P = 3/9 = 1/3$. Shortcut: Simplify the final fraction.

Q420 ? Approximation ? Placement

A server failure case uses the quantities in this question. Determine the requested result using the approximation model: A bag has 4 red and 8 blue balls. What is the probability of drawing red?

Options: 4/8 | 1/3 | 8/12 | 1/12

Answer: 1/3

Solution: Context: server failure. Favourable = 4; total = 12. $P = 4/12 = 1/3$. Shortcut: Simplify the final fraction.

PRACTICE ? solutions hidden below**Q421 ? Options elimination ? Beginner**

A candidate choice case uses the quantities in this question. Determine the requested result using the options elimination model: A bag has 5 red and 3 blue

balls. What is the probability of drawing red?
Options: $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{5}{8}$ | $\frac{5}{3}$

Q422 ? Boundary case ? Easy

A medical screening case uses the quantities in this question. Determine the requested result using the boundary case model: A bag has 6 red and 5 blue balls. What is the probability of drawing red?
Options: $\frac{6}{5}$ | $\frac{5}{11}$ | $\frac{6}{11}$ | $\frac{1}{11}$

Q423 ? Case split ? Medium

A urn experiment case uses the quantities in this question. Determine the requested result using the case split model: A bag has 7 red and 7 blue balls. What is the probability of drawing red?
Options: $\frac{1}{2}$ | $\frac{7}{14}$ | $\frac{7}{7}$ | $\frac{1}{14}$

Q424 ? Contradiction check ? Hard

A risk estimate case uses the quantities in this question. Determine the requested result using the contradiction check model: A bag has 2 red and 9 blue balls. What is the probability of drawing red?
Options: $\frac{2}{11}$ | $\frac{9}{11}$ | $\frac{2}{9}$ | $\frac{1}{11}$

Q425 ? Statement I test ? Placement

A quality inspection case uses the quantities in this question. Determine the requested result using the statement i test model: A bag has 3 red and 4 blue balls. What is the probability of drawing red?
Options: $\frac{3}{4}$ | $\frac{3}{7}$ | $\frac{4}{7}$ | $\frac{1}{7}$

Q426 ? Statement II test ? Beginner

A card draw case uses the quantities in this question. Determine the requested result using the statement ii test model: A bag has 4 red and 6 blue balls. What is the probability of drawing red?
Options: $\frac{2}{5}$ | $\frac{6}{10}$ | $\frac{4}{6}$ | $\frac{1}{10}$

Q427 ? Combined statements ? Easy

A lottery selection case uses the quantities in this question. Determine the requested result using the combined statements model: A bag has 5 red and 8 blue balls. What is the probability of drawing red?
Options: $\frac{1}{13}$ | $\frac{5}{8}$ | $\frac{8}{13}$ | $\frac{5}{13}$

Q428 ? Counting restriction ? Medium

A server failure case uses the quantities in this question. Determine the requested result using the counting restriction model: A bag has 6 red and 3 blue balls. What is the probability of drawing red?
Options: $\frac{1}{9}$ | $\frac{2}{3}$ | $\frac{3}{9}$ | $\frac{6}{3}$

Q429 ? Circular arrangement ? Hard

A candidate choice case uses the quantities in this question. Determine the requested result using the circular arrangement model: A bag has 7 red and 5 blue balls. What is the probability of drawing red?
Options: $\frac{5}{12}$ | $\frac{1}{12}$ | $\frac{7}{5}$ | $\frac{7}{12}$

Q430 ? Complement method ? Placement

A medical screening case uses the quantities in this question. Determine the requested result using the complement method model: A bag has 2 red and 7 blue balls. What is the probability of drawing red?
Options: $\frac{2}{7}$ | $\frac{2}{9}$ | $\frac{7}{9}$ | $\frac{1}{9}$

Q431 ? Pattern: differences ? Beginner

A urn experiment case uses the quantities in this question. Determine the requested result using the pattern: differences model: A bag has 3 red and 9 blue balls. What is the probability of drawing red?
Options: $\frac{1}{4}$ | $\frac{9}{12}$ | $\frac{1}{12}$ | $\frac{3}{9}$

Q432 ? Pattern: ratios ? Easy

A risk estimate case uses the quantities in this question. Determine the requested result using the pattern: ratios model: A bag has 4 red and 4 blue balls. What is the probability of drawing red?
Options: $\frac{4}{4}$ | $\frac{4}{8}$ | $\frac{1}{8}$ | $\frac{1}{2}$

Q433 ? Pattern: alternating ? Medium

A quality inspection case uses the quantities in this question. Determine the requested result using the pattern: alternating model: A bag has 5 red and 6 blue balls. What is the probability of drawing red?
Options: $\frac{5}{11}$ | $\frac{6}{11}$ | $\frac{5}{6}$ | $\frac{1}{11}$

Q434 ? Pattern: squares / cubes ? Hard

A card draw case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: A bag has 6 red and 8 blue balls. What is the probability of drawing red?
Options: $\frac{3}{7}$ | $\frac{8}{14}$ | $\frac{6}{8}$ | $\frac{1}{14}$

Q435 ? Inference from facts ? Placement

A lottery selection case uses the quantities in this question. Determine the requested result using the inference from facts model: A bag has 7 red and 3 blue balls. What is the probability of drawing red?
Options: $\frac{1}{10}$ | $\frac{3}{10}$ | $\frac{7}{10}$ | $\frac{7}{3}$

Q436 ? Assumption check ? Beginner

A server failure case uses the quantities in this question. Determine the requested result using the assumption check model: A bag has 2 red and 5 blue balls. What is the probability of drawing red?
Options: $\frac{5}{7}$ | $\frac{1}{7}$ | $\frac{2}{5}$ | $\frac{2}{7}$

Q437 ? Cause vs correlation ? Easy

A candidate choice case uses the quantities in this question. Determine the requested result using the cause vs correlation model: A bag has 3 red and 7 blue balls. What is the probability of drawing red?

Options: $\frac{3}{7}$ | $\frac{3}{10}$ | $\frac{7}{10}$ | $\frac{1}{10}$

Q438 ? Grammar agreement ? Medium

A medical screening case uses the quantities in this question. Determine the requested result using the grammar agreement model: A bag has 4 red and 9 blue balls. What is the probability of drawing red?

Options: $\frac{9}{13}$ | $\frac{1}{13}$ | $\frac{4}{13}$ | $\frac{4}{9}$

Q439 ? Vocabulary in context ? Hard

A urn experiment case uses the quantities in this question. Determine the requested result using the vocabulary in context model: A bag has 5 red and 4 blue balls. What is the probability of drawing red?

Options: $\frac{4}{9}$ | $\frac{5}{4}$ | $\frac{5}{9}$ | $\frac{1}{9}$

Q440 ? Mixed placement challenge ? Placement

A risk estimate case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: A bag has 6 red and 6 blue balls. What is the probability of drawing red?

Options: $\frac{1}{12}$ | $\frac{1}{2}$ | $\frac{6}{12}$ | $\frac{6}{6}$

HCF & LCM ? 40 questions

WORKED EXAMPLES

Q441 ? Direct computation ? Beginner

A maintenance cycle case uses the quantities in this question. Determine the requested result using the direct computation model: Find the HCF of 14 and 21.

Options: 6 | 9 | 8 | 7

Answer: 7

Solution: Context: maintenance cycle. By prime factors or Euclid's method, $HCF(14, 21) = 7$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q442 ? Reverse value ? Easy

A bell schedule case uses the quantities in this question. Determine the requested result using the reverse value model: Find the HCF of 16 and 24.

Options: 9 | 8 | 7 | 10

Answer: 8

Solution: Context: bell schedule. By prime factors or Euclid's method, $HCF(16, 24) = 8$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q443 ? Missing base ? Medium

A tile packing case uses the quantities in this question. Determine the requested result using the missing base model: Find the HCF of 18 and 27.

Options: 9 | 10 | 8 | 11

Answer: 9

Solution: Context: tile packing. By prime factors or Euclid's method, $HCF(18, 27) = 9$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q444 ? Successive change ? Hard

A gear rotation case uses the quantities in this question. Determine the requested result using the successive change model: Find the HCF of 20 and 30.

Options: 12 | 11 | 10 | 9

Answer: 10

Solution: Context: gear rotation. By prime factors or Euclid's method, $HCF(20, 30) = 10$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q445 ? Comparison trap ? Placement

A batch size case uses the quantities in this question. Determine the requested result using the comparison trap model: Find the HCF of 22 and 33.

Options: 13 | 10 | 12 | 11

Answer: 11

Solution: Context: batch size. By prime factors or Euclid's method, $HCF(22, 33) = 11$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q446 ? Ratio form ? Beginner

A traffic signals case uses the quantities in this question. Determine the requested result using the ratio form model: Find the HCF of 24 and 36.

Options: 12 | 13 | 11 | 14

Answer: 12

Solution: Context: traffic signals. By prime factors or Euclid's method, $HCF(24, 36) = 12$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q447 ? Weighted result ? Easy

A medicine dosage case uses the quantities in this question. Determine the requested result using the weighted result model: Find the HCF of 26 and 39.

Options: 13 | 14 | 12 | 15

Answer: 13

Solution: Context: medicine dosage. By prime factors or Euclid's method, $HCF(26, 39) = 13$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q448 ? Replacement cycle ? Medium

A event recurrence case uses the quantities in this question. Determine the requested result using the replacement cycle model: Find the HCF of 28 and 42.

Options: 16 | 13 | 15 | 14

Answer: 14

Solution: Context: event recurrence. By prime factors or Euclid's method, $HCF(28, 42) = 14$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q449 ? Unit conversion ? Hard

A maintenance cycle case uses the quantities in this question. Determine the requested result using the unit conversion model: Find the HCF of 30 and 45.

Options: 15 | 17 | 13 | 19

Answer: 15

Solution: Context: maintenance cycle. By prime factors or Euclid's method, $HCF(30, 45) = 15$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q450 ? Work-rate model ? Placement

A bell schedule case uses the quantities in this question. Determine the requested result using the work-rate model model: Find the HCF of 32 and 48.

Options: 16 | 18 | 14 | 20

Answer: 16

Solution: Context: bell schedule. By prime factors or Euclid's method, $HCF(32, 48) = 16$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q451 ? Speed-time model ? Beginner

A tile packing case uses the quantities in this question. Determine the requested result using the speed-time model model: Find the HCF of 34 and 51.

Options: 15 | 19 | 17 | 21

Answer: 17

Solution: Context: tile packing. By prime factors or Euclid's method, $HCF(34, 51) = 17$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q452 ? Table lookup ? Easy

A gear rotation case uses the quantities in this question. Determine the requested result using the table lookup model: Find the HCF of 36 and 54.

Options: 22 | 16 | 20 | 18

Answer: 18

Solution: Context: gear rotation. By prime factors or Euclid's method, $\text{HCF}(36, 54) = 18$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q453 ? Chart comparison ? Medium

A batch size case uses the quantities in this question. Determine the requested result using the chart comparison model: Find the HCF of 38 and 57.

Options: 19 | 21 | 17 | 23

Answer: 19

Solution: Context: batch size. By prime factors or Euclid's method, $\text{HCF}(38, 57) = 19$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q454 ? Multi-step word problem ? Hard

A traffic signals case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Find the HCF of 40 and 60.

Options: 18 | 20 | 22 | 24

Answer: 20

Solution: Context: traffic signals. By prime factors or Euclid's method, $\text{HCF}(40, 60) = 20$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q455 ? Maximum / minimum ? Placement

A medicine dosage case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Find the HCF of 42 and 63.

Options: 25 | 19 | 23 | 21

Answer: 21

Solution: Context: medicine dosage. By prime factors or Euclid's method, $\text{HCF}(42, 63) = 21$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q456 ? Least / greatest value ? Beginner

A event recurrence case uses the quantities in this question. Determine the requested result using the least / greatest value model: Find the HCF of 44 and 66.

Options: 26 | 20 | 24 | 22

Answer: 22

Solution: Context: event recurrence. By prime factors or Euclid's method, $\text{HCF}(44, 66) = 22$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q457 ? Remainder cycle ? Easy

A maintenance cycle case uses the quantities in this question. Determine the requested result using the remainder cycle model: Find the HCF of 46 and 69.

Options: 23 | 25 | 21 | 27

Answer: 23

Solution: Context: maintenance cycle. By prime factors or Euclid's method, $\text{HCF}(46, 69) = 23$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q458 ? Divisibility test ? Medium

A bell schedule case uses the quantities in this question. Determine the requested result using the divisibility test model: Find the HCF of 48 and 72.

Options: 24 | 26 | 22 | 28

Answer: 24

Solution: Context: bell schedule. By prime factors or Euclid's method, $\text{HCF}(48, 72) = 24$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q459 ? Factorisation ? Hard

A tile packing case uses the quantities in this question. Determine the requested result using the factorisation model: Find the HCF of 50 and 75.

Options: 31 | 22 | 28 | 25

Answer: 25

Solution: Context: tile packing. By prime factors or Euclid's method, $\text{HCF}(50, 75) = 25$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

Q460 ? Approximation ? Placement

A gear rotation case uses the quantities in this question. Determine the requested result using the approximation model: Find the HCF of 52 and 78.

Options: 26 | 29 | 23 | 32

Answer: 26

Solution: Context: gear rotation. By prime factors or Euclid's method, $\text{HCF}(52, 78) = 26$. Shortcut: Euclid repeatedly replaces the larger number by the remainder.

PRACTICE ? solutions hidden below**Q461 ? Options elimination ? Beginner**

A batch size case uses the quantities in this question. Determine the requested result using the options elimination model: Find the HCF of 54 and 81.

Options: 27 | 24 | 30 | 33

Q462 ? Boundary case ? Easy

A traffic signals case uses the quantities in this question. Determine the requested result using the boundary case model: Find the HCF of 56 and 84.

Options: 31 | 34 | 25 | 28

Q463 ? Case split ? Medium

A medicine dosage case uses the quantities in this question. Determine the requested result using the case split model: Find the HCF of 58 and 87.

Options: 35 | 26 | 32 | 29

Q464 ? Contradiction check ? Hard

A event recurrence case uses the quantities in this question. Determine the requested result using the contradiction check model: Find the HCF of 60 and 90.

Options: 30 | 33 | 27 | 36

Q465 ? Statement I test ? Placement

A maintenance cycle case uses the quantities in this question. Determine the requested result using the statement i test model: Find the HCF of 62 and 93.
Options: 37 | 34 | 31 | 28

Q466 ? Statement II test ? Beginner

A bell schedule case uses the quantities in this question. Determine the requested result using the statement ii test model: Find the HCF of 64 and 96.
Options: 38 | 29 | 35 | 32

Q467 ? Combined statements ? Easy

A tile packing case uses the quantities in this question. Determine the requested result using the combined statements model: Find the HCF of 66 and 99.
Options: 33 | 36 | 30 | 39

Q468 ? Counting restriction ? Medium

A gear rotation case uses the quantities in this question. Determine the requested result using the counting restriction model: Find the HCF of 68 and 102.
Options: 34 | 37 | 31 | 40

Q469 ? Circular arrangement ? Hard

A batch size case uses the quantities in this question. Determine the requested result using the circular arrangement model: Find the HCF of 70 and 105.
Options: 35 | 39 | 31 | 43

Q470 ? Complement method ? Placement

A traffic signals case uses the quantities in this question. Determine the requested result using the complement method model: Find the HCF of 72 and 108.
Options: 44 | 32 | 40 | 36

Q471 ? Pattern: differences ? Beginner

A medicine dosage case uses the quantities in this question. Determine the requested result using the pattern: differences model: Find the HCF of 74 and 111.
Options: 37 | 41 | 33 | 45

Q472 ? Pattern: ratios ? Easy

A event recurrence case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Find the HCF of 76 and 114.
Options: 34 | 38 | 46 | 42

Q473 ? Pattern: alternating ? Medium

A maintenance cycle case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Find the HCF of 78 and 117.
Options: 43 | 47 | 35 | 39

Q474 ? Pattern: squares / cubes ? Hard

A bell schedule case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Find the HCF of 80 and 120.
Options: 48 | 44 | 40 | 36

Q475 ? Inference from facts ? Placement

A tile packing case uses the quantities in this question. Determine the requested result using the inference from facts model: Find the HCF of 82 and 123.
Options: 37 | 41 | 45 | 49

Q476 ? Assumption check ? Beginner

A gear rotation case uses the quantities in this question. Determine the requested result using the assumption check model: Find the HCF of 84 and 126.
Options: 42 | 46 | 38 | 50

Q477 ? Cause vs correlation ? Easy

A batch size case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Find the HCF of 86 and 129.
Options: 51 | 39 | 47 | 43

Q478 ? Grammar agreement ? Medium

A traffic signals case uses the quantities in this question. Determine the requested result using the grammar agreement model: Find the HCF of 88 and 132.
Options: 44 | 48 | 40 | 52

Q479 ? Vocabulary in context ? Hard

A medicine dosage case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Find the HCF of 90 and 135.
Options: 40 | 45 | 55 | 50

Q480 ? Mixed placement challenge ? Placement

A event recurrence case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Find the HCF of 92 and 138.
Options: 46 | 51 | 41 | 56

Quadratic Equations ? 40 questions

WORKED EXAMPLES

Q481 ? Direct computation ? Beginner

A projectile path case uses the quantities in this question. Determine the requested result using the direct computation model: Which is a positive root of $x^2 + 1x - 12 = 0$?

Options: 3 | 5 | 4 | 2

Answer: 3

Solution: Context: projectile path. Substitute $x=3$: $9+1\times 3-12=0$, so 3 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q482 ? Reverse value ? Easy

A area constraint case uses the quantities in this question. Determine the requested result using the reverse value model: Which is a positive root of $x^2 + 1x - 20 = 0$?

Options: 5 | 6 | 3 | 4

Answer: 4

Solution: Context: area constraint. Substitute $x=4$: $16+1\times 4-20=0$, so 4 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q483 ? Missing base ? Medium

A profit maximum case uses the quantities in this question. Determine the requested result using the missing base model: Which is a positive root of $x^2 + 1x - 30 = 0$?

Options: 7 | 4 | 6 | 5

Answer: 5

Solution: Context: profit maximum. Substitute $x=5$: $25+1\times 5-30=0$, so 5 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q484 ? Successive change ? Hard

A root condition case uses the quantities in this question. Determine the requested result using the successive change model: Which is a positive root of $x^2 + 1x - 42 = 0$?

Options: 5 | 6 | 7 | 8

Answer: 6

Solution: Context: root condition. Substitute $x=6$: $36+1\times 6-42=0$, so 6 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q485 ? Comparison trap ? Placement

A rectangular field case uses the quantities in this question. Determine the requested result using the comparison trap model: Which is a positive root of $x^2 + 1x - 56 = 0$?

Options: 9 | 8 | 7 | 6

Answer: 7

Solution: Context: rectangular field. Substitute $x=7$: $49+1\times 7-56=0$, so 7 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q486 ? Ratio form ? Beginner

A number pair case uses the quantities in this question. Determine the requested result using the ratio form model: Which is a positive root of $x^2 + 1x - 72 = 0$?

Options: 8 | 9 | 7 | 10

Answer: 8

Solution: Context: number pair. Substitute $x=8$: $64+1\times 8-72=0$, so 8 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q487 ? Weighted result ? Easy

A motion model case uses the quantities in this question. Determine the requested result using the weighted result model: Which is a positive root of $x^2 + -6x - 27 = 0$?

Options: 10 | 8 | 11 | 9

Answer: 9

Solution: Context: motion model. Substitute $x=9$: $81+-6\times 9-27=0$, so 9 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q488 ? Replacement cycle ? Medium

A equation roots case uses the quantities in this question. Determine the requested result using the replacement cycle model: Which is a positive root of $x^2 + -6x - 40 = 0$?

Options: 12 | 9 | 11 | 10

Answer: 10

Solution: Context: equation roots. Substitute $x=10$: $100+-6\times 10-40=0$, so 10 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q489 ? Unit conversion ? Hard

A projectile path case uses the quantities in this question. Determine the requested result using the unit conversion model: Which is a positive root of $x^2 + 3x - 10 = 0$?

Options: 4 | 3 | 2 | 1

Answer: 2

Solution: Context: projectile path. Substitute $x=2$: $4+3\times 2-10=0$, so 2 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q490 ? Work-rate model ? Placement

A area constraint case uses the quantities in this question. Determine the requested result using the work-rate model model: Which is a positive root of $x^2 + 3x - 18 = 0$?

Options: 3 | 2 | 4 | 5

Answer: 3

Solution: Context: area constraint. Substitute $x=3$: $9+3\times 3-18=0$, so 3 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q491 ? Speed-time model ? Beginner

A profit maximum case uses the quantities in this question. Determine the requested result using the speed-time model model: Which is a positive root of $x^2 + 3x - 28 = 0$?

Options: 3 | 4 | 5 | 6

Answer: 4

Solution: Context: profit maximum. Substitute $x=4$: $16+3\times 4-28=0$, so 4 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q492 ? Table lookup ? Easy

A root condition case uses the quantities in this question. Determine the requested result using the table lookup model: Which is a positive root of $x^2 + 3x - 40 = 0$?

Options: 5 | 6 | 4 | 7

Answer: 5

Solution: Context: root condition. Substitute $x=5$: $25+3\times 5-40=0$, so 5 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q493 ? Chart comparison ? Medium

A rectangular field case uses the quantities in this question. Determine the requested result using the chart comparison model: Which is a positive root of $x^2 + 3x - 54 = 0$?

Options: 6 | 7 | 8 | 5

Answer: 6

Solution: Context: rectangular field. Substitute $x=6$: $36+3\times 6-54=0$, so 6 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q494 ? Multi-step word problem ? Hard

A number pair case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Which is a positive root of $x^2 + -4x - 21 = 0$?

Options: 8 | 9 | 6 | 7

Answer: 7

Solution: Context: number pair. Substitute $x=7$: $49+-4\times 7-21=0$, so 7 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q495 ? Maximum / minimum ? Placement

A motion model case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Which is a positive root of $x^2 + -4x - 32 = 0$?

Options: 10 | 7 | 9 | 8

Answer: 8

Solution: Context: motion model. Substitute $x=8$: $64+-4\times 8-32=0$, so 8 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q496 ? Least / greatest value ? Beginner

A equation roots case uses the quantities in this question. Determine the requested result using the least / greatest value model: Which is a positive root of $x^2 + -4x - 45 = 0$?

Options: 9 | 10 | 8 | 11

Answer: 9

Solution: Context: equation roots. Substitute $x=9$: $81+-4\times 9-45=0$, so 9 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q497 ? Remainder cycle ? Easy

A projectile path case uses the quantities in this question. Determine the requested result using the remainder cycle model: Which is a positive root of $x^2 + -4x - 60 = 0$?

Options: 10 | 11 | 9 | 12

Answer: 10

Solution: Context: projectile path. Substitute $x=10$: $100+-4\times 10-60=0$, so 10 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q498 ? Divisibility test ? Medium

A area constraint case uses the quantities in this question. Determine the requested result using the divisibility test model: Which is a positive root of $x^2 + 5x - 14 = 0$?

Options: 2 | 3 | 4 | 1

Answer: 2

Solution: Context: area constraint. Substitute $x=2$: $4+5\times 2-14=0$, so 2 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q499 ? Factorisation ? Hard

A profit maximum case uses the quantities in this question. Determine the requested result using the factorisation model: Which is a positive root of $x^2 + 5x - 24 = 0$?

Options: 4 | 2 | 5 | 3

Answer: 3

Solution: Context: profit maximum. Substitute $x=3$: $9+5\times 3-24=0$, so 3 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

Q500 ? Approximation ? Placement

A root condition case uses the quantities in this question. Determine the requested result using the approximation model: Which is a positive root of $x^2 + 5x - 36 = 0$?

Options: 6 | 5 | 4 | 3

Answer: 4

Solution: Context: root condition. Substitute $x=4$: $16+5\times 4-36=0$, so 4 is a root. Shortcut: Option substitution is often faster than the quadratic formula.

PRACTICE ? solutions hidden below**Q501 ? Options elimination ? Beginner**

A rectangular field case uses the quantities in this question. Determine the requested result using the options elimination model: Which is a positive root of $x^2 +$

$-2x - 15 = 0$?
Options: 4 | 5 | 6 | 7

Q502 ? Boundary case ? Easy

A number pair case uses the quantities in this question. Determine the requested result using the boundary case model: Which is a positive root of $x^2 + -2x - 24 = 0$?
Options: 5 | 8 | 7 | 6

Q503 ? Case split ? Medium

A motion model case uses the quantities in this question. Determine the requested result using the case split model: Which is a positive root of $x^2 + -2x - 35 = 0$?
Options: 9 | 7 | 8 | 6

Q504 ? Contradiction check ? Hard

A equation roots case uses the quantities in this question. Determine the requested result using the contradiction check model: Which is a positive root of $x^2 + -2x - 48 = 0$?
Options: 8 | 7 | 9 | 10

Q505 ? Statement I test ? Placement

A projectile path case uses the quantities in this question. Determine the requested result using the statement i test model: Which is a positive root of $x^2 + -2x - 63 = 0$?
Options: 9 | 11 | 8 | 10

Q506 ? Statement II test ? Beginner

A area constraint case uses the quantities in this question. Determine the requested result using the statement ii test model: Which is a positive root of $x^2 + -2x - 80 = 0$?
Options: 12 | 9 | 11 | 10

Q507 ? Combined statements ? Easy

A profit maximum case uses the quantities in this question. Determine the requested result using the combined statements model: Which is a positive root of $x^2 + 7x - 18 = 0$?
Options: 4 | 2 | 3 | 1

Q508 ? Counting restriction ? Medium

A root condition case uses the quantities in this question. Determine the requested result using the counting restriction model: Which is a positive root of $x^2 + 0x - 9 = 0$?
Options: 2 | 3 | 4 | 5

Q509 ? Circular arrangement ? Hard

A rectangular field case uses the quantities in this question. Determine the requested result using the circular arrangement model: Which is a positive root of $x^2 + 0x - 16 = 0$?
Options: 5 | 4 | 3 | 6

Q510 ? Complement method ? Placement

A number pair case uses the quantities in this question. Determine the requested result using the complement method model: Which is a positive root of $x^2 + 0x - 25 = 0$?
Options: 5 | 6 | 7 | 4

Q511 ? Pattern: differences ? Beginner

A motion model case uses the quantities in this question. Determine the requested result using the pattern: differences model: Which is a positive root of $x^2 + 0x - 36 = 0$?
Options: 7 | 8 | 6 | 5

Q512 ? Pattern: ratios ? Easy

A equation roots case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Which is a positive root of $x^2 + 0x - 49 = 0$?
Options: 6 | 7 | 8 | 9

Q513 ? Pattern: alternating ? Medium

A projectile path case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Which is a positive root of $x^2 + 0x - 64 = 0$?
Options: 10 | 7 | 9 | 8

Q514 ? Pattern: squares / cubes ? Hard

A area constraint case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Which is a positive root of $x^2 + 0x - 81 = 0$?
Options: 9 | 10 | 8 | 11

Q515 ? Inference from facts ? Placement

A profit maximum case uses the quantities in this question. Determine the requested result using the inference from facts model: Which is a positive root of $x^2 + -7x - 30 = 0$?
Options: 10 | 11 | 12 | 9

Q516 ? Assumption check ? Beginner

A root condition case uses the quantities in this question. Determine the requested result using the assumption check model: Which is a positive root of $x^2 + 2x - 8 = 0$?
Options: 3 | 1 | 2 | 4

Q517 ? Cause vs correlation ? Easy

A rectangular field case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Which is a positive root of $x^2 + 2x - 15 = 0$?

Options: 5 | 2 | 4 | 3

Q518 ? Grammar agreement ? Medium

A number pair case uses the quantities in this question. Determine the requested result using the grammar agreement model: Which is a positive root of $x^2 + 2x - 24 = 0$?

Options: 6 | 4 | 3 | 5

Q519 ? Vocabulary in context ? Hard

A motion model case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Which is a positive root of $x^2 + 2x - 35 = 0$?

Options: 4 | 5 | 6 | 7

Q520 ? Mixed placement challenge ? Placement

A equation roots case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Which is a positive root of $x^2 + 2x - 48 = 0$?

Options: 8 | 7 | 6 | 5

Mensuration ? 40 questions

WORKED EXAMPLES

Q521 ? Direct computation ? Beginner

A tank capacity case uses the quantities in this question. Determine the requested result using the direct computation model: Find the area of a rectangle 9 cm long and 5 cm broad.

Options: 55 | 50 | 45 | 40

Answer: 45

Solution: Context: tank capacity. Area = $l \times b = 9 \times 5 = 45 \text{ cm}^2$. Shortcut: Area always uses square units.

Q522 ? Reverse value ? Easy

A floor tiling case uses the quantities in this question. Determine the requested result using the reverse value model: Find the area of a rectangle 10 cm long and 6 cm broad.

Options: 60 | 66 | 54 | 72

Answer: 60

Solution: Context: floor tiling. Area = $l \times b = 10 \times 6 = 60 \text{ cm}^2$. Shortcut: Area always uses square units.

Q523 ? Missing base ? Medium

A cylindrical drum case uses the quantities in this question. Determine the requested result using the missing base model: Find the area of a rectangle 11 cm long and 7 cm broad.

Options: 77 | 93 | 69 | 85

Answer: 77

Solution: Context: cylindrical drum. Area = $l \times b = 11 \times 7 = 77 \text{ cm}^2$. Shortcut: Area always uses square units.

Q524 ? Successive change ? Hard

A circular track case uses the quantities in this question. Determine the requested result using the successive change model: Find the area of a rectangle 12 cm long and 8 cm broad.

Options: 96 | 116 | 86 | 106

Answer: 96

Solution: Context: circular track. Area = $l \times b = 12 \times 8 = 96 \text{ cm}^2$. Shortcut: Area always uses square units.

Q525 ? Comparison trap ? Placement

A paint coverage case uses the quantities in this question. Determine the requested result using the comparison trap model: Find the area of a rectangle 13 cm long and 9 cm broad.

Options: 141 | 105 | 129 | 117

Answer: 117

Solution: Context: paint coverage. Area = $l \times b = 13 \times 9 = 117 \text{ cm}^2$. Shortcut: Area always uses square units.

Q526 ? Ratio form ? Beginner

A warehouse volume case uses the quantities in this question. Determine the requested result using the ratio form model: Find the area of a rectangle 14 cm long and 10 cm broad.

Options: 140 | 154 | 126 | 168

Answer: 140

Solution: Context: warehouse volume. Area = $l \times b = 14 \times 10 = 140 \text{ cm}^2$. Shortcut: Area always uses square units.

Q527 ? Weighted result ? Easy

A cone container case uses the quantities in this question. Determine the requested result using the weighted result model: Find the area of a rectangle 15 cm long and 4 cm broad.

Options: 54 | 60 | 66 | 72

Answer: 60

Solution: Context: cone container. Area = $l \times b = 15 \times 4 = 60 \text{ cm}^2$. Shortcut: Area always uses square units.

Q528 ? Replacement cycle ? Medium

A sphere packing case uses the quantities in this question. Determine the requested result using the replacement cycle model: Find the area of a rectangle 16 cm long and 5 cm broad.

Options: 72 | 96 | 88 | 80

Answer: 80

Solution: Context: sphere packing. Area = $l \times b = 16 \times 5 = 80 \text{ cm}^2$. Shortcut: Area always uses square units.

Q529 ? Unit conversion ? Hard

A tank capacity case uses the quantities in this question. Determine the requested result using the unit conversion model: Find the area of a rectangle 17 cm long and 6 cm broad.

Options: 92 | 102 | 112 | 122

Answer: 102

Solution: Context: tank capacity. Area = $l \times b = 17 \times 6 = 102 \text{ cm}^2$. Shortcut: Area always uses square units.

Q530 ? Work-rate model ? Placement

A floor tiling case uses the quantities in this question. Determine the requested result using the work-rate model model: Find the area of a rectangle 18 cm long and 7 cm broad.

Options: 126 | 139 | 113 | 152

Answer: 126

Solution: Context: floor tiling. Area = $l \times b = 18 \times 7 = 126 \text{ cm}^2$. Shortcut: Area always uses square units.

Q531 ? Speed-time model ? Beginner

A cylindrical drum case uses the quantities in this question. Determine the requested result using the speed-time model model: Find the area of a rectangle 19 cm long and 8 cm broad.

Options: 182 | 137 | 167 | 152

Answer: 152

Solution: Context: cylindrical drum. Area = $l \times b = 19 \times 8 = 152 \text{ cm}^2$. Shortcut: Area always uses square units.

Q532 ? Table lookup ? Easy

A circular track case uses the quantities in this question. Determine the requested result using the table lookup model: Find the area of a rectangle 20 cm long and 9 cm broad.

Options: 216 | 162 | 198 | 180

Answer: 180

Solution: Context: circular track. Area = $l \times b = 20 \times 9 = 180 \text{ cm}^2$. Shortcut: Area always uses square units.

Q533 ? Chart comparison ? Medium

A paint coverage case uses the quantities in this question. Determine the requested result using the chart comparison model: Find the area of a rectangle 21 cm long and 10 cm broad.

Options: 210 | 231 | 189 | 252

Answer: 210

Solution: Context: paint coverage. Area = $l \times b = 21 \times 10 = 210 \text{ cm}^2$. Shortcut: Area always uses square units.

Q534 ? Multi-step word problem ? Hard

A warehouse volume case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Find the area of a rectangle 22 cm long and 4 cm broad.

Options: 88 | 106 | 97 | 79

Answer: 88

Solution: Context: warehouse volume. Area = $l \times b = 22 \times 4 = 88 \text{ cm}^2$. Shortcut: Area always uses square units.

Q535 ? Maximum / minimum ? Placement

A cone container case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Find the area of a rectangle 23 cm long and 5 cm broad.

Options: 139 | 103 | 127 | 115

Answer: 115

Solution: Context: cone container. Area = $l \times b = 23 \times 5 = 115 \text{ cm}^2$. Shortcut: Area always uses square units.

Q536 ? Least / greatest value ? Beginner

A sphere packing case uses the quantities in this question. Determine the requested result using the least / greatest value model: Find the area of a rectangle 24 cm long and 6 cm broad.

Options: 144 | 158 | 130 | 172

Answer: 144

Solution: Context: sphere packing. Area = $l \times b = 24 \times 6 = 144 \text{ cm}^2$. Shortcut: Area always uses square units.

Q537 ? Remainder cycle ? Easy

A tank capacity case uses the quantities in this question. Determine the requested result using the remainder cycle model: Find the area of a rectangle 25 cm long and 7 cm broad.

Options: 175 | 193 | 157 | 211

Answer: 175

Solution: Context: tank capacity. Area = $l \times b = 25 \times 7 = 175 \text{ cm}^2$. Shortcut: Area always uses square units.

Q538 ? Divisibility test ? Medium

A floor tiling case uses the quantities in this question. Determine the requested result using the divisibility test model: Find the area of a rectangle 26 cm long and 8 cm broad.

Options: 208 | 229 | 187 | 250

Answer: 208

Solution: Context: floor tiling. Area = $l \times b = 26 \times 8 = 208 \text{ cm}^2$. Shortcut: Area always uses square units.

Q539 ? Factorisation ? Hard

A cylindrical drum case uses the quantities in this question. Determine the requested result using the factorisation model: Find the area of a rectangle 27 cm long and 9 cm broad.

Options: 291 | 219 | 267 | 243

Answer: 243

Solution: Context: cylindrical drum. Area = $l \times b = 27 \times 9 = 243 \text{ cm}^2$. Shortcut: Area always uses square units.

Q540 ? Approximation ? Placement

A circular track case uses the quantities in this question. Determine the requested result using the approximation model: Find the area of a rectangle 28 cm long and 10 cm broad.

Options: 280 | 308 | 252 | 336

Answer: 280

Solution: Context: circular track. Area = $l \times b = 28 \times 10 = 280 \text{ cm}^2$. Shortcut: Area always uses square units.

PRACTICE ? solutions hidden below**Q541 ? Options elimination ? Beginner**

A paint coverage case uses the quantities in this question. Determine the requested result using the options elimination model: Find the area of a rectangle 29

cm long and 4 cm broad.
Options: 116 | 128 | 104 | 140

Q542 ? Boundary case ? Easy

A warehouse volume case uses the quantities in this question. Determine the requested result using the boundary case model: Find the area of a rectangle 30 cm long and 5 cm broad.
Options: 180 | 135 | 165 | 150

Q543 ? Case split ? Medium

A cone container case uses the quantities in this question. Determine the requested result using the case split model: Find the area of a rectangle 31 cm long and 6 cm broad.
Options: 224 | 205 | 186 | 167

Q544 ? Contradiction check ? Hard

A sphere packing case uses the quantities in this question. Determine the requested result using the contradiction check model: Find the area of a rectangle 32 cm long and 7 cm broad.
Options: 224 | 246 | 202 | 268

Q545 ? Statement I test ? Placement

A tank capacity case uses the quantities in this question. Determine the requested result using the statement i test model: Find the area of a rectangle 33 cm long and 8 cm broad.
Options: 264 | 290 | 238 | 316

Q546 ? Statement II test ? Beginner

A floor tiling case uses the quantities in this question. Determine the requested result using the statement ii test model: Find the area of a rectangle 34 cm long and 9 cm broad.
Options: 368 | 275 | 337 | 306

Q547 ? Combined statements ? Easy

A cylindrical drum case uses the quantities in this question. Determine the requested result using the combined statements model: Find the area of a rectangle 35 cm long and 10 cm broad.
Options: 420 | 350 | 385 | 315

Q548 ? Counting restriction ? Medium

A circular track case uses the quantities in this question. Determine the requested result using the counting restriction model: Find the area of a rectangle 36 cm long and 4 cm broad.
Options: 144 | 158 | 130 | 172

Q549 ? Circular arrangement ? Hard

A paint coverage case uses the quantities in this question. Determine the requested result using the circular arrangement model: Find the area of a rectangle 37 cm long and 5 cm broad.
Options: 223 | 166 | 204 | 185

Q550 ? Complement method ? Placement

A warehouse volume case uses the quantities in this question. Determine the requested result using the complement method model: Find the area of a rectangle 38 cm long and 6 cm broad.
Options: 274 | 205 | 251 | 228

Q551 ? Pattern: differences ? Beginner

A cone container case uses the quantities in this question. Determine the requested result using the pattern: differences model: Find the area of a rectangle 39 cm long and 7 cm broad.
Options: 273 | 300 | 246 | 327

Q552 ? Pattern: ratios ? Easy

A sphere packing case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Find the area of a rectangle 40 cm long and 8 cm broad.
Options: 384 | 352 | 320 | 288

Q553 ? Pattern: alternating ? Medium

A tank capacity case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Find the area of a rectangle 41 cm long and 9 cm broad.
Options: 406 | 443 | 332 | 369

Q554 ? Pattern: squares / cubes ? Hard

A floor tiling case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Find the area of a rectangle 42 cm long and 10 cm broad.
Options: 504 | 462 | 420 | 378

Q555 ? Inference from facts ? Placement

A cylindrical drum case uses the quantities in this question. Determine the requested result using the inference from facts model: Find the area of a rectangle 43 cm long and 4 cm broad.
Options: 172 | 189 | 155 | 206

Q556 ? Assumption check ? Beginner

A circular track case uses the quantities in this question. Determine the requested result using the assumption check model: Find the area of a rectangle 44 cm long and 5 cm broad.
Options: 264 | 198 | 242 | 220

Q557 ? Cause vs correlation ? Easy

A paint coverage case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Find the area of a rectangle 45 cm long and 6 cm broad.

Options: 324 | 243 | 297 | 270

Q558 ? Grammar agreement ? Medium

A warehouse volume case uses the quantities in this question. Determine the requested result using the grammar agreement model: Find the area of a rectangle 46 cm long and 7 cm broad.

Options: 322 | 354 | 290 | 386

Q559 ? Vocabulary in context ? Hard

A cone container case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Find the area of a rectangle 47 cm long and 8 cm broad.

Options: 338 | 376 | 452 | 414

Q560 ? Mixed placement challenge ? Placement

A sphere packing case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Find the area of a rectangle 48 cm long and 9 cm broad.

Options: 432 | 475 | 518 | 389

Series ? 40 questions

WORKED EXAMPLES

Q561 ? Direct computation ? Beginner

A code sequence case uses the quantities in this question. Determine the requested result using the direct computation model: Find the next term: 2, 5, 8, 11, ?

Options: 16 | 13 | 15 | 14

Answer: 14

Solution: Context: code sequence. The common difference is 3; next = $11 + 3 = 14$. Shortcut: Subtract adjacent terms before trying complex rules.

Q562 ? Reverse value ? Easy

A production cycle case uses the quantities in this question. Determine the requested result using the reverse value model: Find the next term: 3, 7, 11, 15, ?

Options: 23 | 21 | 19 | 17

Answer: 19

Solution: Context: production cycle. The common difference is 4; next = $15 + 4 = 19$. Shortcut: Subtract adjacent terms before trying complex rules.

Q563 ? Missing base ? Medium

A number pattern case uses the quantities in this question. Determine the requested result using the missing base model: Find the next term: 4, 9, 14, 19, ?

Options: 24 | 26 | 22 | 28

Answer: 24

Solution: Context: number pattern. The common difference is 5; next = $19 + 5 = 24$. Shortcut: Subtract adjacent terms before trying complex rules.

Q564 ? Successive change ? Hard

A daily visitors case uses the quantities in this question. Determine the requested result using the successive change model: Find the next term: 5, 11, 17, 23, ?

Options: 29 | 32 | 26 | 35

Answer: 29

Solution: Context: daily visitors. The common difference is 6; next = $23 + 6 = 29$. Shortcut: Subtract adjacent terms before trying complex rules.

Q565 ? Comparison trap ? Placement

A stock movement case uses the quantities in this question. Determine the requested result using the comparison trap model: Find the next term: 6, 13, 20, 27, ?

Options: 40 | 31 | 37 | 34

Answer: 34

Solution: Context: stock movement. The common difference is 7; next = $27 + 7 = 34$. Shortcut: Subtract adjacent terms before trying complex rules.

Q566 ? Ratio form ? Beginner

A machine output case uses the quantities in this question. Determine the requested result using the ratio form model: Find the next term: 7, 9, 11, 13, ?

Options: 15 | 17 | 13 | 19

Answer: 15

Solution: Context: machine output. The common difference is 2; next = $13 + 2 = 15$. Shortcut: Subtract adjacent terms before trying complex rules.

Q567 ? Weighted result ? Easy

A calendar code case uses the quantities in this question. Determine the requested result using the weighted result model: Find the next term: 8, 11, 14, 17, ?

Options: 18 | 20 | 22 | 24

Answer: 20

Solution: Context: calendar code. The common difference is 3; next = $17 + 3 = 20$. Shortcut: Subtract adjacent terms before trying complex rules.

Q568 ? Replacement cycle ? Medium

A exam puzzle case uses the quantities in this question. Determine the requested result using the replacement cycle model: Find the next term: 9, 13, 17, 21, ?

Options: 31 | 22 | 28 | 25

Answer: 25

Solution: Context: exam puzzle. The common difference is 4; next = $21 + 4 = 25$. Shortcut: Subtract adjacent terms before trying complex rules.

Q569 ? Unit conversion ? Hard

A code sequence case uses the quantities in this question. Determine the requested result using the unit conversion model: Find the next term: 10, 15, 20, 25, ?

Options: 36 | 27 | 33 | 30

Answer: 30

Solution: Context: code sequence. The common difference is 5; next = $25 + 5 = 30$. Shortcut: Subtract adjacent terms before trying complex rules.

Q570 ? Work-rate model ? Placement

A production cycle case uses the quantities in this question. Determine the requested result using the work-rate model model: Find the next term: 11, 17, 23, 29, ?

Options: 35 | 39 | 31 | 43

Answer: 35

Solution: Context: production cycle. The common difference is 6; next = $29 + 6 = 35$. Shortcut: Subtract adjacent terms before trying complex rules.

Q571 ? Speed-time model ? Beginner

A number pattern case uses the quantities in this question. Determine the requested result using the speed-time model model: Find the next term: 12, 19, 26, 33, ?

Options: 36 | 48 | 44 | 40

Answer: 40

Solution: Context: number pattern. The common difference is 7; next = $33 + 7 = 40$. Shortcut: Subtract adjacent terms before trying complex rules.

Q572 ? Table lookup ? Easy

A daily visitors case uses the quantities in this question. Determine the requested result using the table lookup model: Find the next term: 13, 15, 17, 19, ?

Options: 25 | 19 | 23 | 21

Answer: 21

Solution: Context: daily visitors. The common difference is 2; next = $19 + 2 = 21$. Shortcut: Subtract adjacent terms before trying complex rules.

Q573 ? Chart comparison ? Medium

A stock movement case uses the quantities in this question. Determine the requested result using the chart comparison model: Find the next term: 14, 17, 20, 23, ?

Options: 26 | 29 | 23 | 32

Answer: 26

Solution: Context: stock movement. The common difference is 3; next = $23 + 3 = 26$. Shortcut: Subtract adjacent terms before trying complex rules.

Q574 ? Multi-step word problem ? Hard

A machine output case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Find the next term: 15, 19, 23, 27, ?

Options: 28 | 31 | 34 | 37

Answer: 31

Solution: Context: machine output. The common difference is 4; next = $27 + 4 = 31$. Shortcut: Subtract adjacent terms before trying complex rules.

Q575 ? Maximum / minimum ? Placement

A calendar code case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Find the next term: 16, 21, 26, 31, ?

Options: 36 | 40 | 32 | 44

Answer: 36

Solution: Context: calendar code. The common difference is 5; next = $31 + 5 = 36$. Shortcut: Subtract adjacent terms before trying complex rules.

Q576 ? Least / greatest value ? Beginner

A exam puzzle case uses the quantities in this question. Determine the requested result using the least / greatest value model: Find the next term: 17, 23, 29, 35, ?

Options: 49 | 37 | 45 | 41

Answer: 41

Solution: Context: exam puzzle. The common difference is 6; next = $35 + 6 = 41$. Shortcut: Subtract adjacent terms before trying complex rules.

Q577 ? Remainder cycle ? Easy

A code sequence case uses the quantities in this question. Determine the requested result using the remainder cycle model: Find the next term: 18, 25, 32, 39, ?

Options: 46 | 51 | 41 | 56

Answer: 46

Solution: Context: code sequence. The common difference is 7; next = $39 + 7 = 46$. Shortcut: Subtract adjacent terms before trying complex rules.

Q578 ? Divisibility test ? Medium

A production cycle case uses the quantities in this question. Determine the requested result using the divisibility test model: Find the next term: 19, 21, 23, 25, ?

Options: 27 | 30 | 24 | 33

Answer: 27

Solution: Context: production cycle. The common difference is 2; next = $25 + 2 = 27$. Shortcut: Subtract adjacent terms before trying complex rules.

Q579 ? Factorisation ? Hard

A number pattern case uses the quantities in this question. Determine the requested result using the factorisation model: Find the next term: 20, 23, 26, 29, ?

Options: 38 | 29 | 35 | 32

Answer: 32

Solution: Context: number pattern. The common difference is 3; next = $29 + 3 = 32$. Shortcut: Subtract adjacent terms before trying complex rules.

Q580 ? Approximation ? Placement

A daily visitors case uses the quantities in this question. Determine the requested result using the approximation model: Find the next term: 21, 25, 29, 33, ?

Options: 45 | 41 | 37 | 33

Answer: 37

Solution: Context: daily visitors. The common difference is 4; next = $33 + 4 = 37$. Shortcut: Subtract adjacent terms before trying complex rules.

PRACTICE ? solutions hidden below

Q581 ? Options elimination ? Beginner

A stock movement case uses the quantities in this question. Determine the requested result using the options elimination model: Find the next term: 22, 27, 32, 37, ?

Options: 38 | 42 | 46 | 50

Q582 ? Boundary case ? Easy

A machine output case uses the quantities in this question. Determine the requested result using the boundary case model: Find the next term: 23, 29, 35, 41, ?

Options: 57 | 52 | 47 | 42

Q583 ? Case split ? Medium

A calendar code case uses the quantities in this question. Determine the requested result using the case split model: Find the next term: 24, 31, 38, 45, ?
Options: 52 | 57 | 62 | 47

Q584 ? Contradiction check ? Hard

A exam puzzle case uses the quantities in this question. Determine the requested result using the contradiction check model: Find the next term: 25, 27, 29, 31, ?
Options: 33 | 36 | 30 | 39

Q585 ? Statement I test ? Placement

A code sequence case uses the quantities in this question. Determine the requested result using the statement i test model: Find the next term: 26, 29, 32, 35, ?
Options: 34 | 38 | 46 | 42

Q586 ? Statement II test ? Beginner

A production cycle case uses the quantities in this question. Determine the requested result using the statement ii test model: Find the next term: 27, 31, 35, 39, ?
Options: 43 | 47 | 51 | 39

Q587 ? Combined statements ? Easy

A number pattern case uses the quantities in this question. Determine the requested result using the combined statements model: Find the next term: 28, 33, 38, 43, ?
Options: 58 | 43 | 53 | 48

Q588 ? Counting restriction ? Medium

A daily visitors case uses the quantities in this question. Determine the requested result using the counting restriction model: Find the next term: 29, 35, 41, 47, ?
Options: 48 | 53 | 58 | 63

Q589 ? Circular arrangement ? Hard

A stock movement case uses the quantities in this question. Determine the requested result using the circular arrangement model: Find the next term: 30, 37, 44, 51, ?
Options: 52 | 70 | 64 | 58

Q590 ? Complement method ? Placement

A machine output case uses the quantities in this question. Determine the requested result using the complement method model: Find the next term: 31, 33, 35, 37, ?
Options: 47 | 35 | 43 | 39

Q591 ? Pattern: differences ? Beginner

A calendar code case uses the quantities in this question. Determine the requested result using the pattern: differences model: Find the next term: 32, 35, 38, 41, ?
Options: 44 | 48 | 40 | 52

Q592 ? Pattern: ratios ? Easy

A exam puzzle case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Find the next term: 33, 37, 41, 45, ?
Options: 44 | 49 | 59 | 54

Q593 ? Pattern: alternating ? Medium

A code sequence case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Find the next term: 34, 39, 44, 49, ?
Options: 54 | 59 | 49 | 64

Q594 ? Pattern: squares / cubes ? Hard

A production cycle case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Find the next term: 35, 41, 47, 53, ?
Options: 65 | 71 | 53 | 59

Q595 ? Inference from facts ? Placement

A number pattern case uses the quantities in this question. Determine the requested result using the inference from facts model: Find the next term: 36, 43, 50, 57, ?
Options: 58 | 64 | 70 | 76

Q596 ? Assumption check ? Beginner

A daily visitors case uses the quantities in this question. Determine the requested result using the assumption check model: Find the next term: 37, 39, 41, 43, ?
Options: 55 | 50 | 45 | 40

Q597 ? Cause vs correlation ? Easy

A stock movement case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Find the next term: 38, 41, 44, 47, ?
Options: 50 | 55 | 60 | 45

Q598 ? Grammar agreement ? Medium

A machine output case uses the quantities in this question. Determine the requested result using the grammar agreement model: Find the next term: 39, 43, 47, 51, ?
Options: 67 | 49 | 61 | 55

Q599 ? Vocabulary in context ? Hard

A calendar code case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Find the next term: 40, 45, 50, 55, ?

Options: 54 | 66 | 60 | 72

Q600 ? Mixed placement challenge ? Placement

A exam puzzle case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Find the next term: 41, 47, 53, 59, ?

Options: 79 | 72 | 65 | 58

Direction Sense ? 40 questions

WORKED EXAMPLES

Q601 ? Direct computation ? Beginner

A campus navigation case uses the quantities in this question. Determine the requested result using the direct computation model: Riya walks 4 km east, then 5 km north. How far is she from the start?

Options: 5.4 | 8.4 | 7.4 | 6.4

Answer: 6.4

Solution: Context: campus navigation. Distance = $\sqrt{(4^2+5^2)} = 6.4$ km. Shortcut: Track final x and y coordinates.

Q602 ? Reverse value ? Easy

A drone route case uses the quantities in this question. Determine the requested result using the reverse value model: Riya walks 5 km east, then 6 km north. How far is she from the start?

Options: 9.81 | 7.81 | 8.81 | 6.81

Answer: 7.81

Solution: Context: drone route. Distance = $\sqrt{(5^2+6^2)} = 7.81$ km. Shortcut: Track final x and y coordinates.

Q603 ? Missing base ? Medium

A delivery map case uses the quantities in this question. Determine the requested result using the missing base model: Riya walks 6 km east, then 7 km north. How far is she from the start?

Options: 9.22 | 10.22 | 8.22 | 11.22

Answer: 9.22

Solution: Context: delivery map. Distance = $\sqrt{(6^2+7^2)} = 9.22$ km. Shortcut: Track final x and y coordinates.

Q604 ? Successive change ? Hard

A city grid case uses the quantities in this question. Determine the requested result using the successive change model: Riya walks 7 km east, then 8 km north. How far is she from the start?

Options: 12.63 | 11.63 | 10.63 | 9.63

Answer: 10.63

Solution: Context: city grid. Distance = $\sqrt{(7^2+8^2)} = 10.63$ km. Shortcut: Track final x and y coordinates.

Q605 ? Comparison trap ? Placement

A treasure hunt case uses the quantities in this question. Determine the requested result using the comparison trap model: Riya walks 8 km east, then 4 km north. How far is she from the start?

Options: 10.94 | 7.94 | 9.94 | 8.94

Answer: 8.94

Solution: Context: treasure hunt. Distance = $\sqrt{(8^2+4^2)} = 8.94$ km. Shortcut: Track final x and y coordinates.

Q606 ? Ratio form ? Beginner

A robot movement case uses the quantities in this question. Determine the requested result using the ratio form model: Riya walks 9 km east, then 5 km north. How far is she from the start?

Options: 10.3 | 11.3 | 9.3 | 12.3

Answer: 10.3

Solution: Context: robot movement. Distance = $\sqrt{(9^2+5^2)} = 10.3$ km. Shortcut: Track final x and y coordinates.

Q607 ? Weighted result ? Easy

A warehouse aisle case uses the quantities in this question. Determine the requested result using the weighted result model: Riya walks 10 km east, then 6 km north. How far is she from the start?

Options: 11.66 | 12.66 | 10.66 | 13.66

Answer: 11.66

Solution: Context: warehouse aisle. Distance = $\sqrt{(10^2+6^2)} = 11.66$ km. Shortcut: Track final x and y coordinates.

Q608 ? Replacement cycle ? Medium

A emergency response case uses the quantities in this question. Determine the requested result using the replacement cycle model: Riya walks 11 km east, then 7 km north. How far is she from the start?

Options: 15.04 | 12.04 | 14.04 | 13.04

Answer: 13.04

Solution: Context: emergency response. Distance = $\sqrt{(11^2+7^2)} = 13.04$ km. Shortcut: Track final x and y coordinates.

Q609 ? Unit conversion ? Hard

A campus navigation case uses the quantities in this question. Determine the requested result using the unit conversion model: Riya walks 12 km east, then 8 km north. How far is she from the start?

Options: 14.42 | 15.42 | 13.42 | 16.42

Answer: 14.42

Solution: Context: campus navigation. Distance = $\sqrt{(12^2+8^2)} = 14.42$ km. Shortcut: Track final x and y coordinates.

Q610 ? Work-rate model ? Placement

A drone route case uses the quantities in this question. Determine the requested result using the work-rate model model: Riya walks 13 km east, then 4 km north. How far is she from the start?

Options: 13.6 | 14.6 | 12.6 | 15.6

Answer: 13.6

Solution: Context: drone route. Distance = $\sqrt{(13^2+4^2)} = 13.6$ km. Shortcut: Track final x and y coordinates.

Q611 ? Speed-time model ? Beginner

A delivery map case uses the quantities in this question. Determine the requested result using the speed-time model model: Riya walks 14 km east, then 5 km north. How far is she from the start?

Options: 16.87 | 13.87 | 15.87 | 14.87

Answer: 14.87

Solution: Context: delivery map. Distance = $\sqrt{(14^2+5^2)} = 14.87$ km. Shortcut: Track final x and y coordinates.

Q612 ? Table lookup ? Easy

A city grid case uses the quantities in this question. Determine the requested result using the table lookup model: Riya walks 15 km east, then 6 km north. How far is she from the start?

Options: 18.16 | 15.16 | 17.16 | 16.16

Answer: 16.16

Solution: Context: city grid. Distance = $\sqrt{(15^2+6^2)} = 16.16$ km. Shortcut: Track final x and y coordinates.

Q613 ? Chart comparison ? Medium

A treasure hunt case uses the quantities in this question. Determine the requested result using the chart comparison model: Riya walks 16 km east, then 7 km north. How far is she from the start?

Options: 17.46 | 18.46 | 16.46 | 19.46

Answer: 17.46

Solution: Context: treasure hunt. Distance = $\sqrt{(16^2+7^2)} = 17.46$ km. Shortcut: Track final x and y coordinates.

Q614 ? Multi-step word problem ? Hard

A robot movement case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Riya walks 17 km east, then 8 km north. How far is she from the start?

Options: 18.79 | 19.79 | 17.79 | 20.79

Answer: 18.79

Solution: Context: robot movement. Distance = $\sqrt{(17^2+8^2)} = 18.79$ km. Shortcut: Track final x and y coordinates.

Q615 ? Maximum / minimum ? Placement

A warehouse aisle case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Riya walks 18 km east, then 4 km north. How far is she from the start?

Options: 20.44 | 17.44 | 19.44 | 18.44

Answer: 18.44

Solution: Context: warehouse aisle. Distance = $\sqrt{(18^2+4^2)} = 18.44$ km. Shortcut: Track final x and y coordinates.

Q616 ? Least / greatest value ? Beginner

A emergency response case uses the quantities in this question. Determine the requested result using the least / greatest value model: Riya walks 19 km east, then 5 km north. How far is she from the start?

Options: 21.65 | 20.65 | 19.65 | 18.65

Answer: 19.65

Solution: Context: emergency response. Distance = $\sqrt{(19^2+5^2)} = 19.65$ km. Shortcut: Track final x and y coordinates.

Q617 ? Remainder cycle ? Easy

A campus navigation case uses the quantities in this question. Determine the requested result using the remainder cycle model: Riya walks 20 km east, then 6 km north. How far is she from the start?

Options: 20.88 | 21.88 | 19.88 | 22.88

Answer: 20.88

Solution: Context: campus navigation. Distance = $\sqrt{(20^2+6^2)} = 20.88$ km. Shortcut: Track final x and y coordinates.

Q618 ? Divisibility test ? Medium

A drone route case uses the quantities in this question. Determine the requested result using the divisibility test model: Riya walks 21 km east, then 7 km north. How far is she from the start?

Options: 22.14 | 23.14 | 21.14 | 24.14

Answer: 22.14

Solution: Context: drone route. Distance = $\sqrt{(21^2+7^2)} = 22.14$ km. Shortcut: Track final x and y coordinates.

Q619 ? Factorisation ? Hard

A delivery map case uses the quantities in this question. Determine the requested result using the factorisation model: Riya walks 22 km east, then 8 km north. How far is she from the start?

Options: 25.41 | 22.41 | 24.41 | 23.41

Answer: 23.41

Solution: Context: delivery map. Distance = $\sqrt{(22^2+8^2)} = 23.41$ km. Shortcut: Track final x and y coordinates.

Q620 ? Approximation ? Placement

A city grid case uses the quantities in this question. Determine the requested result using the approximation model: Riya walks 23 km east, then 4 km north. How far is she from the start?

Options: 23.35 | 24.35 | 22.35 | 25.35

Answer: 23.35

Solution: Context: city grid. Distance = $\sqrt{(23^2+4^2)} = 23.35$ km. Shortcut: Track final x and y coordinates.

PRACTICE ? solutions hidden below**Q621 ? Options elimination ? Beginner**

A treasure hunt case uses the quantities in this question. Determine the requested result using the options elimination model: Riya walks 24 km east, then 5

km north. How far is she from the start?
Options: 24.52 | 25.52 | 23.52 | 26.52

Q622 ? Boundary case ? Easy

A robot movement case uses the quantities in this question. Determine the requested result using the boundary case model: Riya walks 25 km east, then 6 km north. How far is she from the start?
Options: 27.71 | 24.71 | 26.71 | 25.71

Q623 ? Case split ? Medium

A warehouse aisle case uses the quantities in this question. Determine the requested result using the case split model: Riya walks 26 km east, then 7 km north. How far is she from the start?
Options: 28.93 | 25.93 | 27.93 | 26.93

Q624 ? Contradiction check ? Hard

A emergency response case uses the quantities in this question. Determine the requested result using the contradiction check model: Riya walks 27 km east, then 8 km north. How far is she from the start?
Options: 28.16 | 29.16 | 27.16 | 30.16

Q625 ? Statement I test ? Placement

A campus navigation case uses the quantities in this question. Determine the requested result using the statement i test model: Riya walks 28 km east, then 4 km north. How far is she from the start?
Options: 28.28 | 29.28 | 27.28 | 30.28

Q626 ? Statement II test ? Beginner

A drone route case uses the quantities in this question. Determine the requested result using the statement ii test model: Riya walks 29 km east, then 5 km north. How far is she from the start?
Options: 31.43 | 28.43 | 30.43 | 29.43

Q627 ? Combined statements ? Easy

A delivery map case uses the quantities in this question. Determine the requested result using the combined statements model: Riya walks 30 km east, then 6 km north. How far is she from the start?
Options: 30.59 | 31.59 | 29.59 | 32.59

Q628 ? Counting restriction ? Medium

A city grid case uses the quantities in this question. Determine the requested result using the counting restriction model: Riya walks 31 km east, then 7 km north. How far is she from the start?
Options: 31.78 | 32.78 | 30.78 | 33.78

Q629 ? Circular arrangement ? Hard

A treasure hunt case uses the quantities in this question. Determine the requested result using the circular arrangement model: Riya walks 32 km east, then 8 km north. How far is she from the start?
Options: 32.98 | 33.98 | 31.98 | 34.98

Q630 ? Complement method ? Placement

A robot movement case uses the quantities in this question. Determine the requested result using the complement method model: Riya walks 33 km east, then 4 km north. How far is she from the start?
Options: 35.24 | 32.24 | 34.24 | 33.24

Q631 ? Pattern: differences ? Beginner

A warehouse aisle case uses the quantities in this question. Determine the requested result using the pattern: differences model: Riya walks 34 km east, then 5 km north. How far is she from the start?
Options: 34.37 | 35.37 | 33.37 | 36.37

Q632 ? Pattern: ratios ? Easy

A emergency response case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Riya walks 35 km east, then 6 km north. How far is she from the start?
Options: 37.51 | 34.51 | 36.51 | 35.51

Q633 ? Pattern: alternating ? Medium

A campus navigation case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Riya walks 36 km east, then 7 km north. How far is she from the start?
Options: 38.67 | 35.67 | 37.67 | 36.67

Q634 ? Pattern: squares / cubes ? Hard

A drone route case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Riya walks 37 km east, then 8 km north. How far is she from the start?
Options: 37.85 | 38.85 | 36.85 | 39.85

Q635 ? Inference from facts ? Placement

A delivery map case uses the quantities in this question. Determine the requested result using the inference from facts model: Riya walks 38 km east, then 4 km north. How far is she from the start?
Options: 38.21 | 39.21 | 37.21 | 40.21

Q636 ? Assumption check ? Beginner

A city grid case uses the quantities in this question. Determine the requested result using the assumption check model: Riya walks 39 km east, then 5 km north. How far is she from the start?
Options: 39.32 | 40.32 | 38.32 | 41.32

Q637 ? Cause vs correlation ? Easy

A treasure hunt case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Riya walks 40 km east, then 6 km north. How far is she from the start?

Options: 42.45 | 39.45 | 41.45 | 40.45

Q638 ? Grammar agreement ? Medium

A robot movement case uses the quantities in this question. Determine the requested result using the grammar agreement model: Riya walks 41 km east, then 7 km north. How far is she from the start?

Options: 41.59 | 42.59 | 40.59 | 43.59

Q639 ? Vocabulary in context ? Hard

A warehouse aisle case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Riya walks 42 km east, then 8 km north. How far is she from the start?

Options: 44.76 | 41.76 | 43.76 | 42.76

Q640 ? Mixed placement challenge ? Placement

A emergency response case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Riya walks 43 km east, then 4 km north. How far is she from the start?

Options: 43.19 | 44.19 | 42.19 | 45.19

Coding–Decoding ? 40 questions

WORKED EXAMPLES

Q641 ? Direct computation ? Beginner

A secure token case uses the quantities in this question. Determine the requested result using the direct computation model: Every letter moves 2 places forward. How is CAT coded?

Options: DBU | CAT | 13 | ECV

Answer: ECV

Solution: Context: secure token. $C \rightarrow E$, $A \rightarrow C$, $T \rightarrow V$; answer ECV. Shortcut: Use alphabet positions when wrapping after Z.

Q642 ? Reverse value ? Easy

A letter shift case uses the quantities in this question. Determine the requested result using the reverse value model: Every letter moves 3 places forward. How is CAT coded?

Options: FDW | CAT | DBU | ECV

Answer: FDW

Solution: Context: letter shift. $C \rightarrow F$, $A \rightarrow D$, $T \rightarrow W$; answer FDW. Shortcut: Use alphabet positions when wrapping after Z.

Q643 ? Missing base ? Medium

A phone keypad case uses the quantities in this question. Determine the requested result using the missing base model: Every letter moves 4 places forward. How is CAT coded?

Options: ECV | DBU | GEX | CAT

Answer: GEX

Solution: Context: phone keypad. $C \rightarrow G$, $A \rightarrow E$, $T \rightarrow X$; answer GEX. Shortcut: Use alphabet positions when wrapping after Z.

Q644 ? Successive change ? Hard

A encrypted label case uses the quantities in this question. Determine the requested result using the successive change model: Every letter moves 5 places forward. How is CAT coded?

Options: HFY | CAT | DBU | ECV

Answer: HFY

Solution: Context: encrypted label. $C \rightarrow H$, $A \rightarrow F$, $T \rightarrow Y$; answer HFY. Shortcut: Use alphabet positions when wrapping after Z.

Q645 ? Comparison trap ? Placement

A pattern code case uses the quantities in this question. Determine the requested result using the comparison trap model: Every letter moves 1 place forward. How is CAT coded?

Options: 13 | ECV | DBU | CAT

Answer: DBU

Solution: Context: pattern code. $C \rightarrow D$, $A \rightarrow B$, $T \rightarrow U$; answer DBU. Shortcut: Use alphabet positions when wrapping after Z.

Q646 ? Ratio form ? Beginner

A symbol language case uses the quantities in this question. Determine the requested result using the ratio form model: Every letter moves 2 places forward. How is CAT coded?

Options: ECV | DBU | 13 | CAT

Answer: ECV

Solution: Context: symbol language. $C \rightarrow E$, $A \rightarrow C$, $T \rightarrow V$; answer ECV. Shortcut: Use alphabet positions when wrapping after Z.

Q647 ? Weighted result ? Easy

A ID transformation case uses the quantities in this question. Determine the requested result using the weighted result model: Every letter moves 3 places forward. How is CAT coded?

Options: FDW | CAT | DBU | ECV

Answer: FDW

Solution: Context: ID transformation. $C \rightarrow F$, $A \rightarrow D$, $T \rightarrow W$; answer FDW. Shortcut: Use alphabet positions when wrapping after Z.

Q648 ? Replacement cycle ? Medium

A message protocol case uses the quantities in this question. Determine the requested result using the replacement cycle model: Every letter moves 4 places forward. How is CAT coded?

Options: ECV | DBU | CAT | GEX

Answer: GEX

Solution: Context: message protocol. $C \rightarrow G$, $A \rightarrow E$, $T \rightarrow X$; answer GEX. Shortcut: Use alphabet positions when wrapping after Z.

Q649 ? Unit conversion ? Hard

A secure token case uses the quantities in this question. Determine the requested result using the unit conversion model: Every letter moves 5 places forward. How is CAT coded?

Options: CAT | DBU | HFY | ECV

Answer: HFY

Solution: Context: secure token. $C \rightarrow H$, $A \rightarrow F$, $T \rightarrow Y$; answer HFY. Shortcut: Use alphabet positions when wrapping after Z.

Q650 ? Work-rate model ? Placement

A letter shift case uses the quantities in this question. Determine the requested result using the work-rate model model: Every letter moves 1 place forward. How is CAT coded?

Options: ECV | 13 | DBU | CAT

Answer: DBU

Solution: Context: letter shift. $C \rightarrow D$, $A \rightarrow B$, $T \rightarrow U$; answer DBU. Shortcut: Use alphabet positions when wrapping after Z.

Q651 ? Speed-time model ? Beginner

A phone keypad case uses the quantities in this question. Determine the requested result using the speed-time model model: Every letter moves 2 places forward. How is CAT coded?

Options: CAT | DBU | ECV | 13

Answer: ECV

Solution: Context: phone keypad. C→E, A→C, T→V; answer ECV. Shortcut: Use alphabet positions when wrapping after Z.

Q652 ? Table lookup ? Easy

A encrypted label case uses the quantities in this question. Determine the requested result using the table lookup model: Every letter moves 3 places forward. How is CAT coded?

Options: ECV | FDW | CAT | DBU

Answer: FDW

Solution: Context: encrypted label. C→F, A→D, T→W; answer FDW. Shortcut: Use alphabet positions when wrapping after Z.

Q653 ? Chart comparison ? Medium

A pattern code case uses the quantities in this question. Determine the requested result using the chart comparison model: Every letter moves 4 places forward. How is CAT coded?

Options: DBU | ECV | CAT | GEX

Answer: GEX

Solution: Context: pattern code. C→G, A→E, T→X; answer GEX. Shortcut: Use alphabet positions when wrapping after Z.

Q654 ? Multi-step word problem ? Hard

A symbol language case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Every letter moves 5 places forward. How is CAT coded?

Options: HFY | CAT | DBU | ECV

Answer: HFY

Solution: Context: symbol language. C→H, A→F, T→Y; answer HFY. Shortcut: Use alphabet positions when wrapping after Z.

Q655 ? Maximum / minimum ? Placement

A ID transformation case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Every letter moves 1 place forward. How is CAT coded?

Options: CAT | 13 | ECV | DBU

Answer: DBU

Solution: Context: ID transformation. C→D, A→B, T→U; answer DBU. Shortcut: Use alphabet positions when wrapping after Z.

Q656 ? Least / greatest value ? Beginner

A message protocol case uses the quantities in this question. Determine the requested result using the least / greatest value model: Every letter moves 2 places forward. How is CAT coded?

Options: ECV | CAT | DBU | 13

Answer: ECV

Solution: Context: message protocol. C→E, A→C, T→V; answer ECV. Shortcut: Use alphabet positions when wrapping after Z.

Q657 ? Remainder cycle ? Easy

A secure token case uses the quantities in this question. Determine the requested result using the remainder cycle model: Every letter moves 3 places forward. How is CAT coded?

Options: DBU | FDW | ECV | CAT

Answer: FDW

Solution: Context: secure token. C→F, A→D, T→W; answer FDW. Shortcut: Use alphabet positions when wrapping after Z.

Q658 ? Divisibility test ? Medium

A letter shift case uses the quantities in this question. Determine the requested result using the divisibility test model: Every letter moves 4 places forward. How is CAT coded?

Options: GEX | CAT | DBU | ECV

Answer: GEX

Solution: Context: letter shift. C→G, A→E, T→X; answer GEX. Shortcut: Use alphabet positions when wrapping after Z.

Q659 ? Factorisation ? Hard

A phone keypad case uses the quantities in this question. Determine the requested result using the factorisation model: Every letter moves 5 places forward. How is CAT coded?

Options: ECV | CAT | HFY | DBU

Answer: HFY

Solution: Context: phone keypad. C→H, A→F, T→Y; answer HFY. Shortcut: Use alphabet positions when wrapping after Z.

Q660 ? Approximation ? Placement

A encrypted label case uses the quantities in this question. Determine the requested result using the approximation model: Every letter moves 1 place forward. How is CAT coded?

Options: DBU | ECV | 13 | CAT

Answer: DBU

Solution: Context: encrypted label. C→D, A→B, T→U; answer DBU. Shortcut: Use alphabet positions when wrapping after Z.

PRACTICE ? solutions hidden below**Q661 ? Options elimination ? Beginner**

A pattern code case uses the quantities in this question. Determine the requested result using the options elimination model: Every letter moves 2 places

forward. How is CAT coded?
Options: 13 | CAT | DBU | ECV

Q662 ? Boundary case ? Easy

A symbol language case uses the quantities in this question. Determine the requested result using the boundary case model: Every letter moves 3 places forward. How is CAT coded?
Options: ECV | DBU | CAT | FDW

Q663 ? Case split ? Medium

A ID transformation case uses the quantities in this question. Determine the requested result using the case split model: Every letter moves 4 places forward. How is CAT coded?
Options: CAT | GEX | DBU | ECV

Q664 ? Contradiction check ? Hard

A message protocol case uses the quantities in this question. Determine the requested result using the contradiction check model: Every letter moves 5 places forward. How is CAT coded?
Options: ECV | DBU | HFY | CAT

Q665 ? Statement I test ? Placement

A secure token case uses the quantities in this question. Determine the requested result using the statement i test model: Every letter moves 1 place forward. How is CAT coded?
Options: CAT | DBU | ECV | 13

Q666 ? Statement II test ? Beginner

A letter shift case uses the quantities in this question. Determine the requested result using the statement ii test model: Every letter moves 2 places forward. How is CAT coded?
Options: ECV | CAT | 13 | DBU

Q667 ? Combined statements ? Easy

A phone keypad case uses the quantities in this question. Determine the requested result using the combined statements model: Every letter moves 3 places forward. How is CAT coded?
Options: DBU | ECV | CAT | FDW

Q668 ? Counting restriction ? Medium

A encrypted label case uses the quantities in this question. Determine the requested result using the counting restriction model: Every letter moves 4 places forward. How is CAT coded?
Options: GEX | CAT | DBU | ECV

Q669 ? Circular arrangement ? Hard

A pattern code case uses the quantities in this question. Determine the requested result using the circular arrangement model: Every letter moves 5 places forward. How is CAT coded?
Options: HFY | ECV | DBU | CAT

Q670 ? Complement method ? Placement

A symbol language case uses the quantities in this question. Determine the requested result using the complement method model: Every letter moves 1 place forward. How is CAT coded?
Options: DBU | CAT | ECV | 13

Q671 ? Pattern: differences ? Beginner

A ID transformation case uses the quantities in this question. Determine the requested result using the pattern: differences model: Every letter moves 2 places forward. How is CAT coded?
Options: ECV | 13 | DBU | CAT

Q672 ? Pattern: ratios ? Easy

A message protocol case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Every letter moves 3 places forward. How is CAT coded?
Options: CAT | DBU | ECV | FDW

Q673 ? Pattern: alternating ? Medium

A secure token case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Every letter moves 4 places forward. How is CAT coded?
Options: ECV | GEX | CAT | DBU

Q674 ? Pattern: squares / cubes ? Hard

A letter shift case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Every letter moves 5 places forward. How is CAT coded?
Options: DBU | ECV | CAT | HFY

Q675 ? Inference from facts ? Placement

A phone keypad case uses the quantities in this question. Determine the requested result using the inference from facts model: Every letter moves 1 place forward. How is CAT coded?
Options: 13 | CAT | DBU | ECV

Q676 ? Assumption check ? Beginner

A encrypted label case uses the quantities in this question. Determine the requested result using the assumption check model: Every letter moves 2 places forward. How is CAT coded?
Options: DBU | CAT | 13 | ECV

Q677 ? Cause vs correlation ? Easy

A pattern code case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Every letter moves 3 places forward. How is CAT coded?

Options: FDW | CAT | DBU | ECV

Q678 ? Grammar agreement ? Medium

A symbol language case uses the quantities in this question. Determine the requested result using the grammar agreement model: Every letter moves 4 places forward. How is CAT coded?

Options: ECV | DBU | GEX | CAT

Q679 ? Vocabulary in context ? Hard

A ID transformation case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Every letter moves 5 places forward. How is CAT coded?

Options: HFY | CAT | DBU | ECV

Q680 ? Mixed placement challenge ? Placement

A message protocol case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Every letter moves 1 place forward. How is CAT coded?

Options: 13 | ECV | DBU | CAT

Data Interpretation ? 40 questions

WORKED EXAMPLES

Q681 ? Direct computation ? Beginner

A quarterly dashboard case uses the quantities in this question. Determine the requested result using the direct computation model: Sales: Q1 ■125L, Q2 ■155L, Q3 ■185L, Q4 ■215L. Percentage increase from Q1 to Q4?

Options: 67 | 82 | 77 | 72

Answer: 72

Solution: Context: quarterly dashboard. Increase = 90. Increase% = $90/125 \times 100 = 72\%$. Shortcut: Estimate the fraction before exact division.

Q682 ? Reverse value ? Easy

A placement report case uses the quantities in this question. Determine the requested result using the reverse value model: Sales: Q1 ■130L, Q2 ■160L, Q3 ■190L, Q4 ■220L. Percentage increase from Q1 to Q4?

Options: 69.23 | 74.23 | 64.23 | 79.23

Answer: 69.23

Solution: Context: placement report. Increase = 90. Increase% = $90/130 \times 100 = 69.23\%$. Shortcut: Estimate the fraction before exact division.

Q683 ? Missing base ? Medium

A sales chart case uses the quantities in this question. Determine the requested result using the missing base model: Sales: Q1 ■135L, Q2 ■165L, Q3 ■195L, Q4 ■225L. Percentage increase from Q1 to Q4?

Options: 71.67 | 76.67 | 61.67 | 66.67

Answer: 66.67

Solution: Context: sales chart. Increase = 90. Increase% = $90/135 \times 100 = 66.67\%$. Shortcut: Estimate the fraction before exact division.

Q684 ? Successive change ? Hard

A attendance table case uses the quantities in this question. Determine the requested result using the successive change model: Sales: Q1 ■140L, Q2 ■170L, Q3 ■200L, Q4 ■230L. Percentage increase from Q1 to Q4?

Options: 59.29 | 69.29 | 64.29 | 74.29

Answer: 64.29

Solution: Context: attendance table. Increase = 90. Increase% = $90/140 \times 100 = 64.29\%$. Shortcut: Estimate the fraction before exact division.

Q685 ? Comparison trap ? Placement

A budget sheet case uses the quantities in this question. Determine the requested result using the comparison trap model: Sales: Q1 ■145L, Q2 ■175L, Q3 ■205L, Q4 ■235L. Percentage increase from Q1 to Q4?

Options: 72.07 | 62.07 | 67.07 | 57.07

Answer: 62.07

Solution: Context: budget sheet. Increase = 90. Increase% = $90/145 \times 100 = 62.07\%$. Shortcut: Estimate the fraction before exact division.

Q686 ? Ratio form ? Beginner

A product funnel case uses the quantities in this question. Determine the requested result using the ratio form model: Sales: Q1 ■150L, Q2 ■180L, Q3 ■210L, Q4 ■240L. Percentage increase from Q1 to Q4?

Options: 60 | 65 | 70 | 55

Answer: 60

Solution: Context: product funnel. Increase = 90. Increase% = $90/150 \times 100 = 60\%$. Shortcut: Estimate the fraction before exact division.

Q687 ? Weighted result ? Easy

A survey graph case uses the quantities in this question. Determine the requested result using the weighted result model: Sales: Q1 ■155L, Q2 ■185L, Q3 ■215L, Q4 ■245L. Percentage increase from Q1 to Q4?

Options: 58.06 | 53.06 | 63.06 | 68.06

Answer: 58.06

Solution: Context: survey graph. Increase = 90. Increase% = $90/155 \times 100 = 58.06\%$. Shortcut: Estimate the fraction before exact division.

Q688 ? Replacement cycle ? Medium

A regional KPI case uses the quantities in this question. Determine the requested result using the replacement cycle model: Sales: Q1 ■160L, Q2 ■190L, Q3 ■220L, Q4 ■250L. Percentage increase from Q1 to Q4?

Options: 66.25 | 51.25 | 61.25 | 56.25

Answer: 56.25

Solution: Context: regional KPI. Increase = 90. Increase% = $90/160 \times 100 = 56.25\%$. Shortcut: Estimate the fraction before exact division.

Q689 ? Unit conversion ? Hard

A quarterly dashboard case uses the quantities in this question. Determine the requested result using the unit conversion model: Sales: Q1 ■165L, Q2 ■195L, Q3 ■225L, Q4 ■255L. Percentage increase from Q1 to Q4?

Options: 54.55 | 59.55 | 64.55 | 49.55

Answer: 54.55

Solution: Context: quarterly dashboard. Increase = 90. Increase% = $90/165 \times 100 = 54.55\%$. Shortcut: Estimate the fraction before exact division.

Q690 ? Work-rate model ? Placement

A placement report case uses the quantities in this question. Determine the requested result using the work-rate model model: Sales: Q1 ■170L, Q2 ■200L, Q3 ■230L, Q4 ■260L. Percentage increase from Q1 to Q4?

Options: 62.94 | 52.94 | 57.94 | 47.94

Answer: 52.94

Solution: Context: placement report. Increase = 90. Increase% = $90/170 \times 100 = 52.94\%$. Shortcut: Estimate the fraction before exact division.

Q691 ? Speed-time model ? Beginner

A sales chart case uses the quantities in this question. Determine the requested result using the speed-time model model: Sales: Q1 ■175L, Q2 ■205L, Q3 ■235L, Q4 ■265L. Percentage increase from Q1 to Q4?

Options: 46.43 | 56.43 | 51.43 | 61.43

Answer: 51.43

Solution: Context: sales chart. Increase = 90. Increase% = $90/175 \times 100 = 51.43\%$. Shortcut: Estimate the fraction before exact division.

Q692 ? Table lookup ? Easy

A attendance table case uses the quantities in this question. Determine the requested result using the table lookup model: Sales: Q1 ■180L, Q2 ■210L, Q3 ■240L, Q4 ■270L. Percentage increase from Q1 to Q4?

Options: 50 | 55 | 45 | 60

Answer: 50

Solution: Context: attendance table. Increase = 90. Increase% = $90/180 \times 100 = 50\%$. Shortcut: Estimate the fraction before exact division.

Q693 ? Chart comparison ? Medium

A budget sheet case uses the quantities in this question. Determine the requested result using the chart comparison model: Sales: Q1 ■185L, Q2 ■215L, Q3 ■245L, Q4 ■275L. Percentage increase from Q1 to Q4?

Options: 58.65 | 43.65 | 53.65 | 48.65

Answer: 48.65

Solution: Context: budget sheet. Increase = 90. Increase% = $90/185 \times 100 = 48.65\%$. Shortcut: Estimate the fraction before exact division.

Q694 ? Multi-step word problem ? Hard

A product funnel case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Sales: Q1 ■190L, Q2 ■220L, Q3 ■250L, Q4 ■280L. Percentage increase from Q1 to Q4?

Options: 47.37 | 42.37 | 52.37 | 57.37

Answer: 47.37

Solution: Context: product funnel. Increase = 90. Increase% = $90/190 \times 100 = 47.37\%$. Shortcut: Estimate the fraction before exact division.

Q695 ? Maximum / minimum ? Placement

A survey graph case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Sales: Q1 ■195L, Q2 ■225L, Q3 ■255L, Q4 ■285L. Percentage increase from Q1 to Q4?

Options: 56.15 | 51.15 | 46.15 | 41.15

Answer: 46.15

Solution: Context: survey graph. Increase = 90. Increase% = $90/195 \times 100 = 46.15\%$. Shortcut: Estimate the fraction before exact division.

Q696 ? Least / greatest value ? Beginner

A regional KPI case uses the quantities in this question. Determine the requested result using the least / greatest value model: Sales: Q1 ■200L, Q2 ■230L, Q3 ■260L, Q4 ■290L. Percentage increase from Q1 to Q4?

Options: 50 | 55 | 40 | 45

Answer: 45

Solution: Context: regional KPI. Increase = 90. Increase% = $90/200 \times 100 = 45\%$. Shortcut: Estimate the fraction before exact division.

Q697 ? Remainder cycle ? Easy

A quarterly dashboard case uses the quantities in this question. Determine the requested result using the remainder cycle model: Sales: Q1 ■205L, Q2 ■235L, Q3 ■265L, Q4 ■295L. Percentage increase from Q1 to Q4?

Options: 43.9 | 48.9 | 38.9 | 53.9

Answer: 43.9

Solution: Context: quarterly dashboard. Increase = 90. Increase% = $90/205 \times 100 = 43.9\%$. Shortcut: Estimate the fraction before exact division.

Q698 ? Divisibility test ? Medium

A placement report case uses the quantities in this question. Determine the requested result using the divisibility test model: Sales: Q1 ■210L, Q2 ■240L, Q3 ■270L, Q4 ■300L. Percentage increase from Q1 to Q4?

Options: 37.86 | 47.86 | 42.86 | 52.86

Answer: 42.86

Solution: Context: placement report. Increase = 90. Increase% = $90/210 \times 100 = 42.86\%$. Shortcut: Estimate the fraction before exact division.

Q699 ? Factorisation ? Hard

A sales chart case uses the quantities in this question. Determine the requested result using the factorisation model: Sales: Q1 ■215L, Q2 ■245L, Q3 ■275L, Q4 ■305L. Percentage increase from Q1 to Q4?

Options: 41.86 | 46.86 | 51.86 | 36.86

Answer: 41.86

Solution: Context: sales chart. Increase = 90. Increase% = $90/215 \times 100 = 41.86\%$. Shortcut: Estimate the fraction before exact division.

Q700 ? Approximation ? Placement

A attendance table case uses the quantities in this question. Determine the requested result using the approximation model: Sales: Q1 ■220L, Q2 ■250L, Q3 ■280L, Q4 ■310L. Percentage increase from Q1 to Q4?

Options: 50.91 | 45.91 | 40.91 | 35.91

Answer: 40.91

Solution: Context: attendance table. Increase = 90. Increase% = $90/220 \times 100 = 40.91\%$. Shortcut: Estimate the fraction before exact division.

PRACTICE ? solutions hidden below**Q701 ? Options elimination ? Beginner**

A budget sheet case uses the quantities in this question. Determine the requested result using the options elimination model: Sales: Q1 ■225L, Q2 ■255L, Q3

■285L, Q4 ■315L. Percentage increase from Q1 to Q4?
Options: 35 | 40 | 45 | 50

Q702 ? Boundary case ? Easy

A product funnel case uses the quantities in this question. Determine the requested result using the boundary case model: Sales: Q1 ■230L, Q2 ■260L, Q3 ■290L, Q4 ■320L. Percentage increase from Q1 to Q4?
Options: 39.13 | 44.13 | 34.13 | 49.13

Q703 ? Case split ? Medium

A survey graph case uses the quantities in this question. Determine the requested result using the case split model: Sales: Q1 ■235L, Q2 ■265L, Q3 ■295L, Q4 ■325L. Percentage increase from Q1 to Q4?
Options: 48.3 | 33.3 | 43.3 | 38.3

Q704 ? Contradiction check ? Hard

A regional KPI case uses the quantities in this question. Determine the requested result using the contradiction check model: Sales: Q1 ■240L, Q2 ■270L, Q3 ■300L, Q4 ■330L. Percentage increase from Q1 to Q4?
Options: 37.5 | 42.5 | 32.5 | 47.5

Q705 ? Statement I test ? Placement

A quarterly dashboard case uses the quantities in this question. Determine the requested result using the statement i test model: Sales: Q1 ■245L, Q2 ■275L, Q3 ■305L, Q4 ■335L. Percentage increase from Q1 to Q4?
Options: 31.73 | 36.73 | 46.73 | 41.73

Q706 ? Statement II test ? Beginner

A placement report case uses the quantities in this question. Determine the requested result using the statement ii test model: Sales: Q1 ■250L, Q2 ■280L, Q3 ■310L, Q4 ■340L. Percentage increase from Q1 to Q4?
Options: 41 | 46 | 31 | 36

Q707 ? Combined statements ? Easy

A sales chart case uses the quantities in this question. Determine the requested result using the combined statements model: Sales: Q1 ■255L, Q2 ■285L, Q3 ■315L, Q4 ■345L. Percentage increase from Q1 to Q4?
Options: 45.29 | 40.29 | 35.29 | 30.29

Q708 ? Counting restriction ? Medium

A attendance table case uses the quantities in this question. Determine the requested result using the counting restriction model: Sales: Q1 ■260L, Q2 ■290L, Q3 ■320L, Q4 ■350L. Percentage increase from Q1 to Q4?
Options: 29.62 | 34.62 | 39.62 | 44.62

Q709 ? Circular arrangement ? Hard

A budget sheet case uses the quantities in this question. Determine the requested result using the circular arrangement model: Sales: Q1 ■265L, Q2 ■295L, Q3 ■325L, Q4 ■355L. Percentage increase from Q1 to Q4?
Options: 33.96 | 38.96 | 28.96 | 43.96

Q710 ? Complement method ? Placement

A product funnel case uses the quantities in this question. Determine the requested result using the complement method model: Sales: Q1 ■270L, Q2 ■300L, Q3 ■330L, Q4 ■360L. Percentage increase from Q1 to Q4?
Options: 43.33 | 28.33 | 38.33 | 33.33

Q711 ? Pattern: differences ? Beginner

A survey graph case uses the quantities in this question. Determine the requested result using the pattern: differences model: Sales: Q1 ■275L, Q2 ■305L, Q3 ■335L, Q4 ■365L. Percentage increase from Q1 to Q4?
Options: 32.73 | 37.73 | 27.73 | 42.73

Q712 ? Pattern: ratios ? Easy

A regional KPI case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Sales: Q1 ■280L, Q2 ■310L, Q3 ■340L, Q4 ■370L. Percentage increase from Q1 to Q4?
Options: 37.14 | 32.14 | 27.14 | 42.14

Q713 ? Pattern: alternating ? Medium

A quarterly dashboard case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Sales: Q1 ■285L, Q2 ■315L, Q3 ■345L, Q4 ■375L. Percentage increase from Q1 to Q4?
Options: 41.58 | 26.58 | 36.58 | 31.58

Q714 ? Pattern: squares / cubes ? Hard

A placement report case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Sales: Q1 ■290L, Q2 ■320L, Q3 ■350L, Q4 ■380L. Percentage increase from Q1 to Q4?
Options: 41.03 | 31.03 | 36.03 | 26.03

Q715 ? Inference from facts ? Placement

A sales chart case uses the quantities in this question. Determine the requested result using the inference from facts model: Sales: Q1 ■295L, Q2 ■325L, Q3 ■355L, Q4 ■385L. Percentage increase from Q1 to Q4?
Options: 25.51 | 30.51 | 35.51 | 40.51

Q716 ? Assumption check ? Beginner

A attendance table case uses the quantities in this question. Determine the requested result using the assumption check model: Sales: Q1 ■300L, Q2 ■330L, Q3 ■360L, Q4 ■390L. Percentage increase from Q1 to Q4?
Options: 30 | 35 | 25 | 40

Q717 ? Cause vs correlation ? Easy

A budget sheet case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Sales: Q1 ■305L, Q2 ■335L, Q3 ■365L, Q4 ■395L. Percentage increase from Q1 to Q4?

Options: 39.51 | 24.51 | 34.51 | 29.51

Q718 ? Grammar agreement ? Medium

A product funnel case uses the quantities in this question. Determine the requested result using the grammar agreement model: Sales: Q1 ■310L, Q2 ■340L, Q3 ■370L, Q4 ■400L. Percentage increase from Q1 to Q4?

Options: 29.03 | 34.03 | 24.03 | 39.03

Q719 ? Vocabulary in context ? Hard

A survey graph case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Sales: Q1 ■315L, Q2 ■345L, Q3 ■375L, Q4 ■405L. Percentage increase from Q1 to Q4?

Options: 28.57 | 33.57 | 23.57 | 38.57

Q720 ? Mixed placement challenge ? Placement

A regional KPI case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Sales: Q1 ■320L, Q2 ■350L, Q3 ■380L, Q4 ■410L. Percentage increase from Q1 to Q4?

Options: 38.13 | 23.13 | 33.13 | 28.13

Clocks ? 40 questions

WORKED EXAMPLES

Q721 ? Direct computation ? Beginner

A analog dial case uses the quantities in this question. Determine the requested result using the direct computation model: Find the smaller angle between the hands at 2:10.

Options: 5 | 15 | 0 | 10

Answer: 5

Solution: Context: analog dial. Angle = $|30H - 5.5M| = 5^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q722 ? Reverse value ? Easy

A watch calibration case uses the quantities in this question. Determine the requested result using the reverse value model: Find the smaller angle between the hands at 3:20.

Options: 30 | 25 | 20 | 15

Answer: 20

Solution: Context: watch calibration. Angle = $|30H - 5.5M| = 20^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q723 ? Missing base ? Medium

A factory shift case uses the quantities in this question. Determine the requested result using the missing base model: Find the smaller angle between the hands at 4:30.

Options: 45 | 50 | 40 | 55

Answer: 45

Solution: Context: factory shift. Angle = $|30H - 5.5M| = 45^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q724 ? Successive change ? Hard

A clock-angle puzzle case uses the quantities in this question. Determine the requested result using the successive change model: Find the smaller angle between the hands at 5:40.

Options: 65 | 75 | 70 | 80

Answer: 70

Solution: Context: clock-angle puzzle. Angle = $|30H - 5.5M| = 70^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q725 ? Comparison trap ? Placement

A time-zone display case uses the quantities in this question. Determine the requested result using the comparison trap model: Find the smaller angle between the hands at 6:50.

Options: 105 | 90 | 100 | 95

Answer: 95

Solution: Context: time-zone display. Angle = $|30H - 5.5M| = 95^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q726 ? Ratio form ? Beginner

A race timer case uses the quantities in this question. Determine the requested result using the ratio form model: Find the smaller angle between the hands at 7:00.

Options: 150 | 155 | 145 | 160

Answer: 150

Solution: Context: race timer. Angle = $|30H - 5.5M| = 150^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q727 ? Weighted result ? Easy

A office clock case uses the quantities in this question. Determine the requested result using the weighted result model: Find the smaller angle between the hands at 8:10.

Options: 175 | 180 | 170 | 185

Answer: 175

Solution: Context: office clock. Angle = $|30H - 5.5M| = 175^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q728 ? Replacement cycle ? Medium

A gain-loss audit case uses the quantities in this question. Determine the requested result using the replacement cycle model: Find the smaller angle between the hands at 9:20.

Options: 170 | 155 | 165 | 160

Answer: 160

Solution: Context: gain-loss audit. Angle = $|30H - 5.5M| = 160^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q729 ? Unit conversion ? Hard

A analog dial case uses the quantities in this question. Determine the requested result using the unit conversion model: Find the smaller angle between the hands at 10:30.

Options: 135 | 140 | 130 | 145

Answer: 135

Solution: Context: analog dial. Angle = $|30H - 5.5M| = 135^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q730 ? Work-rate model ? Placement

A watch calibration case uses the quantities in this question. Determine the requested result using the work-rate model model: Find the smaller angle between the hands at 11:40.

Options: 110 | 115 | 105 | 120

Answer: 110

Solution: Context: watch calibration. Angle = $|30H - 5.5M| = 110^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q731 ? Speed-time model ? Beginner

A factory shift case uses the quantities in this question. Determine the requested result using the speed-time model model: Find the smaller angle between the hands at 1:50.

Options: 125 | 110 | 120 | 115

Answer: 115

Solution: Context: factory shift. Angle = $|30H - 5.5M| = 115^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q732 ? Table lookup ? Easy

A clock-angle puzzle case uses the quantities in this question. Determine the requested result using the table lookup model: Find the smaller angle between the hands at 2:00.

Options: 70 | 60 | 65 | 55

Answer: 60

Solution: Context: clock-angle puzzle. Angle = $|30H - 5.5M| = 60^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q733 ? Chart comparison ? Medium

A time-zone display case uses the quantities in this question. Determine the requested result using the chart comparison model: Find the smaller angle between the hands at 3:10.

Options: 45 | 40 | 35 | 30

Answer: 35

Solution: Context: time-zone display. Angle = $|30H - 5.5M| = 35^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q734 ? Multi-step word problem ? Hard

A race timer case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Find the smaller angle between the hands at 4:20.

Options: 10 | 15 | 20 | 5

Answer: 10

Solution: Context: race timer. Angle = $|30H - 5.5M| = 10^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q735 ? Maximum / minimum ? Placement

A office clock case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Find the smaller angle between the hands at 5:30.

Options: 25 | 10 | 20 | 15

Answer: 15

Solution: Context: office clock. Angle = $|30H - 5.5M| = 15^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q736 ? Least / greatest value ? Beginner

A gain-loss audit case uses the quantities in this question. Determine the requested result using the least / greatest value model: Find the smaller angle between the hands at 6:40.

Options: 50 | 35 | 45 | 40

Answer: 40

Solution: Context: gain-loss audit. Angle = $|30H - 5.5M| = 40^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q737 ? Remainder cycle ? Easy

A analog dial case uses the quantities in this question. Determine the requested result using the remainder cycle model: Find the smaller angle between the hands at 7:50.

Options: 70 | 75 | 60 | 65

Answer: 65

Solution: Context: analog dial. Angle = $|30H - 5.5M| = 65^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q738 ? Divisibility test ? Medium

A watch calibration case uses the quantities in this question. Determine the requested result using the divisibility test model: Find the smaller angle between the hands at 8:00.

Options: 130 | 115 | 125 | 120

Answer: 120

Solution: Context: watch calibration. Angle = $|30H - 5.5M| = 120^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q739 ? Factorisation ? Hard

A factory shift case uses the quantities in this question. Determine the requested result using the factorisation model: Find the smaller angle between the hands at 9:10.

Options: 155 | 140 | 150 | 145

Answer: 145

Solution: Context: factory shift. Angle = $|30H - 5.5M| = 145^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

Q740 ? Approximation ? Placement

A clock-angle puzzle case uses the quantities in this question. Determine the requested result using the approximation model: Find the smaller angle between the hands at 10:20.

Options: 170 | 175 | 165 | 180

Answer: 170

Solution: Context: clock-angle puzzle. Angle = $|30H - 5.5M| = 170^\circ$. Shortcut: If angle exceeds 180° , subtract from 360° .

PRACTICE ? solutions hidden below**Q741 ? Options elimination ? Beginner**

A time-zone display case uses the quantities in this question. Determine the requested result using the options elimination model: Find the smaller angle

between the hands at 11:30.
Options: 165 | 170 | 160 | 175

Q742 ? Boundary case ? Easy

A race timer case uses the quantities in this question. Determine the requested result using the boundary case model: Find the smaller angle between the hands at 1:40.
Options: 180 | 165 | 175 | 170

Q743 ? Case split ? Medium

A office clock case uses the quantities in this question. Determine the requested result using the case split model: Find the smaller angle between the hands at 2:50.
Options: 145 | 150 | 140 | 155

Q744 ? Contradiction check ? Hard

A gain-loss audit case uses the quantities in this question. Determine the requested result using the contradiction check model: Find the smaller angle between the hands at 3:00.
Options: 90 | 95 | 85 | 100

Q745 ? Statement I test ? Placement

A analog dial case uses the quantities in this question. Determine the requested result using the statement i test model: Find the smaller angle between the hands at 4:10.
Options: 60 | 65 | 75 | 70

Q746 ? Statement II test ? Beginner

A watch calibration case uses the quantities in this question. Determine the requested result using the statement ii test model: Find the smaller angle between the hands at 5:20.
Options: 40 | 45 | 50 | 35

Q747 ? Combined statements ? Easy

A factory shift case uses the quantities in this question. Determine the requested result using the combined statements model: Find the smaller angle between the hands at 6:30.
Options: 15 | 20 | 10 | 25

Q748 ? Counting restriction ? Medium

A clock-angle puzzle case uses the quantities in this question. Determine the requested result using the counting restriction model: Find the smaller angle between the hands at 7:40.
Options: 10 | 15 | 20 | 5

Q749 ? Circular arrangement ? Hard

A time-zone display case uses the quantities in this question. Determine the requested result using the circular arrangement model: Find the smaller angle between the hands at 8:50.
Options: 35 | 40 | 30 | 45

Q750 ? Complement method ? Placement

A race timer case uses the quantities in this question. Determine the requested result using the complement method model: Find the smaller angle between the hands at 9:00.
Options: 95 | 90 | 85 | 100

Q751 ? Pattern: differences ? Beginner

A office clock case uses the quantities in this question. Determine the requested result using the pattern: differences model: Find the smaller angle between the hands at 10:10.
Options: 115 | 120 | 110 | 125

Q752 ? Pattern: ratios ? Easy

A gain-loss audit case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Find the smaller angle between the hands at 11:20.
Options: 150 | 135 | 145 | 140

Q753 ? Pattern: alternating ? Medium

A analog dial case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Find the smaller angle between the hands at 1:30.
Options: 145 | 130 | 140 | 135

Q754 ? Pattern: squares / cubes ? Hard

A watch calibration case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Find the smaller angle between the hands at 2:40.
Options: 160 | 165 | 155 | 170

Q755 ? Inference from facts ? Placement

A factory shift case uses the quantities in this question. Determine the requested result using the inference from facts model: Find the smaller angle between the hands at 3:50.
Options: 175 | 180 | 170 | 185

Q756 ? Assumption check ? Beginner

A clock-angle puzzle case uses the quantities in this question. Determine the requested result using the assumption check model: Find the smaller angle between the hands at 4:00.
Options: 130 | 115 | 125 | 120

Q757 ? Cause vs correlation ? Easy

A time-zone display case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Find the smaller angle between the hands at 5:10.

Options: 90 | 95 | 100 | 105

Q758 ? Grammar agreement ? Medium

A race timer case uses the quantities in this question. Determine the requested result using the grammar agreement model: Find the smaller angle between the hands at 6:20.

Options: 70 | 75 | 80 | 65

Q759 ? Vocabulary in context ? Hard

A office clock case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Find the smaller angle between the hands at 7:30.

Options: 40 | 45 | 55 | 50

Q760 ? Mixed placement challenge ? Placement

A gain-loss audit case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Find the smaller angle between the hands at 8:40.

Options: 30 | 15 | 25 | 20

Verbal Reasoning ? 40 questions

WORKED EXAMPLES

Q761 ? Direct computation ? Beginner

A grammar audit case uses the quantities in this question. Determine the requested result using the direct computation model: Choose the grammatically correct sentence.

Options: She is senior than me. | Each of the players have a locker. | Neither answers are correct. | Neither answer is correct.

Answer: Neither answer is correct.

Solution: Context: grammar audit. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q762 ? Reverse value ? Easy

A sentence correction case uses the quantities in this question. Determine the requested result using the reverse value model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | She is senior to me. | Each of the players have a locker. | She is senior than me.

Answer: She is senior to me.

Solution: Context: sentence correction. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q763 ? Missing base ? Medium

A usage choice case uses the quantities in this question. Determine the requested result using the missing base model: Choose the grammatically correct sentence.

Options: The committee has submitted its report. | Each of the players have a locker. | She is senior than me. | Neither answers are correct.

Answer: The committee has submitted its report.

Solution: Context: usage choice. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q764 ? Successive change ? Hard

A error spotting case uses the quantities in this question. Determine the requested result using the successive change model: Choose the grammatically correct sentence.

Options: She is senior than me. | Each of the players have a locker. | Each of the players has a locker. | Neither answers are correct.

Answer: Each of the players has a locker.

Solution: Context: error spotting. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q765 ? Comparison trap ? Placement

A cloze test case uses the quantities in this question. Determine the requested result using the comparison trap model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | She is senior than me. | Each of the players have a locker. | Neither answer is correct.

Answer: Neither answer is correct.

Solution: Context: cloze test. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q766 ? Ratio form ? Beginner

A sentence completion case uses the quantities in this question. Determine the requested result using the ratio form model: Choose the grammatically correct sentence.

Options: She is senior to me. | Each of the players have a locker. | She is senior than me. | Neither answers are correct.

Answer: She is senior to me.

Solution: Context: sentence completion. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q767 ? Weighted result ? Easy

A parallelism case uses the quantities in this question. Determine the requested result using the weighted result model: Choose the grammatically correct sentence.

Options: The committee has submitted its report. | Neither answers are correct. | Each of the players have a locker. | She is senior than me.

Answer: The committee has submitted its report.

Solution: Context: parallelism. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q768 ? Replacement cycle ? Medium

A pronoun reference case uses the quantities in this question. Determine the requested result using the replacement cycle model: Choose the grammatically correct sentence.

Options: She is senior than me. | Each of the players have a locker. | Each of the players has a locker. | Neither answers are correct.

Answer: Each of the players has a locker.

Solution: Context: pronoun reference. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q769 ? Unit conversion ? Hard

A grammar audit case uses the quantities in this question. Determine the requested result using the unit conversion model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | Neither answer is correct. | Each of the players have a locker. | She is senior than me.

Answer: Neither answer is correct.

Solution: Context: grammar audit. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q770 ? Work-rate model ? Placement

A sentence correction case uses the quantities in this question. Determine the requested result using the work-rate model model: Choose the grammatically correct sentence.

Options: She is senior to me. | Each of the players have a locker. | She is senior than me. | Neither answers are correct.

Answer: She is senior to me.

Solution: Context: sentence correction. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q771 ? Speed-time model ? Beginner

A usage choice case uses the quantities in this question. Determine the requested result using the speed-time model model: Choose the grammatically correct sentence.

Options: She is senior than me. | Each of the players have a locker. | The committee has submitted its report. | Neither answers are correct.

Answer: The committee has submitted its report.

Solution: Context: usage choice. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q772 ? Table lookup ? Easy

A error spotting case uses the quantities in this question. Determine the requested result using the table lookup model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | She is senior than me. | Each of the players have a locker. | Each of the players has a locker.

Answer: Each of the players has a locker.

Solution: Context: error spotting. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q773 ? Chart comparison ? Medium

A cloze test case uses the quantities in this question. Determine the requested result using the chart comparison model: Choose the grammatically correct sentence.

Options: Neither answer is correct. | Each of the players have a locker. | She is senior than me. | Neither answers are correct.

Answer: Neither answer is correct.

Solution: Context: cloze test. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q774 ? Multi-step word problem ? Hard

A sentence completion case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | She is senior to me. | Each of the players have a locker. | She is senior than me.

Answer: She is senior to me.

Solution: Context: sentence completion. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q775 ? Maximum / minimum ? Placement

A parallelism case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Choose the grammatically correct sentence.

Options: She is senior than me. | Each of the players have a locker. | Neither answers are correct. | The committee has submitted its report.

Answer: The committee has submitted its report.

Solution: Context: parallelism. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q776 ? Least / greatest value ? Beginner

A pronoun reference case uses the quantities in this question. Determine the requested result using the least / greatest value model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | Each of the players has a locker. | Each of the players have a locker. | She is senior than me.

Answer: Each of the players has a locker.

Solution: Context: pronoun reference. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q777 ? Remainder cycle ? Easy

A grammar audit case uses the quantities in this question. Determine the requested result using the remainder cycle model: Choose the grammatically correct sentence.

Options: Each of the players have a locker. | She is senior than me. | Neither answer is correct. | Neither answers are correct.

Answer: Neither answer is correct.

Solution: Context: grammar audit. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q778 ? Divisibility test ? Medium

A sentence correction case uses the quantities in this question. Determine the requested result using the divisibility test model: Choose the grammatically correct sentence.

Options: She is senior than me. | Each of the players have a locker. | She is senior to me. | Neither answers are correct.

Answer: She is senior to me.

Solution: Context: sentence correction. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q779 ? Factorisation ? Hard

A usage choice case uses the quantities in this question. Determine the requested result using the factorisation model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | She is senior than me. | Each of the players have a locker. | The committee has submitted its report.

Answer: The committee has submitted its report.

Solution: Context: usage choice. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

Q780 ? Approximation ? Placement

A error spotting case uses the quantities in this question. Determine the requested result using the approximation model: Choose the grammatically correct sentence.

Options: Each of the players has a locker. | Each of the players have a locker. | She is senior than me. | Neither answers are correct.

Answer: Each of the players has a locker.

Solution: Context: error spotting. This option follows standard agreement, pronoun, and preposition rules. Shortcut: Eliminate obvious agreement errors.

PRACTICE ? solutions hidden below**Q781 ? Options elimination ? Beginner**

A cloze test case uses the quantities in this question. Determine the requested result using the options elimination model: Choose the grammatically correct

sentence.

Options: Neither answer is correct. | Neither answers are correct. | Each of the players have a locker. | She is senior than me.

Q782 ? Boundary case ? Easy

A sentence completion case uses the quantities in this question. Determine the requested result using the boundary case model: Choose the grammatically correct sentence.

Options: She is senior than me. | Each of the players have a locker. | She is senior to me. | Neither answers are correct.

Q783 ? Case split ? Medium

A parallelism case uses the quantities in this question. Determine the requested result using the case split model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | The committee has submitted its report. | Each of the players have a locker. | She is senior than me.

Q784 ? Contradiction check ? Hard

A pronoun reference case uses the quantities in this question. Determine the requested result using the contradiction check model: Choose the grammatically correct sentence.

Options: Each of the players has a locker. | Each of the players have a locker. | She is senior than me. | Neither answers are correct.

Q785 ? Statement I test ? Placement

A grammar audit case uses the quantities in this question. Determine the requested result using the statement i test model: Choose the grammatically correct sentence.

Options: Neither answer is correct. | She is senior than me. | Each of the players have a locker. | Neither answers are correct.

Q786 ? Statement II test ? Beginner

A sentence correction case uses the quantities in this question. Determine the requested result using the statement ii test model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | She is senior than me. | Each of the players have a locker. | She is senior to me.

Q787 ? Combined statements ? Easy

A usage choice case uses the quantities in this question. Determine the requested result using the combined statements model: Choose the grammatically correct sentence.

Options: The committee has submitted its report. | Each of the players have a locker. | She is senior than me. | Neither answers are correct.

Q788 ? Counting restriction ? Medium

A error spotting case uses the quantities in this question. Determine the requested result using the counting restriction model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | Each of the players has a locker. | Each of the players have a locker. | She is senior than me.

Q789 ? Circular arrangement ? Hard

A cloze test case uses the quantities in this question. Determine the requested result using the circular arrangement model: Choose the grammatically correct sentence.

Options: She is senior than me. | Each of the players have a locker. | Neither answers are correct. | Neither answer is correct.

Q790 ? Complement method ? Placement

A sentence completion case uses the quantities in this question. Determine the requested result using the complement method model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | She is senior to me. | Each of the players have a locker. | She is senior than me.

Q791 ? Pattern: differences ? Beginner

A parallelism case uses the quantities in this question. Determine the requested result using the pattern: differences model: Choose the grammatically correct sentence.

Options: The committee has submitted its report. | Each of the players have a locker. | She is senior than me. | Neither answers are correct.

Q792 ? Pattern: ratios ? Easy

A pronoun reference case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Choose the grammatically correct sentence.

Options: She is senior than me. | Each of the players have a locker. | Each of the players has a locker. | Neither answers are correct.

Q793 ? Pattern: alternating ? Medium

A grammar audit case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | She is senior than me. | Each of the players have a locker. | Neither answer is correct.

Q794 ? Pattern: squares / cubes ? Hard

A sentence correction case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Choose the grammatically correct sentence.

Options: She is senior to me. | Each of the players have a locker. | She is senior than me. | Neither answers are correct.

Q795 ? Inference from facts ? Placement

A usage choice case uses the quantities in this question. Determine the requested result using the inference from facts model: Choose the grammatically correct sentence.

Options: The committee has submitted its report. | Neither answers are correct. | Each of the players have a locker. | She is senior than me.

Q796 ? Assumption check ? Beginner

A error spotting case uses the quantities in this question. Determine the requested result using the assumption check model: Choose the grammatically correct sentence.

Options: She is senior than me. | Each of the players have a locker. | Each of the players has a locker. | Neither answers are correct.

Q797 ? Cause vs correlation ? Easy

A cloze test case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | Neither answer is correct. | Each of the players have a locker. | She is senior than me.

Q798 ? Grammar agreement ? Medium

A sentence completion case uses the quantities in this question. Determine the requested result using the grammar agreement model: Choose the grammatically correct sentence.

Options: She is senior to me. | Each of the players have a locker. | She is senior than me. | Neither answers are correct.

Q799 ? Vocabulary in context ? Hard

A parallelism case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Choose the grammatically correct sentence.

Options: She is senior than me. | Each of the players have a locker. | The committee has submitted its report. | Neither answers are correct.

Q800 ? Mixed placement challenge ? Placement

A pronoun reference case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Choose the grammatically correct sentence.

Options: Neither answers are correct. | She is senior than me. | Each of the players have a locker. | Each of the players has a locker.

Partnership ? 40 questions

WORKED EXAMPLES

Q801 ? Direct computation ? Beginner

A startup equity split case uses the quantities in this question. Determine the requested result using the direct computation model: A invests ■11000 for 12 months and B ■9000 for 7 months. Divide a ■10000 profit: find A's share.

Options: 6969.23 | 6669.23 | 6869.23 | 6769.23

Answer: 6769.23

Solution: Context: startup equity split. Money-month ratio=132:63. A's share= $10000 \times 132 / 195 = \blacksquare 6769.23$. Shortcut: Compare money-months.

Q802 ? Reverse value ? Easy

A consulting venture case uses the quantities in this question. Determine the requested result using the reverse value model: A invests ■12000 for 12 months and B ■10000 for 8 months. Divide a ■10000 profit: find A's share.

Options: 6428.57 | 6528.57 | 6328.57 | 6628.57

Answer: 6428.57

Solution: Context: consulting venture. Money-month ratio=144:80. A's share= $10000 \times 144 / 224 = \blacksquare 6428.57$. Shortcut: Compare money-months.

Q803 ? Missing base ? Medium

A restaurant partners case uses the quantities in this question. Determine the requested result using the missing base model: A invests ■13000 for 12 months and B ■11000 for 9 months. Divide a ■10000 profit: find A's share.

Options: 6317.65 | 6017.65 | 6217.65 | 6117.65

Answer: 6117.65

Solution: Context: restaurant partners. Money-month ratio=156:99. A's share= $10000 \times 156 / 255 = \blacksquare 6117.65$. Shortcut: Compare money-months.

Q804 ? Successive change ? Hard

A campus project case uses the quantities in this question. Determine the requested result using the successive change model: A invests ■14000 for 12 months and B ■12000 for 10 months. Divide a ■10000 profit: find A's share.

Options: 5733.33 | 5933.33 | 5833.33 | 6033.33

Answer: 5833.33

Solution: Context: campus project. Money-month ratio=168:120. A's share= $10000 \times 168 / 288 = \blacksquare 5833.33$. Shortcut: Compare money-months.

Q805 ? Comparison trap ? Placement

A agency ownership case uses the quantities in this question. Determine the requested result using the comparison trap model: A invests ■15000 for 12 months and B ■13000 for 11 months. Divide a ■10000 profit: find A's share.

Options: 5572.76 | 5672.76 | 5472.76 | 5772.76

Answer: 5572.76

Solution: Context: agency ownership. Money-month ratio=180:143. A's share= $10000 \times 180 / 323 = \blacksquare 5572.76$. Shortcut: Compare money-months.

Q806 ? Ratio form ? Beginner

A franchise investment case uses the quantities in this question. Determine the requested result using the ratio form model: A invests ■16000 for 12 months and B ■14000 for 12 months. Divide a ■10000 profit: find A's share.

Options: 5533.33 | 5233.33 | 5433.33 | 5333.33

Answer: 5333.33

Solution: Context: franchise investment. Money-month ratio=192:168. A's share= $10000 \times 192 / 360 = \blacksquare 5333.33$. Shortcut: Compare money-months.

Q807 ? Weighted result ? Easy

A joint research case uses the quantities in this question. Determine the requested result using the weighted result model: A invests ■17000 for 12 months and B ■15000 for 6 months. Divide a ■10000 profit: find A's share.

Options: 6938.78 | 6838.78 | 7038.78 | 7138.78

Answer: 6938.78

Solution: Context: joint research. Money-month ratio=204:90. A's share= $10000 \times 204 / 294 = \blacksquare 6938.78$. Shortcut: Compare money-months.

Q808 ? Replacement cycle ? Medium

A co-founder exit case uses the quantities in this question. Determine the requested result using the replacement cycle model: A invests ■18000 for 12 months and B ■16000 for 7 months. Divide a ■10000 profit: find A's share.

Options: 6785.37 | 6485.37 | 6685.37 | 6585.37

Answer: 6585.37

Solution: Context: co-founder exit. Money-month ratio=216:112. A's share= $10000 \times 216 / 328 = \blacksquare 6585.37$. Shortcut: Compare money-months.

Q809 ? Unit conversion ? Hard

A startup equity split case uses the quantities in this question. Determine the requested result using the unit conversion model: A invests ■19000 for 12 months and B ■8000 for 8 months. Divide a ■10000 profit: find A's share.

Options: 7808.22 | 7908.22 | 7708.22 | 8008.22

Answer: 7808.22

Solution: Context: startup equity split. Money-month ratio=228:64. A's share= $10000 \times 228 / 292 = \blacksquare 7808.22$. Shortcut: Compare money-months.

Q810 ? Work-rate model ? Placement

A consulting venture case uses the quantities in this question. Determine the requested result using the work-rate model model: A invests ■20000 for 12 months and B ■9000 for 9 months. Divide a ■10000 profit: find A's share.

Options: 7476.64 | 7576.64 | 7376.64 | 7676.64

Answer: 7476.64

Solution: Context: consulting venture. Money-month ratio=240:81. A's share= $10000 \times 240 / 321 = \blacksquare 7476.64$. Shortcut: Compare money-months.

Q811 ? Speed-time model ? Beginner

A restaurant partners case uses the quantities in this question. Determine the requested result using the speed-time model model: A invests ■21000 for 12 months and B ■10000 for 10 months. Divide a ■10000 profit: find A's share.

Options: 7359.09 | 7059.09 | 7259.09 | 7159.09

Answer: 7159.09

Solution: Context: restaurant partners. Money-month ratio=252:100. A's share= $10000 \times 252 / 352 = \blacksquare 7159.09$. Shortcut: Compare money-months.

Q812 ? Table lookup ? Easy

A campus project case uses the quantities in this question. Determine the requested result using the table lookup model: A invests ■22000 for 12 months and B ■11000 for 11 months. Divide a ■10000 profit: find A's share.

Options: 7057.14 | 6857.14 | 6957.14 | 6757.14

Answer: 6857.14

Solution: Context: campus project. Money-month ratio=264:121. A's share= $10000 \times 264 / 385 = \blacksquare 6857.14$. Shortcut: Compare money-months.

Q813 ? Chart comparison ? Medium

A agency ownership case uses the quantities in this question. Determine the requested result using the chart comparison model: A invests ■23000 for 12 months and B ■12000 for 12 months. Divide a ■10000 profit: find A's share.

Options: 6571.43 | 6671.43 | 6471.43 | 6771.43

Answer: 6571.43

Solution: Context: agency ownership. Money-month ratio=276:144. A's share= $10000 \times 276 / 420 = \blacksquare 6571.43$. Shortcut: Compare money-months.

Q814 ? Multi-step word problem ? Hard

A franchise investment case uses the quantities in this question. Determine the requested result using the multi-step word problem model: A invests ■24000 for 12 months and B ■13000 for 6 months. Divide a ■10000 profit: find A's share.

Options: 8068.85 | 7868.85 | 7968.85 | 7768.85

Answer: 7868.85

Solution: Context: franchise investment. Money-month ratio=288:78. A's share= $10000 \times 288 / 366 = \blacksquare 7868.85$. Shortcut: Compare money-months.

Q815 ? Maximum / minimum ? Placement

A joint research case uses the quantities in this question. Determine the requested result using the maximum / minimum model: A invests ■25000 for 12 months and B ■14000 for 7 months. Divide a ■10000 profit: find A's share.

Options: 7737.69 | 7437.69 | 7637.69 | 7537.69

Answer: 7537.69

Solution: Context: joint research. Money-month ratio=300:98. A's share= $10000 \times 300 / 398 = \blacksquare 7537.69$. Shortcut: Compare money-months.

Q816 ? Least / greatest value ? Beginner

A co-founder exit case uses the quantities in this question. Determine the requested result using the least / greatest value model: A invests ■26000 for 12 months and B ■15000 for 8 months. Divide a ■10000 profit: find A's share.

Options: 7222.22 | 7322.22 | 7122.22 | 7422.22

Answer: 7222.22

Solution: Context: co-founder exit. Money-month ratio=312:120. A's share= $10000 \times 312 / 432 = \blacksquare 7222.22$. Shortcut: Compare money-months.

Q817 ? Remainder cycle ? Easy

A startup equity split case uses the quantities in this question. Determine the requested result using the remainder cycle model: A invests ■27000 for 12 months and B ■16000 for 9 months. Divide a ■10000 profit: find A's share.

Options: 7023.08 | 7123.08 | 6823.08 | 6923.08

Answer: 6923.08

Solution: Context: startup equity split. Money-month ratio=324:144. A's share= $10000 \times 324 / 468 = \blacksquare 6923.08$. Shortcut: Compare money-months.

Q818 ? Divisibility test ? Medium

A consulting venture case uses the quantities in this question. Determine the requested result using the divisibility test model: A invests ■28000 for 12 months and B ■8000 for 10 months. Divide a ■10000 profit: find A's share.

Options: 7976.92 | 8276.92 | 8176.92 | 8076.92

Answer: 8076.92

Solution: Context: consulting venture. Money-month ratio=336:80. A's share= $10000 \times 336 / 416 = \blacksquare 8076.92$. Shortcut: Compare money-months.

Q819 ? Factorisation ? Hard

A restaurant partners case uses the quantities in this question. Determine the requested result using the factorisation model: A invests ■29000 for 12 months and B ■9000 for 11 months. Divide a ■10000 profit: find A's share.

Options: 7985.23 | 7685.23 | 7885.23 | 7785.23

Answer: 7785.23

Solution: Context: restaurant partners. Money-month ratio=348:99. A's share= $10000 \times 348 / 447 = \blacksquare 7785.23$. Shortcut: Compare money-months.

Q820 ? Approximation ? Placement

A campus project case uses the quantities in this question. Determine the requested result using the approximation model: A invests ■30000 for 12 months and B ■10000 for 12 months. Divide a ■10000 profit: find A's share.

Options: 7500 | 7600 | 7400 | 7700

Answer: 7500

Solution: Context: campus project. Money-month ratio=360:120. A's share= $10000 \times 360 / 480 = \blacksquare 7500$. Shortcut: Compare money-months.

PRACTICE ? solutions hidden below**Q821 ? Options elimination ? Beginner**

A agency ownership case uses the quantities in this question. Determine the requested result using the options elimination model: A invests ■31000 for 12

months and B ■11000 for 6 months. Divide a ■10000 profit: find A's share.
Options: 8493.15 | 8593.15 | 8393.15 | 8693.15

Q822 ? Boundary case ? Easy

A franchise investment case uses the quantities in this question. Determine the requested result using the boundary case model: A invests ■32000 for 12 months and B ■12000 for 7 months. Divide a ■10000 profit: find A's share.
Options: 8405.13 | 8105.13 | 8305.13 | 8205.13

Q823 ? Case split ? Medium

A joint research case uses the quantities in this question. Determine the requested result using the case split model: A invests ■33000 for 12 months and B ■13000 for 8 months. Divide a ■10000 profit: find A's share.
Options: 7920 | 8020 | 7820 | 8120

Q824 ? Contradiction check ? Hard

A co-founder exit case uses the quantities in this question. Determine the requested result using the contradiction check model: A invests ■34000 for 12 months and B ■14000 for 9 months. Divide a ■10000 profit: find A's share.
Options: 7640.45 | 7740.45 | 7540.45 | 7840.45

Q825 ? Statement I test ? Placement

A startup equity split case uses the quantities in this question. Determine the requested result using the statement i test model: A invests ■35000 for 12 months and B ■15000 for 10 months. Divide a ■10000 profit: find A's share.
Options: 7568.42 | 7268.42 | 7468.42 | 7368.42

Q826 ? Statement II test ? Beginner

A consulting venture case uses the quantities in this question. Determine the requested result using the statement ii test model: A invests ■36000 for 12 months and B ■16000 for 11 months. Divide a ■10000 profit: find A's share.
Options: 7305.26 | 7005.26 | 7205.26 | 7105.26

Q827 ? Combined statements ? Easy

A restaurant partners case uses the quantities in this question. Determine the requested result using the combined statements model: A invests ■37000 for 12 months and B ■8000 for 12 months. Divide a ■10000 profit: find A's share.
Options: 8222.22 | 8322.22 | 8122.22 | 8422.22

Q828 ? Counting restriction ? Medium

A campus project case uses the quantities in this question. Determine the requested result using the counting restriction model: A invests ■38000 for 12 months and B ■9000 for 6 months. Divide a ■10000 profit: find A's share.
Options: 8941.18 | 8841.18 | 9041.18 | 9141.18

Q829 ? Circular arrangement ? Hard

A agency ownership case uses the quantities in this question. Determine the requested result using the circular arrangement model: A invests ■39000 for 12 months and B ■10000 for 7 months. Divide a ■10000 profit: find A's share.
Options: 8898.88 | 8598.88 | 8798.88 | 8698.88

Q830 ? Complement method ? Placement

A franchise investment case uses the quantities in this question. Determine the requested result using the complement method model: A invests ■40000 for 12 months and B ■11000 for 8 months. Divide a ■10000 profit: find A's share.
Options: 8450.7 | 8550.7 | 8350.7 | 8650.7

Q831 ? Pattern: differences ? Beginner

A joint research case uses the quantities in this question. Determine the requested result using the pattern: differences model: A invests ■41000 for 12 months and B ■12000 for 9 months. Divide a ■10000 profit: find A's share.
Options: 8200 | 8300 | 8100 | 8400

Q832 ? Pattern: ratios ? Easy

A co-founder exit case uses the quantities in this question. Determine the requested result using the pattern: ratios model: A invests ■42000 for 12 months and B ■13000 for 10 months. Divide a ■10000 profit: find A's share.
Options: 7849.53 | 7949.53 | 8149.53 | 8049.53

Q833 ? Pattern: alternating ? Medium

A startup equity split case uses the quantities in this question. Determine the requested result using the pattern: alternating model: A invests ■43000 for 12 months and B ■14000 for 11 months. Divide a ■10000 profit: find A's share.
Options: 7901.49 | 7601.49 | 7801.49 | 7701.49

Q834 ? Pattern: squares / cubes ? Hard

A consulting venture case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: A invests ■44000 for 12 months and B ■15000 for 12 months. Divide a ■10000 profit: find A's share.
Options: 7457.63 | 7557.63 | 7357.63 | 7657.63

Q835 ? Inference from facts ? Placement

A restaurant partners case uses the quantities in this question. Determine the requested result using the inference from facts model: A invests ■45000 for 12 months and B ■16000 for 6 months. Divide a ■10000 profit: find A's share.
Options: 8490.57 | 8590.57 | 8390.57 | 8690.57

Q836 ? Assumption check ? Beginner

A campus project case uses the quantities in this question. Determine the requested result using the assumption check model: A invests ■46000 for 12 months and B ■8000 for 7 months. Divide a ■10000 profit: find A's share.
Options: 9278.95 | 8978.95 | 9178.95 | 9078.95

Q837 ? Cause vs correlation ? Easy

A agency ownership case uses the quantities in this question. Determine the requested result using the cause vs correlation model: A invests ■47000 for 12 months and B ■9000 for 8 months. Divide a ■10000 profit: find A's share.

Options: 9067.92 | 8867.92 | 8967.92 | 8767.92

Q838 ? Grammar agreement ? Medium

A franchise investment case uses the quantities in this question. Determine the requested result using the grammar agreement model: A invests ■48000 for 12 months and B ■10000 for 9 months. Divide a ■10000 profit: find A's share.

Options: 8648.65 | 8748.65 | 8548.65 | 8848.65

Q839 ? Vocabulary in context ? Hard

A joint research case uses the quantities in this question. Determine the requested result using the vocabulary in context model: A invests ■49000 for 12 months and B ■11000 for 10 months. Divide a ■10000 profit: find A's share.

Options: 8624.07 | 8324.07 | 8524.07 | 8424.07

Q840 ? Mixed placement challenge ? Placement

A co-founder exit case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: A invests ■50000 for 12 months and B ■12000 for 11 months. Divide a ■10000 profit: find A's share.

Options: 8396.72 | 8096.72 | 8296.72 | 8196.72

Mixture & Alligation ? 40 questions

WORKED EXAMPLES

Q841 ? Direct computation ? Beginner

A fuel blend case uses the quantities in this question. Determine the requested result using the direct computation model: 3 kg at ■31/kg is mixed with 4 kg at ■61/kg. Find mixture price/kg.

Options: 52.14 | 50.14 | 48.14 | 46.14

Answer: 48.14

Solution: Context: fuel blend. Total cost=■337; quantity=7 kg; price=■48.14/kg. Shortcut: Total cost ÷ total quantity.

Q842 ? Reverse value ? Easy

A coffee recipe case uses the quantities in this question. Determine the requested result using the reverse value model: 4 kg at ■32/kg is mixed with 5 kg at ■62/kg. Find mixture price/kg.

Options: 50.67 | 52.67 | 46.67 | 48.67

Answer: 48.67

Solution: Context: coffee recipe. Total cost=■438; quantity=9 kg; price=■48.67/kg. Shortcut: Total cost ÷ total quantity.

Q843 ? Missing base ? Medium

A metal alloy case uses the quantities in this question. Determine the requested result using the missing base model: 5 kg at ■33/kg is mixed with 6 kg at ■63/kg. Find mixture price/kg.

Options: 49.36 | 51.36 | 47.36 | 53.36

Answer: 49.36

Solution: Context: metal alloy. Total cost=■543; quantity=11 kg; price=■49.36/kg. Shortcut: Total cost ÷ total quantity.

Q844 ? Successive change ? Hard

A chemical concentration case uses the quantities in this question. Determine the requested result using the successive change model: 2 kg at ■34/kg is mixed with 7 kg at ■64/kg. Find mixture price/kg.

Options: 55.33 | 59.33 | 57.33 | 61.33

Answer: 57.33

Solution: Context: chemical concentration. Total cost=■516; quantity=9 kg; price=■57.33/kg. Shortcut: Total cost ÷ total quantity.

Q845 ? Comparison trap ? Placement

A food-grade mix case uses the quantities in this question. Determine the requested result using the comparison trap model: 3 kg at ■35/kg is mixed with 3 kg at ■65/kg. Find mixture price/kg.

Options: 50 | 52 | 48 | 54

Answer: 50

Solution: Context: food-grade mix. Total cost=■300; quantity=6 kg; price=■50/kg. Shortcut: Total cost ÷ total quantity.

Q846 ? Ratio form ? Beginner

A paint formulation case uses the quantities in this question. Determine the requested result using the ratio form model: 4 kg at ■36/kg is mixed with 4 kg at ■66/kg. Find mixture price/kg.

Options: 55 | 49 | 53 | 51

Answer: 51

Solution: Context: paint formulation. Total cost=■408; quantity=8 kg; price=■51/kg. Shortcut: Total cost ÷ total quantity.

Q847 ? Weighted result ? Easy

A feed composition case uses the quantities in this question. Determine the requested result using the weighted result model: 5 kg at ■37/kg is mixed with 5 kg at ■67/kg. Find mixture price/kg.

Options: 52 | 54 | 50 | 56

Answer: 52

Solution: Context: feed composition. Total cost=■520; quantity=10 kg; price=■52/kg. Shortcut: Total cost ÷ total quantity.

Q848 ? Replacement cycle ? Medium

A water treatment case uses the quantities in this question. Determine the requested result using the replacement cycle model: 2 kg at ■38/kg is mixed with 6 kg at ■68/kg. Find mixture price/kg.

Options: 64.5 | 58.5 | 62.5 | 60.5

Answer: 60.5

Solution: Context: water treatment. Total cost=■484; quantity=8 kg; price=■60.5/kg. Shortcut: Total cost ÷ total quantity.

Q849 ? Unit conversion ? Hard

A fuel blend case uses the quantities in this question. Determine the requested result using the unit conversion model: 3 kg at ■39/kg is mixed with 7 kg at ■69/kg. Find mixture price/kg.

Options: 60 | 62 | 58 | 64

Answer: 60

Solution: Context: fuel blend. Total cost=■600; quantity=10 kg; price=■60/kg. Shortcut: Total cost ÷ total quantity.

Q850 ? Work-rate model ? Placement

A coffee recipe case uses the quantities in this question. Determine the requested result using the work-rate model model: 4 kg at ■40/kg is mixed with 3 kg at ■70/kg. Find mixture price/kg.

Options: 52.86 | 54.86 | 50.86 | 56.86

Answer: 52.86

Solution: Context: coffee recipe. Total cost=■370; quantity=7 kg; price=■52.86/kg. Shortcut: Total cost ÷ total quantity.

Q851 ? Speed-time model ? Beginner

A metal alloy case uses the quantities in this question. Determine the requested result using the speed-time model model: 5 kg at ■41/kg is mixed with 4 kg at ■71/kg. Find mixture price/kg.

Options: 58.33 | 52.33 | 56.33 | 54.33

Answer: 54.33

Solution: Context: metal alloy. Total cost=■489; quantity=9 kg; price=■54.33/kg. Shortcut: Total cost ÷ total quantity.

Q852 ? Table lookup ? Easy

A chemical concentration case uses the quantities in this question. Determine the requested result using the table lookup model: 2 kg at ■42/kg is mixed with 5 kg at ■72/kg. Find mixture price/kg.

Options: 63.43 | 65.43 | 61.43 | 67.43

Answer: 63.43

Solution: Context: chemical concentration. Total cost=■444; quantity=7 kg; price=■63.43/kg. Shortcut: Total cost ÷ total quantity.

Q853 ? Chart comparison ? Medium

A food-grade mix case uses the quantities in this question. Determine the requested result using the chart comparison model: 3 kg at ■43/kg is mixed with 6 kg at ■73/kg. Find mixture price/kg.

Options: 63 | 65 | 61 | 67

Answer: 63

Solution: Context: food-grade mix. Total cost=■567; quantity=9 kg; price=■63/kg. Shortcut: Total cost ÷ total quantity.

Q854 ? Multi-step word problem ? Hard

A paint formulation case uses the quantities in this question. Determine the requested result using the multi-step word problem model: 4 kg at ■44/kg is mixed with 7 kg at ■74/kg. Find mixture price/kg.

Options: 63.09 | 65.09 | 61.09 | 67.09

Answer: 63.09

Solution: Context: paint formulation. Total cost=■694; quantity=11 kg; price=■63.09/kg. Shortcut: Total cost ÷ total quantity.

Q855 ? Maximum / minimum ? Placement

A feed composition case uses the quantities in this question. Determine the requested result using the maximum / minimum model: 5 kg at ■30/kg is mixed with 3 kg at ■75/kg. Find mixture price/kg.

Options: 50.88 | 46.88 | 48.88 | 44.88

Answer: 46.88

Solution: Context: feed composition. Total cost=■375; quantity=8 kg; price=■46.88/kg. Shortcut: Total cost ÷ total quantity.

Q856 ? Least / greatest value ? Beginner

A water treatment case uses the quantities in this question. Determine the requested result using the least / greatest value model: 2 kg at ■31/kg is mixed with 4 kg at ■76/kg. Find mixture price/kg.

Options: 61 | 63 | 59 | 65

Answer: 61

Solution: Context: water treatment. Total cost=■366; quantity=6 kg; price=■61/kg. Shortcut: Total cost ÷ total quantity.

Q857 ? Remainder cycle ? Easy

A fuel blend case uses the quantities in this question. Determine the requested result using the remainder cycle model: 3 kg at ■32/kg is mixed with 5 kg at ■77/kg. Find mixture price/kg.

Options: 64.13 | 62.13 | 60.13 | 58.13

Answer: 60.13

Solution: Context: fuel blend. Total cost=■481; quantity=8 kg; price=■60.13/kg. Shortcut: Total cost ÷ total quantity.

Q858 ? Divisibility test ? Medium

A coffee recipe case uses the quantities in this question. Determine the requested result using the divisibility test model: 4 kg at ■33/kg is mixed with 6 kg at ■78/kg. Find mixture price/kg.

Options: 58 | 64 | 62 | 60

Answer: 60

Solution: Context: coffee recipe. Total cost=■600; quantity=10 kg; price=■60/kg. Shortcut: Total cost ÷ total quantity.

Q859 ? Factorisation ? Hard

A metal alloy case uses the quantities in this question. Determine the requested result using the factorisation model: 5 kg at ■34/kg is mixed with 7 kg at ■79/kg. Find mixture price/kg.

Options: 60.25 | 62.25 | 58.25 | 64.25

Answer: 60.25

Solution: Context: metal alloy. Total cost=■723; quantity=12 kg; price=■60.25/kg. Shortcut: Total cost ÷ total quantity.

Q860 ? Approximation ? Placement

A chemical concentration case uses the quantities in this question. Determine the requested result using the approximation model: 2 kg at ■35/kg is mixed with 3 kg at ■60/kg. Find mixture price/kg.

Options: 54 | 48 | 52 | 50

Answer: 50

Solution: Context: chemical concentration. Total cost=■250; quantity=5 kg; price=■50/kg. Shortcut: Total cost ÷ total quantity.

PRACTICE ? solutions hidden below**Q861 ? Options elimination ? Beginner**

A food-grade mix case uses the quantities in this question. Determine the requested result using the options elimination model: 3 kg at ■36/kg is mixed with 4

kg at ■61/kg. Find mixture price/kg.
Options: 48.29 | 50.29 | 52.29 | 54.29

Q862 ? Boundary case ? Easy

A paint formulation case uses the quantities in this question. Determine the requested result using the boundary case model: 4 kg at ■37/kg is mixed with 5 kg at ■62/kg. Find mixture price/kg.
Options: 54.89 | 52.89 | 50.89 | 48.89

Q863 ? Case split ? Medium

A feed composition case uses the quantities in this question. Determine the requested result using the case split model: 5 kg at ■38/kg is mixed with 6 kg at ■63/kg. Find mixture price/kg.
Options: 51.64 | 53.64 | 55.64 | 49.64

Q864 ? Contradiction check ? Hard

A water treatment case uses the quantities in this question. Determine the requested result using the contradiction check model: 2 kg at ■39/kg is mixed with 7 kg at ■64/kg. Find mixture price/kg.
Options: 62.44 | 60.44 | 58.44 | 56.44

Q865 ? Statement I test ? Placement

A fuel blend case uses the quantities in this question. Determine the requested result using the statement i test model: 3 kg at ■40/kg is mixed with 3 kg at ■65/kg. Find mixture price/kg.
Options: 56.5 | 50.5 | 54.5 | 52.5

Q866 ? Statement II test ? Beginner

A coffee recipe case uses the quantities in this question. Determine the requested result using the statement ii test model: 4 kg at ■41/kg is mixed with 4 kg at ■66/kg. Find mixture price/kg.
Options: 53.5 | 55.5 | 51.5 | 57.5

Q867 ? Combined statements ? Easy

A metal alloy case uses the quantities in this question. Determine the requested result using the combined statements model: 5 kg at ■42/kg is mixed with 5 kg at ■67/kg. Find mixture price/kg.
Options: 58.5 | 52.5 | 56.5 | 54.5

Q868 ? Counting restriction ? Medium

A chemical concentration case uses the quantities in this question. Determine the requested result using the counting restriction model: 2 kg at ■43/kg is mixed with 6 kg at ■68/kg. Find mixture price/kg.
Options: 59.75 | 61.75 | 63.75 | 65.75

Q869 ? Circular arrangement ? Hard

A food-grade mix case uses the quantities in this question. Determine the requested result using the circular arrangement model: 3 kg at ■44/kg is mixed with 7 kg at ■69/kg. Find mixture price/kg.
Options: 65.5 | 59.5 | 63.5 | 61.5

Q870 ? Complement method ? Placement

A paint formulation case uses the quantities in this question. Determine the requested result using the complement method model: 4 kg at ■30/kg is mixed with 3 kg at ■70/kg. Find mixture price/kg.
Options: 51.14 | 45.14 | 49.14 | 47.14

Q871 ? Pattern: differences ? Beginner

A feed composition case uses the quantities in this question. Determine the requested result using the pattern: differences model: 5 kg at ■31/kg is mixed with 4 kg at ■71/kg. Find mixture price/kg.
Options: 48.78 | 50.78 | 46.78 | 52.78

Q872 ? Pattern: ratios ? Easy

A water treatment case uses the quantities in this question. Determine the requested result using the pattern: ratios model: 2 kg at ■32/kg is mixed with 5 kg at ■72/kg. Find mixture price/kg.
Options: 58.57 | 64.57 | 62.57 | 60.57

Q873 ? Pattern: alternating ? Medium

A fuel blend case uses the quantities in this question. Determine the requested result using the pattern: alternating model: 3 kg at ■33/kg is mixed with 6 kg at ■73/kg. Find mixture price/kg.
Options: 59.67 | 61.67 | 57.67 | 63.67

Q874 ? Pattern: squares / cubes ? Hard

A coffee recipe case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: 4 kg at ■34/kg is mixed with 7 kg at ■74/kg. Find mixture price/kg.
Options: 61.45 | 63.45 | 57.45 | 59.45

Q875 ? Inference from facts ? Placement

A metal alloy case uses the quantities in this question. Determine the requested result using the inference from facts model: 5 kg at ■35/kg is mixed with 3 kg at ■75/kg. Find mixture price/kg.
Options: 48 | 50 | 52 | 54

Q876 ? Assumption check ? Beginner

A chemical concentration case uses the quantities in this question. Determine the requested result using the assumption check model: 2 kg at ■36/kg is mixed with 4 kg at ■76/kg. Find mixture price/kg.
Options: 66.67 | 60.67 | 64.67 | 62.67

Q877 ? Cause vs correlation ? Easy

A food-grade mix case uses the quantities in this question. Determine the requested result using the cause vs correlation model: 3 kg at ■37/kg is mixed with 5 kg at ■77/kg. Find mixture price/kg.

Options: 62 | 64 | 60 | 66

Q878 ? Grammar agreement ? Medium

A paint formulation case uses the quantities in this question. Determine the requested result using the grammar agreement model: 4 kg at ■38/kg is mixed with 6 kg at ■78/kg. Find mixture price/kg.

Options: 66 | 60 | 64 | 62

Q879 ? Vocabulary in context ? Hard

A feed composition case uses the quantities in this question. Determine the requested result using the vocabulary in context model: 5 kg at ■39/kg is mixed with 7 kg at ■79/kg. Find mixture price/kg.

Options: 66.33 | 60.33 | 64.33 | 62.33

Q880 ? Mixed placement challenge ? Placement

A water treatment case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: 2 kg at ■40/kg is mixed with 3 kg at ■60/kg. Find mixture price/kg.

Options: 52 | 54 | 50 | 56

Pipes & Cisterns ? 40 questions

WORKED EXAMPLES

Q881 ? Direct computation ? Beginner

A reservoir controls case uses the quantities in this question. Determine the requested result using the direct computation model: Inlets fill a tank in 11 and 16 hours. Find their combined time.

Options: 5.52 | 8.52 | 7.52 | 6.52

Answer: 6.52

Solution: Context: reservoir controls. Rate= $1/11+1/16$; time= $176/27=6.52$ hours. Shortcut: For two inlets use $ab/(a+b)$.

Q882 ? Reverse value ? Easy

A swimming pool case uses the quantities in this question. Determine the requested result using the reverse value model: Inlets fill a tank in 12 and 17 hours. Find their combined time.

Options: 9.03 | 7.03 | 8.03 | 6.03

Answer: 7.03

Solution: Context: swimming pool. Rate= $1/12+1/17$; time= $204/29=7.03$ hours. Shortcut: For two inlets use $ab/(a+b)$.

Q883 ? Missing base ? Medium

A server upload queue case uses the quantities in this question. Determine the requested result using the missing base model: Inlets fill a tank in 13 and 18 hours. Find their combined time.

Options: 7.55 | 9.55 | 6.55 | 8.55

Answer: 7.55

Solution: Context: server upload queue. Rate= $1/13+1/18$; time= $234/31=7.55$ hours. Shortcut: For two inlets use $ab/(a+b)$.

Q884 ? Successive change ? Hard

A irrigation tank case uses the quantities in this question. Determine the requested result using the successive change model: Inlets fill a tank in 14 and 19 hours. Find their combined time.

Options: 8.06 | 9.06 | 7.06 | 10.06

Answer: 8.06

Solution: Context: irrigation tank. Rate= $1/14+1/19$; time= $266/33=8.06$ hours. Shortcut: For two inlets use $ab/(a+b)$.

Q885 ? Comparison trap ? Placement

A production vessel case uses the quantities in this question. Determine the requested result using the comparison trap model: Inlets fill a tank in 15 and 20 hours. Find their combined time.

Options: 10.57 | 7.57 | 9.57 | 8.57

Answer: 8.57

Solution: Context: production vessel. Rate= $1/15+1/20$; time= $300/35=8.57$ hours. Shortcut: For two inlets use $ab/(a+b)$.

Q886 ? Ratio form ? Beginner

A water treatment plant case uses the quantities in this question. Determine the requested result using the ratio form model: Inlets fill a tank in 16 and 21 hours. Find their combined time.

Options: 9.08 | 10.08 | 8.08 | 11.08

Answer: 9.08

Solution: Context: water treatment plant. Rate= $1/16+1/21$; time= $336/37=9.08$ hours. Shortcut: For two inlets use $ab/(a+b)$.

Q887 ? Weighted result ? Easy

A battery charge cycle case uses the quantities in this question. Determine the requested result using the weighted result model: Inlets fill a tank in 17 and 22 hours. Find their combined time.

Options: 10.59 | 8.59 | 11.59 | 9.59

Answer: 9.59

Solution: Context: battery charge cycle. Rate= $1/17+1/22$; time= $374/39=9.59$ hours. Shortcut: For two inlets use $ab/(a+b)$.

Q888 ? Replacement cycle ? Medium

A pipeline network case uses the quantities in this question. Determine the requested result using the replacement cycle model: Inlets fill a tank in 18 and 23 hours. Find their combined time.

Options: 12.1 | 9.1 | 11.1 | 10.1

Answer: 10.1

Solution: Context: pipeline network. Rate= $1/18+1/23$; time= $414/41=10.1$ hours. Shortcut: For two inlets use $ab/(a+b)$.

Q889 ? Unit conversion ? Hard

A reservoir controls case uses the quantities in this question. Determine the requested result using the unit conversion model: Inlets fill a tank in 19 and 24 hours. Find their combined time.

Options: 9.6 | 10.6 | 11.6 | 12.6

Answer: 10.6

Solution: Context: reservoir controls. Rate= $1/19+1/24$; time= $456/43=10.6$ hours. Shortcut: For two inlets use $ab/(a+b)$.

Q890 ? Work-rate model ? Placement

A swimming pool case uses the quantities in this question. Determine the requested result using the work-rate model model: Inlets fill a tank in 10 and 25 hours. Find their combined time.

Options: 7.14 | 9.14 | 6.14 | 8.14

Answer: 7.14

Solution: Context: swimming pool. Rate= $1/10+1/25$; time= $250/35=7.14$ hours. Shortcut: For two inlets use $ab/(a+b)$.

Q891 ? Speed-time model ? Beginner

A server upload queue case uses the quantities in this question. Determine the requested result using the speed-time model model: Inlets fill a tank in 11 and 26 hours. Find their combined time.

Options: 6.73 | 7.73 | 8.73 | 9.73

Answer: 7.73

Solution: Context: server upload queue. Rate= $\frac{1}{11}+\frac{1}{26}$; time= $\frac{286}{37}=7.73$ hours. Shortcut: For two inlets use $\frac{ab}{(a+b)}$.

Q892 ? Table lookup ? Easy

A irrigation tank case uses the quantities in this question. Determine the requested result using the table lookup model: Inlets fill a tank in 12 and 15 hours. Find their combined time.

Options: 8.67 | 7.67 | 6.67 | 5.67

Answer: 6.67

Solution: Context: irrigation tank. Rate= $\frac{1}{12}+\frac{1}{15}$; time= $\frac{180}{27}=6.67$ hours. Shortcut: For two inlets use $\frac{ab}{(a+b)}$.

Q893 ? Chart comparison ? Medium

A production vessel case uses the quantities in this question. Determine the requested result using the chart comparison model: Inlets fill a tank in 13 and 16 hours. Find their combined time.

Options: 7.17 | 8.17 | 6.17 | 9.17

Answer: 7.17

Solution: Context: production vessel. Rate= $\frac{1}{13}+\frac{1}{16}$; time= $\frac{208}{29}=7.17$ hours. Shortcut: For two inlets use $\frac{ab}{(a+b)}$.

Q894 ? Multi-step word problem ? Hard

A water treatment plant case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Inlets fill a tank in 14 and 17 hours. Find their combined time.

Options: 8.68 | 9.68 | 6.68 | 7.68

Answer: 7.68

Solution: Context: water treatment plant. Rate= $\frac{1}{14}+\frac{1}{17}$; time= $\frac{238}{31}=7.68$ hours. Shortcut: For two inlets use $\frac{ab}{(a+b)}$.

Q895 ? Maximum / minimum ? Placement

A battery charge cycle case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Inlets fill a tank in 15 and 18 hours. Find their combined time.

Options: 10.18 | 7.18 | 9.18 | 8.18

Answer: 8.18

Solution: Context: battery charge cycle. Rate= $\frac{1}{15}+\frac{1}{18}$; time= $\frac{270}{33}=8.18$ hours. Shortcut: For two inlets use $\frac{ab}{(a+b)}$.

Q896 ? Least / greatest value ? Beginner

A pipeline network case uses the quantities in this question. Determine the requested result using the least / greatest value model: Inlets fill a tank in 16 and 19 hours. Find their combined time.

Options: 9.69 | 8.69 | 7.69 | 10.69

Answer: 8.69

Solution: Context: pipeline network. Rate= $\frac{1}{16}+\frac{1}{19}$; time= $\frac{304}{35}=8.69$ hours. Shortcut: For two inlets use $\frac{ab}{(a+b)}$.

Q897 ? Remainder cycle ? Easy

A reservoir controls case uses the quantities in this question. Determine the requested result using the remainder cycle model: Inlets fill a tank in 17 and 20 hours. Find their combined time.

Options: 9.19 | 10.19 | 8.19 | 11.19

Answer: 9.19

Solution: Context: reservoir controls. Rate= $\frac{1}{17}+\frac{1}{20}$; time= $\frac{340}{37}=9.19$ hours. Shortcut: For two inlets use $\frac{ab}{(a+b)}$.

Q898 ? Divisibility test ? Medium

A swimming pool case uses the quantities in this question. Determine the requested result using the divisibility test model: Inlets fill a tank in 18 and 21 hours. Find their combined time.

Options: 9.69 | 11.69 | 8.69 | 10.69

Answer: 9.69

Solution: Context: swimming pool. Rate= $\frac{1}{18}+\frac{1}{21}$; time= $\frac{378}{39}=9.69$ hours. Shortcut: For two inlets use $\frac{ab}{(a+b)}$.

Q899 ? Factorisation ? Hard

A server upload queue case uses the quantities in this question. Determine the requested result using the factorisation model: Inlets fill a tank in 19 and 22 hours. Find their combined time.

Options: 12.2 | 9.2 | 11.2 | 10.2

Answer: 10.2

Solution: Context: server upload queue. Rate= $\frac{1}{19}+\frac{1}{22}$; time= $\frac{418}{41}=10.2$ hours. Shortcut: For two inlets use $\frac{ab}{(a+b)}$.

Q900 ? Approximation ? Placement

A irrigation tank case uses the quantities in this question. Determine the requested result using the approximation model: Inlets fill a tank in 10 and 23 hours. Find their combined time.

Options: 6.97 | 7.97 | 8.97 | 5.97

Answer: 6.97

Solution: Context: irrigation tank. Rate= $\frac{1}{10}+\frac{1}{23}$; time= $\frac{230}{33}=6.97$ hours. Shortcut: For two inlets use $\frac{ab}{(a+b)}$.

PRACTICE ? solutions hidden below**Q901 ? Options elimination ? Beginner**

A production vessel case uses the quantities in this question. Determine the requested result using the options elimination model: Inlets fill a tank in 11 and 24

hours. Find their combined time.
Options: 8.54 | 9.54 | 6.54 | 7.54

Q902 ? Boundary case ? Easy

A water treatment plant case uses the quantities in this question. Determine the requested result using the boundary case model: Inlets fill a tank in 12 and 25 hours. Find their combined time.
Options: 10.11 | 7.11 | 9.11 | 8.11

Q903 ? Case split ? Medium

A battery charge cycle case uses the quantities in this question. Determine the requested result using the case split model: Inlets fill a tank in 13 and 26 hours. Find their combined time.
Options: 9.67 | 8.67 | 7.67 | 10.67

Q904 ? Contradiction check ? Hard

A pipeline network case uses the quantities in this question. Determine the requested result using the contradiction check model: Inlets fill a tank in 14 and 15 hours. Find their combined time.
Options: 7.24 | 9.24 | 6.24 | 8.24

Q905 ? Statement I test ? Placement

A reservoir controls case uses the quantities in this question. Determine the requested result using the statement i test model: Inlets fill a tank in 15 and 16 hours. Find their combined time.
Options: 6.74 | 7.74 | 8.74 | 9.74

Q906 ? Statement II test ? Beginner

A swimming pool case uses the quantities in this question. Determine the requested result using the statement ii test model: Inlets fill a tank in 16 and 17 hours. Find their combined time.
Options: 10.24 | 7.24 | 9.24 | 8.24

Q907 ? Combined statements ? Easy

A server upload queue case uses the quantities in this question. Determine the requested result using the combined statements model: Inlets fill a tank in 17 and 18 hours. Find their combined time.
Options: 8.74 | 9.74 | 7.74 | 10.74

Q908 ? Counting restriction ? Medium

A irrigation tank case uses the quantities in this question. Determine the requested result using the counting restriction model: Inlets fill a tank in 18 and 19 hours. Find their combined time.
Options: 10.24 | 8.24 | 11.24 | 9.24

Q909 ? Circular arrangement ? Hard

A production vessel case uses the quantities in this question. Determine the requested result using the circular arrangement model: Inlets fill a tank in 19 and 20 hours. Find their combined time.
Options: 11.74 | 8.74 | 10.74 | 9.74

Q910 ? Complement method ? Placement

A water treatment plant case uses the quantities in this question. Determine the requested result using the complement method model: Inlets fill a tank in 10 and 21 hours. Find their combined time.
Options: 6.77 | 7.77 | 5.77 | 8.77

Q911 ? Pattern: differences ? Beginner

A battery charge cycle case uses the quantities in this question. Determine the requested result using the pattern: differences model: Inlets fill a tank in 11 and 22 hours. Find their combined time.
Options: 7.33 | 9.33 | 6.33 | 8.33

Q912 ? Pattern: ratios ? Easy

A pipeline network case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Inlets fill a tank in 12 and 23 hours. Find their combined time.
Options: 6.89 | 7.89 | 8.89 | 9.89

Q913 ? Pattern: alternating ? Medium

A reservoir controls case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Inlets fill a tank in 13 and 24 hours. Find their combined time.
Options: 10.43 | 7.43 | 9.43 | 8.43

Q914 ? Pattern: squares / cubes ? Hard

A swimming pool case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Inlets fill a tank in 14 and 25 hours. Find their combined time.
Options: 8.97 | 9.97 | 7.97 | 10.97

Q915 ? Inference from facts ? Placement

A server upload queue case uses the quantities in this question. Determine the requested result using the inference from facts model: Inlets fill a tank in 15 and 26 hours. Find their combined time.
Options: 10.51 | 8.51 | 11.51 | 9.51

Q916 ? Assumption check ? Beginner

A irrigation tank case uses the quantities in this question. Determine the requested result using the assumption check model: Inlets fill a tank in 16 and 15 hours. Find their combined time.
Options: 6.74 | 9.74 | 8.74 | 7.74

Q917 ? Cause vs correlation ? Easy

A production vessel case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Inlets fill a tank in 17 and 16 hours. Find their combined time.

Options: 9.24 | 8.24 | 7.24 | 10.24

Q918 ? Grammar agreement ? Medium

A water treatment plant case uses the quantities in this question. Determine the requested result using the grammar agreement model: Inlets fill a tank in 18 and 17 hours. Find their combined time.

Options: 8.74 | 7.74 | 9.74 | 10.74

Q919 ? Vocabulary in context ? Hard

A battery charge cycle case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Inlets fill a tank in 19 and 18 hours. Find their combined time.

Options: 9.24 | 11.24 | 8.24 | 10.24

Q920 ? Mixed placement challenge ? Placement

A pipeline network case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Inlets fill a tank in 10 and 19 hours. Find their combined time.

Options: 8.55 | 7.55 | 6.55 | 5.55

Boats & Streams ? 40 questions

WORKED EXAMPLES

Q921 ? Direct computation ? Beginner

A river ferry case uses the quantities in this question. Determine the requested result using the direct computation model: Downstream speed is 13 km/h and upstream is 9 km/h. Find still-water speed.

Options: 13 | 10 | 12 | 11

Answer: 11

Solution: Context: river ferry. Boat speed= $(13+9)/2=11$ km/h. Shortcut: Boat= $(D+U)/2$.

Q922 ? Reverse value ? Easy

A rescue boat case uses the quantities in this question. Determine the requested result using the reverse value model: Downstream speed is 15 km/h and upstream is 9 km/h. Find still-water speed.

Options: 12 | 13 | 11 | 14

Answer: 12

Solution: Context: rescue boat. Boat speed= $(15+9)/2=12$ km/h. Shortcut: Boat= $(D+U)/2$.

Q923 ? Missing base ? Medium

A canal logistics case uses the quantities in this question. Determine the requested result using the missing base model: Downstream speed is 17 km/h and upstream is 9 km/h. Find still-water speed.

Options: 13 | 14 | 12 | 15

Answer: 13

Solution: Context: canal logistics. Boat speed= $(17+9)/2=13$ km/h. Shortcut: Boat= $(D+U)/2$.

Q924 ? Successive change ? Hard

A upstream cargo case uses the quantities in this question. Determine the requested result using the successive change model: Downstream speed is 19 km/h and upstream is 9 km/h. Find still-water speed.

Options: 16 | 13 | 15 | 14

Answer: 14

Solution: Context: upstream cargo. Boat speed= $(19+9)/2=14$ km/h. Shortcut: Boat= $(D+U)/2$.

Q925 ? Comparison trap ? Placement

A lake patrol case uses the quantities in this question. Determine the requested result using the comparison trap model: Downstream speed is 16 km/h and upstream is 14 km/h. Find still-water speed.

Options: 19 | 13 | 17 | 15

Answer: 15

Solution: Context: lake patrol. Boat speed= $(16+14)/2=15$ km/h. Shortcut: Boat= $(D+U)/2$.

Q926 ? Ratio form ? Beginner

A rowing trial case uses the quantities in this question. Determine the requested result using the ratio form model: Downstream speed is 18 km/h and upstream is 14 km/h. Find still-water speed.

Options: 16 | 18 | 14 | 20

Answer: 16

Solution: Context: rowing trial. Boat speed= $(18+14)/2=16$ km/h. Shortcut: Boat= $(D+U)/2$.

Q927 ? Weighted result ? Easy

A stream survey case uses the quantities in this question. Determine the requested result using the weighted result model: Downstream speed is 20 km/h and upstream is 14 km/h. Find still-water speed.

Options: 17 | 19 | 15 | 21

Answer: 17

Solution: Context: stream survey. Boat speed= $(20+14)/2=17$ km/h. Shortcut: Boat= $(D+U)/2$.

Q928 ? Replacement cycle ? Medium

A harbour transfer case uses the quantities in this question. Determine the requested result using the replacement cycle model: Downstream speed is 22 km/h and upstream is 14 km/h. Find still-water speed.

Options: 22 | 16 | 20 | 18

Answer: 18

Solution: Context: harbour transfer. Boat speed= $(22+14)/2=18$ km/h. Shortcut: Boat= $(D+U)/2$.

Q929 ? Unit conversion ? Hard

A river ferry case uses the quantities in this question. Determine the requested result using the unit conversion model: Downstream speed is 24 km/h and upstream is 14 km/h. Find still-water speed.

Options: 23 | 21 | 19 | 17

Answer: 19

Solution: Context: river ferry. Boat speed= $(24+14)/2=19$ km/h. Shortcut: Boat= $(D+U)/2$.

Q930 ? Work-rate model ? Placement

A rescue boat case uses the quantities in this question. Determine the requested result using the work-rate model model: Downstream speed is 21 km/h and upstream is 19 km/h. Find still-water speed.

Options: 20 | 22 | 18 | 24

Answer: 20

Solution: Context: rescue boat. Boat speed= $(21+19)/2=20$ km/h. Shortcut: Boat= $(D+U)/2$.

Q931 ? Speed-time model ? Beginner

A canal logistics case uses the quantities in this question. Determine the requested result using the speed-time model model: Downstream speed is 23 km/h and upstream is 19 km/h. Find still-water speed.

Options: 21 | 23 | 19 | 25

Answer: 21

Solution: Context: canal logistics. Boat speed= $(23+19)/2=21$ km/h. Shortcut: Boat= $(D+U)/2$.

Q932 ? Table lookup ? Easy

A upstream cargo case uses the quantities in this question. Determine the requested result using the table lookup model: Downstream speed is 13 km/h and upstream is 7 km/h. Find still-water speed.

Options: 12 | 9 | 11 | 10

Answer: 10

Solution: Context: upstream cargo. Boat speed= $(13+7)/2=10$ km/h. Shortcut: Boat= $(D+U)/2$.

Q933 ? Chart comparison ? Medium

A lake patrol case uses the quantities in this question. Determine the requested result using the chart comparison model: Downstream speed is 15 km/h and upstream is 7 km/h. Find still-water speed.

Options: 11 | 12 | 10 | 13

Answer: 11

Solution: Context: lake patrol. Boat speed= $(15+7)/2=11$ km/h. Shortcut: Boat= $(D+U)/2$.

Q934 ? Multi-step word problem ? Hard

A rowing trial case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Downstream speed is 17 km/h and upstream is 7 km/h. Find still-water speed.

Options: 12 | 13 | 11 | 14

Answer: 12

Solution: Context: rowing trial. Boat speed= $(17+7)/2=12$ km/h. Shortcut: Boat= $(D+U)/2$.

Q935 ? Maximum / minimum ? Placement

A stream survey case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Downstream speed is 14 km/h and upstream is 12 km/h. Find still-water speed.

Options: 15 | 12 | 14 | 13

Answer: 13

Solution: Context: stream survey. Boat speed= $(14+12)/2=13$ km/h. Shortcut: Boat= $(D+U)/2$.

Q936 ? Least / greatest value ? Beginner

A harbour transfer case uses the quantities in this question. Determine the requested result using the least / greatest value model: Downstream speed is 16 km/h and upstream is 12 km/h. Find still-water speed.

Options: 14 | 15 | 13 | 16

Answer: 14

Solution: Context: harbour transfer. Boat speed= $(16+12)/2=14$ km/h. Shortcut: Boat= $(D+U)/2$.

Q937 ? Remainder cycle ? Easy

A river ferry case uses the quantities in this question. Determine the requested result using the remainder cycle model: Downstream speed is 18 km/h and upstream is 12 km/h. Find still-water speed.

Options: 15 | 17 | 13 | 19

Answer: 15

Solution: Context: river ferry. Boat speed= $(18+12)/2=15$ km/h. Shortcut: Boat= $(D+U)/2$.

Q938 ? Divisibility test ? Medium

A rescue boat case uses the quantities in this question. Determine the requested result using the divisibility test model: Downstream speed is 20 km/h and upstream is 12 km/h. Find still-water speed.

Options: 14 | 18 | 16 | 20

Answer: 16

Solution: Context: rescue boat. Boat speed= $(20+12)/2=16$ km/h. Shortcut: Boat= $(D+U)/2$.

Q939 ? Factorisation ? Hard

A canal logistics case uses the quantities in this question. Determine the requested result using the factorisation model: Downstream speed is 22 km/h and upstream is 12 km/h. Find still-water speed.

Options: 21 | 15 | 19 | 17

Answer: 17

Solution: Context: canal logistics. Boat speed= $(22+12)/2=17$ km/h. Shortcut: Boat= $(D+U)/2$.

Q940 ? Approximation ? Placement

A upstream cargo case uses the quantities in this question. Determine the requested result using the approximation model: Downstream speed is 19 km/h and upstream is 17 km/h. Find still-water speed.

Options: 18 | 20 | 16 | 22

Answer: 18

Solution: Context: upstream cargo. Boat speed= $(19+17)/2=18$ km/h. Shortcut: Boat= $(D+U)/2$.

PRACTICE ? solutions hidden below**Q941 ? Options elimination ? Beginner**

A lake patrol case uses the quantities in this question. Determine the requested result using the options elimination model: Downstream speed is 21 km/h and

upstream is 17 km/h. Find still-water speed.
Options: 19 | 17 | 21 | 23

Q942 ? Boundary case ? Easy

A rowing trial case uses the quantities in this question. Determine the requested result using the boundary case model: Downstream speed is 23 km/h and upstream is 17 km/h. Find still-water speed.
Options: 24 | 18 | 22 | 20

Q943 ? Case split ? Medium

A stream survey case uses the quantities in this question. Determine the requested result using the case split model: Downstream speed is 25 km/h and upstream is 17 km/h. Find still-water speed.
Options: 25 | 23 | 21 | 19

Q944 ? Contradiction check ? Hard

A harbour transfer case uses the quantities in this question. Determine the requested result using the contradiction check model: Downstream speed is 15 km/h and upstream is 5 km/h. Find still-water speed.
Options: 10 | 11 | 9 | 12

Q945 ? Statement I test ? Placement

A river ferry case uses the quantities in this question. Determine the requested result using the statement i test model: Downstream speed is 12 km/h and upstream is 10 km/h. Find still-water speed.
Options: 13 | 10 | 12 | 11

Q946 ? Statement II test ? Beginner

A rescue boat case uses the quantities in this question. Determine the requested result using the statement ii test model: Downstream speed is 14 km/h and upstream is 10 km/h. Find still-water speed.
Options: 14 | 11 | 13 | 12

Q947 ? Combined statements ? Easy

A canal logistics case uses the quantities in this question. Determine the requested result using the combined statements model: Downstream speed is 16 km/h and upstream is 10 km/h. Find still-water speed.
Options: 13 | 14 | 12 | 15

Q948 ? Counting restriction ? Medium

A upstream cargo case uses the quantities in this question. Determine the requested result using the counting restriction model: Downstream speed is 18 km/h and upstream is 10 km/h. Find still-water speed.
Options: 14 | 15 | 13 | 16

Q949 ? Circular arrangement ? Hard

A lake patrol case uses the quantities in this question. Determine the requested result using the circular arrangement model: Downstream speed is 20 km/h and upstream is 10 km/h. Find still-water speed.
Options: 19 | 13 | 17 | 15

Q950 ? Complement method ? Placement

A rowing trial case uses the quantities in this question. Determine the requested result using the complement method model: Downstream speed is 17 km/h and upstream is 15 km/h. Find still-water speed.
Options: 20 | 16 | 18 | 14

Q951 ? Pattern: differences ? Beginner

A stream survey case uses the quantities in this question. Determine the requested result using the pattern: differences model: Downstream speed is 19 km/h and upstream is 15 km/h. Find still-water speed.
Options: 17 | 19 | 15 | 21

Q952 ? Pattern: ratios ? Easy

A harbour transfer case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Downstream speed is 21 km/h and upstream is 15 km/h. Find still-water speed.
Options: 18 | 20 | 16 | 22

Q953 ? Pattern: alternating ? Medium

A river ferry case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Downstream speed is 23 km/h and upstream is 15 km/h. Find still-water speed.
Options: 23 | 17 | 21 | 19

Q954 ? Pattern: squares / cubes ? Hard

A rescue boat case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Downstream speed is 25 km/h and upstream is 15 km/h. Find still-water speed.
Options: 20 | 22 | 18 | 24

Q955 ? Inference from facts ? Placement

A canal logistics case uses the quantities in this question. Determine the requested result using the inference from facts model: Downstream speed is 22 km/h and upstream is 20 km/h. Find still-water speed.
Options: 19 | 21 | 23 | 25

Q956 ? Assumption check ? Beginner

A upstream cargo case uses the quantities in this question. Determine the requested result using the assumption check model: Downstream speed is 12 km/h and upstream is 8 km/h. Find still-water speed.
Options: 12 | 9 | 11 | 10

Q957 ? Cause vs correlation ? Easy

A lake patrol case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Downstream speed is 14 km/h and upstream is 8 km/h. Find still-water speed.

Options: 11 | 12 | 10 | 13

Q958 ? Grammar agreement ? Medium

A rowing trial case uses the quantities in this question. Determine the requested result using the grammar agreement model: Downstream speed is 16 km/h and upstream is 8 km/h. Find still-water speed.

Options: 12 | 13 | 11 | 14

Q959 ? Vocabulary in context ? Hard

A stream survey case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Downstream speed is 18 km/h and upstream is 8 km/h. Find still-water speed.

Options: 15 | 12 | 14 | 13

Q960 ? Mixed placement challenge ? Placement

A harbour transfer case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Downstream speed is 15 km/h and upstream is 13 km/h. Find still-water speed.

Options: 16 | 13 | 15 | 14

Calendar ? 40 questions

WORKED EXAMPLES

Q961 ? Direct computation ? Beginner

A release schedule case uses the quantities in this question. Determine the requested result using the direct computation model: If today is Tuesday, what day is it after 31 days?

Options: Friday | Wednesday | 13 | Thursday

Answer: Friday

Solution: Context: release schedule. $31 \bmod 7=3$; moving 3 days gives Friday. Shortcut: Only odd days matter.

Q962 ? Reverse value ? Easy

A weekday audit case uses the quantities in this question. Determine the requested result using the reverse value model: If today is Wednesday, what day is it after 32 days?

Options: Sunday | Thursday | Friday | Saturday

Answer: Sunday

Solution: Context: weekday audit. $32 \bmod 7=4$; moving 4 days gives Sunday. Shortcut: Only odd days matter.

Q963 ? Missing base ? Medium

A festival planning case uses the quantities in this question. Determine the requested result using the missing base model: If today is Thursday, what day is it after 33 days?

Options: Tuesday | Friday | Saturday | Sunday

Answer: Tuesday

Solution: Context: festival planning. $33 \bmod 7=5$; moving 5 days gives Tuesday. Shortcut: Only odd days matter.

Q964 ? Successive change ? Hard

A exam timetable case uses the quantities in this question. Determine the requested result using the successive change model: If today is Friday, what day is it after 34 days?

Options: Sunday | Saturday | Monday | Thursday

Answer: Thursday

Solution: Context: exam timetable. $34 \bmod 7=6$; moving 6 days gives Thursday. Shortcut: Only odd days matter.

Q965 ? Comparison trap ? Placement

A leap-year check case uses the quantities in this question. Determine the requested result using the comparison trap model: If today is Saturday, what day is it after 35 days?

Options: Tuesday | Monday | Sunday | Saturday

Answer: Saturday

Solution: Context: leap-year check. $35 \bmod 7=0$; moving 0 days gives Saturday. Shortcut: Only odd days matter.

Q966 ? Ratio form ? Beginner

A delivery calendar case uses the quantities in this question. Determine the requested result using the ratio form model: If today is Sunday, what day is it after 36 days?

Options: Wednesday | Monday | Tuesday | 13

Answer: Monday

Solution: Context: delivery calendar. $36 \bmod 7=1$; moving 1 days gives Monday. Shortcut: Only odd days matter.

Q967 ? Weighted result ? Easy

A meeting rotation case uses the quantities in this question. Determine the requested result using the weighted result model: If today is Monday, what day is it after 37 days?

Options: Tuesday | Thursday | 13 | Wednesday

Answer: Wednesday

Solution: Context: meeting rotation. $37 \bmod 7=2$; moving 2 days gives Wednesday. Shortcut: Only odd days matter.

Q968 ? Replacement cycle ? Medium

A roster cycle case uses the quantities in this question. Determine the requested result using the replacement cycle model: If today is Tuesday, what day is it after 38 days?

Options: Friday | Wednesday | 13 | Thursday

Answer: Friday

Solution: Context: roster cycle. $38 \bmod 7=3$; moving 3 days gives Friday. Shortcut: Only odd days matter.

Q969 ? Unit conversion ? Hard

A release schedule case uses the quantities in this question. Determine the requested result using the unit conversion model: If today is Wednesday, what day is it after 39 days?

Options: Sunday | Thursday | Friday | Saturday

Answer: Sunday

Solution: Context: release schedule. $39 \bmod 7=4$; moving 4 days gives Sunday. Shortcut: Only odd days matter.

Q970 ? Work-rate model ? Placement

A weekday audit case uses the quantities in this question. Determine the requested result using the work-rate model model: If today is Thursday, what day is it after 40 days?

Options: Tuesday | Friday | Saturday | Sunday

Answer: Tuesday

Solution: Context: weekday audit. $40 \bmod 7=5$; moving 5 days gives Tuesday. Shortcut: Only odd days matter.

Q971 ? Speed-time model ? Beginner

A festival planning case uses the quantities in this question. Determine the requested result using the speed-time model model: If today is Friday, what day is it after 41 days?

Options: Sunday | Saturday | Monday | Thursday

Answer: Thursday

Solution: Context: festival planning. $41 \bmod 7=6$; moving 6 days gives Thursday. Shortcut: Only odd days matter.

Q972 ? Table lookup ? Easy

A exam timetable case uses the quantities in this question. Determine the requested result using the table lookup model: If today is Saturday, what day is it after 42 days?

Options: Tuesday | Monday | Sunday | Saturday

Answer: Saturday

Solution: Context: exam timetable. $42 \bmod 7=0$; moving 0 days gives Saturday. Shortcut: Only odd days matter.

Q973 ? Chart comparison ? Medium

A leap-year check case uses the quantities in this question. Determine the requested result using the chart comparison model: If today is Sunday, what day is it after 43 days?

Options: Wednesday | Monday | Tuesday | 13

Answer: Monday

Solution: Context: leap-year check. $43 \bmod 7=1$; moving 1 days gives Monday. Shortcut: Only odd days matter.

Q974 ? Multi-step word problem ? Hard

A delivery calendar case uses the quantities in this question. Determine the requested result using the multi-step word problem model: If today is Monday, what day is it after 44 days?

Options: Tuesday | Thursday | 13 | Wednesday

Answer: Wednesday

Solution: Context: delivery calendar. $44 \bmod 7=2$; moving 2 days gives Wednesday. Shortcut: Only odd days matter.

Q975 ? Maximum / minimum ? Placement

A meeting rotation case uses the quantities in this question. Determine the requested result using the maximum / minimum model: If today is Tuesday, what day is it after 45 days?

Options: Friday | Wednesday | 13 | Thursday

Answer: Friday

Solution: Context: meeting rotation. $45 \bmod 7=3$; moving 3 days gives Friday. Shortcut: Only odd days matter.

Q976 ? Least / greatest value ? Beginner

A roster cycle case uses the quantities in this question. Determine the requested result using the least / greatest value model: If today is Wednesday, what day is it after 46 days?

Options: Sunday | Thursday | Friday | Saturday

Answer: Sunday

Solution: Context: roster cycle. $46 \bmod 7=4$; moving 4 days gives Sunday. Shortcut: Only odd days matter.

Q977 ? Remainder cycle ? Easy

A release schedule case uses the quantities in this question. Determine the requested result using the remainder cycle model: If today is Thursday, what day is it after 47 days?

Options: Tuesday | Friday | Saturday | Sunday

Answer: Tuesday

Solution: Context: release schedule. $47 \bmod 7=5$; moving 5 days gives Tuesday. Shortcut: Only odd days matter.

Q978 ? Divisibility test ? Medium

A weekday audit case uses the quantities in this question. Determine the requested result using the divisibility test model: If today is Friday, what day is it after 48 days?

Options: Sunday | Saturday | Monday | Thursday

Answer: Thursday

Solution: Context: weekday audit. $48 \bmod 7=6$; moving 6 days gives Thursday. Shortcut: Only odd days matter.

Q979 ? Factorisation ? Hard

A festival planning case uses the quantities in this question. Determine the requested result using the factorisation model: If today is Saturday, what day is it after 49 days?

Options: Tuesday | Monday | Sunday | Saturday

Answer: Saturday

Solution: Context: festival planning. $49 \bmod 7=0$; moving 0 days gives Saturday. Shortcut: Only odd days matter.

Q980 ? Approximation ? Placement

A exam timetable case uses the quantities in this question. Determine the requested result using the approximation model: If today is Sunday, what day is it after 50 days?

Options: Wednesday | Monday | Tuesday | 13

Answer: Monday

Solution: Context: exam timetable. $50 \bmod 7=1$; moving 1 days gives Monday. Shortcut: Only odd days matter.

PRACTICE ? solutions hidden below**Q981 ? Options elimination ? Beginner**

A leap-year check case uses the quantities in this question. Determine the requested result using the options elimination model: If today is Monday, what day

is it after 51 days?

Options: Tuesday | Thursday | 13 | Wednesday

Q982 ? Boundary case ? Easy

A delivery calendar case uses the quantities in this question. Determine the requested result using the boundary case model: If today is Tuesday, what day is it after 52 days?

Options: Friday | Wednesday | 13 | Thursday

Q983 ? Case split ? Medium

A meeting rotation case uses the quantities in this question. Determine the requested result using the case split model: If today is Wednesday, what day is it after 53 days?

Options: Sunday | Thursday | Friday | Saturday

Q984 ? Contradiction check ? Hard

A roster cycle case uses the quantities in this question. Determine the requested result using the contradiction check model: If today is Thursday, what day is it after 54 days?

Options: Tuesday | Friday | Saturday | Sunday

Q985 ? Statement I test ? Placement

A release schedule case uses the quantities in this question. Determine the requested result using the statement i test model: If today is Friday, what day is it after 55 days?

Options: Sunday | Saturday | Monday | Thursday

Q986 ? Statement II test ? Beginner

A weekday audit case uses the quantities in this question. Determine the requested result using the statement ii test model: If today is Saturday, what day is it after 56 days?

Options: Tuesday | Monday | Sunday | Saturday

Q987 ? Combined statements ? Easy

A festival planning case uses the quantities in this question. Determine the requested result using the combined statements model: If today is Sunday, what day is it after 57 days?

Options: Wednesday | Monday | Tuesday | 13

Q988 ? Counting restriction ? Medium

A exam timetable case uses the quantities in this question. Determine the requested result using the counting restriction model: If today is Monday, what day is it after 58 days?

Options: Tuesday | Thursday | 13 | Wednesday

Q989 ? Circular arrangement ? Hard

A leap-year check case uses the quantities in this question. Determine the requested result using the circular arrangement model: If today is Tuesday, what day is it after 59 days?

Options: Friday | Wednesday | 13 | Thursday

Q990 ? Complement method ? Placement

A delivery calendar case uses the quantities in this question. Determine the requested result using the complement method model: If today is Wednesday, what day is it after 60 days?

Options: Sunday | Thursday | Friday | Saturday

Q991 ? Pattern: differences ? Beginner

A meeting rotation case uses the quantities in this question. Determine the requested result using the pattern: differences model: If today is Thursday, what day is it after 61 days?

Options: Tuesday | Friday | Saturday | Sunday

Q992 ? Pattern: ratios ? Easy

A roster cycle case uses the quantities in this question. Determine the requested result using the pattern: ratios model: If today is Friday, what day is it after 62 days?

Options: Sunday | Saturday | Monday | Thursday

Q993 ? Pattern: alternating ? Medium

A release schedule case uses the quantities in this question. Determine the requested result using the pattern: alternating model: If today is Saturday, what day is it after 63 days?

Options: Tuesday | Monday | Sunday | Saturday

Q994 ? Pattern: squares / cubes ? Hard

A weekday audit case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: If today is Sunday, what day is it after 64 days?

Options: Wednesday | Monday | Tuesday | 13

Q995 ? Inference from facts ? Placement

A festival planning case uses the quantities in this question. Determine the requested result using the inference from facts model: If today is Monday, what day is it after 65 days?

Options: Tuesday | Thursday | 13 | Wednesday

Q996 ? Assumption check ? Beginner

A exam timetable case uses the quantities in this question. Determine the requested result using the assumption check model: If today is Tuesday, what day is it after 66 days?

Options: Friday | Wednesday | 13 | Thursday

Q997 ? Cause vs correlation ? Easy

A leap-year check case uses the quantities in this question. Determine the requested result using the cause vs correlation model: If today is Wednesday, what day is it after 67 days?

Options: Sunday | Thursday | Friday | Saturday

Q998 ? Grammar agreement ? Medium

A delivery calendar case uses the quantities in this question. Determine the requested result using the grammar agreement model: If today is Thursday, what day is it after 68 days?

Options: Tuesday | Friday | Saturday | Sunday

Q999 ? Vocabulary in context ? Hard

A meeting rotation case uses the quantities in this question. Determine the requested result using the vocabulary in context model: If today is Friday, what day is it after 69 days?

Options: Sunday | Saturday | Monday | Thursday

Q1000 ? Mixed placement challenge ? Placement

A roster cycle case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: If today is Saturday, what day is it after 70 days?

Options: Tuesday | Monday | Sunday | Saturday

Permutation ? 40 questions

WORKED EXAMPLES

Q1001 ? Direct computation ? Beginner

A password design case uses the quantities in this question. Determine the requested result using the direct computation model: How many ordered arrangements of 3 objects chosen from 5 distinct objects?

Options: 54 | 72 | 66 | 60

Answer: 60

Solution: Context: password design. $5P3=5!/(2)!=60$. Shortcut: Cancel factorials early.

Q1002 ? Reverse value ? Easy

A podium finish case uses the quantities in this question. Determine the requested result using the reverse value model: How many ordered arrangements of 4 objects chosen from 6 distinct objects?

Options: 432 | 324 | 396 | 360

Answer: 360

Solution: Context: podium finish. $6P4=6!/(2)!=360$. Shortcut: Cancel factorials early.

Q1003 ? Missing base ? Medium

A seat allocation case uses the quantities in this question. Determine the requested result using the missing base model: How many ordered arrangements of 2 objects chosen from 7 distinct objects?

Options: 42 | 46 | 38 | 50

Answer: 42

Solution: Context: seat allocation. $7P2=7!/(5)!=42$. Shortcut: Cancel factorials early.

Q1004 ? Successive change ? Hard

A letter arrangement case uses the quantities in this question. Determine the requested result using the successive change model: How many ordered arrangements of 3 objects chosen from 8 distinct objects?

Options: 302 | 370 | 336 | 404

Answer: 336

Solution: Context: letter arrangement. $8P3=8!/(5)!=336$. Shortcut: Cancel factorials early.

Q1005 ? Comparison trap ? Placement

A team roles case uses the quantities in this question. Determine the requested result using the comparison trap model: How many ordered arrangements of 4 objects chosen from 4 distinct objects?

Options: 28 | 22 | 26 | 24

Answer: 24

Solution: Context: team roles. $4P4=4!/(0)!=24$. Shortcut: Cancel factorials early.

Q1006 ? Ratio form ? Beginner

A route ordering case uses the quantities in this question. Determine the requested result using the ratio form model: How many ordered arrangements of 2 objects chosen from 5 distinct objects?

Options: 20 | 22 | 18 | 24

Answer: 20

Solution: Context: route ordering. $5P2=5!/(3)!=20$. Shortcut: Cancel factorials early.

Q1007 ? Weighted result ? Easy

A task scheduling case uses the quantities in this question. Determine the requested result using the weighted result model: How many ordered arrangements of 3 objects chosen from 6 distinct objects?

Options: 120 | 132 | 108 | 144

Answer: 120

Solution: Context: task scheduling. $6P3=6!/(3)!=120$. Shortcut: Cancel factorials early.

Q1008 ? Replacement cycle ? Medium

A ID generation case uses the quantities in this question. Determine the requested result using the replacement cycle model: How many ordered arrangements of 4 objects chosen from 7 distinct objects?

Options: 1008 | 756 | 924 | 840

Answer: 840

Solution: Context: ID generation. $7P4=7!/(3)!=840$. Shortcut: Cancel factorials early.

Q1009 ? Unit conversion ? Hard

A password design case uses the quantities in this question. Determine the requested result using the unit conversion model: How many ordered arrangements of 2 objects chosen from 8 distinct objects?

Options: 56 | 62 | 50 | 68

Answer: 56

Solution: Context: password design. $8P2=8!/(6)!=56$. Shortcut: Cancel factorials early.

Q1010 ? Work-rate model ? Placement

A podium finish case uses the quantities in this question. Determine the requested result using the work-rate model model: How many ordered arrangements of 3 objects chosen from 4 distinct objects?

Options: 24 | 26 | 22 | 28

Answer: 24

Solution: Context: podium finish. $4P3=4!/(1)!=24$. Shortcut: Cancel factorials early.

Q1011 ? Speed-time model ? Beginner

A seat allocation case uses the quantities in this question. Determine the requested result using the speed-time model model: How many ordered arrangements of 4 objects chosen from 5 distinct objects?

Options: 144 | 108 | 132 | 120

Answer: 120

Solution: Context: seat allocation. $5P4 = 5!/(1!) = 120$. Shortcut: Cancel factorials early.

Q1012 ? Table lookup ? Easy

A letter arrangement case uses the quantities in this question. Determine the requested result using the table lookup model: How many ordered arrangements of 2 objects chosen from 6 distinct objects?

Options: 36 | 27 | 33 | 30

Answer: 30

Solution: Context: letter arrangement. $6P2 = 6!/(4!) = 30$. Shortcut: Cancel factorials early.

Q1013 ? Chart comparison ? Medium

A team roles case uses the quantities in this question. Determine the requested result using the chart comparison model: How many ordered arrangements of 3 objects chosen from 7 distinct objects?

Options: 210 | 231 | 189 | 252

Answer: 210

Solution: Context: team roles. $7P3 = 7!/(4!) = 210$. Shortcut: Cancel factorials early.

Q1014 ? Multi-step word problem ? Hard

A route ordering case uses the quantities in this question. Determine the requested result using the multi-step word problem model: How many ordered arrangements of 4 objects chosen from 8 distinct objects?

Options: 1680 | 1848 | 1512 | 2016

Answer: 1680

Solution: Context: route ordering. $8P4 = 8!/(4!) = 1680$. Shortcut: Cancel factorials early.

Q1015 ? Maximum / minimum ? Placement

A task scheduling case uses the quantities in this question. Determine the requested result using the maximum / minimum model: How many ordered arrangements of 2 objects chosen from 4 distinct objects?

Options: 14 | 11 | 13 | 12

Answer: 12

Solution: Context: task scheduling. $4P2 = 4!/(2!) = 12$. Shortcut: Cancel factorials early.

Q1016 ? Least / greatest value ? Beginner

A ID generation case uses the quantities in this question. Determine the requested result using the least / greatest value model: How many ordered arrangements of 3 objects chosen from 5 distinct objects?

Options: 60 | 66 | 54 | 72

Answer: 60

Solution: Context: ID generation. $5P3 = 5!/(2!) = 60$. Shortcut: Cancel factorials early.

Q1017 ? Remainder cycle ? Easy

A password design case uses the quantities in this question. Determine the requested result using the remainder cycle model: How many ordered arrangements of 4 objects chosen from 6 distinct objects?

Options: 360 | 396 | 324 | 432

Answer: 360

Solution: Context: password design. $6P4 = 6!/(2!) = 360$. Shortcut: Cancel factorials early.

Q1018 ? Divisibility test ? Medium

A podium finish case uses the quantities in this question. Determine the requested result using the divisibility test model: How many ordered arrangements of 2 objects chosen from 7 distinct objects?

Options: 38 | 46 | 42 | 50

Answer: 42

Solution: Context: podium finish. $7P2 = 7!/(5!) = 42$. Shortcut: Cancel factorials early.

Q1019 ? Factorisation ? Hard

A seat allocation case uses the quantities in this question. Determine the requested result using the factorisation model: How many ordered arrangements of 3 objects chosen from 8 distinct objects?

Options: 404 | 302 | 370 | 336

Answer: 336

Solution: Context: seat allocation. $8P3 = 8!/(5!) = 336$. Shortcut: Cancel factorials early.

Q1020 ? Approximation ? Placement

A letter arrangement case uses the quantities in this question. Determine the requested result using the approximation model: How many ordered arrangements of 4 objects chosen from 4 distinct objects?

Options: 24 | 26 | 22 | 28

Answer: 24

Solution: Context: letter arrangement. $4P4 = 4!/(0!) = 24$. Shortcut: Cancel factorials early.

PRACTICE ? solutions hidden below**Q1021 ? Options elimination ? Beginner**

A team roles case uses the quantities in this question. Determine the requested result using the options elimination model: How many ordered arrangements

of 2 objects chosen from 5 distinct objects?

Options: 18 | 20 | 22 | 24

Q1022 ? Boundary case ? Easy

A route ordering case uses the quantities in this question. Determine the requested result using the boundary case model: How many ordered arrangements of 3 objects chosen from 6 distinct objects?

Options: 144 | 108 | 132 | 120

Q1023 ? Case split ? Medium

A task scheduling case uses the quantities in this question. Determine the requested result using the case split model: How many ordered arrangements of 4 objects chosen from 7 distinct objects?

Options: 924 | 840 | 756 | 1008

Q1024 ? Contradiction check ? Hard

A ID generation case uses the quantities in this question. Determine the requested result using the contradiction check model: How many ordered arrangements of 2 objects chosen from 8 distinct objects?

Options: 68 | 62 | 56 | 50

Q1025 ? Statement I test ? Placement

A password design case uses the quantities in this question. Determine the requested result using the statement i test model: How many ordered arrangements of 3 objects chosen from 4 distinct objects?

Options: 24 | 26 | 22 | 28

Q1026 ? Statement II test ? Beginner

A podium finish case uses the quantities in this question. Determine the requested result using the statement ii test model: How many ordered arrangements of 4 objects chosen from 5 distinct objects?

Options: 144 | 108 | 132 | 120

Q1027 ? Combined statements ? Easy

A seat allocation case uses the quantities in this question. Determine the requested result using the combined statements model: How many ordered arrangements of 2 objects chosen from 6 distinct objects?

Options: 30 | 33 | 27 | 36

Q1028 ? Counting restriction ? Medium

A letter arrangement case uses the quantities in this question. Determine the requested result using the counting restriction model: How many ordered arrangements of 3 objects chosen from 7 distinct objects?

Options: 189 | 210 | 231 | 252

Q1029 ? Circular arrangement ? Hard

A team roles case uses the quantities in this question. Determine the requested result using the circular arrangement model: How many ordered arrangements of 4 objects chosen from 8 distinct objects?

Options: 2016 | 1512 | 1848 | 1680

Q1030 ? Complement method ? Placement

A route ordering case uses the quantities in this question. Determine the requested result using the complement method model: How many ordered arrangements of 2 objects chosen from 4 distinct objects?

Options: 12 | 13 | 11 | 14

Q1031 ? Pattern: differences ? Beginner

A task scheduling case uses the quantities in this question. Determine the requested result using the pattern: differences model: How many ordered arrangements of 3 objects chosen from 5 distinct objects?

Options: 72 | 66 | 60 | 54

Q1032 ? Pattern: ratios ? Easy

A ID generation case uses the quantities in this question. Determine the requested result using the pattern: ratios model: How many ordered arrangements of 4 objects chosen from 6 distinct objects?

Options: 324 | 396 | 360 | 432

Q1033 ? Pattern: alternating ? Medium

A password design case uses the quantities in this question. Determine the requested result using the pattern: alternating model: How many ordered arrangements of 2 objects chosen from 7 distinct objects?

Options: 42 | 46 | 50 | 38

Q1034 ? Pattern: squares / cubes ? Hard

A podium finish case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: How many ordered arrangements of 3 objects chosen from 8 distinct objects?

Options: 404 | 336 | 370 | 302

Q1035 ? Inference from facts ? Placement

A seat allocation case uses the quantities in this question. Determine the requested result using the inference from facts model: How many ordered arrangements of 4 objects chosen from 4 distinct objects?

Options: 24 | 26 | 22 | 28

Q1036 ? Assumption check ? Beginner

A letter arrangement case uses the quantities in this question. Determine the requested result using the assumption check model: How many ordered arrangements of 2 objects chosen from 5 distinct objects?

Options: 24 | 18 | 22 | 20

Q1037 ? Cause vs correlation ? Easy

A team roles case uses the quantities in this question. Determine the requested result using the cause vs correlation model: How many ordered arrangements of 3 objects chosen from 6 distinct objects?

Options: 120 | 132 | 108 | 144

Q1038 ? Grammar agreement ? Medium

A route ordering case uses the quantities in this question. Determine the requested result using the grammar agreement model: How many ordered arrangements of 4 objects chosen from 7 distinct objects?

Options: 840 | 756 | 924 | 1008

Q1039 ? Vocabulary in context ? Hard

A task scheduling case uses the quantities in this question. Determine the requested result using the vocabulary in context model: How many ordered arrangements of 2 objects chosen from 8 distinct objects?

Options: 50 | 56 | 68 | 62

Q1040 ? Mixed placement challenge ? Placement

A ID generation case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: How many ordered arrangements of 3 objects chosen from 4 distinct objects?

Options: 28 | 22 | 26 | 24

Combination ? 40 questions

WORKED EXAMPLES

Q1041 ? Direct computation ? Beginner

A committee selection case uses the quantities in this question. Determine the requested result using the direct computation model: How many committees of 3 can be chosen from 6 people?

Options: 24 | 18 | 22 | 20

Answer: 20

Solution: Context: committee selection. ${}^6C_3 = 6!/[3!3!] = 20$. Shortcut: Use $nCr = nC(n-r)$.

Q1042 ? Reverse value ? Easy

A team shortlist case uses the quantities in this question. Determine the requested result using the reverse value model: How many committees of 4 can be chosen from 7 people?

Options: 43 | 31 | 39 | 35

Answer: 35

Solution: Context: team shortlist. ${}^7C_4 = 7!/[4!3!] = 35$. Shortcut: Use $nCr = nC(n-r)$.

Q1043 ? Missing base ? Medium

A feature selection case uses the quantities in this question. Determine the requested result using the missing base model: How many committees of 2 can be chosen from 8 people?

Options: 28 | 31 | 25 | 34

Answer: 28

Solution: Context: feature selection. ${}^8C_2 = 8!/[2!6!] = 28$. Shortcut: Use $nCr = nC(n-r)$.

Q1044 ? Successive change ? Hard

A panel formation case uses the quantities in this question. Determine the requested result using the successive change model: How many committees of 3 can be chosen from 9 people?

Options: 84 | 92 | 76 | 100

Answer: 84

Solution: Context: panel formation. ${}^9C_3 = 9!/[3!6!] = 84$. Shortcut: Use $nCr = nC(n-r)$.

Q1045 ? Comparison trap ? Placement

A scholarship group case uses the quantities in this question. Determine the requested result using the comparison trap model: How many committees of 4 can be chosen from 10 people?

Options: 252 | 189 | 231 | 210

Answer: 210

Solution: Context: scholarship group. ${}^{10}C_4 = 10!/[4!6!] = 210$. Shortcut: Use $nCr = nC(n-r)$.

Q1046 ? Ratio form ? Beginner

A menu bundle case uses the quantities in this question. Determine the requested result using the ratio form model: How many committees of 2 can be chosen from 5 people?

Options: 10 | 11 | 9 | 12

Answer: 10

Solution: Context: menu bundle. ${}^5C_2 = 5!/[2!3!] = 10$. Shortcut: Use $nCr = nC(n-r)$.

Q1047 ? Weighted result ? Easy

A hiring shortlist case uses the quantities in this question. Determine the requested result using the weighted result model: How many committees of 3 can be chosen from 6 people?

Options: 18 | 20 | 22 | 24

Answer: 20

Solution: Context: hiring shortlist. ${}^6C_3 = 6!/[3!3!] = 20$. Shortcut: Use $nCr = nC(n-r)$.

Q1048 ? Replacement cycle ? Medium

A lottery sample case uses the quantities in this question. Determine the requested result using the replacement cycle model: How many committees of 4 can be chosen from 7 people?

Options: 35 | 39 | 31 | 43

Answer: 35

Solution: Context: lottery sample. ${}^7C_4 = 7!/[4!3!] = 35$. Shortcut: Use $nCr = nC(n-r)$.

Q1049 ? Unit conversion ? Hard

A committee selection case uses the quantities in this question. Determine the requested result using the unit conversion model: How many committees of 2 can be chosen from 8 people?

Options: 34 | 25 | 31 | 28

Answer: 28

Solution: Context: committee selection. ${}^8C_2 = 8!/[2!6!] = 28$. Shortcut: Use $nCr = nC(n-r)$.

Q1050 ? Work-rate model ? Placement

A team shortlist case uses the quantities in this question. Determine the requested result using the work-rate model model: How many committees of 3 can be chosen from 9 people?

Options: 84 | 76 | 92 | 100

Answer: 84

Solution: Context: team shortlist. ${}^9C_3 = 9!/[3!6!] = 84$. Shortcut: Use $nCr = nC(n-r)$.

Q1051 ? Speed-time model ? Beginner

A feature selection case uses the quantities in this question. Determine the requested result using the speed-time model model: How many committees of 4 can be chosen from 10 people?

Options: 210 | 231 | 189 | 252

Answer: 210

Solution: Context: feature selection. $10C4 = 10!/[4!6!] = 210$. Shortcut: Use $nCr = nC(n-r)$.

Q1052 ? Table lookup ? Easy

A panel formation case uses the quantities in this question. Determine the requested result using the table lookup model: How many committees of 2 can be chosen from 5 people?

Options: 12 | 9 | 11 | 10

Answer: 10

Solution: Context: panel formation. $5C2 = 5!/[2!3!] = 10$. Shortcut: Use $nCr = nC(n-r)$.

Q1053 ? Chart comparison ? Medium

A scholarship group case uses the quantities in this question. Determine the requested result using the chart comparison model: How many committees of 3 can be chosen from 6 people?

Options: 20 | 22 | 18 | 24

Answer: 20

Solution: Context: scholarship group. $6C3 = 6!/[3!3!] = 20$. Shortcut: Use $nCr = nC(n-r)$.

Q1054 ? Multi-step word problem ? Hard

A menu bundle case uses the quantities in this question. Determine the requested result using the multi-step word problem model: How many committees of 4 can be chosen from 7 people?

Options: 35 | 39 | 31 | 43

Answer: 35

Solution: Context: menu bundle. $7C4 = 7!/[4!3!] = 35$. Shortcut: Use $nCr = nC(n-r)$.

Q1055 ? Maximum / minimum ? Placement

A hiring shortlist case uses the quantities in this question. Determine the requested result using the maximum / minimum model: How many committees of 2 can be chosen from 8 people?

Options: 31 | 34 | 25 | 28

Answer: 28

Solution: Context: hiring shortlist. $8C2 = 8!/[2!6!] = 28$. Shortcut: Use $nCr = nC(n-r)$.

Q1056 ? Least / greatest value ? Beginner

A lottery sample case uses the quantities in this question. Determine the requested result using the least / greatest value model: How many committees of 3 can be chosen from 9 people?

Options: 92 | 84 | 76 | 100

Answer: 84

Solution: Context: lottery sample. $9C3 = 9!/[3!6!] = 84$. Shortcut: Use $nCr = nC(n-r)$.

Q1057 ? Remainder cycle ? Easy

A committee selection case uses the quantities in this question. Determine the requested result using the remainder cycle model: How many committees of 4 can be chosen from 10 people?

Options: 210 | 231 | 189 | 252

Answer: 210

Solution: Context: committee selection. $10C4 = 10!/[4!6!] = 210$. Shortcut: Use $nCr = nC(n-r)$.

Q1058 ? Divisibility test ? Medium

A team shortlist case uses the quantities in this question. Determine the requested result using the divisibility test model: How many committees of 2 can be chosen from 5 people?

Options: 12 | 11 | 10 | 9

Answer: 10

Solution: Context: team shortlist. $5C2 = 5!/[2!3!] = 10$. Shortcut: Use $nCr = nC(n-r)$.

Q1059 ? Factorisation ? Hard

A feature selection case uses the quantities in this question. Determine the requested result using the factorisation model: How many committees of 3 can be chosen from 6 people?

Options: 24 | 18 | 22 | 20

Answer: 20

Solution: Context: feature selection. $6C3 = 6!/[3!3!] = 20$. Shortcut: Use $nCr = nC(n-r)$.

Q1060 ? Approximation ? Placement

A panel formation case uses the quantities in this question. Determine the requested result using the approximation model: How many committees of 4 can be chosen from 7 people?

Options: 43 | 35 | 39 | 31

Answer: 35

Solution: Context: panel formation. $7C4 = 7!/[4!3!] = 35$. Shortcut: Use $nCr = nC(n-r)$.

PRACTICE ? solutions hidden below**Q1061 ? Options elimination ? Beginner**

A scholarship group case uses the quantities in this question. Determine the requested result using the options elimination model: How many committees of 2

can be chosen from 8 people?
Options: 28 | 25 | 31 | 34

Q1062 ? Boundary case ? Easy

A menu bundle case uses the quantities in this question. Determine the requested result using the boundary case model: How many committees of 3 can be chosen from 9 people?
Options: 100 | 76 | 92 | 84

Q1063 ? Case split ? Medium

A hiring shortlist case uses the quantities in this question. Determine the requested result using the case split model: How many committees of 4 can be chosen from 10 people?
Options: 252 | 231 | 210 | 189

Q1064 ? Contradiction check ? Hard

A lottery sample case uses the quantities in this question. Determine the requested result using the contradiction check model: How many committees of 2 can be chosen from 5 people?
Options: 10 | 11 | 9 | 12

Q1065 ? Statement I test ? Placement

A committee selection case uses the quantities in this question. Determine the requested result using the statement i test model: How many committees of 3 can be chosen from 6 people?
Options: 20 | 22 | 18 | 24

Q1066 ? Statement II test ? Beginner

A team shortlist case uses the quantities in this question. Determine the requested result using the statement ii test model: How many committees of 4 can be chosen from 7 people?
Options: 43 | 31 | 39 | 35

Q1067 ? Combined statements ? Easy

A feature selection case uses the quantities in this question. Determine the requested result using the combined statements model: How many committees of 2 can be chosen from 8 people?
Options: 28 | 31 | 25 | 34

Q1068 ? Counting restriction ? Medium

A panel formation case uses the quantities in this question. Determine the requested result using the counting restriction model: How many committees of 3 can be chosen from 9 people?
Options: 84 | 100 | 92 | 76

Q1069 ? Circular arrangement ? Hard

A scholarship group case uses the quantities in this question. Determine the requested result using the circular arrangement model: How many committees of 4 can be chosen from 10 people?
Options: 252 | 189 | 231 | 210

Q1070 ? Complement method ? Placement

A menu bundle case uses the quantities in this question. Determine the requested result using the complement method model: How many committees of 2 can be chosen from 5 people?
Options: 9 | 10 | 11 | 12

Q1071 ? Pattern: differences ? Beginner

A hiring shortlist case uses the quantities in this question. Determine the requested result using the pattern: differences model: How many committees of 3 can be chosen from 6 people?
Options: 20 | 22 | 18 | 24

Q1072 ? Pattern: ratios ? Easy

A lottery sample case uses the quantities in this question. Determine the requested result using the pattern: ratios model: How many committees of 4 can be chosen from 7 people?
Options: 31 | 39 | 35 | 43

Q1073 ? Pattern: alternating ? Medium

A committee selection case uses the quantities in this question. Determine the requested result using the pattern: alternating model: How many committees of 2 can be chosen from 8 people?
Options: 34 | 25 | 31 | 28

Q1074 ? Pattern: squares / cubes ? Hard

A team shortlist case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: How many committees of 3 can be chosen from 9 people?
Options: 84 | 92 | 76 | 100

Q1075 ? Inference from facts ? Placement

A feature selection case uses the quantities in this question. Determine the requested result using the inference from facts model: How many committees of 4 can be chosen from 10 people?
Options: 189 | 210 | 231 | 252

Q1076 ? Assumption check ? Beginner

A panel formation case uses the quantities in this question. Determine the requested result using the assumption check model: How many committees of 2 can be chosen from 5 people?
Options: 12 | 9 | 11 | 10

Q1077 ? Cause vs correlation ? Easy

A scholarship group case uses the quantities in this question. Determine the requested result using the cause vs correlation model: How many committees of 3 can be chosen from 6 people?

Options: 24 | 22 | 20 | 18

Q1078 ? Grammar agreement ? Medium

A menu bundle case uses the quantities in this question. Determine the requested result using the grammar agreement model: How many committees of 4 can be chosen from 7 people?

Options: 35 | 39 | 31 | 43

Q1079 ? Vocabulary in context ? Hard

A hiring shortlist case uses the quantities in this question. Determine the requested result using the vocabulary in context model: How many committees of 2 can be chosen from 8 people?

Options: 28 | 31 | 25 | 34

Q1080 ? Mixed placement challenge ? Placement

A lottery sample case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: How many committees of 3 can be chosen from 9 people?

Options: 100 | 76 | 92 | 84

Number System ? 40 questions

WORKED EXAMPLES

Q1081 ? Direct computation ? Beginner

A remainder audit case uses the quantities in this question. Determine the requested result using the direct computation model: Find the remainder when 117 is divided by 4.

Options: 3 | 2 | 1 | 4

Answer: 1

Solution: Context: remainder audit. $117=4\times 29+1$; remainder=1. Shortcut: Find the nearest lower multiple.

Q1082 ? Reverse value ? Easy

A digit puzzle case uses the quantities in this question. Determine the requested result using the reverse value model: Find the remainder when 134 is divided by 5.

Options: 0 | 5 | 4 | 1

Answer: 4

Solution: Context: digit puzzle. $134=5\times 26+4$; remainder=4. Shortcut: Find the nearest lower multiple.

Q1083 ? Missing base ? Medium

A number machine case uses the quantities in this question. Determine the requested result using the missing base model: Find the remainder when 151 is divided by 6.

Options: 6 | 1 | 2 | 3

Answer: 1

Solution: Context: number machine. $151=6\times 25+1$; remainder=1. Shortcut: Find the nearest lower multiple.

Q1084 ? Successive change ? Hard

A modular code case uses the quantities in this question. Determine the requested result using the successive change model: Find the remainder when 168 is divided by 7.

Options: 0 | 1 | 2 | 7

Answer: 0

Solution: Context: modular code. $168=7\times 24+0$; remainder=0. Shortcut: Find the nearest lower multiple.

Q1085 ? Comparison trap ? Placement

A billing sequence case uses the quantities in this question. Determine the requested result using the comparison trap model: Find the remainder when 185 is divided by 8.

Options: 8 | 3 | 2 | 1

Answer: 1

Solution: Context: billing sequence. $185=8\times 23+1$; remainder=1. Shortcut: Find the nearest lower multiple.

Q1086 ? Ratio form ? Beginner

An integer game case uses the quantities in this question. Determine the requested result using the ratio form model: Find the remainder when 202 is divided by 9.

Options: 6 | 9 | 5 | 4

Answer: 4

Solution: Context: integer game. $202=9\times 22+4$; remainder=4. Shortcut: Find the nearest lower multiple.

Q1087 ? Weighted result ? Easy

A factor table case uses the quantities in this question. Determine the requested result using the weighted result model: Find the remainder when 219 is divided by 10.

Options: 1 | 10 | 9 | 0

Answer: 9

Solution: Context: factor table. $219=10\times 21+9$; remainder=9. Shortcut: Find the nearest lower multiple.

Q1088 ? Replacement cycle ? Medium

A base conversion case uses the quantities in this question. Determine the requested result using the replacement cycle model: Find the remainder when 236 is divided by 11.

Options: 11 | 7 | 6 | 5

Answer: 5

Solution: Context: base conversion. $236=11\times 21+5$; remainder=5. Shortcut: Find the nearest lower multiple.

Q1089 ? Unit conversion ? Hard

A remainder audit case uses the quantities in this question. Determine the requested result using the unit conversion model: Find the remainder when 253 is divided by 3.

Options: 3 | 2 | 1 | 0

Answer: 1

Solution: Context: remainder audit. $253=3\times 84+1$; remainder=1. Shortcut: Find the nearest lower multiple.

Q1090 ? Work-rate model ? Placement

A digit puzzle case uses the quantities in this question. Determine the requested result using the work-rate model model: Find the remainder when 270 is divided by 4.

Options: 0 | 2 | 3 | 4

Answer: 2

Solution: Context: digit puzzle. $270=4\times 67+2$; remainder=2. Shortcut: Find the nearest lower multiple.

Q1091 ? Speed-time model ? Beginner

A number machine case uses the quantities in this question. Determine the requested result using the speed-time model model: Find the remainder when 287 is divided by 5.

Options: 2 | 3 | 4 | 5

Answer: 2

Solution: Context: number machine. $287=5\times 57+2$; remainder=2. Shortcut: Find the nearest lower multiple.

Q1092 ? Table lookup ? Easy

A modular code case uses the quantities in this question. Determine the requested result using the table lookup model: Find the remainder when 304 is divided by 6.

Options: 4 | 0 | 5 | 6

Answer: 4

Solution: Context: modular code. $304=6\times 50+4$; remainder=4. Shortcut: Find the nearest lower multiple.

Q1093 ? Chart comparison ? Medium

A billing sequence case uses the quantities in this question. Determine the requested result using the chart comparison model: Find the remainder when 321 is divided by 7.

Options: 6 | 0 | 1 | 7

Answer: 6

Solution: Context: billing sequence. $321=7\times 45+6$; remainder=6. Shortcut: Find the nearest lower multiple.

Q1094 ? Multi-step word problem ? Hard

A integer game case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Find the remainder when 338 is divided by 8.

Options: 8 | 2 | 3 | 4

Answer: 2

Solution: Context: integer game. $338=8\times 42+2$; remainder=2. Shortcut: Find the nearest lower multiple.

Q1095 ? Maximum / minimum ? Placement

A factor table case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Find the remainder when 355 is divided by 9.

Options: 6 | 5 | 4 | 9

Answer: 4

Solution: Context: factor table. $355=9\times 39+4$; remainder=4. Shortcut: Find the nearest lower multiple.

Q1096 ? Least / greatest value ? Beginner

A base conversion case uses the quantities in this question. Determine the requested result using the least / greatest value model: Find the remainder when 372 is divided by 10.

Options: 4 | 3 | 2 | 10

Answer: 2

Solution: Context: base conversion. $372=10\times 37+2$; remainder=2. Shortcut: Find the nearest lower multiple.

Q1097 ? Remainder cycle ? Easy

A remainder audit case uses the quantities in this question. Determine the requested result using the remainder cycle model: Find the remainder when 389 is divided by 11.

Options: 11 | 6 | 4 | 5

Answer: 4

Solution: Context: remainder audit. $389=11\times 35+4$; remainder=4. Shortcut: Find the nearest lower multiple.

Q1098 ? Divisibility test ? Medium

A digit puzzle case uses the quantities in this question. Determine the requested result using the divisibility test model: Find the remainder when 406 is divided by 3.

Options: 1 | 2 | 3 | 0

Answer: 1

Solution: Context: digit puzzle. $406=3\times 135+1$; remainder=1. Shortcut: Find the nearest lower multiple.

Q1099 ? Factorisation ? Hard

A number machine case uses the quantities in this question. Determine the requested result using the factorisation model: Find the remainder when 423 is divided by 4.

Options: 1 | 0 | 4 | 3

Answer: 3

Solution: Context: number machine. $423=4\times 105+3$; remainder=3. Shortcut: Find the nearest lower multiple.

Q1100 ? Approximation ? Placement

A modular code case uses the quantities in this question. Determine the requested result using the approximation model: Find the remainder when 440 is divided by 5.

Options: 0 | 5 | 1 | 2

Answer: 0

Solution: Context: modular code. $440=5\times 88+0$; remainder=0. Shortcut: Find the nearest lower multiple.

PRACTICE ? solutions hidden below**Q1101 ? Options elimination ? Beginner**

A billing sequence case uses the quantities in this question. Determine the requested result using the options elimination model: Find the remainder when 457

is divided by 6.

Options: 1 | 2 | 3 | 6

Q1102 ? Boundary case ? Easy

A integer game case uses the quantities in this question. Determine the requested result using the boundary case model: Find the remainder when 474 is divided by 7.

Options: 7 | 0 | 6 | 5

Q1103 ? Case split ? Medium

A factor table case uses the quantities in this question. Determine the requested result using the case split model: Find the remainder when 491 is divided by 8.

Options: 5 | 4 | 3 | 8

Q1104 ? Contradiction check ? Hard

A base conversion case uses the quantities in this question. Determine the requested result using the contradiction check model: Find the remainder when 508 is divided by 9.

Options: 9 | 6 | 4 | 5

Q1105 ? Statement I test ? Placement

A remainder audit case uses the quantities in this question. Determine the requested result using the statement i test model: Find the remainder when 525 is divided by 10.

Options: 5 | 6 | 7 | 10

Q1106 ? Statement II test ? Beginner

A digit puzzle case uses the quantities in this question. Determine the requested result using the statement ii test model: Find the remainder when 542 is divided by 11.

Options: 4 | 5 | 11 | 3

Q1107 ? Combined statements ? Easy

A number machine case uses the quantities in this question. Determine the requested result using the combined statements model: Find the remainder when 559 is divided by 3.

Options: 1 | 2 | 0 | 3

Q1108 ? Counting restriction ? Medium

A modular code case uses the quantities in this question. Determine the requested result using the counting restriction model: Find the remainder when 576 is divided by 4.

Options: 1 | 2 | 4 | 0

Q1109 ? Circular arrangement ? Hard

A billing sequence case uses the quantities in this question. Determine the requested result using the circular arrangement model: Find the remainder when 593 is divided by 5.

Options: 3 | 5 | 4 | 0

Q1110 ? Complement method ? Placement

A integer game case uses the quantities in this question. Determine the requested result using the complement method model: Find the remainder when 610 is divided by 6.

Options: 5 | 0 | 6 | 4

Q1111 ? Pattern: differences ? Beginner

A factor table case uses the quantities in this question. Determine the requested result using the pattern: differences model: Find the remainder when 627 is divided by 7.

Options: 7 | 6 | 4 | 5

Q1112 ? Pattern: ratios ? Easy

A base conversion case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Find the remainder when 644 is divided by 8.

Options: 4 | 5 | 6 | 8

Q1113 ? Pattern: alternating ? Medium

A remainder audit case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Find the remainder when 661 is divided by 9.

Options: 4 | 5 | 6 | 9

Q1114 ? Pattern: squares / cubes ? Hard

A digit puzzle case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Find the remainder when 678 is divided by 10.

Options: 8 | 9 | 0 | 10

Q1115 ? Inference from facts ? Placement

A number machine case uses the quantities in this question. Determine the requested result using the inference from facts model: Find the remainder when 695 is divided by 11.

Options: 11 | 2 | 3 | 4

Q1116 ? Assumption check ? Beginner

A modular code case uses the quantities in this question. Determine the requested result using the assumption check model: Find the remainder when 712 is divided by 3.

Options: 0 | 3 | 2 | 1

Q1117 ? Cause vs correlation ? Easy

A billing sequence case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Find the remainder when 729 is divided by 4.

Options: 4 | 3 | 2 | 1

Q1118 ? Grammar agreement ? Medium

A integer game case uses the quantities in this question. Determine the requested result using the grammar agreement model: Find the remainder when 746 is divided by 5.

Options: 1 | 2 | 3 | 5

Q1119 ? Vocabulary in context ? Hard

A factor table case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Find the remainder when 763 is divided by 6.

Options: 1 | 2 | 3 | 6

Q1120 ? Mixed placement challenge ? Placement

A base conversion case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Find the remainder when 780 is divided by 7.

Options: 4 | 7 | 5 | 3

Divisibility ? 40 questions

WORKED EXAMPLES

Q1121 ? Direct computation ? Beginner

A barcode validation case uses the quantities in this question. Determine the requested result using the direct computation model: Which option is divisible by 4?

Options: 127 | 126 | 125 | 124

Answer: 124

Solution: Context: barcode validation. $124/4=31$, an integer. Shortcut: Test digits before dividing.

Q1122 ? Reverse value ? Easy

A invoice check case uses the quantities in this question. Determine the requested result using the reverse value model: Which option is divisible by 5?

Options: 125 | 126 | 127 | 128

Answer: 125

Solution: Context: invoice check. $125/5=25$, an integer. Shortcut: Test digits before dividing.

Q1123 ? Missing base ? Medium

A seat-number rule case uses the quantities in this question. Determine the requested result using the missing base model: Which option is divisible by 9?

Options: 126 | 127 | 128 | 129

Answer: 126

Solution: Context: seat-number rule. $126/9=14$, an integer. Shortcut: Test digits before dividing.

Q1124 ? Successive change ? Hard

A data checksum case uses the quantities in this question. Determine the requested result using the successive change model: Which option is divisible by 3?

Options: 138 | 137 | 136 | 135

Answer: 135

Solution: Context: data checksum. $135/3=45$, an integer. Shortcut: Test digits before dividing.

Q1125 ? Comparison trap ? Placement

A serial code case uses the quantities in this question. Determine the requested result using the comparison trap model: Which option is divisible by 4?

Options: 143 | 142 | 141 | 140

Answer: 140

Solution: Context: serial code. $140/4=35$, an integer. Shortcut: Test digits before dividing.

Q1126 ? Ratio form ? Beginner

A digit audit case uses the quantities in this question. Determine the requested result using the ratio form model: Which option is divisible by 5?

Options: 140 | 141 | 142 | 143

Answer: 140

Solution: Context: digit audit. $140/5=28$, an integer. Shortcut: Test digits before dividing.

Q1127 ? Weighted result ? Easy

A packet routing case uses the quantities in this question. Determine the requested result using the weighted result model: Which option is divisible by 9?

Options: 144 | 145 | 146 | 147

Answer: 144

Solution: Context: packet routing. $144/9=16$, an integer. Shortcut: Test digits before dividing.

Q1128 ? Replacement cycle ? Medium

A number lock case uses the quantities in this question. Determine the requested result using the replacement cycle model: Which option is divisible by 3?

Options: 153 | 152 | 151 | 150

Answer: 150

Solution: Context: number lock. $150/3=50$, an integer. Shortcut: Test digits before dividing.

Q1129 ? Unit conversion ? Hard

A barcode validation case uses the quantities in this question. Determine the requested result using the unit conversion model: Which option is divisible by 4?

Options: 156 | 157 | 158 | 159

Answer: 156

Solution: Context: barcode validation. $156/4=39$, an integer. Shortcut: Test digits before dividing.

Q1130 ? Work-rate model ? Placement

A invoice check case uses the quantities in this question. Determine the requested result using the work-rate model model: Which option is divisible by 5?

Options: 160 | 161 | 162 | 163

Answer: 160

Solution: Context: invoice check. $160/5=32$, an integer. Shortcut: Test digits before dividing.

Q1131 ? Speed-time model ? Beginner

A seat-number rule case uses the quantities in this question. Determine the requested result using the speed-time model model: Which option is divisible by 9?

Options: 165 | 164 | 163 | 162

Answer: 162

Solution: Context: seat-number rule. $162/9=18$, an integer. Shortcut: Test digits before dividing.

Q1132 ? Table lookup ? Easy

A data checksum case uses the quantities in this question. Determine the requested result using the table lookup model: Which option is divisible by 3?

Options: 171 | 170 | 169 | 168

Answer: 168

Solution: Context: data checksum. $168/3=56$, an integer. Shortcut: Test digits before dividing.

Q1133 ? Chart comparison ? Medium

A serial code case uses the quantities in this question. Determine the requested result using the chart comparison model: Which option is divisible by 4?

Options: 172 | 173 | 174 | 175

Answer: 172

Solution: Context: serial code. $172/4=43$, an integer. Shortcut: Test digits before dividing.

Q1134 ? Multi-step word problem ? Hard

A digit audit case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Which option is divisible by 5?

Options: 175 | 176 | 177 | 178

Answer: 175

Solution: Context: digit audit. $175/5=35$, an integer. Shortcut: Test digits before dividing.

Q1135 ? Maximum / minimum ? Placement

A packet routing case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Which option is divisible by 9?

Options: 183 | 182 | 181 | 180

Answer: 180

Solution: Context: packet routing. $180/9=20$, an integer. Shortcut: Test digits before dividing.

Q1136 ? Least / greatest value ? Beginner

A number lock case uses the quantities in this question. Determine the requested result using the least / greatest value model: Which option is divisible by 3?

Options: 183 | 184 | 185 | 186

Answer: 183

Solution: Context: number lock. $183/3=61$, an integer. Shortcut: Test digits before dividing.

Q1137 ? Remainder cycle ? Easy

A barcode validation case uses the quantities in this question. Determine the requested result using the remainder cycle model: Which option is divisible by 4?

Options: 188 | 189 | 190 | 191

Answer: 188

Solution: Context: barcode validation. $188/4=47$, an integer. Shortcut: Test digits before dividing.

Q1138 ? Divisibility test ? Medium

A invoice check case uses the quantities in this question. Determine the requested result using the divisibility test model: Which option is divisible by 5?

Options: 193 | 192 | 191 | 190

Answer: 190

Solution: Context: invoice check. $190/5=38$, an integer. Shortcut: Test digits before dividing.

Q1139 ? Factorisation ? Hard

A seat-number rule case uses the quantities in this question. Determine the requested result using the factorisation model: Which option is divisible by 9?

Options: 192 | 191 | 190 | 189

Answer: 189

Solution: Context: seat-number rule. $189/9=21$, an integer. Shortcut: Test digits before dividing.

Q1140 ? Approximation ? Placement

A data checksum case uses the quantities in this question. Determine the requested result using the approximation model: Which option is divisible by 3?

Options: 198 | 199 | 200 | 201

Answer: 198

Solution: Context: data checksum. $198/3=66$, an integer. Shortcut: Test digits before dividing.

PRACTICE ? solutions hidden below

Q1141 ? Options elimination ? Beginner

A serial code case uses the quantities in this question. Determine the requested result using the options elimination model: Which option is divisible by 4?

Options: 204 | 205 | 206 | 207

Q1142 ? Boundary case ? Easy

A digit audit case uses the quantities in this question. Determine the requested result using the boundary case model: Which option is divisible by 5?

Options: 208 | 207 | 206 | 205

Q1143 ? Case split ? Medium

A packet routing case uses the quantities in this question. Determine the requested result using the case split model: Which option is divisible by 9?

Options: 210 | 209 | 208 | 207

Q1144 ? Contradiction check ? Hard

A number lock case uses the quantities in this question. Determine the requested result using the contradiction check model: Which option is divisible by 3?

Options: 216 | 217 | 218 | 219

Q1145 ? Statement I test ? Placement

A barcode validation case uses the quantities in this question. Determine the requested result using the statement I test model: Which option is divisible by 4?

Options: 220 | 221 | 222 | 223

Q1146 ? Statement II test ? Beginner

A invoice check case uses the quantities in this question. Determine the requested result using the statement ii test model: Which option is divisible by 5?
Options: 223 | 222 | 221 | 220

Q1147 ? Combined statements ? Easy

A seat-number rule case uses the quantities in this question. Determine the requested result using the combined statements model: Which option is divisible by 9?
Options: 225 | 226 | 227 | 228

Q1148 ? Counting restriction ? Medium

A data checksum case uses the quantities in this question. Determine the requested result using the counting restriction model: Which option is divisible by 3?
Options: 231 | 232 | 233 | 234

Q1149 ? Circular arrangement ? Hard

A serial code case uses the quantities in this question. Determine the requested result using the circular arrangement model: Which option is divisible by 4?
Options: 239 | 238 | 237 | 236

Q1150 ? Complement method ? Placement

A digit audit case uses the quantities in this question. Determine the requested result using the complement method model: Which option is divisible by 5?
Options: 243 | 242 | 241 | 240

Q1151 ? Pattern: differences ? Beginner

A packet routing case uses the quantities in this question. Determine the requested result using the pattern: differences model: Which option is divisible by 9?
Options: 243 | 244 | 245 | 246

Q1152 ? Pattern: ratios ? Easy

A number lock case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Which option is divisible by 3?
Options: 246 | 247 | 248 | 249

Q1153 ? Pattern: alternating ? Medium

A barcode validation case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Which option is divisible by 4?
Options: 255 | 254 | 253 | 252

Q1154 ? Pattern: squares / cubes ? Hard

A invoice check case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Which option is divisible by 5?
Options: 255 | 256 | 257 | 258

Q1155 ? Inference from facts ? Placement

A seat-number rule case uses the quantities in this question. Determine the requested result using the inference from facts model: Which option is divisible by 9?
Options: 252 | 253 | 254 | 255

Q1156 ? Assumption check ? Beginner

A data checksum case uses the quantities in this question. Determine the requested result using the assumption check model: Which option is divisible by 3?
Options: 267 | 266 | 265 | 264

Q1157 ? Cause vs correlation ? Easy

A serial code case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Which option is divisible by 4?
Options: 271 | 270 | 269 | 268

Q1158 ? Grammar agreement ? Medium

A digit audit case uses the quantities in this question. Determine the requested result using the grammar agreement model: Which option is divisible by 5?
Options: 270 | 271 | 272 | 273

Q1159 ? Vocabulary in context ? Hard

A packet routing case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Which option is divisible by 9?
Options: 270 | 271 | 272 | 273

Q1160 ? Mixed placement challenge ? Placement

A number lock case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Which option is divisible by 3?
Options: 282 | 281 | 280 | 279

Simplification ? 40 questions

WORKED EXAMPLES

Q1161 ? Direct computation ? Beginner

A spreadsheet formula case uses the quantities in this question. Determine the requested result using the direct computation model: Simplify $11 + 4 \times 3$.

Options: 27 | 21 | 25 | 23

Answer: 23

Solution: Context: spreadsheet formula. BODMAS: $4 \times 3 = 12$; then $11 + 12 = 23$. Shortcut: Mark higher-priority operations.

Q1162 ? Reverse value ? Easy

A billing expression case uses the quantities in this question. Determine the requested result using the reverse value model: Simplify $12 + 5 \times 4$.

Options: 38 | 29 | 35 | 32

Answer: 32

Solution: Context: billing expression. BODMAS: $5 \times 4 = 20$; then $12 + 20 = 32$. Shortcut: Mark higher-priority operations.

Q1163 ? Missing base ? Medium

A unit-cost calculation case uses the quantities in this question. Determine the requested result using the missing base model: Simplify $13 + 6 \times 5$.

Options: 43 | 47 | 39 | 51

Answer: 43

Solution: Context: unit-cost calculation. BODMAS: $6 \times 5 = 30$; then $13 + 30 = 43$. Shortcut: Mark higher-priority operations.

Q1164 ? Successive change ? Hard

A exam simplification case uses the quantities in this question. Determine the requested result using the successive change model: Simplify $14 + 7 \times 6$.

Options: 50 | 56 | 68 | 62

Answer: 56

Solution: Context: exam simplification. BODMAS: $7 \times 6 = 42$; then $14 + 42 = 56$. Shortcut: Mark higher-priority operations.

Q1165 ? Comparison trap ? Placement

A engineering estimate case uses the quantities in this question. Determine the requested result using the comparison trap model: Simplify $15 + 8 \times 2$.

Options: 37 | 28 | 34 | 31

Answer: 31

Solution: Context: engineering estimate. BODMAS: $8 \times 2 = 16$; then $15 + 16 = 31$. Shortcut: Mark higher-priority operations.

Q1166 ? Ratio form ? Beginner

A invoice arithmetic case uses the quantities in this question. Determine the requested result using the ratio form model: Simplify $16 + 9 \times 3$.

Options: 51 | 47 | 43 | 39

Answer: 43

Solution: Context: invoice arithmetic. BODMAS: $9 \times 3 = 27$; then $16 + 27 = 43$. Shortcut: Mark higher-priority operations.

Q1167 ? Weighted result ? Easy

A nested operation case uses the quantities in this question. Determine the requested result using the weighted result model: Simplify $17 + 3 \times 4$.

Options: 29 | 26 | 32 | 35

Answer: 29

Solution: Context: nested operation. BODMAS: $3 \times 4 = 12$; then $17 + 12 = 29$. Shortcut: Mark higher-priority operations.

Q1168 ? Replacement cycle ? Medium

A mental-math drill case uses the quantities in this question. Determine the requested result using the replacement cycle model: Simplify $18 + 4 \times 5$.

Options: 38 | 42 | 34 | 46

Answer: 38

Solution: Context: mental-math drill. BODMAS: $4 \times 5 = 20$; then $18 + 20 = 38$. Shortcut: Mark higher-priority operations.

Q1169 ? Unit conversion ? Hard

A spreadsheet formula case uses the quantities in this question. Determine the requested result using the unit conversion model: Simplify $19 + 5 \times 6$.

Options: 54 | 59 | 44 | 49

Answer: 49

Solution: Context: spreadsheet formula. BODMAS: $5 \times 6 = 30$; then $19 + 30 = 49$. Shortcut: Mark higher-priority operations.

Q1170 ? Work-rate model ? Placement

A billing expression case uses the quantities in this question. Determine the requested result using the work-rate model model: Simplify $20 + 6 \times 2$.

Options: 32 | 35 | 29 | 38

Answer: 32

Solution: Context: billing expression. BODMAS: $6 \times 2 = 12$; then $20 + 12 = 32$. Shortcut: Mark higher-priority operations.

Q1171 ? Speed-time model ? Beginner

A unit-cost calculation case uses the quantities in this question. Determine the requested result using the speed-time model model: Simplify $21 + 7 \times 3$.

Options: 38 | 46 | 42 | 50

Answer: 42

Solution: Context: unit-cost calculation. BODMAS: $7 \times 3 = 21$; then $21 + 21 = 42$. Shortcut: Mark higher-priority operations.

Q1172 ? Table lookup ? Easy

A exam simplification case uses the quantities in this question. Determine the requested result using the table lookup model: Simplify $22 + 8 \times 4$.

Options: 54 | 59 | 49 | 64

Answer: 54

Solution: Context: exam simplification. BODMAS: $8 \times 4 = 32$; then $22 + 32 = 54$. Shortcut: Mark higher-priority operations.

Q1173 ? Chart comparison ? Medium

A engineering estimate case uses the quantities in this question. Determine the requested result using the chart comparison model: Simplify $23 + 9 \times 5$.

Options: 68 | 75 | 61 | 82

Answer: 68

Solution: Context: engineering estimate. BODMAS: $9 \times 5 = 45$; then $23 + 45 = 68$. Shortcut: Mark higher-priority operations.

Q1174 ? Multi-step word problem ? Hard

A invoice arithmetic case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Simplify $24 + 3 \times 6$.

Options: 38 | 42 | 46 | 50

Answer: 42

Solution: Context: invoice arithmetic. BODMAS: $3 \times 6 = 18$; then $24 + 18 = 42$. Shortcut: Mark higher-priority operations.

Q1175 ? Maximum / minimum ? Placement

A nested operation case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Simplify $25 + 4 \times 2$.

Options: 33 | 36 | 30 | 39

Answer: 33

Solution: Context: nested operation. BODMAS: $4 \times 2 = 8$; then $25 + 8 = 33$. Shortcut: Mark higher-priority operations.

Q1176 ? Least / greatest value ? Beginner

A mental-math drill case uses the quantities in this question. Determine the requested result using the least / greatest value model: Simplify $26 + 5 \times 3$.

Options: 49 | 37 | 45 | 41

Answer: 41

Solution: Context: mental-math drill. BODMAS: $5 \times 3 = 15$; then $26 + 15 = 41$. Shortcut: Mark higher-priority operations.

Q1177 ? Remainder cycle ? Easy

A spreadsheet formula case uses the quantities in this question. Determine the requested result using the remainder cycle model: Simplify $27 + 6 \times 4$.

Options: 61 | 51 | 56 | 46

Answer: 51

Solution: Context: spreadsheet formula. BODMAS: $6 \times 4 = 24$; then $27 + 24 = 51$. Shortcut: Mark higher-priority operations.

Q1178 ? Divisibility test ? Medium

A billing expression case uses the quantities in this question. Determine the requested result using the divisibility test model: Simplify $28 + 7 \times 5$.

Options: 57 | 69 | 63 | 75

Answer: 63

Solution: Context: billing expression. BODMAS: $7 \times 5 = 35$; then $28 + 35 = 63$. Shortcut: Mark higher-priority operations.

Q1179 ? Factorisation ? Hard

A unit-cost calculation case uses the quantities in this question. Determine the requested result using the factorisation model: Simplify $29 + 8 \times 6$.

Options: 93 | 85 | 77 | 69

Answer: 77

Solution: Context: unit-cost calculation. BODMAS: $8 \times 6 = 48$; then $29 + 48 = 77$. Shortcut: Mark higher-priority operations.

Q1180 ? Approximation ? Placement

A exam simplification case uses the quantities in this question. Determine the requested result using the approximation model: Simplify $30 + 9 \times 2$.

Options: 58 | 43 | 53 | 48

Answer: 48

Solution: Context: exam simplification. BODMAS: $9 \times 2 = 18$; then $30 + 18 = 48$. Shortcut: Mark higher-priority operations.

PRACTICE ? solutions hidden below

Q1181 ? Options elimination ? Beginner

A engineering estimate case uses the quantities in this question. Determine the requested result using the options elimination model: Simplify $31 + 3 \times 3$.

Options: 36 | 40 | 44 | 48

Q1182 ? Boundary case ? Easy

A invoice arithmetic case uses the quantities in this question. Determine the requested result using the boundary case model: Simplify $32 + 4 \times 4$.

Options: 58 | 53 | 48 | 43

Q1183 ? Case split ? Medium

A nested operation case uses the quantities in this question. Determine the requested result using the case split model: Simplify $33 + 5 \times 5$.

Options: 58 | 64 | 52 | 70

Q1184 ? Contradiction check ? Hard

A mental-math drill case uses the quantities in this question. Determine the requested result using the contradiction check model: Simplify $34 + 6 \times 6$.

Options: 70 | 77 | 84 | 63

Q1185 ? Statement I test ? Placement

A spreadsheet formula case uses the quantities in this question. Determine the requested result using the statement i test model: Simplify $35 + 7 \times 2$.

Options: 44 | 49 | 59 | 54

Q1186 ? Statement II test ? Beginner

A billing expression case uses the quantities in this question. Determine the requested result using the statement ii test model: Simplify $36 + 8 \times 3$.

Options: 72 | 60 | 66 | 54

Q1187 ? Combined statements ? Easy

A unit-cost calculation case uses the quantities in this question. Determine the requested result using the combined statements model: Simplify $37 + 9 \times 4$.

Options: 73 | 80 | 66 | 87

Q1188 ? Counting restriction ? Medium

A exam simplification case uses the quantities in this question. Determine the requested result using the counting restriction model: Simplify $38 + 3 \times 5$.

Options: 48 | 53 | 58 | 63

Q1189 ? Circular arrangement ? Hard

A engineering estimate case uses the quantities in this question. Determine the requested result using the circular arrangement model: Simplify $39 + 4 \times 6$.

Options: 57 | 75 | 69 | 63

Q1190 ? Complement method ? Placement

A invoice arithmetic case uses the quantities in this question. Determine the requested result using the complement method model: Simplify $40 + 5 \times 2$.

Options: 50 | 55 | 60 | 45

Q1191 ? Pattern: differences ? Beginner

A nested operation case uses the quantities in this question. Determine the requested result using the pattern: differences model: Simplify $41 + 6 \times 3$.

Options: 71 | 65 | 59 | 53

Q1192 ? Pattern: ratios ? Easy

A mental-math drill case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Simplify $42 + 7 \times 4$.

Options: 63 | 77 | 70 | 84

Q1193 ? Pattern: alternating ? Medium

A spreadsheet formula case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Simplify $43 + 8 \times 5$.

Options: 99 | 75 | 91 | 83

Q1194 ? Pattern: squares / cubes ? Hard

A billing expression case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Simplify $44 + 9 \times 6$.

Options: 98 | 108 | 88 | 118

Q1195 ? Inference from facts ? Placement

A unit-cost calculation case uses the quantities in this question. Determine the requested result using the inference from facts model: Simplify $45 + 3 \times 2$.

Options: 46 | 51 | 56 | 61

Q1196 ? Assumption check ? Beginner

A exam simplification case uses the quantities in this question. Determine the requested result using the assumption check model: Simplify $46 + 4 \times 3$.

Options: 52 | 70 | 64 | 58

Q1197 ? Cause vs correlation ? Easy

A engineering estimate case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Simplify $47 + 5 \times 4$.

Options: 67 | 74 | 60 | 81

Q1198 ? Grammar agreement ? Medium

A invoice arithmetic case uses the quantities in this question. Determine the requested result using the grammar agreement model: Simplify $48 + 6 \times 5$.

Options: 78 | 86 | 70 | 94

Q1199 ? Vocabulary in context ? Hard

A nested operation case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Simplify $49 + 7 \times 6$.

Options: 91 | 109 | 82 | 100

Q1200 ? Mixed placement challenge ? Placement

A mental-math drill case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Simplify $50 + 8 \times 2$.

Options: 80 | 73 | 66 | 59

Surds & Indices ? 40 questions

WORKED EXAMPLES

Q1201 ? Direct computation ? Beginner

A square-root measurement case uses the quantities in this question. Determine the requested result using the direct computation model: Simplify $\sqrt{9}$.

Options: 3 | 4 | 9 | 2

Answer: 3

Solution: Context: square-root measurement. $\sqrt{9}=3$, the principal square root. Shortcut: Memorise common squares.

Q1202 ? Reverse value ? Easy

A scaling law case uses the quantities in this question. Determine the requested result using the reverse value model: Simplify $\sqrt{16}$.

Options: 5 | 3 | 4 | 16

Answer: 4

Solution: Context: scaling law. $\sqrt{16}=4$, the principal square root. Shortcut: Memorise common squares.

Q1203 ? Missing base ? Medium

A scientific notation case uses the quantities in this question. Determine the requested result using the missing base model: Simplify $\sqrt{25}$.

Options: 25 | 4 | 6 | 5

Answer: 5

Solution: Context: scientific notation. $\sqrt{25}=5$, the principal square root. Shortcut: Memorise common squares.

Q1204 ? Successive change ? Hard

A geometric growth case uses the quantities in this question. Determine the requested result using the successive change model: Simplify $\sqrt{36}$.

Options: 36 | 5 | 6 | 7

Answer: 6

Solution: Context: geometric growth. $\sqrt{36}=6$, the principal square root. Shortcut: Memorise common squares.

Q1205 ? Comparison trap ? Placement

A signal amplitude case uses the quantities in this question. Determine the requested result using the comparison trap model: Simplify $\sqrt{49}$.

Options: 8 | 7 | 6 | 49

Answer: 7

Solution: Context: signal amplitude. $\sqrt{49}=7$, the principal square root. Shortcut: Memorise common squares.

Q1206 ? Ratio form ? Beginner

A area side-length case uses the quantities in this question. Determine the requested result using the ratio form model: Simplify $\sqrt{64}$.

Options: 8 | 9 | 7 | 64

Answer: 8

Solution: Context: area side-length. $\sqrt{64}=8$, the principal square root. Shortcut: Memorise common squares.

Q1207 ? Weighted result ? Easy

A index comparison case uses the quantities in this question. Determine the requested result using the weighted result model: Simplify $\sqrt{81}$.

Options: 10 | 8 | 81 | 9

Answer: 9

Solution: Context: index comparison. $\sqrt{81}=9$, the principal square root. Shortcut: Memorise common squares.

Q1208 ? Replacement cycle ? Medium

A radical simplification case uses the quantities in this question. Determine the requested result using the replacement cycle model: Simplify $\sqrt{100}$.

Options: 100 | 9 | 11 | 10

Answer: 10

Solution: Context: radical simplification. $\sqrt{100}=10$, the principal square root. Shortcut: Memorise common squares.

Q1209 ? Unit conversion ? Hard

A square-root measurement case uses the quantities in this question. Determine the requested result using the unit conversion model: Simplify $\sqrt{121}$.

Options: 11 | 12 | 10 | 121

Answer: 11

Solution: Context: square-root measurement. $\sqrt{121}=11$, the principal square root. Shortcut: Memorise common squares.

Q1210 ? Work-rate model ? Placement

A scaling law case uses the quantities in this question. Determine the requested result using the work-rate model model: Simplify $\sqrt{144}$.

Options: 12 | 13 | 11 | 144

Answer: 12

Solution: Context: scaling law. $\sqrt{144}=12$, the principal square root. Shortcut: Memorise common squares.

Q1211 ? Speed-time model ? Beginner

A scientific notation case uses the quantities in this question. Determine the requested result using the speed-time model model: Simplify $\sqrt{169}$.

Options: 169 | 12 | 14 | 13

Answer: 13

Solution: Context: scientific notation. $\sqrt{169}=13$, the principal square root. Shortcut: Memorise common squares.

Q1212 ? Table lookup ? Easy

A geometric growth case uses the quantities in this question. Determine the requested result using the table lookup model: Simplify $\sqrt{196}$.

Options: 196 | 13 | 15 | 14

Answer: 14

Solution: Context: geometric growth. $\sqrt{196}=14$, the principal square root. Shortcut: Memorise common squares.

Q1213 ? Chart comparison ? Medium

A signal amplitude case uses the quantities in this question. Determine the requested result using the chart comparison model: Simplify $\sqrt{225}$.

Options: 15 | 16 | 14 | 225

Answer: 15

Solution: Context: signal amplitude. $\sqrt{225}=15$, the principal square root. Shortcut: Memorise common squares.

Q1214 ? Multi-step word problem ? Hard

A area side-length case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Simplify $\sqrt{4}$.

Options: 1 | 2 | 3 | 4

Answer: 2

Solution: Context: area side-length. $\sqrt{4}=2$, the principal square root. Shortcut: Memorise common squares.

Q1215 ? Maximum / minimum ? Placement

A index comparison case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Simplify $\sqrt{9}$.

Options: 3 | 4 | 9 | 2

Answer: 3

Solution: Context: index comparison. $\sqrt{9}=3$, the principal square root. Shortcut: Memorise common squares.

Q1216 ? Least / greatest value ? Beginner

A radical simplification case uses the quantities in this question. Determine the requested result using the least / greatest value model: Simplify $\sqrt{16}$.

Options: 5 | 3 | 4 | 16

Answer: 4

Solution: Context: radical simplification. $\sqrt{16}=4$, the principal square root. Shortcut: Memorise common squares.

Q1217 ? Remainder cycle ? Easy

A square-root measurement case uses the quantities in this question. Determine the requested result using the remainder cycle model: Simplify $\sqrt{25}$.

Options: 25 | 4 | 6 | 5

Answer: 5

Solution: Context: square-root measurement. $\sqrt{25}=5$, the principal square root. Shortcut: Memorise common squares.

Q1218 ? Divisibility test ? Medium

A scaling law case uses the quantities in this question. Determine the requested result using the divisibility test model: Simplify $\sqrt{36}$.

Options: 36 | 5 | 6 | 7

Answer: 6

Solution: Context: scaling law. $\sqrt{36}=6$, the principal square root. Shortcut: Memorise common squares.

Q1219 ? Factorisation ? Hard

A scientific notation case uses the quantities in this question. Determine the requested result using the factorisation model: Simplify $\sqrt{49}$.

Options: 8 | 7 | 6 | 49

Answer: 7

Solution: Context: scientific notation. $\sqrt{49}=7$, the principal square root. Shortcut: Memorise common squares.

Q1220 ? Approximation ? Placement

A geometric growth case uses the quantities in this question. Determine the requested result using the approximation model: Simplify $\sqrt{64}$.

Options: 8 | 9 | 7 | 64

Answer: 8

Solution: Context: geometric growth. $\sqrt{64}=8$, the principal square root. Shortcut: Memorise common squares.

PRACTICE ? solutions hidden below

Q1221 ? Options elimination ? Beginner

A signal amplitude case uses the quantities in this question. Determine the requested result using the options elimination model: Simplify $\sqrt{81}$.

Options: 10 | 8 | 81 | 9

Q1222 ? Boundary case ? Easy

A area side-length case uses the quantities in this question. Determine the requested result using the boundary case model: Simplify $\sqrt{100}$.

Options: 100 | 9 | 11 | 10

Q1223 ? Case split ? Medium

A index comparison case uses the quantities in this question. Determine the requested result using the case split model: Simplify $\sqrt{121}$.

Options: 11 | 12 | 10 | 121

Q1224 ? Contradiction check ? Hard

A radical simplification case uses the quantities in this question. Determine the requested result using the contradiction check model: Simplify $\sqrt{144}$.

Options: 12 | 13 | 11 | 144

Q1225 ? Statement I test ? Placement

A square-root measurement case uses the quantities in this question. Determine the requested result using the statement i test model: Simplify $\sqrt{169}$.

Options: 169 | 12 | 14 | 13

Q1226 ? Statement II test ? Beginner

A scaling law case uses the quantities in this question. Determine the requested result using the statement ii test model: Simplify $\sqrt{196}$.

Options: 196 | 13 | 15 | 14

Q1227 ? Combined statements ? Easy

A scientific notation case uses the quantities in this question. Determine the requested result using the combined statements model: Simplify $\sqrt{225}$.

Options: 15 | 16 | 14 | 225

Q1228 ? Counting restriction ? Medium

A geometric growth case uses the quantities in this question. Determine the requested result using the counting restriction model: Simplify $\sqrt{4}$.

Options: 1 | 2 | 3 | 4

Q1229 ? Circular arrangement ? Hard

A signal amplitude case uses the quantities in this question. Determine the requested result using the circular arrangement model: Simplify $\sqrt{9}$.

Options: 3 | 4 | 9 | 2

Q1230 ? Complement method ? Placement

A area side-length case uses the quantities in this question. Determine the requested result using the complement method model: Simplify $\sqrt{16}$.

Options: 5 | 3 | 4 | 16

Q1231 ? Pattern: differences ? Beginner

A index comparison case uses the quantities in this question. Determine the requested result using the pattern: differences model: Simplify $\sqrt{25}$.

Options: 25 | 4 | 6 | 5

Q1232 ? Pattern: ratios ? Easy

A radical simplification case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Simplify $\sqrt{36}$.

Options: 36 | 5 | 6 | 7

Q1233 ? Pattern: alternating ? Medium

A square-root measurement case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Simplify $\sqrt{49}$.

Options: 8 | 7 | 6 | 49

Q1234 ? Pattern: squares / cubes ? Hard

A scaling law case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Simplify $\sqrt{64}$.

Options: 8 | 9 | 7 | 64

Q1235 ? Inference from facts ? Placement

A scientific notation case uses the quantities in this question. Determine the requested result using the inference from facts model: Simplify $\sqrt{81}$.

Options: 10 | 8 | 81 | 9

Q1236 ? Assumption check ? Beginner

A geometric growth case uses the quantities in this question. Determine the requested result using the assumption check model: Simplify $\sqrt{100}$.

Options: 100 | 9 | 11 | 10

Q1237 ? Cause vs correlation ? Easy

A signal amplitude case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Simplify $\sqrt{121}$.

Options: 11 | 12 | 10 | 121

Q1238 ? Grammar agreement ? Medium

A area side-length case uses the quantities in this question. Determine the requested result using the grammar agreement model: Simplify $\sqrt{144}$.

Options: 12 | 13 | 11 | 144

Q1239 ? Vocabulary in context ? Hard

A index comparison case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Simplify $\sqrt{169}$.

Options: 169 | 12 | 14 | 13

Q1240 ? Mixed placement challenge ? Placement

A radical simplification case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Simplify $\sqrt{196}$.

Options: 196 | 13 | 15 | 14

Logarithms ? 40 questions

WORKED EXAMPLES

Q1241 ? Direct computation ? Beginner

A decibel scale case uses the quantities in this question. Determine the requested result using the direct computation model: Evaluate log base 3 of 27.

Options: 3 | 5 | 4 | 2

Answer: 3

Solution: Context: decibel scale. $3^3=27$, hence $\log_3(27)=3$. Shortcut: Ask "base to what power?"

Q1242 ? Reverse value ? Easy

A pH reading case uses the quantities in this question. Determine the requested result using the reverse value model: Evaluate log base 4 of 256.

Options: 5 | 6 | 3 | 4

Answer: 4

Solution: Context: pH reading. $4^4=256$, hence $\log_4(256)=4$. Shortcut: Ask "base to what power?"

Q1243 ? Missing base ? Medium

A compound growth case uses the quantities in this question. Determine the requested result using the missing base model: Evaluate log base 5 of 3125.

Options: 7 | 4 | 6 | 5

Answer: 5

Solution: Context: compound growth. $5^5=3125$, hence $\log_5(3125)=5$. Shortcut: Ask "base to what power?"

Q1244 ? Successive change ? Hard

A algorithm complexity case uses the quantities in this question. Determine the requested result using the successive change model: Evaluate log base 2 of 4.

Options: 2 | 3 | 4 | 1

Answer: 2

Solution: Context: algorithm complexity. $2^2=4$, hence $\log_2(4)=2$. Shortcut: Ask "base to what power?"

Q1245 ? Comparison trap ? Placement

A earthquake magnitude case uses the quantities in this question. Determine the requested result using the comparison trap model: Evaluate log base 3 of 27.

Options: 4 | 2 | 5 | 3

Answer: 3

Solution: Context: earthquake magnitude. $3^3=27$, hence $\log_3(27)=3$. Shortcut: Ask "base to what power?"

Q1246 ? Ratio form ? Beginner

A financial log-return case uses the quantities in this question. Determine the requested result using the ratio form model: Evaluate log base 4 of 256.

Options: 6 | 5 | 4 | 3

Answer: 4

Solution: Context: financial log-return. $4^4=256$, hence $\log_4(256)=4$. Shortcut: Ask "base to what power?"

Q1247 ? Weighted result ? Easy

A population model case uses the quantities in this question. Determine the requested result using the weighted result model: Evaluate log base 5 of 3125.

Options: 4 | 5 | 6 | 7

Answer: 5

Solution: Context: population model. $5^5=3125$, hence $\log_5(3125)=5$. Shortcut: Ask "base to what power?"

Q1248 ? Replacement cycle ? Medium

A base conversion case uses the quantities in this question. Determine the requested result using the replacement cycle model: Evaluate log base 2 of 4.

Options: 3 | 1 | 2 | 4

Answer: 2

Solution: Context: base conversion. $2^2=4$, hence $\log_2(4)=2$. Shortcut: Ask "base to what power?"

Q1249 ? Unit conversion ? Hard

A decibel scale case uses the quantities in this question. Determine the requested result using the unit conversion model: Evaluate log base 3 of 27.

Options: 5 | 2 | 4 | 3

Answer: 3

Solution: Context: decibel scale. $3^3=27$, hence $\log_3(27)=3$. Shortcut: Ask "base to what power?"

Q1250 ? Work-rate model ? Placement

A pH reading case uses the quantities in this question. Determine the requested result using the work-rate model model: Evaluate log base 4 of 256.

Options: 6 | 4 | 3 | 5

Answer: 4

Solution: Context: pH reading. $4^4=256$, hence $\log_4(256)=4$. Shortcut: Ask "base to what power?"

Q1251 ? Speed-time model ? Beginner

A compound growth case uses the quantities in this question. Determine the requested result using the speed-time model model: Evaluate log base 5 of 3125.

Options: 4 | 5 | 6 | 7

Answer: 5

Solution: Context: compound growth. $5^5=3125$, hence $\log_5(3125)=5$. Shortcut: Ask "base to what power?"

Q1252 ? Table lookup ? Easy

A algorithm complexity case uses the quantities in this question. Determine the requested result using the table lookup model: Evaluate log base 2 of 4.

Options: 1 | 4 | 3 | 2

Answer: 2

Solution: Context: algorithm complexity. $2^2=4$, hence $\log_2(4)=2$. Shortcut: Ask "base to what power?"

Q1253 ? Chart comparison ? Medium

A earthquake magnitude case uses the quantities in this question. Determine the requested result using the chart comparison model: Evaluate log base 3 of 27.

Options: 5 | 4 | 3 | 2

Answer: 3

Solution: Context: earthquake magnitude. $3^3=27$, hence $\log_3(27)=3$. Shortcut: Ask "base to what power?"

Q1254 ? Multi-step word problem ? Hard

A financial log-return case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Evaluate log base 4 of 256.

Options: 3 | 4 | 5 | 6

Answer: 4

Solution: Context: financial log-return. $4^4=256$, hence $\log_4(256)=4$. Shortcut: Ask "base to what power?"

Q1255 ? Maximum / minimum ? Placement

A population model case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Evaluate log base 5 of 3125.

Options: 6 | 5 | 4 | 7

Answer: 5

Solution: Context: population model. $5^5=3125$, hence $\log_5(3125)=5$. Shortcut: Ask "base to what power?"

Q1256 ? Least / greatest value ? Beginner

A base conversion case uses the quantities in this question. Determine the requested result using the least / greatest value model: Evaluate log base 2 of 4.

Options: 4 | 3 | 2 | 1

Answer: 2

Solution: Context: base conversion. $2^2=4$, hence $\log_2(4)=2$. Shortcut: Ask "base to what power?"

Q1257 ? Remainder cycle ? Easy

A decibel scale case uses the quantities in this question. Determine the requested result using the remainder cycle model: Evaluate log base 3 of 27.

Options: 3 | 2 | 4 | 5

Answer: 3

Solution: Context: decibel scale. $3^3=27$, hence $\log_3(27)=3$. Shortcut: Ask "base to what power?"

Q1258 ? Divisibility test ? Medium

A pH reading case uses the quantities in this question. Determine the requested result using the divisibility test model: Evaluate log base 4 of 256.

Options: 3 | 4 | 5 | 6

Answer: 4

Solution: Context: pH reading. $4^4=256$, hence $\log_4(256)=4$. Shortcut: Ask "base to what power?"

Q1259 ? Factorisation ? Hard

A compound growth case uses the quantities in this question. Determine the requested result using the factorisation model: Evaluate log base 5 of 3125.

Options: 5 | 6 | 4 | 7

Answer: 5

Solution: Context: compound growth. $5^5=3125$, hence $\log_5(3125)=5$. Shortcut: Ask "base to what power?"

Q1260 ? Approximation ? Placement

A algorithm complexity case uses the quantities in this question. Determine the requested result using the approximation model: Evaluate log base 2 of 4.

Options: 4 | 2 | 3 | 1

Answer: 2

Solution: Context: algorithm complexity. $2^2=4$, hence $\log_2(4)=2$. Shortcut: Ask "base to what power?"

PRACTICE ? solutions hidden below

Q1261 ? Options elimination ? Beginner

A earthquake magnitude case uses the quantities in this question. Determine the requested result using the options elimination model: Evaluate log base 3 of 27.

Options: 2 | 3 | 4 | 5

Q1262 ? Boundary case ? Easy

A financial log-return case uses the quantities in this question. Determine the requested result using the boundary case model: Evaluate log base 4 of 256.

Options: 5 | 4 | 3 | 6

Q1263 ? Case split ? Medium

A population model case uses the quantities in this question. Determine the requested result using the case split model: Evaluate log base 5 of 3125.

Options: 5 | 6 | 7 | 4

Q1264 ? Contradiction check ? Hard

A base conversion case uses the quantities in this question. Determine the requested result using the contradiction check model: Evaluate log base 2 of 4.

Options: 2 | 1 | 3 | 4

Q1265 ? Statement I test ? Placement

A decibel scale case uses the quantities in this question. Determine the requested result using the statement I test model: Evaluate log base 3 of 27.

Options: 3 | 4 | 2 | 5

Q1266 ? Statement II test ? Beginner

A pH reading case uses the quantities in this question. Determine the requested result using the statement ii test model: Evaluate log base 4 of 256.

Options: 4 | 5 | 3 | 6

Q1267 ? Combined statements ? Easy

A compound growth case uses the quantities in this question. Determine the requested result using the combined statements model: Evaluate log base 5 of 3125.

Options: 6 | 7 | 4 | 5

Q1268 ? Counting restriction ? Medium

A algorithm complexity case uses the quantities in this question. Determine the requested result using the counting restriction model: Evaluate log base 2 of 4.

Options: 1 | 2 | 3 | 4

Q1269 ? Circular arrangement ? Hard

A earthquake magnitude case uses the quantities in this question. Determine the requested result using the circular arrangement model: Evaluate log base 3 of 27.

Options: 3 | 5 | 4 | 2

Q1270 ? Complement method ? Placement

A financial log-return case uses the quantities in this question. Determine the requested result using the complement method model: Evaluate log base 4 of 256.

Options: 5 | 6 | 3 | 4

Q1271 ? Pattern: differences ? Beginner

A population model case uses the quantities in this question. Determine the requested result using the pattern: differences model: Evaluate log base 5 of 3125.

Options: 7 | 4 | 6 | 5

Q1272 ? Pattern: ratios ? Easy

A base conversion case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Evaluate log base 2 of 4.

Options: 2 | 3 | 4 | 1

Q1273 ? Pattern: alternating ? Medium

A decibel scale case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Evaluate log base 3 of 27.

Options: 4 | 2 | 5 | 3

Q1274 ? Pattern: squares / cubes ? Hard

A pH reading case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Evaluate log base 4 of 256.

Options: 6 | 5 | 4 | 3

Q1275 ? Inference from facts ? Placement

A compound growth case uses the quantities in this question. Determine the requested result using the inference from facts model: Evaluate log base 5 of 3125.

Options: 4 | 5 | 6 | 7

Q1276 ? Assumption check ? Beginner

A algorithm complexity case uses the quantities in this question. Determine the requested result using the assumption check model: Evaluate log base 2 of 4.

Options: 3 | 1 | 2 | 4

Q1277 ? Cause vs correlation ? Easy

A earthquake magnitude case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Evaluate log base 3 of 27.

Options: 5 | 2 | 4 | 3

Q1278 ? Grammar agreement ? Medium

A financial log-return case uses the quantities in this question. Determine the requested result using the grammar agreement model: Evaluate log base 4 of 256.

Options: 6 | 4 | 3 | 5

Q1279 ? Vocabulary in context ? Hard

A population model case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Evaluate log base 5 of 3125.

Options: 4 | 5 | 6 | 7

Q1280 ? Mixed placement challenge ? Placement

A base conversion case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Evaluate log base 2 of 4.

Options: 1 | 4 | 3 | 2

Algebra ? 40 questions

WORKED EXAMPLES

Q1281 ? Direct computation ? Beginner

A unknown price case uses the quantities in this question. Determine the requested result using the direct computation model: Solve $3x + 6 = 15$.

Options: 3 | 5 | 4 | 2

Answer: 3

Solution: Context: unknown price. $3x=9$; $x=9/3=3$. Shortcut: Undo addition, then multiplication.

Q1282 ? Reverse value ? Easy

A work-rate equation case uses the quantities in this question. Determine the requested result using the reverse value model: Solve $4x + 7 = 23$.

Options: 5 | 6 | 3 | 4

Answer: 4

Solution: Context: work-rate equation. $4x=16$; $x=16/4=4$. Shortcut: Undo addition, then multiplication.

Q1283 ? Missing base ? Medium

A age variable case uses the quantities in this question. Determine the requested result using the missing base model: Solve $5x + 8 = 33$.

Options: 7 | 4 | 6 | 5

Answer: 5

Solution: Context: age variable. $5x=25$; $x=25/5=5$. Shortcut: Undo addition, then multiplication.

Q1284 ? Successive change ? Hard

A ticket equation case uses the quantities in this question. Determine the requested result using the successive change model: Solve $6x + 9 = 45$.

Options: 5 | 6 | 7 | 8

Answer: 6

Solution: Context: ticket equation. $6x=36$; $x=36/6=6$. Shortcut: Undo addition, then multiplication.

Q1285 ? Comparison trap ? Placement

A inventory balance case uses the quantities in this question. Determine the requested result using the comparison trap model: Solve $2x + 10 = 24$.

Options: 9 | 8 | 7 | 6

Answer: 7

Solution: Context: inventory balance. $2x=14$; $x=14/2=7$. Shortcut: Undo addition, then multiplication.

Q1286 ? Ratio form ? Beginner

A coordinate relation case uses the quantities in this question. Determine the requested result using the ratio form model: Solve $3x + 11 = 35$.

Options: 8 | 9 | 7 | 10

Answer: 8

Solution: Context: coordinate relation. $3x=24$; $x=24/3=8$. Shortcut: Undo addition, then multiplication.

Q1287 ? Weighted result ? Easy

A production target case uses the quantities in this question. Determine the requested result using the weighted result model: Solve $4x + 12 = 48$.

Options: 10 | 8 | 11 | 9

Answer: 9

Solution: Context: production target. $4x=36$; $x=36/4=9$. Shortcut: Undo addition, then multiplication.

Q1288 ? Replacement cycle ? Medium

A linear forecast case uses the quantities in this question. Determine the requested result using the replacement cycle model: Solve $5x + 5 = 55$.

Options: 12 | 9 | 11 | 10

Answer: 10

Solution: Context: linear forecast. $5x=50$; $x=50/5=10$. Shortcut: Undo addition, then multiplication.

Q1289 ? Unit conversion ? Hard

A unknown price case uses the quantities in this question. Determine the requested result using the unit conversion model: Solve $6x + 6 = 72$.

Options: 11 | 12 | 10 | 13

Answer: 11

Solution: Context: unknown price. $6x=66$; $x=66/6=11$. Shortcut: Undo addition, then multiplication.

Q1290 ? Work-rate model ? Placement

A work-rate equation case uses the quantities in this question. Determine the requested result using the work-rate model model: Solve $2x + 7 = 31$.

Options: 12 | 13 | 11 | 14

Answer: 12

Solution: Context: work-rate equation. $2x=24$; $x=24/2=12$. Shortcut: Undo addition, then multiplication.

Q1291 ? Speed-time model ? Beginner

A age variable case uses the quantities in this question. Determine the requested result using the speed-time model model: Solve $3x + 8 = 47$.

Options: 15 | 12 | 14 | 13

Answer: 13

Solution: Context: age variable. $3x=39$; $x=39/3=13$. Shortcut: Undo addition, then multiplication.

Q1292 ? Table lookup ? Easy

A ticket equation case uses the quantities in this question. Determine the requested result using the table lookup model: Solve $4x + 9 = 65$.

Options: 16 | 13 | 15 | 14

Answer: 14

Solution: Context: ticket equation. $4x=56$; $x=56/4=14$. Shortcut: Undo addition, then multiplication.

Q1293 ? Chart comparison ? Medium

A inventory balance case uses the quantities in this question. Determine the requested result using the chart comparison model: Solve $5x + 10 = 85$.

Options: 15 | 17 | 13 | 19

Answer: 15

Solution: Context: inventory balance. $5x=75$; $x=75/5=15$. Shortcut: Undo addition, then multiplication.

Q1294 ? Multi-step word problem ? Hard

A coordinate relation case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Solve $6x + 11 = 107$.

Options: 16 | 18 | 14 | 20

Answer: 16

Solution: Context: coordinate relation. $6x=96$; $x=96/6=16$. Shortcut: Undo addition, then multiplication.

Q1295 ? Maximum / minimum ? Placement

A production target case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Solve $2x + 12 = 46$.

Options: 21 | 15 | 19 | 17

Answer: 17

Solution: Context: production target. $2x=34$; $x=34/2=17$. Shortcut: Undo addition, then multiplication.

Q1296 ? Least / greatest value ? Beginner

A linear forecast case uses the quantities in this question. Determine the requested result using the least / greatest value model: Solve $3x + 5 = 59$.

Options: 22 | 20 | 18 | 16

Answer: 18

Solution: Context: linear forecast. $3x=54$; $x=54/3=18$. Shortcut: Undo addition, then multiplication.

Q1297 ? Remainder cycle ? Easy

A unknown price case uses the quantities in this question. Determine the requested result using the remainder cycle model: Solve $4x + 6 = 82$.

Options: 19 | 21 | 17 | 23

Answer: 19

Solution: Context: unknown price. $4x=76$; $x=76/4=19$. Shortcut: Undo addition, then multiplication.

Q1298 ? Divisibility test ? Medium

A work-rate equation case uses the quantities in this question. Determine the requested result using the divisibility test model: Solve $5x + 7 = 107$.

Options: 20 | 22 | 18 | 24

Answer: 20

Solution: Context: work-rate equation. $5x=100$; $x=100/5=20$. Shortcut: Undo addition, then multiplication.

Q1299 ? Factorisation ? Hard

A age variable case uses the quantities in this question. Determine the requested result using the factorisation model: Solve $6x + 8 = 134$.

Options: 25 | 19 | 23 | 21

Answer: 21

Solution: Context: age variable. $6x=126$; $x=126/6=21$. Shortcut: Undo addition, then multiplication.

Q1300 ? Approximation ? Placement

A ticket equation case uses the quantities in this question. Determine the requested result using the approximation model: Solve $2x + 9 = 53$.

Options: 22 | 24 | 20 | 26

Answer: 22

Solution: Context: ticket equation. $2x=44$; $x=44/2=22$. Shortcut: Undo addition, then multiplication.

PRACTICE ? solutions hidden below

Q1301 ? Options elimination ? Beginner

A inventory balance case uses the quantities in this question. Determine the requested result using the options elimination model: Solve $3x + 10 = 79$.

Options: 23 | 25 | 21 | 27

Q1302 ? Boundary case ? Easy

A coordinate relation case uses the quantities in this question. Determine the requested result using the boundary case model: Solve $4x + 11 = 107$.

Options: 28 | 22 | 26 | 24

Q1303 ? Case split ? Medium

A production target case uses the quantities in this question. Determine the requested result using the case split model: Solve $5x + 12 = 137$.

Options: 31 | 22 | 28 | 25

Q1304 ? Contradiction check ? Hard

A linear forecast case uses the quantities in this question. Determine the requested result using the contradiction check model: Solve $6x + 5 = 161$.

Options: 26 | 29 | 23 | 32

Q1305 ? Statement I test ? Placement

A unknown price case uses the quantities in this question. Determine the requested result using the statement i test model: Solve $2x + 6 = 60$.

Options: 27 | 30 | 24 | 33

Q1306 ? Statement II test ? Beginner

A work-rate equation case uses the quantities in this question. Determine the requested result using the statement ii test model: Solve $3x + 7 = 91$.

Options: 34 | 25 | 31 | 28

Q1307 ? Combined statements ? Easy

A age variable case uses the quantities in this question. Determine the requested result using the combined statements model: Solve $4x + 8 = 124$.

Options: 29 | 32 | 26 | 35

Q1308 ? Counting restriction ? Medium

A ticket equation case uses the quantities in this question. Determine the requested result using the counting restriction model: Solve $5x + 9 = 159$.

Options: 27 | 30 | 33 | 36

Q1309 ? Circular arrangement ? Hard

A inventory balance case uses the quantities in this question. Determine the requested result using the circular arrangement model: Solve $6x + 10 = 196$.

Options: 31 | 34 | 37 | 28

Q1310 ? Complement method ? Placement

A coordinate relation case uses the quantities in this question. Determine the requested result using the complement method model: Solve $2x + 11 = 75$.

Options: 38 | 29 | 35 | 32

Q1311 ? Pattern: differences ? Beginner

A production target case uses the quantities in this question. Determine the requested result using the pattern: differences model: Solve $3x + 12 = 111$.

Options: 33 | 36 | 30 | 39

Q1312 ? Pattern: ratios ? Easy

A linear forecast case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Solve $4x + 5 = 141$.

Options: 31 | 37 | 34 | 40

Q1313 ? Pattern: alternating ? Medium

A unknown price case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Solve $5x + 6 = 181$.

Options: 43 | 31 | 39 | 35

Q1314 ? Pattern: squares / cubes ? Hard

A work-rate equation case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Solve $6x + 7 = 223$.

Options: 44 | 40 | 36 | 32

Q1315 ? Inference from facts ? Placement

A age variable case uses the quantities in this question. Determine the requested result using the inference from facts model: Solve $2x + 8 = 82$.

Options: 37 | 33 | 41 | 45

Q1316 ? Assumption check ? Beginner

A ticket equation case uses the quantities in this question. Determine the requested result using the assumption check model: Solve $3x + 9 = 123$.

Options: 38 | 42 | 34 | 46

Q1317 ? Cause vs correlation ? Easy

A inventory balance case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Solve $4x + 10 = 166$.

Options: 47 | 35 | 43 | 39

Q1318 ? Grammar agreement ? Medium

A coordinate relation case uses the quantities in this question. Determine the requested result using the grammar agreement model: Solve $5x + 11 = 211$.

Options: 40 | 44 | 36 | 48

Q1319 ? Vocabulary in context ? Hard

A production target case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Solve $6x + 12 = 258$.

Options: 37 | 49 | 45 | 41

Q1320 ? Mixed placement challenge ? Placement

A linear forecast case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Solve $2x + 5 = 89$.

Options: 42 | 46 | 50 | 38

Geometry ? 40 questions

WORKED EXAMPLES

Q1321 ? Direct computation ? Beginner

A land survey case uses the quantities in this question. Determine the requested result using the direct computation model: A triangle has angles 31° and 61° . Find the third.

Options: 106 | 79 | 97 | 88

Answer: 88

Solution: Context: land survey. Third angle= $180-31-61=88^\circ$. Shortcut: Subtract their sum from 180° .

Q1322 ? Reverse value ? Easy

A triangular park case uses the quantities in this question. Determine the requested result using the reverse value model: A triangle has angles 32° and 62° . Find the third.

Options: 95 | 86 | 77 | 104

Answer: 86

Solution: Context: triangular park. Third angle= $180-32-62=86^\circ$. Shortcut: Subtract their sum from 180° .

Q1323 ? Missing base ? Medium

A angle of elevation case uses the quantities in this question. Determine the requested result using the missing base model: A triangle has angles 33° and 63° . Find the third.

Options: 84 | 76 | 92 | 100

Answer: 84

Solution: Context: angle of elevation. Third angle= $180-33-63=84^\circ$. Shortcut: Subtract their sum from 180° .

Q1324 ? Successive change ? Hard

A road junction case uses the quantities in this question. Determine the requested result using the successive change model: A triangle has angles 34° and 64° . Find the third.

Options: 82 | 90 | 74 | 98

Answer: 82

Solution: Context: road junction. Third angle= $180-34-64=82^\circ$. Shortcut: Subtract their sum from 180° .

Q1325 ? Comparison trap ? Placement

A polygon design case uses the quantities in this question. Determine the requested result using the comparison trap model: A triangle has angles 35° and 65° . Find the third.

Options: 96 | 72 | 88 | 80

Answer: 80

Solution: Context: polygon design. Third angle= $180-35-65=80^\circ$. Shortcut: Subtract their sum from 180° .

Q1326 ? Ratio form ? Beginner

A coordinate map case uses the quantities in this question. Determine the requested result using the ratio form model: A triangle has angles 36° and 66° . Find the third.

Options: 78 | 86 | 70 | 94

Answer: 78

Solution: Context: coordinate map. Third angle= $180-36-66=78^\circ$. Shortcut: Subtract their sum from 180° .

Q1327 ? Weighted result ? Easy

A architectural plan case uses the quantities in this question. Determine the requested result using the weighted result model: A triangle has angles 37° and 67° . Find the third.

Options: 84 | 92 | 68 | 76

Answer: 76

Solution: Context: architectural plan. Third angle= $180-37-67=76^\circ$. Shortcut: Subtract their sum from 180° .

Q1328 ? Replacement cycle ? Medium

A angle chase case uses the quantities in this question. Determine the requested result using the replacement cycle model: A triangle has angles 38° and 68° . Find the third.

Options: 67 | 88 | 81 | 74

Answer: 74

Solution: Context: angle chase. Third angle= $180-38-68=74^\circ$. Shortcut: Subtract their sum from 180° .

Q1329 ? Unit conversion ? Hard

A land survey case uses the quantities in this question. Determine the requested result using the unit conversion model: A triangle has angles 39° and 69° . Find the third.

Options: 72 | 79 | 65 | 86

Answer: 72

Solution: Context: land survey. Third angle= $180-39-69=72^\circ$. Shortcut: Subtract their sum from 180° .

Q1330 ? Work-rate model ? Placement

A triangular park case uses the quantities in this question. Determine the requested result using the work-rate model model: A triangle has angles 40° and 70° . Find the third.

Options: 70 | 77 | 84 | 63

Answer: 70

Solution: Context: triangular park. Third angle= $180-40-70=70^\circ$. Shortcut: Subtract their sum from 180° .

Q1331 ? Speed-time model ? Beginner

A angle of elevation case uses the quantities in this question. Determine the requested result using the speed-time model model: A triangle has angles 41° and 71° . Find the third.

Options: 61 | 68 | 75 | 82

Answer: 68

Solution: Context: angle of elevation. Third angle= $180-41-71=68^\circ$. Shortcut: Subtract their sum from 180° .

Q1332 ? Table lookup ? Easy

A road junction case uses the quantities in this question. Determine the requested result using the table lookup model: A triangle has angles 42° and 72° . Find the third.

Options: 80 | 73 | 66 | 59

Answer: 66

Solution: Context: road junction. Third angle= $180-42-72=66^\circ$. Shortcut: Subtract their sum from 180° .

Q1333 ? Chart comparison ? Medium

A polygon design case uses the quantities in this question. Determine the requested result using the chart comparison model: A triangle has angles 43° and 73° . Find the third.

Options: 64 | 70 | 76 | 58

Answer: 64

Solution: Context: polygon design. Third angle= $180-43-73=64^\circ$. Shortcut: Subtract their sum from 180° .

Q1334 ? Multi-step word problem ? Hard

A coordinate map case uses the quantities in this question. Determine the requested result using the multi-step word problem model: A triangle has angles 44° and 74° . Find the third.

Options: 56 | 62 | 68 | 74

Answer: 62

Solution: Context: coordinate map. Third angle= $180-44-74=62^\circ$. Shortcut: Subtract their sum from 180° .

Q1335 ? Maximum / minimum ? Placement

A architectural plan case uses the quantities in this question. Determine the requested result using the maximum / minimum model: A triangle has angles 45° and 75° . Find the third.

Options: 54 | 72 | 66 | 60

Answer: 60

Solution: Context: architectural plan. Third angle= $180-45-75=60^\circ$. Shortcut: Subtract their sum from 180° .

Q1336 ? Least / greatest value ? Beginner

A angle chase case uses the quantities in this question. Determine the requested result using the least / greatest value model: A triangle has angles 46° and 76° . Find the third.

Options: 58 | 64 | 52 | 70

Answer: 58

Solution: Context: angle chase. Third angle= $180-46-76=58^\circ$. Shortcut: Subtract their sum from 180° .

Q1337 ? Remainder cycle ? Easy

A land survey case uses the quantities in this question. Determine the requested result using the remainder cycle model: A triangle has angles 47° and 77° . Find the third.

Options: 68 | 62 | 56 | 50

Answer: 56

Solution: Context: land survey. Third angle= $180-47-77=56^\circ$. Shortcut: Subtract their sum from 180° .

Q1338 ? Divisibility test ? Medium

A triangular park case uses the quantities in this question. Determine the requested result using the divisibility test model: A triangle has angles 48° and 78° . Find the third.

Options: 49 | 59 | 54 | 64

Answer: 54

Solution: Context: triangular park. Third angle= $180-48-78=54^\circ$. Shortcut: Subtract their sum from 180° .

Q1339 ? Factorisation ? Hard

A angle of elevation case uses the quantities in this question. Determine the requested result using the factorisation model: A triangle has angles 49° and 79° . Find the third.

Options: 52 | 57 | 47 | 62

Answer: 52

Solution: Context: angle of elevation. Third angle= $180-49-79=52^\circ$. Shortcut: Subtract their sum from 180° .

Q1340 ? Approximation ? Placement

A road junction case uses the quantities in this question. Determine the requested result using the approximation model: A triangle has angles 50° and 80° . Find the third.

Options: 60 | 45 | 55 | 50

Answer: 50

Solution: Context: road junction. Third angle= $180-50-80=50^\circ$. Shortcut: Subtract their sum from 180° .

PRACTICE ? solutions hidden below**Q1341 ? Options elimination ? Beginner**

A polygon design case uses the quantities in this question. Determine the requested result using the options elimination model: A triangle has angles 51° and

81°. Find the third.

Options: 48 | 43 | 53 | 58

Q1342 ? Boundary case ? Easy

A coordinate map case uses the quantities in this question. Determine the requested result using the boundary case model: A triangle has angles 52° and 82°. Find the third.

Options: 56 | 51 | 46 | 41

Q1343 ? Case split ? Medium

A architectural plan case uses the quantities in this question. Determine the requested result using the case split model: A triangle has angles 53° and 83°. Find the third.

Options: 52 | 40 | 48 | 44

Q1344 ? Contradiction check ? Hard

A angle chase case uses the quantities in this question. Determine the requested result using the contradiction check model: A triangle has angles 54° and 84°. Find the third.

Options: 42 | 46 | 38 | 50

Q1345 ? Statement I test ? Placement

A land survey case uses the quantities in this question. Determine the requested result using the statement i test model: A triangle has angles 55° and 85°. Find the third.

Options: 36 | 48 | 44 | 40

Q1346 ? Statement II test ? Beginner

A triangular park case uses the quantities in this question. Determine the requested result using the statement ii test model: A triangle has angles 56° and 86°. Find the third.

Options: 42 | 46 | 34 | 38

Q1347 ? Combined statements ? Easy

A angle of elevation case uses the quantities in this question. Determine the requested result using the combined statements model: A triangle has angles 57° and 87°. Find the third.

Options: 44 | 40 | 36 | 32

Q1348 ? Counting restriction ? Medium

A road junction case uses the quantities in this question. Determine the requested result using the counting restriction model: A triangle has angles 58° and 88°. Find the third.

Options: 34 | 37 | 31 | 40

Q1349 ? Circular arrangement ? Hard

A polygon design case uses the quantities in this question. Determine the requested result using the circular arrangement model: A triangle has angles 59° and 89°. Find the third.

Options: 32 | 35 | 38 | 29

Q1350 ? Complement method ? Placement

A coordinate map case uses the quantities in this question. Determine the requested result using the complement method model: A triangle has angles 60° and 60°. Find the third.

Options: 60 | 66 | 54 | 72

Q1351 ? Pattern: differences ? Beginner

A architectural plan case uses the quantities in this question. Determine the requested result using the pattern: differences model: A triangle has angles 61° and 61°. Find the third.

Options: 70 | 64 | 58 | 52

Q1352 ? Pattern: ratios ? Easy

A angle chase case uses the quantities in this question. Determine the requested result using the pattern: ratios model: A triangle has angles 62° and 62°. Find the third.

Options: 50 | 56 | 68 | 62

Q1353 ? Pattern: alternating ? Medium

A land survey case uses the quantities in this question. Determine the requested result using the pattern: alternating model: A triangle has angles 63° and 63°. Find the third.

Options: 54 | 59 | 49 | 64

Q1354 ? Pattern: squares / cubes ? Hard

A triangular park case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: A triangle has angles 64° and 64°. Find the third.

Options: 62 | 47 | 57 | 52

Q1355 ? Inference from facts ? Placement

A angle of elevation case uses the quantities in this question. Determine the requested result using the inference from facts model: A triangle has angles 65° and 65°. Find the third.

Options: 45 | 50 | 55 | 60

Q1356 ? Assumption check ? Beginner

A road junction case uses the quantities in this question. Determine the requested result using the assumption check model: A triangle has angles 66° and 66°. Find the third.

Options: 58 | 53 | 48 | 43

Q1357 ? Cause vs correlation ? Easy

A polygon design case uses the quantities in this question. Determine the requested result using the cause vs correlation model: A triangle has angles 67° and 67°. Find the third.

Options: 51 | 56 | 41 | 46

Q1358 ? Grammar agreement ? Medium

A coordinate map case uses the quantities in this question. Determine the requested result using the grammar agreement model: A triangle has angles 68° and 68° . Find the third.

Options: 44 | 48 | 40 | 52

Q1359 ? Vocabulary in context ? Hard

A architectural plan case uses the quantities in this question. Determine the requested result using the vocabulary in context model: A triangle has angles 69° and 69° . Find the third.

Options: 38 | 46 | 42 | 50

Q1360 ? Mixed placement challenge ? Placement

A angle chase case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: A triangle has angles 70° and 70° . Find the third.

Options: 40 | 44 | 48 | 36

Data Sufficiency ? 40 questions

WORKED EXAMPLES

Q1361 ? Direct computation ? Beginner

A loan approval case uses the quantities in this question. Determine the requested result using the direct computation model: Find x . I: $x=11$. II: x is positive and below 16.

Options: Statement I alone is sufficient | II alone is sufficient | Both together are required | Data is insufficient

Answer: Statement I alone is sufficient

Solution: Context: loan approval. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1362 ? Reverse value ? Easy

A inventory decision case uses the quantities in this question. Determine the requested result using the reverse value model: Find x . I: $x=12$. II: x is positive and below 17.

Options: Statement I alone is sufficient | Data is insufficient | Both together are required | II alone is sufficient

Answer: Statement I alone is sufficient

Solution: Context: inventory decision. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1363 ? Missing base ? Medium

A candidate eligibility case uses the quantities in this question. Determine the requested result using the missing base model: Find x . I: $x=13$. II: x is positive and below 18.

Options: Statement I alone is sufficient | Data is insufficient | II alone is sufficient | Both together are required

Answer: Statement I alone is sufficient

Solution: Context: candidate eligibility. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1364 ? Successive change ? Hard

A route selection case uses the quantities in this question. Determine the requested result using the successive change model: Find x . I: $x=14$. II: x is positive and below 19.

Options: Both together are required | II alone is sufficient | Data is insufficient | Statement I alone is sufficient

Answer: Statement I alone is sufficient

Solution: Context: route selection. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1365 ? Comparison trap ? Placement

A budget authorization case uses the quantities in this question. Determine the requested result using the comparison trap model: Find x . I: $x=15$. II: x is positive and below 20.

Options: Statement I alone is sufficient | II alone is sufficient | Both together are required | Data is insufficient

Answer: Statement I alone is sufficient

Solution: Context: budget authorization. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1366 ? Ratio form ? Beginner

A quality release case uses the quantities in this question. Determine the requested result using the ratio form model: Find x . I: $x=16$. II: x is positive and below 21.

Options: Data is insufficient | Both together are required | II alone is sufficient | Statement I alone is sufficient

Answer: Statement I alone is sufficient

Solution: Context: quality release. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1367 ? Weighted result ? Easy

A project estimate case uses the quantities in this question. Determine the requested result using the weighted result model: Find x . I: $x=17$. II: x is positive and below 22.

Options: II alone is sufficient | Both together are required | Data is insufficient | Statement I alone is sufficient

Answer: Statement I alone is sufficient

Solution: Context: project estimate. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1368 ? Replacement cycle ? Medium

A schedule decision case uses the quantities in this question. Determine the requested result using the replacement cycle model: Find x . I: $x=18$. II: x is positive and below 23.

Options: Statement I alone is sufficient | II alone is sufficient | Both together are required | Data is insufficient

Answer: Statement I alone is sufficient

Solution: Context: schedule decision. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1369 ? Unit conversion ? Hard

A loan approval case uses the quantities in this question. Determine the requested result using the unit conversion model: Find x . I: $x=19$. II: x is positive and below 24.

Options: Statement I alone is sufficient | Data is insufficient | Both together are required | II alone is sufficient

Answer: Statement I alone is sufficient

Solution: Context: loan approval. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1370 ? Work-rate model ? Placement

A inventory decision case uses the quantities in this question. Determine the requested result using the work-rate model model: Find x . I: $x=20$. II: x is positive and below 25.

Options: Statement I alone is sufficient | Data is insufficient | II alone is sufficient | Both together are required

Answer: Statement I alone is sufficient

Solution: Context: inventory decision. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1371 ? Speed-time model ? Beginner

A candidate eligibility case uses the quantities in this question. Determine the requested result using the speed-time model model: Find x. I: $x=21$. II: x is positive and below 26.

Options: Both together are required | II alone is sufficient | Data is insufficient | Statement I alone is sufficient

Answer: Statement I alone is sufficient

Solution: Context: candidate eligibility. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1372 ? Table lookup ? Easy

A route selection case uses the quantities in this question. Determine the requested result using the table lookup model: Find x. I: $x=22$. II: x is positive and below 27.

Options: Statement I alone is sufficient | II alone is sufficient | Both together are required | Data is insufficient

Answer: Statement I alone is sufficient

Solution: Context: route selection. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1373 ? Chart comparison ? Medium

A budget authorization case uses the quantities in this question. Determine the requested result using the chart comparison model: Find x. I: $x=23$. II: x is positive and below 28.

Options: Data is insufficient | Both together are required | II alone is sufficient | Statement I alone is sufficient

Answer: Statement I alone is sufficient

Solution: Context: budget authorization. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1374 ? Multi-step word problem ? Hard

A quality release case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Find x. I: $x=24$. II: x is positive and below 29.

Options: II alone is sufficient | Both together are required | Data is insufficient | Statement I alone is sufficient

Answer: Statement I alone is sufficient

Solution: Context: quality release. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1375 ? Maximum / minimum ? Placement

A project estimate case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Find x. I: $x=25$. II: x is positive and below 30.

Options: Statement I alone is sufficient | II alone is sufficient | Both together are required | Data is insufficient

Answer: Statement I alone is sufficient

Solution: Context: project estimate. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1376 ? Least / greatest value ? Beginner

A schedule decision case uses the quantities in this question. Determine the requested result using the least / greatest value model: Find x. I: $x=26$. II: x is positive and below 31.

Options: Statement I alone is sufficient | Data is insufficient | Both together are required | II alone is sufficient

Answer: Statement I alone is sufficient

Solution: Context: schedule decision. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1377 ? Remainder cycle ? Easy

A loan approval case uses the quantities in this question. Determine the requested result using the remainder cycle model: Find x. I: $x=27$. II: x is positive and below 32.

Options: Statement I alone is sufficient | Data is insufficient | II alone is sufficient | Both together are required

Answer: Statement I alone is sufficient

Solution: Context: loan approval. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1378 ? Divisibility test ? Medium

A inventory decision case uses the quantities in this question. Determine the requested result using the divisibility test model: Find x. I: $x=28$. II: x is positive and below 33.

Options: Both together are required | II alone is sufficient | Data is insufficient | Statement I alone is sufficient

Answer: Statement I alone is sufficient

Solution: Context: inventory decision. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1379 ? Factorisation ? Hard

A candidate eligibility case uses the quantities in this question. Determine the requested result using the factorisation model: Find x. I: $x=29$. II: x is positive and below 34.

Options: Statement I alone is sufficient | II alone is sufficient | Both together are required | Data is insufficient

Answer: Statement I alone is sufficient

Solution: Context: candidate eligibility. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

Q1380 ? Approximation ? Placement

A route selection case uses the quantities in this question. Determine the requested result using the approximation model: Find x. I: $x=30$. II: x is positive and below 35.

Options: Data is insufficient | Both together are required | II alone is sufficient | Statement I alone is sufficient

Answer: Statement I alone is sufficient

Solution: Context: route selection. I fixes one value, while II permits many; I alone is sufficient. Shortcut: Do not solve beyond sufficiency.

PRACTICE ? solutions hidden below**Q1381 ? Options elimination ? Beginner**

A budget authorization case uses the quantities in this question. Determine the requested result using the options elimination model: Find x. I: $x=31$. II: x is

positive and below 36.

Options: II alone is sufficient | Both together are required | Data is insufficient | Statement I alone is sufficient

Q1382 ? Boundary case ? Easy

A quality release case uses the quantities in this question. Determine the requested result using the boundary case model: Find x. I: $x=32$. II: x is positive and below 37.

Options: Statement I alone is sufficient | II alone is sufficient | Both together are required | Data is insufficient

Q1383 ? Case split ? Medium

A project estimate case uses the quantities in this question. Determine the requested result using the case split model: Find x. I: $x=33$. II: x is positive and below 38.

Options: Statement I alone is sufficient | Data is insufficient | Both together are required | II alone is sufficient

Q1384 ? Contradiction check ? Hard

A schedule decision case uses the quantities in this question. Determine the requested result using the contradiction check model: Find x. I: $x=34$. II: x is positive and below 39.

Options: Statement I alone is sufficient | Data is insufficient | II alone is sufficient | Both together are required

Q1385 ? Statement I test ? Placement

A loan approval case uses the quantities in this question. Determine the requested result using the statement i test model: Find x. I: $x=35$. II: x is positive and below 40.

Options: Both together are required | II alone is sufficient | Data is insufficient | Statement I alone is sufficient

Q1386 ? Statement II test ? Beginner

A inventory decision case uses the quantities in this question. Determine the requested result using the statement ii test model: Find x. I: $x=36$. II: x is positive and below 41.

Options: Statement I alone is sufficient | II alone is sufficient | Both together are required | Data is insufficient

Q1387 ? Combined statements ? Easy

A candidate eligibility case uses the quantities in this question. Determine the requested result using the combined statements model: Find x. I: $x=37$. II: x is positive and below 42.

Options: Data is insufficient | Both together are required | II alone is sufficient | Statement I alone is sufficient

Q1388 ? Counting restriction ? Medium

A route selection case uses the quantities in this question. Determine the requested result using the counting restriction model: Find x. I: $x=38$. II: x is positive and below 43.

Options: II alone is sufficient | Both together are required | Data is insufficient | Statement I alone is sufficient

Q1389 ? Circular arrangement ? Hard

A budget authorization case uses the quantities in this question. Determine the requested result using the circular arrangement model: Find x. I: $x=39$. II: x is positive and below 44.

Options: Statement I alone is sufficient | II alone is sufficient | Both together are required | Data is insufficient

Q1390 ? Complement method ? Placement

A quality release case uses the quantities in this question. Determine the requested result using the complement method model: Find x. I: $x=40$. II: x is positive and below 45.

Options: Statement I alone is sufficient | Data is insufficient | Both together are required | II alone is sufficient

Q1391 ? Pattern: differences ? Beginner

A project estimate case uses the quantities in this question. Determine the requested result using the pattern: differences model: Find x. I: $x=41$. II: x is positive and below 46.

Options: Statement I alone is sufficient | Data is insufficient | II alone is sufficient | Both together are required

Q1392 ? Pattern: ratios ? Easy

A schedule decision case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Find x. I: $x=42$. II: x is positive and below 47.

Options: Both together are required | II alone is sufficient | Data is insufficient | Statement I alone is sufficient

Q1393 ? Pattern: alternating ? Medium

A loan approval case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Find x. I: $x=43$. II: x is positive and below 48.

Options: Statement I alone is sufficient | II alone is sufficient | Both together are required | Data is insufficient

Q1394 ? Pattern: squares / cubes ? Hard

A inventory decision case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Find x. I: $x=44$. II: x is positive and below 49.

Options: Data is insufficient | Both together are required | II alone is sufficient | Statement I alone is sufficient

Q1395 ? Inference from facts ? Placement

A candidate eligibility case uses the quantities in this question. Determine the requested result using the inference from facts model: Find x. I: $x=45$. II: x is positive and below 50.

Options: II alone is sufficient | Both together are required | Data is insufficient | Statement I alone is sufficient

Q1396 ? Assumption check ? Beginner

A route selection case uses the quantities in this question. Determine the requested result using the assumption check model: Find x. I: $x=46$. II: x is positive and below 51.

Options: Statement I alone is sufficient | II alone is sufficient | Both together are required | Data is insufficient

Q1397 ? Cause vs correlation ? Easy

A budget authorization case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Find x. I: $x=47$. II: x is positive and below 52.

Options: Statement I alone is sufficient | Data is insufficient | Both together are required | II alone is sufficient

Q1398 ? Grammar agreement ? Medium

A quality release case uses the quantities in this question. Determine the requested result using the grammar agreement model: Find x. I: $x=48$. II: x is positive and below 53.

Options: Statement I alone is sufficient | Data is insufficient | II alone is sufficient | Both together are required

Q1399 ? Vocabulary in context ? Hard

A project estimate case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Find x. I: $x=49$. II: x is positive and below 54.

Options: Both together are required | II alone is sufficient | Data is insufficient | Statement I alone is sufficient

Q1400 ? Mixed placement challenge ? Placement

A schedule decision case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Find x. I: $x=50$. II: x is positive and below 55.

Options: Statement I alone is sufficient | II alone is sufficient | Both together are required | Data is insufficient

Venn Diagrams ? 40 questions

WORKED EXAMPLES

Q1401 ? Direct computation ? Beginner

A course enrollment case uses the quantities in this question. Determine the requested result using the direct computation model: 41 like Maths, 26 English, 11 both. How many like at least one?

Options: 68 | 50 | 62 | 56

Answer: 56

Solution: Context: course enrollment. Inclusion–exclusion: $41+26-11=56$. Shortcut: $\text{Union}=A+B-\text{intersection}$.

Q1402 ? Reverse value ? Easy

A customer segments case uses the quantities in this question. Determine the requested result using the reverse value model: 42 like Maths, 27 English, 12 both. How many like at least one?

Options: 57 | 63 | 51 | 69

Answer: 57

Solution: Context: customer segments. Inclusion–exclusion: $42+27-12=57$. Shortcut: $\text{Union}=A+B-\text{intersection}$.

Q1403 ? Missing base ? Medium

A skill survey case uses the quantities in this question. Determine the requested result using the missing base model: 43 like Maths, 28 English, 13 both. How many like at least one?

Options: 70 | 64 | 58 | 52

Answer: 58

Solution: Context: skill survey. Inclusion–exclusion: $43+28-13=58$. Shortcut: $\text{Union}=A+B-\text{intersection}$.

Q1404 ? Successive change ? Hard

A club membership case uses the quantities in this question. Determine the requested result using the successive change model: 44 like Maths, 29 English, 14 both. How many like at least one?

Options: 53 | 59 | 65 | 71

Answer: 59

Solution: Context: club membership. Inclusion–exclusion: $44+29-14=59$. Shortcut: $\text{Union}=A+B-\text{intersection}$.

Q1405 ? Comparison trap ? Placement

A product overlap case uses the quantities in this question. Determine the requested result using the comparison trap model: 45 like Maths, 30 English, 15 both. How many like at least one?

Options: 72 | 60 | 66 | 54

Answer: 60

Solution: Context: product overlap. Inclusion–exclusion: $45+30-15=60$. Shortcut: $\text{Union}=A+B-\text{intersection}$.

Q1406 ? Ratio form ? Beginner

A language survey case uses the quantities in this question. Determine the requested result using the ratio form model: 46 like Maths, 31 English, 16 both. How many like at least one?

Options: 61 | 67 | 73 | 55

Answer: 61

Solution: Context: language survey. Inclusion–exclusion: $46+31-16=61$. Shortcut: $\text{Union}=A+B-\text{intersection}$.

Q1407 ? Weighted result ? Easy

A candidate preferences case uses the quantities in this question. Determine the requested result using the weighted result model: 47 like Maths, 32 English, 17 both. How many like at least one?

Options: 56 | 62 | 68 | 74

Answer: 62

Solution: Context: candidate preferences. Inclusion–exclusion: $47+32-17=62$. Shortcut: $\text{Union}=A+B-\text{intersection}$.

Q1408 ? Replacement cycle ? Medium

A market research case uses the quantities in this question. Determine the requested result using the replacement cycle model: 48 like Maths, 33 English, 10 both. How many like at least one?

Options: 64 | 85 | 78 | 71

Answer: 71

Solution: Context: market research. Inclusion–exclusion: $48+33-10=71$. Shortcut: $\text{Union}=A+B-\text{intersection}$.

Q1409 ? Unit conversion ? Hard

A course enrollment case uses the quantities in this question. Determine the requested result using the unit conversion model: 49 like Maths, 34 English, 11 both. How many like at least one?

Options: 72 | 79 | 65 | 86

Answer: 72

Solution: Context: course enrollment. Inclusion–exclusion: $49+34-11=72$. Shortcut: $\text{Union}=A+B-\text{intersection}$.

Q1410 ? Work-rate model ? Placement

A customer segments case uses the quantities in this question. Determine the requested result using the work-rate model model: 50 like Maths, 25 English, 12 both. How many like at least one?

Options: 75 | 69 | 63 | 57

Answer: 63

Solution: Context: customer segments. Inclusion–exclusion: $50+25-12=63$. Shortcut: $\text{Union}=A+B-\text{intersection}$.

Q1411 ? Speed-time model ? Beginner

A skill survey case uses the quantities in this question. Determine the requested result using the speed-time model model: 51 like Maths, 26 English, 13 both. How many like at least one?

Options: 58 | 64 | 76 | 70

Answer: 64

Solution: Context: skill survey. Inclusion–exclusion: $51+26-13=64$. Shortcut: Union= $A+B$ –intersection.

Q1412 ? Table lookup ? Easy

A club membership case uses the quantities in this question. Determine the requested result using the table lookup model: 52 like Maths, 27 English, 14 both. How many like at least one?

Options: 79 | 72 | 65 | 58

Answer: 65

Solution: Context: club membership. Inclusion–exclusion: $52+27-14=65$. Shortcut: Union= $A+B$ –intersection.

Q1413 ? Chart comparison ? Medium

A product overlap case uses the quantities in this question. Determine the requested result using the chart comparison model: 53 like Maths, 28 English, 15 both. How many like at least one?

Options: 66 | 73 | 80 | 59

Answer: 66

Solution: Context: product overlap. Inclusion–exclusion: $53+28-15=66$. Shortcut: Union= $A+B$ –intersection.

Q1414 ? Multi-step word problem ? Hard

A language survey case uses the quantities in this question. Determine the requested result using the multi-step word problem model: 54 like Maths, 29 English, 16 both. How many like at least one?

Options: 81 | 67 | 60 | 74

Answer: 67

Solution: Context: language survey. Inclusion–exclusion: $54+29-16=67$. Shortcut: Union= $A+B$ –intersection.

Q1415 ? Maximum / minimum ? Placement

A candidate preferences case uses the quantities in this question. Determine the requested result using the maximum / minimum model: 55 like Maths, 30 English, 17 both. How many like at least one?

Options: 61 | 82 | 75 | 68

Answer: 68

Solution: Context: candidate preferences. Inclusion–exclusion: $55+30-17=68$. Shortcut: Union= $A+B$ –intersection.

Q1416 ? Least / greatest value ? Beginner

A market research case uses the quantities in this question. Determine the requested result using the least / greatest value model: 56 like Maths, 31 English, 10 both. How many like at least one?

Options: 93 | 77 | 85 | 69

Answer: 77

Solution: Context: market research. Inclusion–exclusion: $56+31-10=77$. Shortcut: Union= $A+B$ –intersection.

Q1417 ? Remainder cycle ? Easy

A course enrollment case uses the quantities in this question. Determine the requested result using the remainder cycle model: 57 like Maths, 32 English, 11 both. How many like at least one?

Options: 78 | 86 | 70 | 94

Answer: 78

Solution: Context: course enrollment. Inclusion–exclusion: $57+32-11=78$. Shortcut: Union= $A+B$ –intersection.

Q1418 ? Divisibility test ? Medium

A customer segments case uses the quantities in this question. Determine the requested result using the divisibility test model: 58 like Maths, 33 English, 12 both. How many like at least one?

Options: 71 | 79 | 87 | 95

Answer: 79

Solution: Context: customer segments. Inclusion–exclusion: $58+33-12=79$. Shortcut: Union= $A+B$ –intersection.

Q1419 ? Factorisation ? Hard

A skill survey case uses the quantities in this question. Determine the requested result using the factorisation model: 59 like Maths, 34 English, 13 both. How many like at least one?

Options: 96 | 72 | 88 | 80

Answer: 80

Solution: Context: skill survey. Inclusion–exclusion: $59+34-13=80$. Shortcut: Union= $A+B$ –intersection.

Q1420 ? Approximation ? Placement

A club membership case uses the quantities in this question. Determine the requested result using the approximation model: 60 like Maths, 25 English, 14 both. How many like at least one?

Options: 71 | 78 | 64 | 85

Answer: 71

Solution: Context: club membership. Inclusion–exclusion: $60+25-14=71$. Shortcut: Union= $A+B$ –intersection.

PRACTICE ? solutions hidden below**Q1421 ? Options elimination ? Beginner**

A product overlap case uses the quantities in this question. Determine the requested result using the options elimination model: 61 like Maths, 26 English, 15

both. How many like at least one?
Options: 86 | 65 | 72 | 79

Q1422 ? Boundary case ? Easy

A language survey case uses the quantities in this question. Determine the requested result using the boundary case model: 62 like Maths, 27 English, 16 both. How many like at least one?
Options: 66 | 87 | 80 | 73

Q1423 ? Case split ? Medium

A candidate preferences case uses the quantities in this question. Determine the requested result using the case split model: 63 like Maths, 28 English, 17 both. How many like at least one?
Options: 74 | 81 | 67 | 88

Q1424 ? Contradiction check ? Hard

A market research case uses the quantities in this question. Determine the requested result using the contradiction check model: 64 like Maths, 29 English, 10 both. How many like at least one?
Options: 83 | 75 | 91 | 99

Q1425 ? Statement I test ? Placement

A course enrollment case uses the quantities in this question. Determine the requested result using the statement i test model: 65 like Maths, 30 English, 11 both. How many like at least one?
Options: 84 | 92 | 76 | 100

Q1426 ? Statement II test ? Beginner

A customer segments case uses the quantities in this question. Determine the requested result using the statement ii test model: 66 like Maths, 31 English, 12 both. How many like at least one?
Options: 103 | 76 | 94 | 85

Q1427 ? Combined statements ? Easy

A skill survey case uses the quantities in this question. Determine the requested result using the combined statements model: 67 like Maths, 32 English, 13 both. How many like at least one?
Options: 86 | 95 | 77 | 104

Q1428 ? Counting restriction ? Medium

A club membership case uses the quantities in this question. Determine the requested result using the counting restriction model: 68 like Maths, 33 English, 14 both. How many like at least one?
Options: 87 | 105 | 96 | 78

Q1429 ? Circular arrangement ? Hard

A product overlap case uses the quantities in this question. Determine the requested result using the circular arrangement model: 69 like Maths, 34 English, 15 both. How many like at least one?
Options: 106 | 79 | 97 | 88

Q1430 ? Complement method ? Placement

A language survey case uses the quantities in this question. Determine the requested result using the complement method model: 70 like Maths, 25 English, 16 both. How many like at least one?
Options: 95 | 79 | 87 | 71

Q1431 ? Pattern: differences ? Beginner

A candidate preferences case uses the quantities in this question. Determine the requested result using the pattern: differences model: 71 like Maths, 26 English, 17 both. How many like at least one?
Options: 80 | 88 | 72 | 96

Q1432 ? Pattern: ratios ? Easy

A market research case uses the quantities in this question. Determine the requested result using the pattern: ratios model: 72 like Maths, 27 English, 10 both. How many like at least one?
Options: 89 | 98 | 107 | 80

Q1433 ? Pattern: alternating ? Medium

A course enrollment case uses the quantities in this question. Determine the requested result using the pattern: alternating model: 73 like Maths, 28 English, 11 both. How many like at least one?
Options: 108 | 81 | 99 | 90

Q1434 ? Pattern: squares / cubes ? Hard

A customer segments case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: 74 like Maths, 29 English, 12 both. How many like at least one?
Options: 91 | 100 | 82 | 109

Q1435 ? Inference from facts ? Placement

A skill survey case uses the quantities in this question. Determine the requested result using the inference from facts model: 75 like Maths, 30 English, 13 both. How many like at least one?
Options: 101 | 83 | 110 | 92

Q1436 ? Assumption check ? Beginner

A club membership case uses the quantities in this question. Determine the requested result using the assumption check model: 76 like Maths, 31 English, 14 both. How many like at least one?
Options: 111 | 84 | 102 | 93

Q1437 ? Cause vs correlation ? Easy

A product overlap case uses the quantities in this question. Determine the requested result using the cause vs correlation model: 77 like Maths, 32 English, 15 both. How many like at least one?

Options: 94 | 103 | 85 | 112

Q1438 ? Grammar agreement ? Medium

A language survey case uses the quantities in this question. Determine the requested result using the grammar agreement model: 78 like Maths, 33 English, 16 both. How many like at least one?

Options: 95 | 105 | 85 | 115

Q1439 ? Vocabulary in context ? Hard

A candidate preferences case uses the quantities in this question. Determine the requested result using the vocabulary in context model: 79 like Maths, 34 English, 17 both. How many like at least one?

Options: 96 | 116 | 86 | 106

Q1440 ? Mixed placement challenge ? Placement

A market research case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: 80 like Maths, 25 English, 10 both. How many like at least one?

Options: 115 | 85 | 105 | 95

Blood Relations ? 40 questions

WORKED EXAMPLES

Q1441 ? Direct computation ? Beginner

A family photograph case uses the quantities in this question. Determine the requested result using the direct computation model: K is the mother of M. What is K's relation to M?

Options: mother | cousin | sibling | grandparent

Answer: mother

Solution: Context: family photograph. The statement directly says K is M's mother. Shortcut: Draw generations and gender.

Q1442 ? Reverse value ? Easy

A genealogy chart case uses the quantities in this question. Determine the requested result using the reverse value model: K is the brother of M. What is K's relation to M?

Options: grandparent | sibling | cousin | brother

Answer: brother

Solution: Context: genealogy chart. The statement directly says K is M's brother. Shortcut: Draw generations and gender.

Q1443 ? Missing base ? Medium

A seating clue case uses the quantities in this question. Determine the requested result using the missing base model: K is the uncle of M. What is K's relation to M?

Options: grandparent | cousin | sibling | uncle

Answer: uncle

Solution: Context: seating clue. The statement directly says K is M's uncle. Shortcut: Draw generations and gender.

Q1444 ? Successive change ? Hard

A inheritance record case uses the quantities in this question. Determine the requested result using the successive change model: K is the father of M. What is K's relation to M?

Options: father | cousin | sibling | grandparent

Answer: father

Solution: Context: inheritance record. The statement directly says K is M's father. Shortcut: Draw generations and gender.

Q1445 ? Comparison trap ? Placement

A team relation case uses the quantities in this question. Determine the requested result using the comparison trap model: K is the mother of M. What is K's relation to M?

Options: mother | sibling | cousin | grandparent

Answer: mother

Solution: Context: team relation. The statement directly says K is M's mother. Shortcut: Draw generations and gender.

Q1446 ? Ratio form ? Beginner

A hospital form case uses the quantities in this question. Determine the requested result using the ratio form model: K is the brother of M. What is K's relation to M?

Options: sibling | brother | grandparent | cousin

Answer: brother

Solution: Context: hospital form. The statement directly says K is M's brother. Shortcut: Draw generations and gender.

Q1447 ? Weighted result ? Easy

A family invitation case uses the quantities in this question. Determine the requested result using the weighted result model: K is the uncle of M. What is K's relation to M?

Options: cousin | sibling | uncle | grandparent

Answer: uncle

Solution: Context: family invitation. The statement directly says K is M's uncle. Shortcut: Draw generations and gender.

Q1448 ? Replacement cycle ? Medium

A tree diagram case uses the quantities in this question. Determine the requested result using the replacement cycle model: K is the father of M. What is K's relation to M?

Options: father | cousin | sibling | grandparent

Answer: father

Solution: Context: tree diagram. The statement directly says K is M's father. Shortcut: Draw generations and gender.

Q1449 ? Unit conversion ? Hard

A family photograph case uses the quantities in this question. Determine the requested result using the unit conversion model: K is the mother of M. What is K's relation to M?

Options: mother | grandparent | sibling | cousin

Answer: mother

Solution: Context: family photograph. The statement directly says K is M's mother. Shortcut: Draw generations and gender.

Q1450 ? Work-rate model ? Placement

A genealogy chart case uses the quantities in this question. Determine the requested result using the work-rate model model: K is the brother of M. What is K's relation to M?

Options: grandparent | brother | cousin | sibling

Answer: brother

Solution: Context: genealogy chart. The statement directly says K is M's brother. Shortcut: Draw generations and gender.

Q1451 ? Speed-time model ? Beginner

A seating clue case uses the quantities in this question. Determine the requested result using the speed-time model model: K is the uncle of M. What is K's relation to M?

Options: uncle | cousin | sibling | grandparent

Answer: uncle

Solution: Context: seating clue. The statement directly says K is M's uncle. Shortcut: Draw generations and gender.

Q1452 ? Table lookup ? Easy

A inheritance record case uses the quantities in this question. Determine the requested result using the table lookup model: K is the father of M. What is K's relation to M?

Options: father | sibling | cousin | grandparent

Answer: father

Solution: Context: inheritance record. The statement directly says K is M's father. Shortcut: Draw generations and gender.

Q1453 ? Chart comparison ? Medium

A team relation case uses the quantities in this question. Determine the requested result using the chart comparison model: K is the mother of M. What is K's relation to M?

Options: grandparent | sibling | cousin | mother

Answer: mother

Solution: Context: team relation. The statement directly says K is M's mother. Shortcut: Draw generations and gender.

Q1454 ? Multi-step word problem ? Hard

A hospital form case uses the quantities in this question. Determine the requested result using the multi-step word problem model: K is the brother of M. What is K's relation to M?

Options: brother | cousin | sibling | grandparent

Answer: brother

Solution: Context: hospital form. The statement directly says K is M's brother. Shortcut: Draw generations and gender.

Q1455 ? Maximum / minimum ? Placement

A family invitation case uses the quantities in this question. Determine the requested result using the maximum / minimum model: K is the uncle of M. What is K's relation to M?

Options: uncle | cousin | sibling | grandparent

Answer: uncle

Solution: Context: family invitation. The statement directly says K is M's uncle. Shortcut: Draw generations and gender.

Q1456 ? Least / greatest value ? Beginner

A tree diagram case uses the quantities in this question. Determine the requested result using the least / greatest value model: K is the father of M. What is K's relation to M?

Options: father | grandparent | sibling | cousin

Answer: father

Solution: Context: tree diagram. The statement directly says K is M's father. Shortcut: Draw generations and gender.

Q1457 ? Remainder cycle ? Easy

A family photograph case uses the quantities in this question. Determine the requested result using the remainder cycle model: K is the mother of M. What is K's relation to M?

Options: grandparent | cousin | mother | sibling

Answer: mother

Solution: Context: family photograph. The statement directly says K is M's mother. Shortcut: Draw generations and gender.

Q1458 ? Divisibility test ? Medium

A genealogy chart case uses the quantities in this question. Determine the requested result using the divisibility test model: K is the brother of M. What is K's relation to M?

Options: brother | cousin | sibling | grandparent

Answer: brother

Solution: Context: genealogy chart. The statement directly says K is M's brother. Shortcut: Draw generations and gender.

Q1459 ? Factorisation ? Hard

A seating clue case uses the quantities in this question. Determine the requested result using the factorisation model: K is the uncle of M. What is K's relation to M?

Options: uncle | sibling | cousin | grandparent

Answer: uncle

Solution: Context: seating clue. The statement directly says K is M's uncle. Shortcut: Draw generations and gender.

Q1460 ? Approximation ? Placement

A inheritance record case uses the quantities in this question. Determine the requested result using the approximation model: K is the father of M. What is K's relation to M?

Options: grandparent | sibling | cousin | father

Answer: father

Solution: Context: inheritance record. The statement directly says K is M's father. Shortcut: Draw generations and gender.

PRACTICE ? solutions hidden below**Q1461 ? Options elimination ? Beginner**

A team relation case uses the quantities in this question. Determine the requested result using the options elimination model: K is the mother of M. What is K's

relation to M?

Options: cousin | sibling | mother | grandparent

Q1462 ? Boundary case ? Easy

A hospital form case uses the quantities in this question. Determine the requested result using the boundary case model: K is the brother of M. What is K's relation to M?

Options: grandparent | cousin | brother | sibling

Q1463 ? Case split ? Medium

A family invitation case uses the quantities in this question. Determine the requested result using the case split model: K is the uncle of M. What is K's relation to M?

Options: grandparent | sibling | cousin | uncle

Q1464 ? Contradiction check ? Hard

A tree diagram case uses the quantities in this question. Determine the requested result using the contradiction check model: K is the father of M. What is K's relation to M?

Options: grandparent | cousin | father | sibling

Q1465 ? Statement I test ? Placement

A family photograph case uses the quantities in this question. Determine the requested result using the statement i test model: K is the mother of M. What is K's relation to M?

Options: mother | cousin | sibling | grandparent

Q1466 ? Statement II test ? Beginner

A genealogy chart case uses the quantities in this question. Determine the requested result using the statement ii test model: K is the brother of M. What is K's relation to M?

Options: sibling | cousin | brother | grandparent

Q1467 ? Combined statements ? Easy

A seating clue case uses the quantities in this question. Determine the requested result using the combined statements model: K is the uncle of M. What is K's relation to M?

Options: uncle | grandparent | sibling | cousin

Q1468 ? Counting restriction ? Medium

A inheritance record case uses the quantities in this question. Determine the requested result using the counting restriction model: K is the father of M. What is K's relation to M?

Options: cousin | sibling | father | grandparent

Q1469 ? Circular arrangement ? Hard

A team relation case uses the quantities in this question. Determine the requested result using the circular arrangement model: K is the mother of M. What is K's relation to M?

Options: mother | cousin | sibling | grandparent

Q1470 ? Complement method ? Placement

A hospital form case uses the quantities in this question. Determine the requested result using the complement method model: K is the brother of M. What is K's relation to M?

Options: grandparent | sibling | cousin | brother

Q1471 ? Pattern: differences ? Beginner

A family invitation case uses the quantities in this question. Determine the requested result using the pattern: differences model: K is the uncle of M. What is K's relation to M?

Options: grandparent | cousin | sibling | uncle

Q1472 ? Pattern: ratios ? Easy

A tree diagram case uses the quantities in this question. Determine the requested result using the pattern: ratios model: K is the father of M. What is K's relation to M?

Options: father | cousin | sibling | grandparent

Q1473 ? Pattern: alternating ? Medium

A family photograph case uses the quantities in this question. Determine the requested result using the pattern: alternating model: K is the mother of M. What is K's relation to M?

Options: mother | sibling | cousin | grandparent

Q1474 ? Pattern: squares / cubes ? Hard

A genealogy chart case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: K is the brother of M. What is K's relation to M?

Options: sibling | brother | grandparent | cousin

Q1475 ? Inference from facts ? Placement

A seating clue case uses the quantities in this question. Determine the requested result using the inference from facts model: K is the uncle of M. What is K's relation to M?

Options: cousin | sibling | uncle | grandparent

Q1476 ? Assumption check ? Beginner

A inheritance record case uses the quantities in this question. Determine the requested result using the assumption check model: K is the father of M. What is K's relation to M?

Options: father | cousin | sibling | grandparent

Q1477 ? Cause vs correlation ? Easy

A team relation case uses the quantities in this question. Determine the requested result using the cause vs correlation model: K is the mother of M. What is K's relation to M?

Options: mother | grandparent | sibling | cousin

Q1478 ? Grammar agreement ? Medium

A hospital form case uses the quantities in this question. Determine the requested result using the grammar agreement model: K is the brother of M. What is K's relation to M?

Options: grandparent | brother | cousin | sibling

Q1479 ? Vocabulary in context ? Hard

A family invitation case uses the quantities in this question. Determine the requested result using the vocabulary in context model: K is the uncle of M. What is K's relation to M?

Options: uncle | cousin | sibling | grandparent

Q1480 ? Mixed placement challenge ? Placement

A tree diagram case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: K is the father of M. What is K's relation to M?

Options: father | sibling | cousin | grandparent

Analogy ? 40 questions

WORKED EXAMPLES

Q1481 ? Direct computation ? Beginner

A tool-function pair case uses the quantities in this question. Determine the requested result using the direct computation model: Doctor:Hospital :: Teacher:?

Options: River | Forest | School | Office

Answer: School

Solution: Context: tool-function pair. Doctor relates to Hospital in the same way Teacher relates to School. Shortcut: Match relationships, not appearance.

Q1482 ? Reverse value ? Easy

A profession-workplace case uses the quantities in this question. Determine the requested result using the reverse value model: Puppy:Dog :: Kitten:?

Options: Cat | River | Office | Forest

Answer: Cat

Solution: Context: profession-workplace. Puppy relates to Dog in the same way Kitten relates to Cat. Shortcut: Match relationships, not appearance.

Q1483 ? Missing base ? Medium

A animal-home case uses the quantities in this question. Determine the requested result using the missing base model: Book:Read :: Song:?

Options: Listen | River | Forest | Office

Answer: Listen

Solution: Context: animal-home. Book relates to Read in the same way Song relates to Listen. Shortcut: Match relationships, not appearance.

Q1484 ? Successive change ? Hard

A object-material case uses the quantities in this question. Determine the requested result using the successive change model: Bird:Nest :: Bee:?

Options: Office | Hive | River | Forest

Answer: Hive

Solution: Context: object-material. Bird relates to Nest in the same way Bee relates to Hive. Shortcut: Match relationships, not appearance.

Q1485 ? Comparison trap ? Placement

A cause-effect case uses the quantities in this question. Determine the requested result using the comparison trap model: Doctor:Hospital :: Teacher:?

Options: Office | Forest | School | River

Answer: School

Solution: Context: cause-effect. Doctor relates to Hospital in the same way Teacher relates to School. Shortcut: Match relationships, not appearance.

Q1486 ? Ratio form ? Beginner

A part-whole case uses the quantities in this question. Determine the requested result using the ratio form model: Puppy:Dog :: Kitten:?

Options: Forest | River | Cat | Office

Answer: Cat

Solution: Context: part-whole. Puppy relates to Dog in the same way Kitten relates to Cat. Shortcut: Match relationships, not appearance.

Q1487 ? Weighted result ? Easy

A synonym relation case uses the quantities in this question. Determine the requested result using the weighted result model: Book:Read :: Song:?

Options: Forest | Office | River | Listen

Answer: Listen

Solution: Context: synonym relation. Book relates to Read in the same way Song relates to Listen. Shortcut: Match relationships, not appearance.

Q1488 ? Replacement cycle ? Medium

A process-output case uses the quantities in this question. Determine the requested result using the replacement cycle model: Bird:Nest :: Bee:?

Options: Hive | Forest | Office | River

Answer: Hive

Solution: Context: process-output. Bird relates to Nest in the same way Bee relates to Hive. Shortcut: Match relationships, not appearance.

Q1489 ? Unit conversion ? Hard

A tool-function pair case uses the quantities in this question. Determine the requested result using the unit conversion model: Doctor:Hospital :: Teacher:?

Options: Office | School | Forest | River

Answer: School

Solution: Context: tool-function pair. Doctor relates to Hospital in the same way Teacher relates to School. Shortcut: Match relationships, not appearance.

Q1490 ? Work-rate model ? Placement

A profession-workplace case uses the quantities in this question. Determine the requested result using the work-rate model model: Puppy:Dog :: Kitten:?

Options: River | Forest | Office | Cat

Answer: Cat

Solution: Context: profession-workplace. Puppy relates to Dog in the same way Kitten relates to Cat. Shortcut: Match relationships, not appearance.

Q1491 ? Speed-time model ? Beginner

A animal-home case uses the quantities in this question. Determine the requested result using the speed-time model model: Book:Read :: Song:?

Options: Office | River | Listen | Forest

Answer: Listen

Solution: Context: animal-home. Book relates to Read in the same way Song relates to Listen. Shortcut: Match relationships, not appearance.

Q1492 ? Table lookup ? Easy

A object-material case uses the quantities in this question. Determine the requested result using the table lookup model: Bird:Nest :: Bee:?

Options: Office | Forest | Hive | River

Answer: Hive

Solution: Context: object-material. Bird relates to Nest in the same way Bee relates to Hive. Shortcut: Match relationships, not appearance.

Q1493 ? Chart comparison ? Medium

A cause-effect case uses the quantities in this question. Determine the requested result using the chart comparison model: Doctor:Hospital :: Teacher:?

Options: School | Forest | River | Office

Answer: School

Solution: Context: cause-effect. Doctor relates to Hospital in the same way Teacher relates to School. Shortcut: Match relationships, not appearance.

Q1494 ? Multi-step word problem ? Hard

A part-whole case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Puppy:Dog :: Kitten:?

Options: Forest | Office | Cat | River

Answer: Cat

Solution: Context: part-whole. Puppy relates to Dog in the same way Kitten relates to Cat. Shortcut: Match relationships, not appearance.

Q1495 ? Maximum / minimum ? Placement

A synonym relation case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Book:Read :: Song:?

Options: River | Forest | Listen | Office

Answer: Listen

Solution: Context: synonym relation. Book relates to Read in the same way Song relates to Listen. Shortcut: Match relationships, not appearance.

Q1496 ? Least / greatest value ? Beginner

A process-output case uses the quantities in this question. Determine the requested result using the least / greatest value model: Bird:Nest :: Bee:?

Options: River | Office | Hive | Forest

Answer: Hive

Solution: Context: process-output. Bird relates to Nest in the same way Bee relates to Hive. Shortcut: Match relationships, not appearance.

Q1497 ? Remainder cycle ? Easy

A tool-function pair case uses the quantities in this question. Determine the requested result using the remainder cycle model: Doctor:Hospital :: Teacher:?

Options: School | River | Forest | Office

Answer: School

Solution: Context: tool-function pair. Doctor relates to Hospital in the same way Teacher relates to School. Shortcut: Match relationships, not appearance.

Q1498 ? Divisibility test ? Medium

A profession-workplace case uses the quantities in this question. Determine the requested result using the divisibility test model: Puppy:Dog :: Kitten:?

Options: Office | Cat | River | Forest

Answer: Cat

Solution: Context: profession-workplace. Puppy relates to Dog in the same way Kitten relates to Cat. Shortcut: Match relationships, not appearance.

Q1499 ? Factorisation ? Hard

A animal-home case uses the quantities in this question. Determine the requested result using the factorisation model: Book:Read :: Song:?

Options: Office | Forest | Listen | River

Answer: Listen

Solution: Context: animal-home. Book relates to Read in the same way Song relates to Listen. Shortcut: Match relationships, not appearance.

Q1500 ? Approximation ? Placement

A object-material case uses the quantities in this question. Determine the requested result using the approximation model: Bird:Nest :: Bee:?

Options: Hive | Forest | River | Office

Answer: Hive

Solution: Context: object-material. Bird relates to Nest in the same way Bee relates to Hive. Shortcut: Match relationships, not appearance.

PRACTICE ? solutions hidden below

Q1501 ? Options elimination ? Beginner

A cause-effect case uses the quantities in this question. Determine the requested result using the options elimination model: Doctor:Hospital :: Teacher:?

Options: Forest | Office | River | School

Q1502 ? Boundary case ? Easy

A part-whole case uses the quantities in this question. Determine the requested result using the boundary case model: Puppy:Dog :: Kitten:?

Options: Cat | Forest | Office | River

Q1503 ? Case split ? Medium

A synonym relation case uses the quantities in this question. Determine the requested result using the case split model: Book:Read :: Song:?

Options: Office | Listen | Forest | River

Q1504 ? Contradiction check ? Hard

A process-output case uses the quantities in this question. Determine the requested result using the contradiction check model: Bird:Nest :: Bee:?

Options: River | Forest | Hive | Office

Q1505 ? Statement I test ? Placement

A tool-function pair case uses the quantities in this question. Determine the requested result using the statement i test model: Doctor:Hospital :: Teacher:?

Options: Office | River | School | Forest

Q1506 ? Statement II test ? Beginner

A profession-workplace case uses the quantities in this question. Determine the requested result using the statement ii test model: Puppy:Dog :: Kitten:?

Options: Cat | Office | Forest | River

Q1507 ? Combined statements ? Easy

A animal-home case uses the quantities in this question. Determine the requested result using the combined statements model: Book:Read :: Song:?

Options: Listen | Forest | River | Office

Q1508 ? Counting restriction ? Medium

A object-material case uses the quantities in this question. Determine the requested result using the counting restriction model: Bird:Nest :: Bee:?

Options: Forest | Hive | Office | River

Q1509 ? Circular arrangement ? Hard

A cause-effect case uses the quantities in this question. Determine the requested result using the circular arrangement model: Doctor:Hospital :: Teacher:?

Options: River | Forest | School | Office

Q1510 ? Complement method ? Placement

A part-whole case uses the quantities in this question. Determine the requested result using the complement method model: Puppy:Dog :: Kitten:?

Options: Cat | River | Office | Forest

Q1511 ? Pattern: differences ? Beginner

A synonym relation case uses the quantities in this question. Determine the requested result using the pattern: differences model: Book:Read :: Song:?

Options: Listen | River | Forest | Office

Q1512 ? Pattern: ratios ? Easy

A process-output case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Bird:Nest :: Bee:?

Options: Office | Hive | River | Forest

Q1513 ? Pattern: alternating ? Medium

A tool-function pair case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Doctor:Hospital :: Teacher:?

Options: Office | Forest | School | River

Q1514 ? Pattern: squares / cubes ? Hard

A profession-workplace case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Puppy:Dog :: Kitten:?

Options: Forest | River | Cat | Office

Q1515 ? Inference from facts ? Placement

A animal-home case uses the quantities in this question. Determine the requested result using the inference from facts model: Book:Read :: Song:?

Options: Forest | Office | River | Listen

Q1516 ? Assumption check ? Beginner

A object-material case uses the quantities in this question. Determine the requested result using the assumption check model: Bird:Nest :: Bee:?

Options: Hive | Forest | Office | River

Q1517 ? Cause vs correlation ? Easy

A cause-effect case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Doctor:Hospital :: Teacher:?

Options: Office | School | Forest | River

Q1518 ? Grammar agreement ? Medium

A part-whole case uses the quantities in this question. Determine the requested result using the grammar agreement model: Puppy:Dog :: Kitten:?

Options: River | Forest | Office | Cat

Q1519 ? Vocabulary in context ? Hard

A synonym relation case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Book:Read :: Song:?

Options: Office | River | Listen | Forest

Q1520 ? Mixed placement challenge ? Placement

A process-output case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Bird:Nest :: Bee:?

Options: Office | Forest | Hive | River

Classification ? 40 questions

WORKED EXAMPLES

Q1521 ? Direct computation ? Beginner

A odd number case uses the quantities in this question. Determine the requested result using the direct computation model: Find the odd one: Triangle, Square, Circle, Cube.

Options: Cube | Circle | Square | Triangle

Answer: Cube

Solution: Context: odd number. Cube does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1522 ? Reverse value ? Easy

A odd word case uses the quantities in this question. Determine the requested result using the reverse value model: Find the odd one: Rose, Lily, Lotus, Mango.

Options: Rose | Lily | Lotus | Mango

Answer: Mango

Solution: Context: odd word. Mango does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1523 ? Missing base ? Medium

A odd shape case uses the quantities in this question. Determine the requested result using the missing base model: Find the odd one: January, March, July, Monday.

Options: March | Monday | January | July

Answer: Monday

Solution: Context: odd shape. Monday does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1524 ? Successive change ? Hard

A odd category case uses the quantities in this question. Determine the requested result using the successive change model: Find the odd one: 2, 4, 6, 9.

Options: 2 | 4 | 6 | 9

Answer: 9

Solution: Context: odd category. 9 does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1525 ? Comparison trap ? Placement

A odd process case uses the quantities in this question. Determine the requested result using the comparison trap model: Find the odd one: Triangle, Square, Circle, Cube.

Options: Square | Triangle | Circle | Cube

Answer: Cube

Solution: Context: odd process. Cube does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1526 ? Ratio form ? Beginner

A odd relation case uses the quantities in this question. Determine the requested result using the ratio form model: Find the odd one: Rose, Lily, Lotus, Mango.

Options: Rose | Lily | Lotus | Mango

Answer: Mango

Solution: Context: odd relation. Mango does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1527 ? Weighted result ? Easy

A odd symbol case uses the quantities in this question. Determine the requested result using the weighted result model: Find the odd one: January, March, July, Monday.

Options: March | Monday | January | July

Answer: Monday

Solution: Context: odd symbol. Monday does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1528 ? Replacement cycle ? Medium

A odd object case uses the quantities in this question. Determine the requested result using the replacement cycle model: Find the odd one: 2, 4, 6, 9.

Options: 2 | 4 | 6 | 9

Answer: 9

Solution: Context: odd object. 9 does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1529 ? Unit conversion ? Hard

A odd number case uses the quantities in this question. Determine the requested result using the unit conversion model: Find the odd one: Triangle, Square, Circle, Cube.

Options: Triangle | Square | Circle | Cube

Answer: Cube

Solution: Context: odd number. Cube does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1530 ? Work-rate model ? Placement

A odd word case uses the quantities in this question. Determine the requested result using the work-rate model model: Find the odd one: Rose, Lily, Lotus, Mango.

Options: Lily | Lotus | Mango | Rose

Answer: Mango

Solution: Context: odd word. Mango does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1531 ? Speed-time model ? Beginner

A odd shape case uses the quantities in this question. Determine the requested result using the speed-time model model: Find the odd one: January, March,

July, Monday.

Options: July | January | Monday | March

Answer: Monday

Solution: Context: odd shape. Monday does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1532 ? Table lookup ? Easy

A odd category case uses the quantities in this question. Determine the requested result using the table lookup model: Find the odd one: 2, 4, 6, 9.

Options: 4 | 9 | 2 | 6

Answer: 9

Solution: Context: odd category. 9 does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1533 ? Chart comparison ? Medium

A odd process case uses the quantities in this question. Determine the requested result using the chart comparison model: Find the odd one: Triangle, Square, Circle, Cube.

Options: Triangle | Square | Cube | Circle

Answer: Cube

Solution: Context: odd process. Cube does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1534 ? Multi-step word problem ? Hard

A odd relation case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Find the odd one: Rose, Lily, Lotus, Mango.

Options: Mango | Rose | Lily | Lotus

Answer: Mango

Solution: Context: odd relation. Mango does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1535 ? Maximum / minimum ? Placement

A odd symbol case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Find the odd one: January, March, July, Monday.

Options: July | January | Monday | March

Answer: Monday

Solution: Context: odd symbol. Monday does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1536 ? Least / greatest value ? Beginner

A odd object case uses the quantities in this question. Determine the requested result using the least / greatest value model: Find the odd one: 2, 4, 6, 9.

Options: 9 | 6 | 4 | 2

Answer: 9

Solution: Context: odd object. 9 does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1537 ? Remainder cycle ? Easy

A odd number case uses the quantities in this question. Determine the requested result using the remainder cycle model: Find the odd one: Triangle, Square, Circle, Cube.

Options: Triangle | Square | Circle | Cube

Answer: Cube

Solution: Context: odd number. Cube does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1538 ? Divisibility test ? Medium

A odd word case uses the quantities in this question. Determine the requested result using the divisibility test model: Find the odd one: Rose, Lily, Lotus, Mango.

Options: Rose | Lotus | Lily | Mango

Answer: Mango

Solution: Context: odd word. Mango does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1539 ? Factorisation ? Hard

A odd shape case uses the quantities in this question. Determine the requested result using the factorisation model: Find the odd one: January, March, July, Monday.

Options: January | March | July | Monday

Answer: Monday

Solution: Context: odd shape. Monday does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

Q1540 ? Approximation ? Placement

A odd category case uses the quantities in this question. Determine the requested result using the approximation model: Find the odd one: 2, 4, 6, 9.

Options: 6 | 4 | 2 | 9

Answer: 9

Solution: Context: odd category. 9 does not share the defining property of the other three. Shortcut: Check category, parity, or dimension.

PRACTICE ? solutions hidden below

Q1541 ? Options elimination ? Beginner

A odd process case uses the quantities in this question. Determine the requested result using the options elimination model: Find the odd one: Triangle, Square, Circle, Cube.

Options: Triangle | Circle | Cube | Square

Q1542 ? Boundary case ? Easy

A odd relation case uses the quantities in this question. Determine the requested result using the boundary case model: Find the odd one: Rose, Lily, Lotus, Mango.

Options: Mango | Lotus | Lily | Rose

Q1543 ? Case split ? Medium

A odd symbol case uses the quantities in this question. Determine the requested result using the case split model: Find the odd one: January, March, July, Monday.

Options: January | March | July | Monday

Q1544 ? Contradiction check ? Hard

A odd object case uses the quantities in this question. Determine the requested result using the contradiction check model: Find the odd one: 2, 4, 6, 9.

Options: 6 | 2 | 9 | 4

Q1545 ? Statement I test ? Placement

A odd number case uses the quantities in this question. Determine the requested result using the statement i test model: Find the odd one: Triangle, Square, Circle, Cube.

Options: Cube | Circle | Square | Triangle

Q1546 ? Statement II test ? Beginner

A odd word case uses the quantities in this question. Determine the requested result using the statement ii test model: Find the odd one: Rose, Lily, Lotus, Mango.

Options: Mango | Lotus | Lily | Rose

Q1547 ? Combined statements ? Easy

A odd shape case uses the quantities in this question. Determine the requested result using the combined statements model: Find the odd one: January, March, July, Monday.

Options: Monday | July | March | January

Q1548 ? Counting restriction ? Medium

A odd category case uses the quantities in this question. Determine the requested result using the counting restriction model: Find the odd one: 2, 4, 6, 9.

Options: 2 | 9 | 4 | 6

Q1549 ? Circular arrangement ? Hard

A odd process case uses the quantities in this question. Determine the requested result using the circular arrangement model: Find the odd one: Triangle, Square, Circle, Cube.

Options: Cube | Circle | Square | Triangle

Q1550 ? Complement method ? Placement

A odd relation case uses the quantities in this question. Determine the requested result using the complement method model: Find the odd one: Rose, Lily, Lotus, Mango.

Options: Rose | Lily | Lotus | Mango

Q1551 ? Pattern: differences ? Beginner

A odd symbol case uses the quantities in this question. Determine the requested result using the pattern: differences model: Find the odd one: January, March, July, Monday.

Options: March | Monday | January | July

Q1552 ? Pattern: ratios ? Easy

A odd object case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Find the odd one: 2, 4, 6, 9.

Options: 2 | 4 | 6 | 9

Q1553 ? Pattern: alternating ? Medium

A odd number case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Find the odd one: Triangle, Square, Circle, Cube.

Options: Square | Triangle | Circle | Cube

Q1554 ? Pattern: squares / cubes ? Hard

A odd word case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Find the odd one: Rose, Lily, Lotus, Mango.

Options: Rose | Lily | Lotus | Mango

Q1555 ? Inference from facts ? Placement

A odd shape case uses the quantities in this question. Determine the requested result using the inference from facts model: Find the odd one: January, March, July, Monday.

Options: March | Monday | January | July

Q1556 ? Assumption check ? Beginner

A odd category case uses the quantities in this question. Determine the requested result using the assumption check model: Find the odd one: 2, 4, 6, 9.

Options: 2 | 4 | 6 | 9

Q1557 ? Cause vs correlation ? Easy

A odd process case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Find the odd one: Triangle, Square, Circle, Cube.

Options: Triangle | Square | Circle | Cube

Q1558 ? Grammar agreement ? Medium

A odd relation case uses the quantities in this question. Determine the requested result using the grammar agreement model: Find the odd one: Rose, Lily, Lotus, Mango.

Options: Lily | Lotus | Mango | Rose

Q1559 ? Vocabulary in context ? Hard

A odd symbol case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Find the odd one: January, March, July, Monday.

Options: July | January | Monday | March

Q1560 ? Mixed placement challenge ? Placement

A odd object case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Find the odd one: 2, 4, 6, 9.

Options: 4 | 9 | 2 | 6

Ranking ? 40 questions

WORKED EXAMPLES

Q1561 ? Direct computation ? Beginner

A leaderboard case uses the quantities in this question. Determine the requested result using the direct computation model: Among 31 students, A is 6th from top. Rank from bottom?

Options: 32 | 23 | 29 | 26

Answer: 26

Solution: Context: leaderboard. $31 - 6 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1562 ? Reverse value ? Easy

A exam rank case uses the quantities in this question. Determine the requested result using the reverse value model: Among 32 students, A is 7th from top. Rank from bottom?

Options: 32 | 23 | 29 | 26

Answer: 26

Solution: Context: exam rank. $32 - 7 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1563 ? Missing base ? Medium

A race finish case uses the quantities in this question. Determine the requested result using the missing base model: Among 33 students, A is 8th from top. Rank from bottom?

Options: 26 | 29 | 23 | 32

Answer: 26

Solution: Context: race finish. $33 - 8 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1564 ? Successive change ? Hard

A salary order case uses the quantities in this question. Determine the requested result using the successive change model: Among 34 students, A is 9th from top. Rank from bottom?

Options: 26 | 29 | 23 | 32

Answer: 26

Solution: Context: salary order. $34 - 9 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1565 ? Comparison trap ? Placement

A queue position case uses the quantities in this question. Determine the requested result using the comparison trap model: Among 35 students, A is 10th from top. Rank from bottom?

Options: 32 | 23 | 29 | 26

Answer: 26

Solution: Context: queue position. $35 - 10 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1566 ? Ratio form ? Beginner

A height order case uses the quantities in this question. Determine the requested result using the ratio form model: Among 36 students, A is 11th from top. Rank from bottom?

Options: 26 | 29 | 23 | 32

Answer: 26

Solution: Context: height order. $36 - 11 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1567 ? Weighted result ? Easy

A performance list case uses the quantities in this question. Determine the requested result using the weighted result model: Among 37 students, A is 12th from top. Rank from bottom?

Options: 26 | 29 | 23 | 32

Answer: 26

Solution: Context: performance list. $37 - 12 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1568 ? Replacement cycle ? Medium

A selection order case uses the quantities in this question. Determine the requested result using the replacement cycle model: Among 38 students, A is 13th from top. Rank from bottom?

Options: 32 | 23 | 29 | 26

Answer: 26

Solution: Context: selection order. $38 - 13 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1569 ? Unit conversion ? Hard

A leaderboard case uses the quantities in this question. Determine the requested result using the unit conversion model: Among 39 students, A is 14th from top. Rank from bottom?

Options: 32 | 23 | 29 | 26

Answer: 26

Solution: Context: leaderboard. $39 - 14 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1570 ? Work-rate model ? Placement

A exam rank case uses the quantities in this question. Determine the requested result using the work-rate model model: Among 40 students, A is 15th from top. Rank from bottom?

Options: 26 | 29 | 23 | 32

Answer: 26

Solution: Context: exam rank. $40 - 15 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1571 ? Speed-time model ? Beginner

A race finish case uses the quantities in this question. Determine the requested result using the speed-time model model: Among 41 students, A is 16th from top. Rank from bottom?

Options: 26 | 29 | 23 | 32

Answer: 26

Solution: Context: race finish. $41 - 16 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1572 ? Table lookup ? Easy

A salary order case uses the quantities in this question. Determine the requested result using the table lookup model: Among 42 students, A is 17th from top. Rank from bottom?

Options: 32 | 23 | 29 | 26

Answer: 26

Solution: Context: salary order. $42 - 17 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1573 ? Chart comparison ? Medium

A queue position case uses the quantities in this question. Determine the requested result using the chart comparison model: Among 43 students, A is 18th from top. Rank from bottom?

Options: 26 | 29 | 23 | 32

Answer: 26

Solution: Context: queue position. $43 - 18 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1574 ? Multi-step word problem ? Hard

A height order case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Among 44 students, A is 19th from top. Rank from bottom?

Options: 26 | 29 | 23 | 32

Answer: 26

Solution: Context: height order. $44 - 19 + 1 = 26$. Shortcut: Opposite rank = total - rank + 1.

Q1575 ? Maximum / minimum ? Placement

A performance list case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Among 45 students, A is 5th from top. Rank from bottom?

Options: 41 | 45 | 37 | 49

Answer: 41

Solution: Context: performance list. $45 - 5 + 1 = 41$. Shortcut: Opposite rank = total - rank + 1.

Q1576 ? Least / greatest value ? Beginner

A selection order case uses the quantities in this question. Determine the requested result using the least / greatest value model: Among 46 students, A is 6th from top. Rank from bottom?

Options: 49 | 37 | 45 | 41

Answer: 41

Solution: Context: selection order. $46 - 6 + 1 = 41$. Shortcut: Opposite rank = total - rank + 1.

Q1577 ? Remainder cycle ? Easy

A leaderboard case uses the quantities in this question. Determine the requested result using the remainder cycle model: Among 47 students, A is 7th from top. Rank from bottom?

Options: 41 | 45 | 37 | 49

Answer: 41

Solution: Context: leaderboard. $47 - 7 + 1 = 41$. Shortcut: Opposite rank = total - rank + 1.

Q1578 ? Divisibility test ? Medium

A exam rank case uses the quantities in this question. Determine the requested result using the divisibility test model: Among 48 students, A is 8th from top. Rank from bottom?

Options: 37 | 49 | 45 | 41

Answer: 41

Solution: Context: exam rank. $48 - 8 + 1 = 41$. Shortcut: Opposite rank = total - rank + 1.

Q1579 ? Factorisation ? Hard

A race finish case uses the quantities in this question. Determine the requested result using the factorisation model: Among 49 students, A is 9th from top. Rank from bottom?

Options: 41 | 45 | 49 | 37

Answer: 41

Solution: Context: race finish. $49 - 9 + 1 = 41$. Shortcut: Opposite rank = total - rank + 1.

Q1580 ? Approximation ? Placement

A salary order case uses the quantities in this question. Determine the requested result using the approximation model: Among 50 students, A is 10th from top. Rank from bottom?

Options: 49 | 45 | 41 | 37

Answer: 41

Solution: Context: salary order. $50 - 10 + 1 = 41$. Shortcut: Opposite rank = total - rank + 1.

PRACTICE ? solutions hidden below**Q1581 ? Options elimination ? Beginner**

A queue position case uses the quantities in this question. Determine the requested result using the options elimination model: Among 51 students, A is 11th

from top. Rank from bottom?
Options: 37 | 41 | 45 | 49

Q1582 ? Boundary case ? Easy

A height order case uses the quantities in this question. Determine the requested result using the boundary case model: Among 52 students, A is 12th from top. Rank from bottom?
Options: 41 | 45 | 37 | 49

Q1583 ? Case split ? Medium

A performance list case uses the quantities in this question. Determine the requested result using the case split model: Among 53 students, A is 13th from top. Rank from bottom?
Options: 49 | 37 | 45 | 41

Q1584 ? Contradiction check ? Hard

A selection order case uses the quantities in this question. Determine the requested result using the contradiction check model: Among 54 students, A is 14th from top. Rank from bottom?
Options: 41 | 45 | 37 | 49

Q1585 ? Statement I test ? Placement

A leaderboard case uses the quantities in this question. Determine the requested result using the statement i test model: Among 55 students, A is 15th from top. Rank from bottom?
Options: 37 | 49 | 45 | 41

Q1586 ? Statement II test ? Beginner

A exam rank case uses the quantities in this question. Determine the requested result using the statement ii test model: Among 56 students, A is 16th from top. Rank from bottom?
Options: 41 | 45 | 49 | 37

Q1587 ? Combined statements ? Easy

A race finish case uses the quantities in this question. Determine the requested result using the combined statements model: Among 57 students, A is 17th from top. Rank from bottom?
Options: 49 | 45 | 41 | 37

Q1588 ? Counting restriction ? Medium

A salary order case uses the quantities in this question. Determine the requested result using the counting restriction model: Among 58 students, A is 18th from top. Rank from bottom?
Options: 37 | 41 | 45 | 49

Q1589 ? Circular arrangement ? Hard

A queue position case uses the quantities in this question. Determine the requested result using the circular arrangement model: Among 59 students, A is 19th from top. Rank from bottom?
Options: 41 | 45 | 37 | 49

Q1590 ? Complement method ? Placement

A height order case uses the quantities in this question. Determine the requested result using the complement method model: Among 60 students, A is 5th from top. Rank from bottom?
Options: 56 | 62 | 50 | 68

Q1591 ? Pattern: differences ? Beginner

A performance list case uses the quantities in this question. Determine the requested result using the pattern: differences model: Among 61 students, A is 6th from top. Rank from bottom?
Options: 68 | 62 | 56 | 50

Q1592 ? Pattern: ratios ? Easy

A selection order case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Among 62 students, A is 7th from top. Rank from bottom?
Options: 50 | 56 | 68 | 62

Q1593 ? Pattern: alternating ? Medium

A leaderboard case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Among 63 students, A is 8th from top. Rank from bottom?
Options: 56 | 62 | 50 | 68

Q1594 ? Pattern: squares / cubes ? Hard

A exam rank case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Among 64 students, A is 9th from top. Rank from bottom?
Options: 62 | 68 | 50 | 56

Q1595 ? Inference from facts ? Placement

A race finish case uses the quantities in this question. Determine the requested result using the inference from facts model: Among 65 students, A is 10th from top. Rank from bottom?
Options: 56 | 50 | 62 | 68

Q1596 ? Assumption check ? Beginner

A salary order case uses the quantities in this question. Determine the requested result using the assumption check model: Among 66 students, A is 11th from top. Rank from bottom?
Options: 68 | 50 | 62 | 56

Q1597 ? Cause vs correlation ? Easy

A queue position case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Among 67 students, A is 12th from top. Rank from bottom?

Options: 56 | 62 | 50 | 68

Q1598 ? Grammar agreement ? Medium

A height order case uses the quantities in this question. Determine the requested result using the grammar agreement model: Among 68 students, A is 13th from top. Rank from bottom?

Options: 68 | 62 | 56 | 50

Q1599 ? Vocabulary in context ? Hard

A performance list case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Among 69 students, A is 14th from top. Rank from bottom?

Options: 50 | 56 | 68 | 62

Q1600 ? Mixed placement challenge ? Placement

A selection order case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Among 70 students, A is 15th from top. Rank from bottom?

Options: 56 | 62 | 50 | 68

Seating Arrangement ? 40 questions

WORKED EXAMPLES

Q1601 ? Direct computation ? Beginner

A round table case uses the quantities in this question. Determine the requested result using the direct computation model: How many circular arrangements of 5 distinct people?

Options: 28 | 22 | 26 | 24

Answer: 24

Solution: Context: round table. Circular arrangements= $(5-1)!=24$. Shortcut: Arrange the remaining $n-1$ people.

Q1602 ? Reverse value ? Easy

A theatre row case uses the quantities in this question. Determine the requested result using the reverse value model: How many circular arrangements of 6 distinct people?

Options: 120 | 132 | 108 | 144

Answer: 120

Solution: Context: theatre row. Circular arrangements= $(6-1)!=120$. Shortcut: Arrange the remaining $n-1$ people.

Q1603 ? Missing base ? Medium

A office desks case uses the quantities in this question. Determine the requested result using the missing base model: How many circular arrangements of 7 distinct people?

Options: 720 | 792 | 864 | 648

Answer: 720

Solution: Context: office desks. Circular arrangements= $(7-1)!=720$. Shortcut: Arrange the remaining $n-1$ people.

Q1604 ? Successive change ? Hard

A conference table case uses the quantities in this question. Determine the requested result using the successive change model: How many circular arrangements of 8 distinct people?

Options: 4536 | 5544 | 5040 | 6048

Answer: 5040

Solution: Context: conference table. Circular arrangements= $(8-1)!=5040$. Shortcut: Arrange the remaining $n-1$ people.

Q1605 ? Comparison trap ? Placement

A circular committee case uses the quantities in this question. Determine the requested result using the comparison trap model: How many circular arrangements of 4 distinct people?

Options: 8 | 7 | 6 | 5

Answer: 6

Solution: Context: circular committee. Circular arrangements= $(4-1)!=6$. Shortcut: Arrange the remaining $n-1$ people.

Q1606 ? Ratio form ? Beginner

A bench arrangement case uses the quantities in this question. Determine the requested result using the ratio form model: How many circular arrangements of 5 distinct people?

Options: 24 | 26 | 22 | 28

Answer: 24

Solution: Context: bench arrangement. Circular arrangements= $(5-1)!=24$. Shortcut: Arrange the remaining $n-1$ people.

Q1607 ? Weighted result ? Easy

A classroom seats case uses the quantities in this question. Determine the requested result using the weighted result model: How many circular arrangements of 6 distinct people?

Options: 120 | 132 | 108 | 144

Answer: 120

Solution: Context: classroom seats. Circular arrangements= $(6-1)!=120$. Shortcut: Arrange the remaining $n-1$ people.

Q1608 ? Replacement cycle ? Medium

A dinner seating case uses the quantities in this question. Determine the requested result using the replacement cycle model: How many circular arrangements of 7 distinct people?

Options: 648 | 864 | 792 | 720

Answer: 720

Solution: Context: dinner seating. Circular arrangements= $(7-1)!=720$. Shortcut: Arrange the remaining $n-1$ people.

Q1609 ? Unit conversion ? Hard

A round table case uses the quantities in this question. Determine the requested result using the unit conversion model: How many circular arrangements of 8 distinct people?

Options: 5040 | 5544 | 6048 | 4536

Answer: 5040

Solution: Context: round table. Circular arrangements= $(8-1)!=5040$. Shortcut: Arrange the remaining $n-1$ people.

Q1610 ? Work-rate model ? Placement

A theatre row case uses the quantities in this question. Determine the requested result using the work-rate model model: How many circular arrangements of 4 distinct people?

Options: 7 | 8 | 6 | 5

Answer: 6

Solution: Context: theatre row. Circular arrangements= $(4-1)!=6$. Shortcut: Arrange the remaining $n-1$ people.

Q1611 ? Speed-time model ? Beginner

A office desks case uses the quantities in this question. Determine the requested result using the speed-time model model: How many circular arrangements of 5 distinct people?

Options: 24 | 26 | 22 | 28

Answer: 24

Solution: Context: office desks. Circular arrangements= $(5-1)!=24$. Shortcut: Arrange the remaining $n-1$ people.

Q1612 ? Table lookup ? Easy

A conference table case uses the quantities in this question. Determine the requested result using the table lookup model: How many circular arrangements of 6 distinct people?

Options: 144 | 108 | 132 | 120

Answer: 120

Solution: Context: conference table. Circular arrangements= $(6-1)!=120$. Shortcut: Arrange the remaining $n-1$ people.

Q1613 ? Chart comparison ? Medium

A circular committee case uses the quantities in this question. Determine the requested result using the chart comparison model: How many circular arrangements of 7 distinct people?

Options: 720 | 792 | 648 | 864

Answer: 720

Solution: Context: circular committee. Circular arrangements= $(7-1)!=720$. Shortcut: Arrange the remaining $n-1$ people.

Q1614 ? Multi-step word problem ? Hard

A bench arrangement case uses the quantities in this question. Determine the requested result using the multi-step word problem model: How many circular arrangements of 8 distinct people?

Options: 4536 | 5040 | 5544 | 6048

Answer: 5040

Solution: Context: bench arrangement. Circular arrangements= $(8-1)!=5040$. Shortcut: Arrange the remaining $n-1$ people.

Q1615 ? Maximum / minimum ? Placement

A classroom seats case uses the quantities in this question. Determine the requested result using the maximum / minimum model: How many circular arrangements of 4 distinct people?

Options: 5 | 8 | 7 | 6

Answer: 6

Solution: Context: classroom seats. Circular arrangements= $(4-1)!=6$. Shortcut: Arrange the remaining $n-1$ people.

Q1616 ? Least / greatest value ? Beginner

A dinner seating case uses the quantities in this question. Determine the requested result using the least / greatest value model: How many circular arrangements of 5 distinct people?

Options: 28 | 22 | 26 | 24

Answer: 24

Solution: Context: dinner seating. Circular arrangements= $(5-1)!=24$. Shortcut: Arrange the remaining $n-1$ people.

Q1617 ? Remainder cycle ? Easy

A round table case uses the quantities in this question. Determine the requested result using the remainder cycle model: How many circular arrangements of 6 distinct people?

Options: 120 | 132 | 108 | 144

Answer: 120

Solution: Context: round table. Circular arrangements= $(6-1)!=120$. Shortcut: Arrange the remaining $n-1$ people.

Q1618 ? Divisibility test ? Medium

A theatre row case uses the quantities in this question. Determine the requested result using the divisibility test model: How many circular arrangements of 7 distinct people?

Options: 648 | 792 | 720 | 864

Answer: 720

Solution: Context: theatre row. Circular arrangements= $(7-1)!=720$. Shortcut: Arrange the remaining $n-1$ people.

Q1619 ? Factorisation ? Hard

A office desks case uses the quantities in this question. Determine the requested result using the factorisation model: How many circular arrangements of 8 distinct people?

Options: 5040 | 5544 | 4536 | 6048

Answer: 5040

Solution: Context: office desks. Circular arrangements= $(8-1)!=5040$. Shortcut: Arrange the remaining $n-1$ people.

Q1620 ? Approximation ? Placement

A conference table case uses the quantities in this question. Determine the requested result using the approximation model: How many circular arrangements of 4 distinct people?

Options: 6 | 7 | 8 | 5

Answer: 6

Solution: Context: conference table. Circular arrangements= $(4-1)!=6$. Shortcut: Arrange the remaining $n-1$ people.

PRACTICE ? solutions hidden below**Q1621 ? Options elimination ? Beginner**

A circular committee case uses the quantities in this question. Determine the requested result using the options elimination model: How many circular

arrangements of 5 distinct people?
Options: 24 | 26 | 22 | 28

Q1622 ? Boundary case ? Easy

A bench arrangement case uses the quantities in this question. Determine the requested result using the boundary case model: How many circular arrangements of 6 distinct people?
Options: 144 | 108 | 132 | 120

Q1623 ? Case split ? Medium

A classroom seats case uses the quantities in this question. Determine the requested result using the case split model: How many circular arrangements of 7 distinct people?
Options: 720 | 792 | 648 | 864

Q1624 ? Contradiction check ? Hard

A dinner seating case uses the quantities in this question. Determine the requested result using the contradiction check model: How many circular arrangements of 8 distinct people?
Options: 6048 | 5040 | 5544 | 4536

Q1625 ? Statement I test ? Placement

A round table case uses the quantities in this question. Determine the requested result using the statement i test model: How many circular arrangements of 4 distinct people?
Options: 5 | 6 | 7 | 8

Q1626 ? Statement II test ? Beginner

A theatre row case uses the quantities in this question. Determine the requested result using the statement ii test model: How many circular arrangements of 5 distinct people?
Options: 28 | 22 | 26 | 24

Q1627 ? Combined statements ? Easy

A office desks case uses the quantities in this question. Determine the requested result using the combined statements model: How many circular arrangements of 6 distinct people?
Options: 120 | 132 | 108 | 144

Q1628 ? Counting restriction ? Medium

A conference table case uses the quantities in this question. Determine the requested result using the counting restriction model: How many circular arrangements of 7 distinct people?
Options: 864 | 648 | 720 | 792

Q1629 ? Circular arrangement ? Hard

A circular committee case uses the quantities in this question. Determine the requested result using the circular arrangement model: How many circular arrangements of 8 distinct people?
Options: 6048 | 5544 | 5040 | 4536

Q1630 ? Complement method ? Placement

A bench arrangement case uses the quantities in this question. Determine the requested result using the complement method model: How many circular arrangements of 4 distinct people?
Options: 6 | 7 | 5 | 8

Q1631 ? Pattern: differences ? Beginner

A classroom seats case uses the quantities in this question. Determine the requested result using the pattern: differences model: How many circular arrangements of 5 distinct people?
Options: 24 | 26 | 22 | 28

Q1632 ? Pattern: ratios ? Easy

A dinner seating case uses the quantities in this question. Determine the requested result using the pattern: ratios model: How many circular arrangements of 6 distinct people?
Options: 144 | 108 | 132 | 120

Q1633 ? Pattern: alternating ? Medium

A round table case uses the quantities in this question. Determine the requested result using the pattern: alternating model: How many circular arrangements of 7 distinct people?
Options: 864 | 792 | 720 | 648

Q1634 ? Pattern: squares / cubes ? Hard

A theatre row case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: How many circular arrangements of 8 distinct people?
Options: 6048 | 4536 | 5544 | 5040

Q1635 ? Inference from facts ? Placement

A office desks case uses the quantities in this question. Determine the requested result using the inference from facts model: How many circular arrangements of 4 distinct people?
Options: 8 | 5 | 6 | 7

Q1636 ? Assumption check ? Beginner

A conference table case uses the quantities in this question. Determine the requested result using the assumption check model: How many circular arrangements of 5 distinct people?
Options: 28 | 22 | 26 | 24

Q1637 ? Cause vs correlation ? Easy

A circular committee case uses the quantities in this question. Determine the requested result using the cause vs correlation model: How many circular arrangements of 6 distinct people?

Options: 120 | 132 | 108 | 144

Q1638 ? Grammar agreement ? Medium

A bench arrangement case uses the quantities in this question. Determine the requested result using the grammar agreement model: How many circular arrangements of 7 distinct people?

Options: 720 | 792 | 864 | 648

Q1639 ? Vocabulary in context ? Hard

A classroom seats case uses the quantities in this question. Determine the requested result using the vocabulary in context model: How many circular arrangements of 8 distinct people?

Options: 4536 | 5544 | 5040 | 6048

Q1640 ? Mixed placement challenge ? Placement

A dinner seating case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: How many circular arrangements of 4 distinct people?

Options: 8 | 7 | 6 | 5

Puzzle Solving ? 40 questions

WORKED EXAMPLES

Q1641 ? Direct computation ? Beginner

A floor puzzle case uses the quantities in this question. Determine the requested result using the direct computation model: Complete: 3, 7, 11, 15, ?

Options: 23 | 17 | 21 | 19

Answer: 19

Solution: Context: floor puzzle. Common increase=4; next=19. Shortcut: Test the simplest consistent pattern first.

Q1642 ? Reverse value ? Easy

A box allocation case uses the quantities in this question. Determine the requested result using the reverse value model: Complete: 4, 9, 14, 19, ?

Options: 28 | 22 | 26 | 24

Answer: 24

Solution: Context: box allocation. Common increase=5; next=24. Shortcut: Test the simplest consistent pattern first.

Q1643 ? Missing base ? Medium

A day scheduling case uses the quantities in this question. Determine the requested result using the missing base model: Complete: 5, 11, 17, 23, ?

Options: 29 | 32 | 26 | 35

Answer: 29

Solution: Context: day scheduling. Common increase=6; next=29. Shortcut: Test the simplest consistent pattern first.

Q1644 ? Successive change ? Hard

A team assignment case uses the quantities in this question. Determine the requested result using the successive change model: Complete: 6, 13, 20, 27, ?

Options: 31 | 37 | 34 | 40

Answer: 34

Solution: Context: team assignment. Common increase=7; next=34. Shortcut: Test the simplest consistent pattern first.

Q1645 ? Comparison trap ? Placement

A room distribution case uses the quantities in this question. Determine the requested result using the comparison trap model: Complete: 7, 10, 13, 16, ?

Options: 23 | 17 | 21 | 19

Answer: 19

Solution: Context: room distribution. Common increase=3; next=19. Shortcut: Test the simplest consistent pattern first.

Q1646 ? Ratio form ? Beginner

A logic grid case uses the quantities in this question. Determine the requested result using the ratio form model: Complete: 8, 12, 16, 20, ?

Options: 24 | 26 | 22 | 28

Answer: 24

Solution: Context: logic grid. Common increase=4; next=24. Shortcut: Test the simplest consistent pattern first.

Q1647 ? Weighted result ? Easy

A interview order case uses the quantities in this question. Determine the requested result using the weighted result model: Complete: 9, 14, 19, 24, ?

Options: 29 | 26 | 32 | 35

Answer: 29

Solution: Context: interview order. Common increase=5; next=29. Shortcut: Test the simplest consistent pattern first.

Q1648 ? Replacement cycle ? Medium

A constraint puzzle case uses the quantities in this question. Determine the requested result using the replacement cycle model: Complete: 10, 16, 22, 28, ?

Options: 34 | 37 | 31 | 40

Answer: 34

Solution: Context: constraint puzzle. Common increase=6; next=34. Shortcut: Test the simplest consistent pattern first.

Q1649 ? Unit conversion ? Hard

A floor puzzle case uses the quantities in this question. Determine the requested result using the unit conversion model: Complete: 11, 18, 25, 32, ?

Options: 47 | 35 | 43 | 39

Answer: 39

Solution: Context: floor puzzle. Common increase=7; next=39. Shortcut: Test the simplest consistent pattern first.

Q1650 ? Work-rate model ? Placement

A box allocation case uses the quantities in this question. Determine the requested result using the work-rate model model: Complete: 12, 15, 18, 21, ?

Options: 24 | 26 | 22 | 28

Answer: 24

Solution: Context: box allocation. Common increase=3; next=24. Shortcut: Test the simplest consistent pattern first.

Q1651 ? Speed-time model ? Beginner

A day scheduling case uses the quantities in this question. Determine the requested result using the speed-time model model: Complete: 13, 17, 21, 25, ?

Options: 29 | 32 | 26 | 35

Answer: 29

Solution: Context: day scheduling. Common increase=4; next=29. Shortcut: Test the simplest consistent pattern first.

Q1652 ? Table lookup ? Easy

A team assignment case uses the quantities in this question. Determine the requested result using the table lookup model: Complete: 14, 19, 24, 29, ?

Options: 40 | 31 | 37 | 34

Answer: 34

Solution: Context: team assignment. Common increase=5; next=34. Shortcut: Test the simplest consistent pattern first.

Q1653 ? Chart comparison ? Medium

A room distribution case uses the quantities in this question. Determine the requested result using the chart comparison model: Complete: 15, 21, 27, 33, ?
Options: 47 | 43 | 39 | 35

Answer: 39

Solution: Context: room distribution. Common increase=6; next=39. Shortcut: Test the simplest consistent pattern first.

Q1654 ? Multi-step word problem ? Hard

A logic grid case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Complete: 16, 23, 30, 37, ?
Options: 44 | 48 | 40 | 52

Answer: 44

Solution: Context: logic grid. Common increase=7; next=44. Shortcut: Test the simplest consistent pattern first.

Q1655 ? Maximum / minimum ? Placement

A interview order case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Complete: 17, 20, 23, 26, ?
Options: 32 | 35 | 26 | 29

Answer: 29

Solution: Context: interview order. Common increase=3; next=29. Shortcut: Test the simplest consistent pattern first.

Q1656 ? Least / greatest value ? Beginner

A constraint puzzle case uses the quantities in this question. Determine the requested result using the least / greatest value model: Complete: 18, 22, 26, 30, ?
Options: 40 | 31 | 37 | 34

Answer: 34

Solution: Context: constraint puzzle. Common increase=4; next=34. Shortcut: Test the simplest consistent pattern first.

Q1657 ? Remainder cycle ? Easy

A floor puzzle case uses the quantities in this question. Determine the requested result using the remainder cycle model: Complete: 19, 24, 29, 34, ?
Options: 39 | 43 | 35 | 47

Answer: 39

Solution: Context: floor puzzle. Common increase=5; next=39. Shortcut: Test the simplest consistent pattern first.

Q1658 ? Divisibility test ? Medium

A box allocation case uses the quantities in this question. Determine the requested result using the divisibility test model: Complete: 20, 26, 32, 38, ?
Options: 40 | 48 | 44 | 52

Answer: 44

Solution: Context: box allocation. Common increase=6; next=44. Shortcut: Test the simplest consistent pattern first.

Q1659 ? Factorisation ? Hard

A day scheduling case uses the quantities in this question. Determine the requested result using the factorisation model: Complete: 21, 28, 35, 42, ?
Options: 49 | 54 | 44 | 59

Answer: 49

Solution: Context: day scheduling. Common increase=7; next=49. Shortcut: Test the simplest consistent pattern first.

Q1660 ? Approximation ? Placement

A team assignment case uses the quantities in this question. Determine the requested result using the approximation model: Complete: 22, 25, 28, 31, ?
Options: 40 | 34 | 37 | 31

Answer: 34

Solution: Context: team assignment. Common increase=3; next=34. Shortcut: Test the simplest consistent pattern first.

PRACTICE ? solutions hidden below

Q1661 ? Options elimination ? Beginner

A room distribution case uses the quantities in this question. Determine the requested result using the options elimination model: Complete: 23, 27, 31, 35, ?
Options: 39 | 35 | 43 | 47

Q1662 ? Boundary case ? Easy

A logic grid case uses the quantities in this question. Determine the requested result using the boundary case model: Complete: 24, 29, 34, 39, ?
Options: 52 | 44 | 48 | 40

Q1663 ? Case split ? Medium

A interview order case uses the quantities in this question. Determine the requested result using the case split model: Complete: 25, 31, 37, 43, ?
Options: 54 | 59 | 44 | 49

Q1664 ? Contradiction check ? Hard

A constraint puzzle case uses the quantities in this question. Determine the requested result using the contradiction check model: Complete: 26, 33, 40, 47, ?
Options: 64 | 54 | 59 | 49

Q1665 ? Statement I test ? Placement

A floor puzzle case uses the quantities in this question. Determine the requested result using the statement i test model: Complete: 27, 30, 33, 36, ?
Options: 35 | 39 | 47 | 43

Q1666 ? Statement II test ? Beginner

A box allocation case uses the quantities in this question. Determine the requested result using the statement ii test model: Complete: 28, 32, 36, 40, ?

Options: 44 | 48 | 40 | 52

Q1667 ? Combined statements ? Easy

A day scheduling case uses the quantities in this question. Determine the requested result using the combined statements model: Complete: 29, 34, 39, 44, ?
Options: 59 | 44 | 54 | 49

Q1668 ? Counting restriction ? Medium

A team assignment case uses the quantities in this question. Determine the requested result using the counting restriction model: Complete: 30, 36, 42, 48, ?
Options: 49 | 54 | 59 | 64

Q1669 ? Circular arrangement ? Hard

A room distribution case uses the quantities in this question. Determine the requested result using the circular arrangement model: Complete: 31, 38, 45, 52, ?
Options: 53 | 71 | 65 | 59

Q1670 ? Complement method ? Placement

A logic grid case uses the quantities in this question. Determine the requested result using the complement method model: Complete: 32, 35, 38, 41, ?
Options: 52 | 40 | 48 | 44

Q1671 ? Pattern: differences ? Beginner

An interview order case uses the quantities in this question. Determine the requested result using the pattern: differences model: Complete: 33, 37, 41, 45, ?
Options: 49 | 54 | 44 | 59

Q1672 ? Pattern: ratios ? Easy

A constraint puzzle case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Complete: 34, 39, 44, 49, ?
Options: 49 | 59 | 54 | 64

Q1673 ? Pattern: alternating ? Medium

A floor puzzle case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Complete: 35, 41, 47, 53, ?
Options: 71 | 59 | 65 | 53

Q1674 ? Pattern: squares / cubes ? Hard

A box allocation case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Complete: 36, 43, 50, 57, ?
Options: 64 | 70 | 76 | 58

Q1675 ? Inference from facts ? Placement

A day scheduling case uses the quantities in this question. Determine the requested result using the inference from facts model: Complete: 37, 40, 43, 46, ?
Options: 49 | 44 | 54 | 59

Q1676 ? Assumption check ? Beginner

A team assignment case uses the quantities in this question. Determine the requested result using the assumption check model: Complete: 38, 42, 46, 50, ?
Options: 64 | 59 | 54 | 49

Q1677 ? Cause vs correlation ? Easy

A room distribution case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Complete: 39, 44, 49, 54, ?
Options: 59 | 65 | 53 | 71

Q1678 ? Grammar agreement ? Medium

A logic grid case uses the quantities in this question. Determine the requested result using the grammar agreement model: Complete: 40, 46, 52, 58, ?
Options: 70 | 76 | 64 | 58

Q1679 ? Vocabulary in context ? Hard

An interview order case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Complete: 41, 48, 55, 62, ?
Options: 62 | 69 | 76 | 83

Q1680 ? Mixed placement challenge ? Placement

A constraint puzzle case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Complete: 42, 45, 48, 51, ?
Options: 54 | 59 | 49 | 64

Logical Reasoning ? 40 questions

WORKED EXAMPLES

Q1681 ? Direct computation ? Beginner

A syllogism case uses the quantities in this question. Determine the requested result using the direct computation model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | All analysts are engineers | No analyst is a graduate

Answer: Some analysts are graduates

Solution: Context: syllogism. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1682 ? Reverse value ? Easy

A statement conclusion case uses the quantities in this question. Determine the requested result using the reverse value model: All engineers are graduates. Some analysts are engineers. What follows?

Options: All analysts are engineers | No analyst is a graduate | All graduates are analysts | Some analysts are graduates

Answer: Some analysts are graduates

Solution: Context: statement conclusion. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1683 ? Missing base ? Medium

A assumption test case uses the quantities in this question. Determine the requested result using the missing base model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | No analyst is a graduate | All analysts are engineers

Answer: Some analysts are graduates

Solution: Context: assumption test. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1684 ? Successive change ? Hard

A cause-effect case uses the quantities in this question. Determine the requested result using the successive change model: All engineers are graduates. Some analysts are engineers. What follows?

Options: All analysts are engineers | All graduates are analysts | Some analysts are graduates | No analyst is a graduate

Answer: Some analysts are graduates

Solution: Context: cause-effect. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1685 ? Comparison trap ? Placement

A decision rule case uses the quantities in this question. Determine the requested result using the comparison trap model: All engineers are graduates. Some analysts are engineers. What follows?

Options: No analyst is a graduate | Some analysts are graduates | All analysts are engineers | All graduates are analysts

Answer: Some analysts are graduates

Solution: Context: decision rule. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1686 ? Ratio form ? Beginner

A course of action case uses the quantities in this question. Determine the requested result using the ratio form model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | No analyst is a graduate | All analysts are engineers

Answer: Some analysts are graduates

Solution: Context: course of action. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1687 ? Weighted result ? Easy

A inference chain case uses the quantities in this question. Determine the requested result using the weighted result model: All engineers are graduates. Some analysts are engineers. What follows?

Options: No analyst is a graduate | All graduates are analysts | All analysts are engineers | Some analysts are graduates

Answer: Some analysts are graduates

Solution: Context: inference chain. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1688 ? Replacement cycle ? Medium

A propositional logic case uses the quantities in this question. Determine the requested result using the replacement cycle model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | All analysts are engineers | No analyst is a graduate

Answer: Some analysts are graduates

Solution: Context: propositional logic. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1689 ? Unit conversion ? Hard

A syllogism case uses the quantities in this question. Determine the requested result using the unit conversion model: All engineers are graduates. Some analysts are engineers. What follows?

Options: All analysts are engineers | No analyst is a graduate | All graduates are analysts | Some analysts are graduates

Answer: Some analysts are graduates

Solution: Context: syllogism. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1690 ? Work-rate model ? Placement

A statement conclusion case uses the quantities in this question. Determine the requested result using the work-rate model model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | No analyst is a graduate | All analysts are engineers

Answer: Some analysts are graduates

Solution: Context: statement conclusion. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1691 ? Speed-time model ? Beginner

A assumption test case uses the quantities in this question. Determine the requested result using the speed-time model model: All engineers are graduates. Some analysts are engineers. What follows?

Options: All analysts are engineers | All graduates are analysts | Some analysts are graduates | No analyst is a graduate

Answer: Some analysts are graduates

Solution: Context: assumption test. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1692 ? Table lookup ? Easy

A cause-effect case uses the quantities in this question. Determine the requested result using the table lookup model: All engineers are graduates. Some analysts are engineers. What follows?

Options: No analyst is a graduate | Some analysts are graduates | All analysts are engineers | All graduates are analysts

Answer: Some analysts are graduates

Solution: Context: cause-effect. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1693 ? Chart comparison ? Medium

A decision rule case uses the quantities in this question. Determine the requested result using the chart comparison model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | No analyst is a graduate | All analysts are engineers

Answer: Some analysts are graduates

Solution: Context: decision rule. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1694 ? Multi-step word problem ? Hard

A course of action case uses the quantities in this question. Determine the requested result using the multi-step word problem model: All engineers are graduates. Some analysts are engineers. What follows?

Options: No analyst is a graduate | All graduates are analysts | All analysts are engineers | Some analysts are graduates

Answer: Some analysts are graduates

Solution: Context: course of action. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1695 ? Maximum / minimum ? Placement

A inference chain case uses the quantities in this question. Determine the requested result using the maximum / minimum model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | All analysts are engineers | No analyst is a graduate

Answer: Some analysts are graduates

Solution: Context: inference chain. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1696 ? Least / greatest value ? Beginner

A propositional logic case uses the quantities in this question. Determine the requested result using the least / greatest value model: All engineers are graduates. Some analysts are engineers. What follows?

Options: All analysts are engineers | No analyst is a graduate | All graduates are analysts | Some analysts are graduates

Answer: Some analysts are graduates

Solution: Context: propositional logic. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1697 ? Remainder cycle ? Easy

A syllogism case uses the quantities in this question. Determine the requested result using the remainder cycle model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | No analyst is a graduate | All analysts are engineers

Answer: Some analysts are graduates

Solution: Context: syllogism. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1698 ? Divisibility test ? Medium

A statement conclusion case uses the quantities in this question. Determine the requested result using the divisibility test model: All engineers are graduates. Some analysts are engineers. What follows?

Options: All analysts are engineers | All graduates are analysts | Some analysts are graduates | No analyst is a graduate

Answer: Some analysts are graduates

Solution: Context: statement conclusion. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1699 ? Factorisation ? Hard

A assumption test case uses the quantities in this question. Determine the requested result using the factorisation model: All engineers are graduates. Some analysts are engineers. What follows?

Options: No analyst is a graduate | Some analysts are graduates | All analysts are engineers | All graduates are analysts

Answer: Some analysts are graduates

Solution: Context: assumption test. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

Q1700 ? Approximation ? Placement

A cause-effect case uses the quantities in this question. Determine the requested result using the approximation model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | No analyst is a graduate | All analysts are engineers

Answer: Some analysts are graduates

Solution: Context: cause-effect. Analysts who are engineers belong to the graduate set, so some analysts are graduates. Shortcut: Draw nested sets; never reverse "all".

PRACTICE ? solutions hidden below**Q1701 ? Options elimination ? Beginner**

A decision rule case uses the quantities in this question. Determine the requested result using the options elimination model: All engineers are graduates. Some analysts are engineers. What follows?

Options: No analyst is a graduate | All graduates are analysts | All analysts are engineers | Some analysts are graduates

Q1702 ? Boundary case ? Easy

A course of action case uses the quantities in this question. Determine the requested result using the boundary case model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | All analysts are engineers | No analyst is a graduate

Q1703 ? Case split ? Medium

A inference chain case uses the quantities in this question. Determine the requested result using the case split model: All engineers are graduates. Some analysts are engineers. What follows?

Options: All analysts are engineers | No analyst is a graduate | All graduates are analysts | Some analysts are graduates

Q1704 ? Contradiction check ? Hard

A propositional logic case uses the quantities in this question. Determine the requested result using the contradiction check model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | No analyst is a graduate | All analysts are engineers

Q1705 ? Statement I test ? Placement

A syllogism case uses the quantities in this question. Determine the requested result using the statement i test model: All engineers are graduates. Some analysts are engineers. What follows?

Options: All analysts are engineers | All graduates are analysts | Some analysts are graduates | No analyst is a graduate

Q1706 ? Statement II test ? Beginner

A statement conclusion case uses the quantities in this question. Determine the requested result using the statement ii test model: All engineers are graduates. Some analysts are engineers. What follows?

Options: No analyst is a graduate | Some analysts are graduates | All analysts are engineers | All graduates are analysts

Q1707 ? Combined statements ? Easy

A assumption test case uses the quantities in this question. Determine the requested result using the combined statements model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | No analyst is a graduate | All analysts are engineers

Q1708 ? Counting restriction ? Medium

A cause-effect case uses the quantities in this question. Determine the requested result using the counting restriction model: All engineers are graduates. Some analysts are engineers. What follows?

Options: No analyst is a graduate | All graduates are analysts | All analysts are engineers | Some analysts are graduates

Q1709 ? Circular arrangement ? Hard

A decision rule case uses the quantities in this question. Determine the requested result using the circular arrangement model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | All analysts are engineers | No analyst is a graduate

Q1710 ? Complement method ? Placement

A course of action case uses the quantities in this question. Determine the requested result using the complement method model: All engineers are graduates. Some analysts are engineers. What follows?

Options: All analysts are engineers | No analyst is a graduate | All graduates are analysts | Some analysts are graduates

Q1711 ? Pattern: differences ? Beginner

A inference chain case uses the quantities in this question. Determine the requested result using the pattern: differences model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | No analyst is a graduate | All analysts are engineers

Q1712 ? Pattern: ratios ? Easy

A propositional logic case uses the quantities in this question. Determine the requested result using the pattern: ratios model: All engineers are graduates. Some analysts are engineers. What follows?

Options: All analysts are engineers | All graduates are analysts | Some analysts are graduates | No analyst is a graduate

Q1713 ? Pattern: alternating ? Medium

A syllogism case uses the quantities in this question. Determine the requested result using the pattern: alternating model: All engineers are graduates. Some

analysts are engineers. What follows?

Options: No analyst is a graduate | Some analysts are graduates | All analysts are engineers | All graduates are analysts

Q1714 ? Pattern: squares / cubes ? Hard

A statement conclusion case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | No analyst is a graduate | All analysts are engineers

Q1715 ? Inference from facts ? Placement

A assumption test case uses the quantities in this question. Determine the requested result using the inference from facts model: All engineers are graduates. Some analysts are engineers. What follows?

Options: No analyst is a graduate | All graduates are analysts | All analysts are engineers | Some analysts are graduates

Q1716 ? Assumption check ? Beginner

A cause-effect case uses the quantities in this question. Determine the requested result using the assumption check model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | All analysts are engineers | No analyst is a graduate

Q1717 ? Cause vs correlation ? Easy

A decision rule case uses the quantities in this question. Determine the requested result using the cause vs correlation model: All engineers are graduates. Some analysts are engineers. What follows?

Options: All analysts are engineers | No analyst is a graduate | All graduates are analysts | Some analysts are graduates

Q1718 ? Grammar agreement ? Medium

A course of action case uses the quantities in this question. Determine the requested result using the grammar agreement model: All engineers are graduates. Some analysts are engineers. What follows?

Options: Some analysts are graduates | All graduates are analysts | No analyst is a graduate | All analysts are engineers

Q1719 ? Vocabulary in context ? Hard

A inference chain case uses the quantities in this question. Determine the requested result using the vocabulary in context model: All engineers are graduates. Some analysts are engineers. What follows?

Options: All analysts are engineers | All graduates are analysts | Some analysts are graduates | No analyst is a graduate

Q1720 ? Mixed placement challenge ? Placement

A propositional logic case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: All engineers are graduates. Some analysts are engineers. What follows?

Options: No analyst is a graduate | Some analysts are graduates | All analysts are engineers | All graduates are analysts

Non-Verbal Reasoning ? 40 questions

WORKED EXAMPLES

Q1721 ? Direct computation ? Beginner

A figure rotation case uses the quantities in this question. Determine the requested result using the direct computation model: A shape sequence adds one side each step. What follows a 4-sided shape?

Options: 6 | 5 | 4 | 7

Answer: 5

Solution: Context: figure rotation. The visual rule adds one side: $4+1=5$. Shortcut: Check sides, rotation, shade, and position.

Q1722 ? Reverse value ? Easy

A mirror image case uses the quantities in this question. Determine the requested result using the reverse value model: A shape sequence adds one side each step. What follows a 5-sided shape?

Options: 6 | 7 | 5 | 8

Answer: 6

Solution: Context: mirror image. The visual rule adds one side: $5+1=6$. Shortcut: Check sides, rotation, shade, and position.

Q1723 ? Missing base ? Medium

A paper fold case uses the quantities in this question. Determine the requested result using the missing base model: A shape sequence adds one side each step. What follows a 6-sided shape?

Options: 7 | 9 | 6 | 8

Answer: 7

Solution: Context: paper fold. The visual rule adds one side: $6+1=7$. Shortcut: Check sides, rotation, shade, and position.

Q1724 ? Successive change ? Hard

A cube net case uses the quantities in this question. Determine the requested result using the successive change model: A shape sequence adds one side each step. What follows a 7-sided shape?

Options: 8 | 9 | 7 | 10

Answer: 8

Solution: Context: cube net. The visual rule adds one side: $7+1=8$. Shortcut: Check sides, rotation, shade, and position.

Q1725 ? Comparison trap ? Placement

A visual series case uses the quantities in this question. Determine the requested result using the comparison trap model: A shape sequence adds one side each step. What follows a 3-sided shape?

Options: 4 | 5 | 3 | 6

Answer: 4

Solution: Context: visual series. The visual rule adds one side: $3+1=4$. Shortcut: Check sides, rotation, shade, and position.

Q1726 ? Ratio form ? Beginner

A pattern matrix case uses the quantities in this question. Determine the requested result using the ratio form model: A shape sequence adds one side each step. What follows a 4-sided shape?

Options: 6 | 7 | 4 | 5

Answer: 5

Solution: Context: pattern matrix. The visual rule adds one side: $4+1=5$. Shortcut: Check sides, rotation, shade, and position.

Q1727 ? Weighted result ? Easy

A shape counting case uses the quantities in this question. Determine the requested result using the weighted result model: A shape sequence adds one side each step. What follows a 5-sided shape?

Options: 8 | 5 | 6 | 7

Answer: 6

Solution: Context: shape counting. The visual rule adds one side: $5+1=6$. Shortcut: Check sides, rotation, shade, and position.

Q1728 ? Replacement cycle ? Medium

A spatial transformation case uses the quantities in this question. Determine the requested result using the replacement cycle model: A shape sequence adds one side each step. What follows a 6-sided shape?

Options: 6 | 9 | 8 | 7

Answer: 7

Solution: Context: spatial transformation. The visual rule adds one side: $6+1=7$. Shortcut: Check sides, rotation, shade, and position.

Q1729 ? Unit conversion ? Hard

A figure rotation case uses the quantities in this question. Determine the requested result using the unit conversion model: A shape sequence adds one side each step. What follows a 7-sided shape?

Options: 9 | 8 | 7 | 10

Answer: 8

Solution: Context: figure rotation. The visual rule adds one side: $7+1=8$. Shortcut: Check sides, rotation, shade, and position.

Q1730 ? Work-rate model ? Placement

A mirror image case uses the quantities in this question. Determine the requested result using the work-rate model model: A shape sequence adds one side each step. What follows a 3-sided shape?

Options: 6 | 4 | 3 | 5

Answer: 4

Solution: Context: mirror image. The visual rule adds one side: $3+1=4$. Shortcut: Check sides, rotation, shade, and position.

Q1731 ? Speed-time model ? Beginner

A paper fold case uses the quantities in this question. Determine the requested result using the speed-time model model: A shape sequence adds one side each step. What follows a 4-sided shape?

Options: 4 | 5 | 6 | 7

Answer: 5

Solution: Context: paper fold. The visual rule adds one side: $4+1=5$. Shortcut: Check sides, rotation, shade, and position.

Q1732 ? Table lookup ? Easy

A cube net case uses the quantities in this question. Determine the requested result using the table lookup model: A shape sequence adds one side each step. What follows a 5-sided shape?

Options: 8 | 7 | 6 | 5

Answer: 6

Solution: Context: cube net. The visual rule adds one side: $5+1=6$. Shortcut: Check sides, rotation, shade, and position.

Q1733 ? Chart comparison ? Medium

A visual series case uses the quantities in this question. Determine the requested result using the chart comparison model: A shape sequence adds one side each step. What follows a 6-sided shape?

Options: 7 | 8 | 6 | 9

Answer: 7

Solution: Context: visual series. The visual rule adds one side: $6+1=7$. Shortcut: Check sides, rotation, shade, and position.

Q1734 ? Multi-step word problem ? Hard

A pattern matrix case uses the quantities in this question. Determine the requested result using the multi-step word problem model: A shape sequence adds one side each step. What follows a 7-sided shape?

Options: 8 | 10 | 9 | 7

Answer: 8

Solution: Context: pattern matrix. The visual rule adds one side: $7+1=8$. Shortcut: Check sides, rotation, shade, and position.

Q1735 ? Maximum / minimum ? Placement

A shape counting case uses the quantities in this question. Determine the requested result using the maximum / minimum model: A shape sequence adds one side each step. What follows a 3-sided shape?

Options: 5 | 4 | 3 | 6

Answer: 4

Solution: Context: shape counting. The visual rule adds one side: $3+1=4$. Shortcut: Check sides, rotation, shade, and position.

Q1736 ? Least / greatest value ? Beginner

A spatial transformation case uses the quantities in this question. Determine the requested result using the least / greatest value model: A shape sequence adds one side each step. What follows a 4-sided shape?

Options: 5 | 6 | 7 | 4

Answer: 5

Solution: Context: spatial transformation. The visual rule adds one side: $4+1=5$. Shortcut: Check sides, rotation, shade, and position.

Q1737 ? Remainder cycle ? Easy

A figure rotation case uses the quantities in this question. Determine the requested result using the remainder cycle model: A shape sequence adds one side each step. What follows a 5-sided shape?

Options: 7 | 8 | 6 | 5

Answer: 6

Solution: Context: figure rotation. The visual rule adds one side: $5+1=6$. Shortcut: Check sides, rotation, shade, and position.

Q1738 ? Divisibility test ? Medium

A mirror image case uses the quantities in this question. Determine the requested result using the divisibility test model: A shape sequence adds one side each step. What follows a 6-sided shape?

Options: 6 | 7 | 8 | 9

Answer: 7

Solution: Context: mirror image. The visual rule adds one side: $6+1=7$. Shortcut: Check sides, rotation, shade, and position.

Q1739 ? Factorisation ? Hard

A paper fold case uses the quantities in this question. Determine the requested result using the factorisation model: A shape sequence adds one side each step. What follows a 7-sided shape?

Options: 10 | 7 | 9 | 8

Answer: 8

Solution: Context: paper fold. The visual rule adds one side: $7+1=8$. Shortcut: Check sides, rotation, shade, and position.

Q1740 ? Approximation ? Placement

A cube net case uses the quantities in this question. Determine the requested result using the approximation model: A shape sequence adds one side each step. What follows a 3-sided shape?

Options: 6 | 5 | 4 | 3

Answer: 4

Solution: Context: cube net. The visual rule adds one side: $3+1=4$. Shortcut: Check sides, rotation, shade, and position.

PRACTICE ? solutions hidden below**Q1741 ? Options elimination ? Beginner**

A visual series case uses the quantities in this question. Determine the requested result using the options elimination model: A shape sequence adds one side

each step. What follows a 4-sided shape?

Options: 4 | 5 | 6 | 7

Q1742 ? Boundary case ? Easy

A pattern matrix case uses the quantities in this question. Determine the requested result using the boundary case model: A shape sequence adds one side each step. What follows a 5-sided shape?

Options: 5 | 8 | 7 | 6

Q1743 ? Case split ? Medium

A shape counting case uses the quantities in this question. Determine the requested result using the case split model: A shape sequence adds one side each step. What follows a 6-sided shape?

Options: 9 | 7 | 8 | 6

Q1744 ? Contradiction check ? Hard

A spatial transformation case uses the quantities in this question. Determine the requested result using the contradiction check model: A shape sequence adds one side each step. What follows a 7-sided shape?

Options: 8 | 7 | 9 | 10

Q1745 ? Statement I test ? Placement

A figure rotation case uses the quantities in this question. Determine the requested result using the statement i test model: A shape sequence adds one side each step. What follows a 3-sided shape?

Options: 3 | 4 | 5 | 6

Q1746 ? Statement II test ? Beginner

A mirror image case uses the quantities in this question. Determine the requested result using the statement ii test model: A shape sequence adds one side each step. What follows a 4-sided shape?

Options: 5 | 6 | 4 | 7

Q1747 ? Combined statements ? Easy

A paper fold case uses the quantities in this question. Determine the requested result using the combined statements model: A shape sequence adds one side each step. What follows a 5-sided shape?

Options: 6 | 7 | 8 | 5

Q1748 ? Counting restriction ? Medium

A cube net case uses the quantities in this question. Determine the requested result using the counting restriction model: A shape sequence adds one side each step. What follows a 6-sided shape?

Options: 8 | 9 | 6 | 7

Q1749 ? Circular arrangement ? Hard

A visual series case uses the quantities in this question. Determine the requested result using the circular arrangement model: A shape sequence adds one side each step. What follows a 7-sided shape?

Options: 10 | 7 | 9 | 8

Q1750 ? Complement method ? Placement

A pattern matrix case uses the quantities in this question. Determine the requested result using the complement method model: A shape sequence adds one side each step. What follows a 3-sided shape?

Options: 5 | 6 | 3 | 4

Q1751 ? Pattern: differences ? Beginner

A shape counting case uses the quantities in this question. Determine the requested result using the pattern: differences model: A shape sequence adds one side each step. What follows a 4-sided shape?

Options: 7 | 4 | 6 | 5

Q1752 ? Pattern: ratios ? Easy

A spatial transformation case uses the quantities in this question. Determine the requested result using the pattern: ratios model: A shape sequence adds one side each step. What follows a 5-sided shape?

Options: 5 | 6 | 7 | 8

Q1753 ? Pattern: alternating ? Medium

A figure rotation case uses the quantities in this question. Determine the requested result using the pattern: alternating model: A shape sequence adds one side each step. What follows a 6-sided shape?

Options: 9 | 8 | 7 | 6

Q1754 ? Pattern: squares / cubes ? Hard

A mirror image case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: A shape sequence adds one side each step. What follows a 7-sided shape?

Options: 8 | 9 | 7 | 10

Q1755 ? Inference from facts ? Placement

A paper fold case uses the quantities in this question. Determine the requested result using the inference from facts model: A shape sequence adds one side each step. What follows a 3-sided shape?

Options: 3 | 4 | 5 | 6

Q1756 ? Assumption check ? Beginner

A cube net case uses the quantities in this question. Determine the requested result using the assumption check model: A shape sequence adds one side each step. What follows a 4-sided shape?

Options: 6 | 5 | 4 | 7

Q1757 ? Cause vs correlation ? Easy

A visual series case uses the quantities in this question. Determine the requested result using the cause vs correlation model: A shape sequence adds one side each step. What follows a 5-sided shape?

Options: 6 | 7 | 5 | 8

Q1758 ? Grammar agreement ? Medium

A pattern matrix case uses the quantities in this question. Determine the requested result using the grammar agreement model: A shape sequence adds one side each step. What follows a 6-sided shape?

Options: 7 | 9 | 6 | 8

Q1759 ? Vocabulary in context ? Hard

A shape counting case uses the quantities in this question. Determine the requested result using the vocabulary in context model: A shape sequence adds one side each step. What follows a 7-sided shape?

Options: 8 | 9 | 7 | 10

Q1760 ? Mixed placement challenge ? Placement

A spatial transformation case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: A shape sequence adds one side each step. What follows a 3-sided shape?

Options: 4 | 5 | 3 | 6

Critical Reasoning ? 40 questions

WORKED EXAMPLES

Q1761 ? Direct computation ? Beginner

A business claim case uses the quantities in this question. Determine the requested result using the direct computation model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | Advertising always raises sales | Sales never change naturally | The campaign definitely caused it

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: business claim. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1762 ? Reverse value ? Easy

A research conclusion case uses the quantities in this question. Determine the requested result using the reverse value model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Advertising always raises sales | The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Sales never change naturally

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: research conclusion. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1763 ? Missing base ? Medium

A advertising study case uses the quantities in this question. Determine the requested result using the missing base model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Sales never change naturally | Advertising always raises sales

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: advertising study. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1764 ? Successive change ? Hard

A policy proposal case uses the quantities in this question. Determine the requested result using the successive change model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Advertising always raises sales | Sales never change naturally | The campaign definitely caused it | The campaign correlates with the rise but causation is unproven

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: policy proposal. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1765 ? Comparison trap ? Placement

A survey inference case uses the quantities in this question. Determine the requested result using the comparison trap model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Sales never change naturally | The campaign definitely caused it | Advertising always raises sales | The campaign correlates with the rise but causation is unproven

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: survey inference. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1766 ? Ratio form ? Beginner

A argument flaw case uses the quantities in this question. Determine the requested result using the ratio form model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | Advertising always raises sales | The campaign definitely caused it | Sales never change naturally

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: argument flaw. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1767 ? Weighted result ? Easy

A evidence test case uses the quantities in this question. Determine the requested result using the weighted result model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Advertising always raises sales | Sales never change naturally

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: evidence test. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1768 ? Replacement cycle ? Medium

A assumption audit case uses the quantities in this question. Determine the requested result using the replacement cycle model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | Advertising always raises sales | Sales never change naturally | The campaign definitely caused it

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: assumption audit. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1769 ? Unit conversion ? Hard

A business claim case uses the quantities in this question. Determine the requested result using the unit conversion model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Advertising always raises sales | The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Sales never change naturally

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: business claim. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1770 ? Work-rate model ? Placement

A research conclusion case uses the quantities in this question. Determine the requested result using the work-rate model model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Sales never change naturally | Advertising always raises sales

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: research conclusion. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1771 ? Speed-time model ? Beginner

A advertising study case uses the quantities in this question. Determine the requested result using the speed-time model model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Advertising always raises sales | Sales never change naturally | The campaign definitely caused it | The campaign correlates with the rise but causation is unproven

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: advertising study. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1772 ? Table lookup ? Easy

A policy proposal case uses the quantities in this question. Determine the requested result using the table lookup model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Sales never change naturally | The campaign definitely caused it | Advertising always raises sales | The campaign correlates with the rise but causation is unproven

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: policy proposal. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1773 ? Chart comparison ? Medium

A survey inference case uses the quantities in this question. Determine the requested result using the chart comparison model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | Advertising always raises sales | The campaign definitely caused it | Sales never change naturally

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: survey inference. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1774 ? Multi-step word problem ? Hard

A argument flaw case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Advertising always raises sales | Sales never change naturally

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: argument flaw. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1775 ? Maximum / minimum ? Placement

A evidence test case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | Advertising always raises sales | Sales never change naturally | The campaign definitely caused it

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: evidence test. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1776 ? Least / greatest value ? Beginner

A assumption audit case uses the quantities in this question. Determine the requested result using the least / greatest value model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Advertising always raises sales | The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Sales never change naturally

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: assumption audit. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1777 ? Remainder cycle ? Easy

A business claim case uses the quantities in this question. Determine the requested result using the remainder cycle model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Sales never change naturally | Advertising always raises sales

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: business claim. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1778 ? Divisibility test ? Medium

A research conclusion case uses the quantities in this question. Determine the requested result using the divisibility test model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Advertising always raises sales | Sales never change naturally | The campaign definitely caused it | The campaign correlates with the rise but causation is unproven

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: research conclusion. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1779 ? Factorisation ? Hard

A advertising study case uses the quantities in this question. Determine the requested result using the factorisation model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Sales never change naturally | The campaign definitely caused it | Advertising always raises sales | The campaign correlates with the rise but causation is unproven

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: advertising study. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

Q1780 ? Approximation ? Placement

A policy proposal case uses the quantities in this question. Determine the requested result using the approximation model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | Advertising always raises sales | The campaign definitely caused it | Sales never change naturally

Answer: The campaign correlates with the rise but causation is unproven

Solution: Context: policy proposal. Timing shows correlation, but other causes remain possible. Shortcut: Test alternative explanations.

PRACTICE ? solutions hidden below**Q1781 ? Options elimination ? Beginner**

A survey inference case uses the quantities in this question. Determine the requested result using the options elimination model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Advertising always raises sales | Sales never change naturally

Q1782 ? Boundary case ? Easy

A argument flaw case uses the quantities in this question. Determine the requested result using the boundary case model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | Advertising always raises sales | Sales never change naturally | The campaign definitely caused it

Q1783 ? Case split ? Medium

A evidence test case uses the quantities in this question. Determine the requested result using the case split model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Advertising always raises sales | The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Sales never change naturally

Q1784 ? Contradiction check ? Hard

A assumption audit case uses the quantities in this question. Determine the requested result using the contradiction check model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Sales never change naturally | Advertising always raises sales

Q1785 ? Statement I test ? Placement

A business claim case uses the quantities in this question. Determine the requested result using the statement i test model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Advertising always raises sales | Sales never change naturally | The campaign definitely caused it | The campaign correlates with the rise but causation is unproven

Q1786 ? Statement II test ? Beginner

A research conclusion case uses the quantities in this question. Determine the requested result using the statement ii test model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Sales never change naturally | The campaign definitely caused it | Advertising always raises sales | The campaign correlates with the rise but causation is unproven

Q1787 ? Combined statements ? Easy

A advertising study case uses the quantities in this question. Determine the requested result using the combined statements model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | Advertising always raises sales | The campaign definitely caused it | Sales never change naturally

Q1788 ? Counting restriction ? Medium

A policy proposal case uses the quantities in this question. Determine the requested result using the counting restriction model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Advertising always raises sales | Sales never change naturally

Q1789 ? Circular arrangement ? Hard

A survey inference case uses the quantities in this question. Determine the requested result using the circular arrangement model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | Advertising always raises sales | Sales never change naturally | The campaign definitely caused it

Q1790 ? Complement method ? Placement

A argument flaw case uses the quantities in this question. Determine the requested result using the complement method model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Advertising always raises sales | The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Sales never change naturally

Q1791 ? Pattern: differences ? Beginner

A evidence test case uses the quantities in this question. Determine the requested result using the pattern: differences model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Sales never change naturally | Advertising always raises sales

Q1792 ? Pattern: ratios ? Easy

A assumption audit case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Advertising always raises sales | Sales never change naturally | The campaign definitely caused it | The campaign correlates with the rise but causation is unproven

Q1793 ? Pattern: alternating ? Medium

A business claim case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Sales never change naturally | The campaign definitely caused it | Advertising always raises sales | The campaign correlates with the rise but causation is unproven

Q1794 ? Pattern: squares / cubes ? Hard

A research conclusion case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | Advertising always raises sales | The campaign definitely caused it | Sales never change naturally

Q1795 ? Inference from facts ? Placement

A advertising study case uses the quantities in this question. Determine the requested result using the inference from facts model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Advertising always raises sales | Sales never change naturally

Q1796 ? Assumption check ? Beginner

A policy proposal case uses the quantities in this question. Determine the requested result using the assumption check model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | Advertising always raises sales | Sales never change naturally | The campaign definitely caused it

Q1797 ? Cause vs correlation ? Easy

A survey inference case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Advertising always raises sales | The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Sales never change naturally

Q1798 ? Grammar agreement ? Medium

A argument flaw case uses the quantities in this question. Determine the requested result using the grammar agreement model: Sales rose after an ad campaign. Which conclusion is safest?

Options: The campaign correlates with the rise but causation is unproven | The campaign definitely caused it | Sales never change naturally | Advertising always raises sales

Q1799 ? Vocabulary in context ? Hard

A evidence test case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Advertising always raises sales | Sales never change naturally | The campaign definitely caused it | The campaign correlates with the rise but causation is unproven

Q1800 ? Mixed placement challenge ? Placement

A assumption audit case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Sales rose after an ad campaign. Which conclusion is safest?

Options: Sales never change naturally | The campaign definitely caused it | Advertising always raises sales | The campaign correlates with the rise but causation is unproven

Quantitative Aptitude ? 40 questions

WORKED EXAMPLES

Q1801 ? Direct computation ? Beginner

A placement screening case uses the quantities in this question. Determine the requested result using the direct computation model: A test gives 21 marks in A and 6 in B. Find total.

Options: 30 | 33 | 24 | 27

Answer: 27

Solution: Context: placement screening. Total=21+6=27. Shortcut: Estimate before adding.

Q1802 ? Reverse value ? Easy

A business arithmetic case uses the quantities in this question. Determine the requested result using the reverse value model: A test gives 22 marks in A and 7 in B. Find total.

Options: 35 | 26 | 32 | 29

Answer: 29

Solution: Context: business arithmetic. Total=22+7=29. Shortcut: Estimate before adding.

Q1803 ? Missing base ? Medium

A campus survey case uses the quantities in this question. Determine the requested result using the missing base model: A test gives 23 marks in A and 8 in B. Find total.

Options: 31 | 34 | 28 | 37

Answer: 31

Solution: Context: campus survey. Total=23+8=31. Shortcut: Estimate before adding.

Q1804 ? Successive change ? Hard

A engineering estimate case uses the quantities in this question. Determine the requested result using the successive change model: A test gives 24 marks in A and 9 in B. Find total.

Options: 39 | 30 | 36 | 33

Answer: 33

Solution: Context: engineering estimate. Total=24+9=33. Shortcut: Estimate before adding.

Q1805 ? Comparison trap ? Placement

A finance case case uses the quantities in this question. Determine the requested result using the comparison trap model: A test gives 25 marks in A and 10 in B. Find total.

Options: 43 | 31 | 39 | 35

Answer: 35

Solution: Context: finance case. Total=25+10=35. Shortcut: Estimate before adding.

Q1806 ? Ratio form ? Beginner

A operations metric case uses the quantities in this question. Determine the requested result using the ratio form model: A test gives 26 marks in A and 11 in B. Find total.

Options: 45 | 41 | 37 | 33

Answer: 37

Solution: Context: operations metric. Total=26+11=37. Shortcut: Estimate before adding.

Q1807 ? Weighted result ? Easy

A exam word problem case uses the quantities in this question. Determine the requested result using the weighted result model: A test gives 27 marks in A and 12 in B. Find total.

Options: 39 | 35 | 43 | 47

Answer: 39

Solution: Context: exam word problem. Total=27+12=39. Shortcut: Estimate before adding.

Q1808 ? Replacement cycle ? Medium

A workplace calculation case uses the quantities in this question. Determine the requested result using the replacement cycle model: A test gives 28 marks in A and 5 in B. Find total.

Options: 33 | 36 | 30 | 39

Answer: 33

Solution: Context: workplace calculation. Total=28+5=33. Shortcut: Estimate before adding.

Q1809 ? Unit conversion ? Hard

A placement screening case uses the quantities in this question. Determine the requested result using the unit conversion model: A test gives 29 marks in A and 6 in B. Find total.

Options: 43 | 31 | 39 | 35

Answer: 35

Solution: Context: placement screening. Total=29+6=35. Shortcut: Estimate before adding.

Q1810 ? Work-rate model ? Placement

A business arithmetic case uses the quantities in this question. Determine the requested result using the work-rate model model: A test gives 30 marks in A and 7 in B. Find total.

Options: 37 | 41 | 33 | 45

Answer: 37

Solution: Context: business arithmetic. Total=30+7=37. Shortcut: Estimate before adding.

Q1811 ? Speed-time model ? Beginner

A campus survey case uses the quantities in this question. Determine the requested result using the speed-time model model: A test gives 31 marks in A and 8 in B. Find total.

Options: 35 | 39 | 47 | 43

Answer: 39

Solution: Context: campus survey. Total=31+8=39. Shortcut: Estimate before adding.

Q1812 ? Table lookup ? Easy

A engineering estimate case uses the quantities in this question. Determine the requested result using the table lookup model: A test gives 32 marks in A and 9 in B. Find total.

Options: 41 | 45 | 49 | 37

Answer: 41

Solution: Context: engineering estimate. Total=32+9=41. Shortcut: Estimate before adding.

Q1813 ? Chart comparison ? Medium

A finance case case uses the quantities in this question. Determine the requested result using the chart comparison model: A test gives 33 marks in A and 10 in B. Find total.

Options: 51 | 47 | 43 | 39

Answer: 43

Solution: Context: finance case. Total=33+10=43. Shortcut: Estimate before adding.

Q1814 ? Multi-step word problem ? Hard

A operations metric case uses the quantities in this question. Determine the requested result using the multi-step word problem model: A test gives 34 marks in A and 11 in B. Find total.

Options: 45 | 40 | 50 | 55

Answer: 45

Solution: Context: operations metric. Total=34+11=45. Shortcut: Estimate before adding.

Q1815 ? Maximum / minimum ? Placement

A exam word problem case uses the quantities in this question. Determine the requested result using the maximum / minimum model: A test gives 35 marks in A and 12 in B. Find total.

Options: 57 | 52 | 47 | 42

Answer: 47

Solution: Context: exam word problem. Total=35+12=47. Shortcut: Estimate before adding.

Q1816 ? Least / greatest value ? Beginner

A workplace calculation case uses the quantities in this question. Determine the requested result using the least / greatest value model: A test gives 36 marks in A and 5 in B. Find total.

Options: 49 | 37 | 45 | 41

Answer: 41

Solution: Context: workplace calculation. Total=36+5=41. Shortcut: Estimate before adding.

Q1817 ? Remainder cycle ? Easy

A placement screening case uses the quantities in this question. Determine the requested result using the remainder cycle model: A test gives 37 marks in A and 6 in B. Find total.

Options: 43 | 47 | 39 | 51

Answer: 43

Solution: Context: placement screening. Total=37+6=43. Shortcut: Estimate before adding.

Q1818 ? Divisibility test ? Medium

A business arithmetic case uses the quantities in this question. Determine the requested result using the divisibility test model: A test gives 38 marks in A and 7 in B. Find total.

Options: 40 | 45 | 55 | 50

Answer: 45

Solution: Context: business arithmetic. Total=38+7=45. Shortcut: Estimate before adding.

Q1819 ? Factorisation ? Hard

A campus survey case uses the quantities in this question. Determine the requested result using the factorisation model: A test gives 39 marks in A and 8 in B. Find total.

Options: 47 | 52 | 42 | 57

Answer: 47

Solution: Context: campus survey. Total=39+8=47. Shortcut: Estimate before adding.

Q1820 ? Approximation ? Placement

A engineering estimate case uses the quantities in this question. Determine the requested result using the approximation model: A test gives 40 marks in A and 9 in B. Find total.

Options: 59 | 44 | 54 | 49

Answer: 49

Solution: Context: engineering estimate. Total=40+9=49. Shortcut: Estimate before adding.

PRACTICE ? solutions hidden below**Q1821 ? Options elimination ? Beginner**

A finance case case uses the quantities in this question. Determine the requested result using the options elimination model: A test gives 41 marks in A and 10

in B. Find total.

Options: 46 | 51 | 56 | 61

Q1822 ? Boundary case ? Easy

A operations metric case uses the quantities in this question. Determine the requested result using the boundary case model: A test gives 42 marks in A and 11 in B. Find total.

Options: 63 | 58 | 53 | 48

Q1823 ? Case split ? Medium

A exam word problem case uses the quantities in this question. Determine the requested result using the case split model: A test gives 43 marks in A and 12 in B. Find total.

Options: 55 | 61 | 67 | 49

Q1824 ? Contradiction check ? Hard

A workplace calculation case uses the quantities in this question. Determine the requested result using the contradiction check model: A test gives 44 marks in A and 5 in B. Find total.

Options: 49 | 54 | 44 | 59

Q1825 ? Statement I test ? Placement

A placement screening case uses the quantities in this question. Determine the requested result using the statement i test model: A test gives 45 marks in A and 6 in B. Find total.

Options: 46 | 56 | 51 | 61

Q1826 ? Statement II test ? Beginner

A business arithmetic case uses the quantities in this question. Determine the requested result using the statement ii test model: A test gives 46 marks in A and 7 in B. Find total.

Options: 53 | 58 | 48 | 63

Q1827 ? Combined statements ? Easy

A campus survey case uses the quantities in this question. Determine the requested result using the combined statements model: A test gives 47 marks in A and 8 in B. Find total.

Options: 61 | 67 | 49 | 55

Q1828 ? Counting restriction ? Medium

A engineering estimate case uses the quantities in this question. Determine the requested result using the counting restriction model: A test gives 48 marks in A and 9 in B. Find total.

Options: 57 | 51 | 63 | 69

Q1829 ? Circular arrangement ? Hard

A finance case case uses the quantities in this question. Determine the requested result using the circular arrangement model: A test gives 49 marks in A and 10 in B. Find total.

Options: 53 | 71 | 65 | 59

Q1830 ? Complement method ? Placement

A operations metric case uses the quantities in this question. Determine the requested result using the complement method model: A test gives 50 marks in A and 11 in B. Find total.

Options: 61 | 67 | 55 | 73

Q1831 ? Pattern: differences ? Beginner

A exam word problem case uses the quantities in this question. Determine the requested result using the pattern: differences model: A test gives 51 marks in A and 12 in B. Find total.

Options: 75 | 69 | 63 | 57

Q1832 ? Pattern: ratios ? Easy

A workplace calculation case uses the quantities in this question. Determine the requested result using the pattern: ratios model: A test gives 52 marks in A and 5 in B. Find total.

Options: 51 | 57 | 69 | 63

Q1833 ? Pattern: alternating ? Medium

A placement screening case uses the quantities in this question. Determine the requested result using the pattern: alternating model: A test gives 53 marks in A and 6 in B. Find total.

Options: 71 | 59 | 65 | 53

Q1834 ? Pattern: squares / cubes ? Hard

A business arithmetic case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: A test gives 54 marks in A and 7 in B. Find total.

Options: 61 | 67 | 73 | 55

Q1835 ? Inference from facts ? Placement

A campus survey case uses the quantities in this question. Determine the requested result using the inference from facts model: A test gives 55 marks in A and 8 in B. Find total.

Options: 57 | 63 | 69 | 75

Q1836 ? Assumption check ? Beginner

A engineering estimate case uses the quantities in this question. Determine the requested result using the assumption check model: A test gives 56 marks in A and 9 in B. Find total.

Options: 58 | 79 | 72 | 65

Q1837 ? Cause vs correlation ? Easy

A finance case case uses the quantities in this question. Determine the requested result using the cause vs correlation model: A test gives 57 marks in A and 10 in B. Find total.

Options: 67 | 74 | 60 | 81

Q1838 ? Grammar agreement ? Medium

A operations metric case uses the quantities in this question. Determine the requested result using the grammar agreement model: A test gives 58 marks in A and 11 in B. Find total.

Options: 76 | 83 | 62 | 69

Q1839 ? Vocabulary in context ? Hard

A exam word problem case uses the quantities in this question. Determine the requested result using the vocabulary in context model: A test gives 59 marks in A and 12 in B. Find total.

Options: 64 | 78 | 71 | 85

Q1840 ? Mixed placement challenge ? Placement

A workplace calculation case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: A test gives 60 marks in A and 5 in B. Find total.

Options: 79 | 72 | 65 | 58

Arithmetic ? 40 questions

WORKED EXAMPLES

Q1841 ? Direct computation ? Beginner

A shopping bill case uses the quantities in this question. Determine the requested result using the direct computation model: Sum the first 11 natural numbers.

Options: 59 | 80 | 73 | 66

Answer: 66

Solution: Context: shopping bill. $n(n+1)/2=11 \times 12/2=66$. Shortcut: Pair first and last.

Q1842 ? Reverse value ? Easy

A travel budget case uses the quantities in this question. Determine the requested result using the reverse value model: Sum the first 12 natural numbers.

Options: 94 | 78 | 86 | 70

Answer: 78

Solution: Context: travel budget. $n(n+1)/2=12 \times 13/2=78$. Shortcut: Pair first and last.

Q1843 ? Missing base ? Medium

A production count case uses the quantities in this question. Determine the requested result using the missing base model: Sum the first 13 natural numbers.

Options: 91 | 100 | 82 | 109

Answer: 91

Solution: Context: production count. $n(n+1)/2=13 \times 14/2=91$. Shortcut: Pair first and last.

Q1844 ? Successive change ? Hard

A rate calculation case uses the quantities in this question. Determine the requested result using the successive change model: Sum the first 14 natural numbers.

Options: 127 | 116 | 105 | 94

Answer: 105

Solution: Context: rate calculation. $n(n+1)/2=14 \times 15/2=105$. Shortcut: Pair first and last.

Q1845 ? Comparison trap ? Placement

A unit economics case uses the quantities in this question. Determine the requested result using the comparison trap model: Sum the first 15 natural numbers.

Options: 144 | 108 | 132 | 120

Answer: 120

Solution: Context: unit economics. $n(n+1)/2=15 \times 16/2=120$. Shortcut: Pair first and last.

Q1846 ? Ratio form ? Beginner

A household expense case uses the quantities in this question. Determine the requested result using the ratio form model: Sum the first 16 natural numbers.

Options: 136 | 150 | 122 | 164

Answer: 136

Solution: Context: household expense. $n(n+1)/2=16 \times 17/2=136$. Shortcut: Pair first and last.

Q1847 ? Weighted result ? Easy

A schedule arithmetic case uses the quantities in this question. Determine the requested result using the weighted result model: Sum the first 17 natural numbers.

Options: 153 | 168 | 138 | 183

Answer: 153

Solution: Context: schedule arithmetic. $n(n+1)/2=17 \times 18/2=153$. Shortcut: Pair first and last.

Q1848 ? Replacement cycle ? Medium

A exam calculation case uses the quantities in this question. Determine the requested result using the replacement cycle model: Sum the first 18 natural numbers.

Options: 205 | 154 | 188 | 171

Answer: 171

Solution: Context: exam calculation. $n(n+1)/2=18 \times 19/2=171$. Shortcut: Pair first and last.

Q1849 ? Unit conversion ? Hard

A shopping bill case uses the quantities in this question. Determine the requested result using the unit conversion model: Sum the first 19 natural numbers.

Options: 228 | 209 | 190 | 171

Answer: 190

Solution: Context: shopping bill. $n(n+1)/2=19 \times 20/2=190$. Shortcut: Pair first and last.

Q1850 ? Work-rate model ? Placement

A travel budget case uses the quantities in this question. Determine the requested result using the work-rate model model: Sum the first 20 natural numbers.

Options: 210 | 231 | 189 | 252

Answer: 210

Solution: Context: travel budget. $n(n+1)/2=20 \times 21/2=210$. Shortcut: Pair first and last.

Q1851 ? Speed-time model ? Beginner

A production count case uses the quantities in this question. Determine the requested result using the speed-time model model: Sum the first 21 natural numbers.

Options: 231 | 254 | 208 | 277

Answer: 231

Solution: Context: production count. $n(n+1)/2=21 \times 22/2=231$. Shortcut: Pair first and last.

Q1852 ? Table lookup ? Easy

A rate calculation case uses the quantities in this question. Determine the requested result using the table lookup model: Sum the first 22 natural numbers.

Options: 303 | 228 | 278 | 253

Answer: 253

Solution: Context: rate calculation. $n(n+1)/2=22 \times 23/2=253$. Shortcut: Pair first and last.

Q1853 ? Chart comparison ? Medium

A unit economics case uses the quantities in this question. Determine the requested result using the chart comparison model: Sum the first 23 natural numbers.

Options: 276 | 304 | 248 | 332

Answer: 276

Solution: Context: unit economics. $n(n+1)/2=23 \times 24/2=276$. Shortcut: Pair first and last.

Q1854 ? Multi-step word problem ? Hard

A household expense case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Sum the first 24 natural numbers.

Options: 270 | 300 | 330 | 360

Answer: 300

Solution: Context: household expense. $n(n+1)/2=24 \times 25/2=300$. Shortcut: Pair first and last.

Q1855 ? Maximum / minimum ? Placement

A schedule arithmetic case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Sum the first 25 natural numbers.

Options: 325 | 358 | 391 | 292

Answer: 325

Solution: Context: schedule arithmetic. $n(n+1)/2=25 \times 26/2=325$. Shortcut: Pair first and last.

Q1856 ? Least / greatest value ? Beginner

A exam calculation case uses the quantities in this question. Determine the requested result using the least / greatest value model: Sum the first 26 natural numbers.

Options: 421 | 316 | 386 | 351

Answer: 351

Solution: Context: exam calculation. $n(n+1)/2=26 \times 27/2=351$. Shortcut: Pair first and last.

Q1857 ? Remainder cycle ? Easy

A shopping bill case uses the quantities in this question. Determine the requested result using the remainder cycle model: Sum the first 27 natural numbers.

Options: 378 | 416 | 340 | 454

Answer: 378

Solution: Context: shopping bill. $n(n+1)/2=27 \times 28/2=378$. Shortcut: Pair first and last.

Q1858 ? Divisibility test ? Medium

A travel budget case uses the quantities in this question. Determine the requested result using the divisibility test model: Sum the first 28 natural numbers.

Options: 365 | 488 | 447 | 406

Answer: 406

Solution: Context: travel budget. $n(n+1)/2=28 \times 29/2=406$. Shortcut: Pair first and last.

Q1859 ? Factorisation ? Hard

A production count case uses the quantities in this question. Determine the requested result using the factorisation model: Sum the first 29 natural numbers.

Options: 435 | 479 | 523 | 391

Answer: 435

Solution: Context: production count. $n(n+1)/2=29 \times 30/2=435$. Shortcut: Pair first and last.

Q1860 ? Approximation ? Placement

A rate calculation case uses the quantities in this question. Determine the requested result using the approximation model: Sum the first 30 natural numbers.

Options: 559 | 418 | 512 | 465

Answer: 465

Solution: Context: rate calculation. $n(n+1)/2=30 \times 31/2=465$. Shortcut: Pair first and last.

PRACTICE ? solutions hidden below**Q1861 ? Options elimination ? Beginner**

A unit economics case uses the quantities in this question. Determine the requested result using the options elimination model: Sum the first 31 natural numbers.

Options: 496 | 446 | 546 | 596

Q1862 ? Boundary case ? Easy

A household expense case uses the quantities in this question. Determine the requested result using the boundary case model: Sum the first 32 natural numbers.

Options: 634 | 581 | 528 | 475

Q1863 ? Case split ? Medium

A schedule arithmetic case uses the quantities in this question. Determine the requested result using the case split model: Sum the first 33 natural numbers.

Options: 561 | 617 | 505 | 673

Q1864 ? Contradiction check ? Hard

A exam calculation case uses the quantities in this question. Determine the requested result using the contradiction check model: Sum the first 34 natural numbers.

Options: 715 | 655 | 595 | 535

Q1865 ? Statement I test ? Placement

A shopping bill case uses the quantities in this question. Determine the requested result using the statement i test model: Sum the first 35 natural numbers.

Options: 567 | 693 | 630 | 756

Q1866 ? Statement II test ? Beginner

A travel budget case uses the quantities in this question. Determine the requested result using the statement ii test model: Sum the first 36 natural numbers.

Options: 800 | 733 | 666 | 599

Q1867 ? Combined statements ? Easy

A production count case uses the quantities in this question. Determine the requested result using the combined statements model: Sum the first 37 natural numbers.

Options: 703 | 773 | 633 | 843

Q1868 ? Counting restriction ? Medium

A rate calculation case uses the quantities in this question. Determine the requested result using the counting restriction model: Sum the first 38 natural numbers.

Options: 815 | 889 | 667 | 741

Q1869 ? Circular arrangement ? Hard

A unit economics case uses the quantities in this question. Determine the requested result using the circular arrangement model: Sum the first 39 natural numbers.

Options: 702 | 936 | 858 | 780

Q1870 ? Complement method ? Placement

A household expense case uses the quantities in this question. Determine the requested result using the complement method model: Sum the first 40 natural numbers.

Options: 984 | 902 | 820 | 738

Q1871 ? Pattern: differences ? Beginner

A schedule arithmetic case uses the quantities in this question. Determine the requested result using the pattern: differences model: Sum the first 41 natural numbers.

Options: 861 | 775 | 947 | 1033

Q1872 ? Pattern: ratios ? Easy

A exam calculation case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Sum the first 42 natural numbers.

Options: 903 | 993 | 1083 | 813

Q1873 ? Pattern: alternating ? Medium

A shopping bill case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Sum the first 43 natural numbers.

Options: 1136 | 851 | 1041 | 946

Q1874 ? Pattern: squares / cubes ? Hard

A travel budget case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Sum the first 44 natural numbers.

Options: 990 | 1089 | 891 | 1188

Q1875 ? Inference from facts ? Placement

A production count case uses the quantities in this question. Determine the requested result using the inference from facts model: Sum the first 45 natural numbers.

Options: 1035 | 1139 | 1243 | 931

Q1876 ? Assumption check ? Beginner

A rate calculation case uses the quantities in this question. Determine the requested result using the assumption check model: Sum the first 46 natural numbers.

Options: 1297 | 973 | 1189 | 1081

Q1877 ? Cause vs correlation ? Easy

A unit economics case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Sum the first 47 natural numbers.

Options: 1128 | 1241 | 1015 | 1354

Q1878 ? Grammar agreement ? Medium

A household expense case uses the quantities in this question. Determine the requested result using the grammar agreement model: Sum the first 48 natural numbers.

Options: 1176 | 1294 | 1058 | 1412

Q1879 ? Vocabulary in context ? Hard

A schedule arithmetic case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Sum the first 49 natural numbers.

Options: 1471 | 1102 | 1348 | 1225

Q1880 ? Mixed placement challenge ? Placement

A exam calculation case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Sum the first 50 natural numbers.

Options: 1531 | 1147 | 1403 | 1275

Modern Math ? 40 questions

WORKED EXAMPLES

Q1881 ? Direct computation ? Beginner

A set selection case uses the quantities in this question. Determine the requested result using the direct computation model: 21 choose X, 16 choose Y, 6 choose both. How many choose X or Y?

Options: 31 | 34 | 37 | 28

Answer: 31

Solution: Context: set selection. $21+16-6=31$. Shortcut: Subtract overlap once.

Q1882 ? Reverse value ? Easy

A progression model case uses the quantities in this question. Determine the requested result using the reverse value model: 22 choose X, 17 choose Y, 7 choose both. How many choose X or Y?

Options: 38 | 29 | 35 | 32

Answer: 32

Solution: Context: progression model. $22+17-7=32$. Shortcut: Subtract overlap once.

Q1883 ? Missing base ? Medium

A counting rule case uses the quantities in this question. Determine the requested result using the missing base model: 23 choose X, 18 choose Y, 8 choose both. How many choose X or Y?

Options: 33 | 36 | 30 | 39

Answer: 33

Solution: Context: counting rule. $23+18-8=33$. Shortcut: Subtract overlap once.

Q1884 ? Successive change ? Hard

A sequence design case uses the quantities in this question. Determine the requested result using the successive change model: 24 choose X, 19 choose Y, 9 choose both. How many choose X or Y?

Options: 31 | 37 | 34 | 40

Answer: 34

Solution: Context: sequence design. $24+19-9=34$. Shortcut: Subtract overlap once.

Q1885 ? Comparison trap ? Placement

A function mapping case uses the quantities in this question. Determine the requested result using the comparison trap model: 25 choose X, 20 choose Y, 10 choose both. How many choose X or Y?

Options: 43 | 31 | 39 | 35

Answer: 35

Solution: Context: function mapping. $25+20-10=35$. Shortcut: Subtract overlap once.

Q1886 ? Ratio form ? Beginner

A distribution problem case uses the quantities in this question. Determine the requested result using the ratio form model: 26 choose X, 21 choose Y, 11 choose both. How many choose X or Y?

Options: 44 | 40 | 36 | 32

Answer: 36

Solution: Context: distribution problem. $26+21-11=36$. Shortcut: Subtract overlap once.

Q1887 ? Weighted result ? Easy

A series application case uses the quantities in this question. Determine the requested result using the weighted result model: 27 choose X, 22 choose Y, 5 choose both. How many choose X or Y?

Options: 44 | 48 | 40 | 52

Answer: 44

Solution: Context: series application. $27+22-5=44$. Shortcut: Subtract overlap once.

Q1888 ? Replacement cycle ? Medium

A set intersection case uses the quantities in this question. Determine the requested result using the replacement cycle model: 28 choose X, 23 choose Y, 6 choose both. How many choose X or Y?

Options: 55 | 50 | 45 | 40

Answer: 45

Solution: Context: set intersection. $28+23-6=45$. Shortcut: Subtract overlap once.

Q1889 ? Unit conversion ? Hard

A set selection case uses the quantities in this question. Determine the requested result using the unit conversion model: 29 choose X, 24 choose Y, 7 choose both. How many choose X or Y?

Options: 51 | 56 | 41 | 46

Answer: 46

Solution: Context: set selection. $29+24-7=46$. Shortcut: Subtract overlap once.

Q1890 ? Work-rate model ? Placement

A progression model case uses the quantities in this question. Determine the requested result using the work-rate model model: 30 choose X, 15 choose Y, 8 choose both. How many choose X or Y?

Options: 37 | 41 | 33 | 45

Answer: 37

Solution: Context: progression model. $30+15-8=37$. Shortcut: Subtract overlap once.

Q1891 ? Speed-time model ? Beginner

A counting rule case uses the quantities in this question. Determine the requested result using the speed-time model model: 31 choose X, 16 choose Y, 9 choose both. How many choose X or Y?

Options: 34 | 38 | 46 | 42

Answer: 38

Solution: Context: counting rule. $31+16-9=38$. Shortcut: Subtract overlap once.

Q1892 ? Table lookup ? Easy

A sequence design case uses the quantities in this question. Determine the requested result using the table lookup model: 32 choose X, 17 choose Y, 10 choose both. How many choose X or Y?

Options: 43 | 47 | 35 | 39

Answer: 39

Solution: Context: sequence design. $32+17-10=39$. Shortcut: Subtract overlap once.

Q1893 ? Chart comparison ? Medium

A function mapping case uses the quantities in this question. Determine the requested result using the chart comparison model: 33 choose X, 18 choose Y, 11 choose both. How many choose X or Y?

Options: 48 | 44 | 40 | 36

Answer: 40

Solution: Context: function mapping. $33+18-11=40$. Shortcut: Subtract overlap once.

Q1894 ? Multi-step word problem ? Hard

A distribution problem case uses the quantities in this question. Determine the requested result using the multi-step word problem model: 34 choose X, 19 choose Y, 5 choose both. How many choose X or Y?

Options: 48 | 43 | 53 | 58

Answer: 48

Solution: Context: distribution problem. $34+19-5=48$. Shortcut: Subtract overlap once.

Q1895 ? Maximum / minimum ? Placement

A series application case uses the quantities in this question. Determine the requested result using the maximum / minimum model: 35 choose X, 20 choose Y, 6 choose both. How many choose X or Y?

Options: 59 | 54 | 49 | 44

Answer: 49

Solution: Context: series application. $35+20-6=49$. Shortcut: Subtract overlap once.

Q1896 ? Least / greatest value ? Beginner

A set intersection case uses the quantities in this question. Determine the requested result using the least / greatest value model: 36 choose X, 21 choose Y, 7 choose both. How many choose X or Y?

Options: 50 | 55 | 60 | 45

Answer: 50

Solution: Context: set intersection. $36+21-7=50$. Shortcut: Subtract overlap once.

Q1897 ? Remainder cycle ? Easy

A set selection case uses the quantities in this question. Determine the requested result using the remainder cycle model: 37 choose X, 22 choose Y, 8 choose both. How many choose X or Y?

Options: 61 | 51 | 56 | 46

Answer: 51

Solution: Context: set selection. $37+22-8=51$. Shortcut: Subtract overlap once.

Q1898 ? Divisibility test ? Medium

A progression model case uses the quantities in this question. Determine the requested result using the divisibility test model: 38 choose X, 23 choose Y, 9 choose both. How many choose X or Y?

Options: 47 | 57 | 52 | 62

Answer: 52

Solution: Context: progression model. $38+23-9=52$. Shortcut: Subtract overlap once.

Q1899 ? Factorisation ? Hard

A counting rule case uses the quantities in this question. Determine the requested result using the factorisation model: 39 choose X, 24 choose Y, 10 choose both. How many choose X or Y?

Options: 53 | 58 | 48 | 63

Answer: 53

Solution: Context: counting rule. $39+24-10=53$. Shortcut: Subtract overlap once.

Q1900 ? Approximation ? Placement

A sequence design case uses the quantities in this question. Determine the requested result using the approximation model: 40 choose X, 15 choose Y, 11 choose both. How many choose X or Y?

Options: 52 | 40 | 48 | 44

Answer: 44

Solution: Context: sequence design. $40+15-11=44$. Shortcut: Subtract overlap once.

PRACTICE ? solutions hidden below**Q1901 ? Options elimination ? Beginner**

A function mapping case uses the quantities in this question. Determine the requested result using the options elimination model: 41 choose X, 16 choose Y, 5

choose both. How many choose X or Y?
Options: 47 | 52 | 57 | 62

Q1902 ? Boundary case ? Easy

A distribution problem case uses the quantities in this question. Determine the requested result using the boundary case model: 42 choose X, 17 choose Y, 6 choose both. How many choose X or Y?
Options: 63 | 58 | 53 | 48

Q1903 ? Case split ? Medium

A series application case uses the quantities in this question. Determine the requested result using the case split model: 43 choose X, 18 choose Y, 7 choose both. How many choose X or Y?
Options: 54 | 59 | 64 | 49

Q1904 ? Contradiction check ? Hard

A set intersection case uses the quantities in this question. Determine the requested result using the contradiction check model: 44 choose X, 19 choose Y, 8 choose both. How many choose X or Y?
Options: 67 | 49 | 61 | 55

Q1905 ? Statement I test ? Placement

A set selection case uses the quantities in this question. Determine the requested result using the statement i test model: 45 choose X, 20 choose Y, 9 choose both. How many choose X or Y?
Options: 50 | 56 | 68 | 62

Q1906 ? Statement II test ? Beginner

A progression model case uses the quantities in this question. Determine the requested result using the statement ii test model: 46 choose X, 21 choose Y, 10 choose both. How many choose X or Y?
Options: 57 | 63 | 51 | 69

Q1907 ? Combined statements ? Easy

A counting rule case uses the quantities in this question. Determine the requested result using the combined statements model: 47 choose X, 22 choose Y, 11 choose both. How many choose X or Y?
Options: 64 | 70 | 52 | 58

Q1908 ? Counting restriction ? Medium

A sequence design case uses the quantities in this question. Determine the requested result using the counting restriction model: 48 choose X, 23 choose Y, 5 choose both. How many choose X or Y?
Options: 80 | 59 | 66 | 73

Q1909 ? Circular arrangement ? Hard

A function mapping case uses the quantities in this question. Determine the requested result using the circular arrangement model: 49 choose X, 24 choose Y, 6 choose both. How many choose X or Y?
Options: 60 | 81 | 74 | 67

Q1910 ? Complement method ? Placement

A distribution problem case uses the quantities in this question. Determine the requested result using the complement method model: 50 choose X, 15 choose Y, 7 choose both. How many choose X or Y?
Options: 58 | 64 | 52 | 70

Q1911 ? Pattern: differences ? Beginner

A series application case uses the quantities in this question. Determine the requested result using the pattern: differences model: 51 choose X, 16 choose Y, 8 choose both. How many choose X or Y?
Options: 71 | 65 | 59 | 53

Q1912 ? Pattern: ratios ? Easy

A set intersection case uses the quantities in this question. Determine the requested result using the pattern: ratios model: 52 choose X, 17 choose Y, 9 choose both. How many choose X or Y?
Options: 54 | 66 | 60 | 72

Q1913 ? Pattern: alternating ? Medium

A set selection case uses the quantities in this question. Determine the requested result using the pattern: alternating model: 53 choose X, 18 choose Y, 10 choose both. How many choose X or Y?
Options: 73 | 61 | 67 | 55

Q1914 ? Pattern: squares / cubes ? Hard

A progression model case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: 54 choose X, 19 choose Y, 11 choose both. How many choose X or Y?
Options: 62 | 68 | 74 | 56

Q1915 ? Inference from facts ? Placement

A counting rule case uses the quantities in this question. Determine the requested result using the inference from facts model: 55 choose X, 20 choose Y, 5 choose both. How many choose X or Y?
Options: 84 | 63 | 70 | 77

Q1916 ? Assumption check ? Beginner

A sequence design case uses the quantities in this question. Determine the requested result using the assumption check model: 56 choose X, 21 choose Y, 6 choose both. How many choose X or Y?
Options: 64 | 85 | 78 | 71

Q1917 ? Cause vs correlation ? Easy

A function mapping case uses the quantities in this question. Determine the requested result using the cause vs correlation model: 57 choose X, 22 choose Y, 7 choose both. How many choose X or Y?

Options: 72 | 79 | 65 | 86

Q1918 ? Grammar agreement ? Medium

A distribution problem case uses the quantities in this question. Determine the requested result using the grammar agreement model: 58 choose X, 23 choose Y, 8 choose both. How many choose X or Y?

Options: 73 | 87 | 66 | 80

Q1919 ? Vocabulary in context ? Hard

A series application case uses the quantities in this question. Determine the requested result using the vocabulary in context model: 59 choose X, 24 choose Y, 9 choose both. How many choose X or Y?

Options: 67 | 74 | 88 | 81

Q1920 ? Mixed placement challenge ? Placement

A set intersection case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: 60 choose X, 15 choose Y, 10 choose both. How many choose X or Y?

Options: 79 | 72 | 65 | 58

Basic Statistics ? 40 questions

WORKED EXAMPLES

Q1921 ? Direct computation ? Beginner

A quality sample case uses the quantities in this question. Determine the requested result using the direct computation model: Find the mean of 1, 3, 5, 7, 9.

Options: 6 | 5 | 4 | 7

Answer: 5

Solution: Context: quality sample. Sum=25; divide by 5 to get 5. Shortcut: Middle equals mean for symmetric data.

Q1922 ? Reverse value ? Easy

A student marks case uses the quantities in this question. Determine the requested result using the reverse value model: Find the mean of 2, 4, 6, 8, 10.

Options: 6 | 7 | 5 | 8

Answer: 6

Solution: Context: student marks. Sum=30; divide by 5 to get 6. Shortcut: Middle equals mean for symmetric data.

Q1923 ? Missing base ? Medium

A sales spread case uses the quantities in this question. Determine the requested result using the missing base model: Find the mean of 3, 5, 7, 9, 11.

Options: 7 | 9 | 6 | 8

Answer: 7

Solution: Context: sales spread. Sum=35; divide by 5 to get 7. Shortcut: Middle equals mean for symmetric data.

Q1924 ? Successive change ? Hard

A sensor readings case uses the quantities in this question. Determine the requested result using the successive change model: Find the mean of 4, 6, 8, 10, 12.

Options: 8 | 9 | 7 | 10

Answer: 8

Solution: Context: sensor readings. Sum=40; divide by 5 to get 8. Shortcut: Middle equals mean for symmetric data.

Q1925 ? Comparison trap ? Placement

A salary data case uses the quantities in this question. Determine the requested result using the comparison trap model: Find the mean of 5, 7, 9, 11, 13.

Options: 11 | 8 | 10 | 9

Answer: 9

Solution: Context: salary data. Sum=45; divide by 5 to get 9. Shortcut: Middle equals mean for symmetric data.

Q1926 ? Ratio form ? Beginner

A delivery times case uses the quantities in this question. Determine the requested result using the ratio form model: Find the mean of 6, 8, 10, 12, 14.

Options: 10 | 11 | 9 | 12

Answer: 10

Solution: Context: delivery times. Sum=50; divide by 5 to get 10. Shortcut: Middle equals mean for symmetric data.

Q1927 ? Weighted result ? Easy

A survey summary case uses the quantities in this question. Determine the requested result using the weighted result model: Find the mean of 7, 9, 11, 13, 15.

Options: 11 | 12 | 10 | 13

Answer: 11

Solution: Context: survey summary. Sum=55; divide by 5 to get 11. Shortcut: Middle equals mean for symmetric data.

Q1928 ? Replacement cycle ? Medium

A performance report case uses the quantities in this question. Determine the requested result using the replacement cycle model: Find the mean of 8, 10, 12, 14, 16.

Options: 14 | 11 | 13 | 12

Answer: 12

Solution: Context: performance report. Sum=60; divide by 5 to get 12. Shortcut: Middle equals mean for symmetric data.

Q1929 ? Unit conversion ? Hard

A quality sample case uses the quantities in this question. Determine the requested result using the unit conversion model: Find the mean of 9, 11, 13, 15, 17.

Options: 13 | 14 | 12 | 15

Answer: 13

Solution: Context: quality sample. Sum=65; divide by 5 to get 13. Shortcut: Middle equals mean for symmetric data.

Q1930 ? Work-rate model ? Placement

A student marks case uses the quantities in this question. Determine the requested result using the work-rate model model: Find the mean of 10, 12, 14, 16, 18.

Options: 14 | 15 | 13 | 16

Answer: 14

Solution: Context: student marks. Sum=70; divide by 5 to get 14. Shortcut: Middle equals mean for symmetric data.

Q1931 ? Speed-time model ? Beginner

A sales spread case uses the quantities in this question. Determine the requested result using the speed-time model model: Find the mean of 11, 13, 15, 17, 19.

Options: 19 | 13 | 17 | 15

Answer: 15

Solution: Context: sales spread. Sum=75; divide by 5 to get 15. Shortcut: Middle equals mean for symmetric data.

Q1932 ? Table lookup ? Easy

A sensor readings case uses the quantities in this question. Determine the requested result using the table lookup model: Find the mean of 12, 14, 16, 18, 20.

Options: 20 | 14 | 18 | 16

Answer: 16

Solution: Context: sensor readings. Sum=80; divide by 5 to get 16. Shortcut: Middle equals mean for symmetric data.

Q1933 ? Chart comparison ? Medium

A salary data case uses the quantities in this question. Determine the requested result using the chart comparison model: Find the mean of 13, 15, 17, 19, 21.

Options: 17 | 19 | 15 | 21

Answer: 17

Solution: Context: salary data. Sum=85; divide by 5 to get 17. Shortcut: Middle equals mean for symmetric data.

Q1934 ? Multi-step word problem ? Hard

A delivery times case uses the quantities in this question. Determine the requested result using the multi-step word problem model: Find the mean of 14, 16, 18, 20, 22.

Options: 18 | 16 | 20 | 22

Answer: 18

Solution: Context: delivery times. Sum=90; divide by 5 to get 18. Shortcut: Middle equals mean for symmetric data.

Q1935 ? Maximum / minimum ? Placement

A survey summary case uses the quantities in this question. Determine the requested result using the maximum / minimum model: Find the mean of 15, 17, 19, 21, 23.

Options: 23 | 17 | 21 | 19

Answer: 19

Solution: Context: survey summary. Sum=95; divide by 5 to get 19. Shortcut: Middle equals mean for symmetric data.

Q1936 ? Least / greatest value ? Beginner

A performance report case uses the quantities in this question. Determine the requested result using the least / greatest value model: Find the mean of 16, 18, 20, 22, 24.

Options: 24 | 22 | 20 | 18

Answer: 20

Solution: Context: performance report. Sum=100; divide by 5 to get 20. Shortcut: Middle equals mean for symmetric data.

Q1937 ? Remainder cycle ? Easy

A quality sample case uses the quantities in this question. Determine the requested result using the remainder cycle model: Find the mean of 17, 19, 21, 23, 25.

Options: 21 | 23 | 19 | 25

Answer: 21

Solution: Context: quality sample. Sum=105; divide by 5 to get 21. Shortcut: Middle equals mean for symmetric data.

Q1938 ? Divisibility test ? Medium

A student marks case uses the quantities in this question. Determine the requested result using the divisibility test model: Find the mean of 18, 20, 22, 24, 26.

Options: 22 | 24 | 20 | 26

Answer: 22

Solution: Context: student marks. Sum=110; divide by 5 to get 22. Shortcut: Middle equals mean for symmetric data.

Q1939 ? Factorisation ? Hard

A sales spread case uses the quantities in this question. Determine the requested result using the factorisation model: Find the mean of 19, 21, 23, 25, 27.

Options: 27 | 21 | 25 | 23

Answer: 23

Solution: Context: sales spread. Sum=115; divide by 5 to get 23. Shortcut: Middle equals mean for symmetric data.

Q1940 ? Approximation ? Placement

A sensor readings case uses the quantities in this question. Determine the requested result using the approximation model: Find the mean of 20, 22, 24, 26, 28.

Options: 24 | 26 | 22 | 28

Answer: 24

Solution: Context: sensor readings. Sum=120; divide by 5 to get 24. Shortcut: Middle equals mean for symmetric data.

PRACTICE ? solutions hidden below**Q1941 ? Options elimination ? Beginner**

A salary data case uses the quantities in this question. Determine the requested result using the options elimination model: Find the mean of 21, 23, 25, 27, 29.

Options: 25 | 28 | 22 | 31

Q1942 ? Boundary case ? Easy

A delivery times case uses the quantities in this question. Determine the requested result using the boundary case model: Find the mean of 22, 24, 26, 28, 30.

Options: 32 | 23 | 29 | 26

Q1943 ? Case split ? Medium

A survey summary case uses the quantities in this question. Determine the requested result using the case split model: Find the mean of 23, 25, 27, 29, 31.

Options: 33 | 24 | 30 | 27

Q1944 ? Contradiction check ? Hard

A performance report case uses the quantities in this question. Determine the requested result using the contradiction check model: Find the mean of 24, 26, 28, 30, 32.

Options: 28 | 31 | 25 | 34

Q1945 ? Statement I test ? Placement

A quality sample case uses the quantities in this question. Determine the requested result using the statement i test model: Find the mean of 25, 27, 29, 31, 33.

Options: 29 | 32 | 26 | 35

Q1946 ? Statement II test ? Beginner

A student marks case uses the quantities in this question. Determine the requested result using the statement ii test model: Find the mean of 26, 28, 30, 32, 34.

Options: 36 | 27 | 33 | 30

Q1947 ? Combined statements ? Easy

A sales spread case uses the quantities in this question. Determine the requested result using the combined statements model: Find the mean of 27, 29, 31, 33, 35.

Options: 31 | 34 | 28 | 37

Q1948 ? Counting restriction ? Medium

A sensor readings case uses the quantities in this question. Determine the requested result using the counting restriction model: Find the mean of 28, 30, 32, 34, 36.

Options: 29 | 32 | 35 | 38

Q1949 ? Circular arrangement ? Hard

A salary data case uses the quantities in this question. Determine the requested result using the circular arrangement model: Find the mean of 29, 31, 33, 35, 37.

Options: 33 | 36 | 30 | 39

Q1950 ? Complement method ? Placement

A delivery times case uses the quantities in this question. Determine the requested result using the complement method model: Find the mean of 30, 32, 34, 36, 38.

Options: 40 | 31 | 37 | 34

Q1951 ? Pattern: differences ? Beginner

A survey summary case uses the quantities in this question. Determine the requested result using the pattern: differences model: Find the mean of 31, 33, 35, 37, 39.

Options: 35 | 39 | 31 | 43

Q1952 ? Pattern: ratios ? Easy

A performance report case uses the quantities in this question. Determine the requested result using the pattern: ratios model: Find the mean of 32, 34, 36, 38, 40.

Options: 32 | 36 | 44 | 40

Q1953 ? Pattern: alternating ? Medium

A quality sample case uses the quantities in this question. Determine the requested result using the pattern: alternating model: Find the mean of 33, 35, 37, 39, 41.

Options: 41 | 45 | 33 | 37

Q1954 ? Pattern: squares / cubes ? Hard

A student marks case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: Find the mean of 34, 36, 38, 40, 42.

Options: 46 | 42 | 38 | 34

Q1955 ? Inference from facts ? Placement

A sales spread case uses the quantities in this question. Determine the requested result using the inference from facts model: Find the mean of 35, 37, 39, 41, 43.

Options: 39 | 35 | 43 | 47

Q1956 ? Assumption check ? Beginner

A sensor readings case uses the quantities in this question. Determine the requested result using the assumption check model: Find the mean of 36, 38, 40, 42, 44.

Options: 40 | 44 | 36 | 48

Q1957 ? Cause vs correlation ? Easy

A salary data case uses the quantities in this question. Determine the requested result using the cause vs correlation model: Find the mean of 37, 39, 41, 43, 45.

Options: 49 | 37 | 45 | 41

Q1958 ? Grammar agreement ? Medium

A delivery times case uses the quantities in this question. Determine the requested result using the grammar agreement model: Find the mean of 38, 40, 42, 44, 46.

Options: 42 | 46 | 38 | 50

Q1959 ? Vocabulary in context ? Hard

A survey summary case uses the quantities in this question. Determine the requested result using the vocabulary in context model: Find the mean of 39, 41, 43, 45, 47.

Options: 39 | 47 | 43 | 51

Q1960 ? Mixed placement challenge ? Placement

A performance report case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: Find the mean of

40, 42, 44, 46, 48.

Options: 44 | 48 | 40 | 52

Frequently Asked ? 40 questions

WORKED EXAMPLES

Q1961 ? Direct computation ? Beginner

A TCS-style screen case uses the quantities in this question. Determine the requested result using the direct computation model: ■110 marked price has 15% discount. Find sale price.

Options: 111.5 | 84.5 | 102.5 | 93.5

Answer: 93.5

Solution: Context: TCS-style screen. $110 \times 85/100 = \blacksquare 93.5$. Shortcut: Multiply by $(100 - \text{discount})/100$.

Q1962 ? Reverse value ? Easy

A banking case case uses the quantities in this question. Determine the requested result using the reverse value model: ■120 marked price has 20% discount. Find sale price.

Options: 96 | 106 | 86 | 116

Answer: 96

Solution: Context: banking case. $120 \times 80/100 = \blacksquare 96$. Shortcut: Multiply by $(100 - \text{discount})/100$.

Q1963 ? Missing base ? Medium

A consulting estimate case uses the quantities in this question. Determine the requested result using the missing base model: ■130 marked price has 25% discount. Find sale price.

Options: 97.5 | 107.5 | 87.5 | 117.5

Answer: 97.5

Solution: Context: consulting estimate. $130 \times 75/100 = \blacksquare 97.5$. Shortcut: Multiply by $(100 - \text{discount})/100$.

Q1964 ? Successive change ? Hard

A retail audit case uses the quantities in this question. Determine the requested result using the successive change model: ■140 marked price has 30% discount. Find sale price.

Options: 98 | 118 | 88 | 108

Answer: 98

Solution: Context: retail audit. $140 \times 70/100 = \blacksquare 98$. Shortcut: Multiply by $(100 - \text{discount})/100$.

Q1965 ? Comparison trap ? Placement

A campus drive case uses the quantities in this question. Determine the requested result using the comparison trap model: ■150 marked price has 35% discount. Find sale price.

Options: 117.5 | 87.5 | 107.5 | 97.5

Answer: 97.5

Solution: Context: campus drive. $150 \times 65/100 = \blacksquare 97.5$. Shortcut: Multiply by $(100 - \text{discount})/100$.

Q1966 ? Ratio form ? Beginner

A operations report case uses the quantities in this question. Determine the requested result using the ratio form model: ■160 marked price has 10% discount. Find sale price.

Options: 144 | 158 | 130 | 172

Answer: 144

Solution: Context: operations report. $160 \times 90/100 = \blacksquare 144$. Shortcut: Multiply by $(100 - \text{discount})/100$.

Q1967 ? Weighted result ? Easy

A finance question case uses the quantities in this question. Determine the requested result using the weighted result model: ■170 marked price has 15% discount. Find sale price.

Options: 144.5 | 158.5 | 130.5 | 172.5

Answer: 144.5

Solution: Context: finance question. $170 \times 85/100 = \blacksquare 144.5$. Shortcut: Multiply by $(100 - \text{discount})/100$.

Q1968 ? Replacement cycle ? Medium

A mixed placement case case uses the quantities in this question. Determine the requested result using the replacement cycle model: ■180 marked price has 20% discount. Find sale price.

Options: 172 | 130 | 158 | 144

Answer: 144

Solution: Context: mixed placement case. $180 \times 80/100 = \blacksquare 144$. Shortcut: Multiply by $(100 - \text{discount})/100$.

Q1969 ? Unit conversion ? Hard

A TCS-style screen case uses the quantities in this question. Determine the requested result using the unit conversion model: ■190 marked price has 25% discount. Find sale price.

Options: 142.5 | 156.5 | 128.5 | 170.5

Answer: 142.5

Solution: Context: TCS-style screen. $190 \times 75/100 = \blacksquare 142.5$. Shortcut: Multiply by $(100 - \text{discount})/100$.

Q1970 ? Work-rate model ? Placement

A banking case case uses the quantities in this question. Determine the requested result using the work-rate model model: ■200 marked price has 30% discount. Find sale price.

Options: 140 | 154 | 126 | 168

Answer: 140

Solution: Context: banking case. $200 \times 70/100 = \blacksquare 140$. Shortcut: Multiply by $(100 - \text{discount})/100$.

Q1971 ? Speed-time model ? Beginner

A consulting estimate case uses the quantities in this question. Determine the requested result using the speed-time model model: ■210 marked price has 35% discount. Find sale price.

Options: 164.5 | 122.5 | 150.5 | 136.5

Answer: 136.5

Solution: Context: consulting estimate. $210 \times 65 / 100 = \blacksquare 136.5$. Shortcut: Multiply by $(100 - \text{discount}) / 100$.

Q1972 ? Table lookup ? Easy

A retail audit case uses the quantities in this question. Determine the requested result using the table lookup model: ■220 marked price has 10% discount. Find sale price.

Options: 238 | 178 | 218 | 198

Answer: 198

Solution: Context: retail audit. $220 \times 90 / 100 = \blacksquare 198$. Shortcut: Multiply by $(100 - \text{discount}) / 100$.

Q1973 ? Chart comparison ? Medium

A campus drive case uses the quantities in this question. Determine the requested result using the chart comparison model: ■230 marked price has 15% discount. Find sale price.

Options: 195.5 | 215.5 | 175.5 | 235.5

Answer: 195.5

Solution: Context: campus drive. $230 \times 85 / 100 = \blacksquare 195.5$. Shortcut: Multiply by $(100 - \text{discount}) / 100$.

Q1974 ? Multi-step word problem ? Hard

A operations report case uses the quantities in this question. Determine the requested result using the multi-step word problem model: ■240 marked price has 20% discount. Find sale price.

Options: 192 | 173 | 211 | 230

Answer: 192

Solution: Context: operations report. $240 \times 80 / 100 = \blacksquare 192$. Shortcut: Multiply by $(100 - \text{discount}) / 100$.

Q1975 ? Maximum / minimum ? Placement

A finance question case uses the quantities in this question. Determine the requested result using the maximum / minimum model: ■250 marked price has 25% discount. Find sale price.

Options: 225.5 | 168.5 | 206.5 | 187.5

Answer: 187.5

Solution: Context: finance question. $250 \times 75 / 100 = \blacksquare 187.5$. Shortcut: Multiply by $(100 - \text{discount}) / 100$.

Q1976 ? Least / greatest value ? Beginner

A mixed placement case case uses the quantities in this question. Determine the requested result using the least / greatest value model: ■260 marked price has 30% discount. Find sale price.

Options: 218 | 200 | 182 | 164

Answer: 182

Solution: Context: mixed placement case. $260 \times 70 / 100 = \blacksquare 182$. Shortcut: Multiply by $(100 - \text{discount}) / 100$.

Q1977 ? Remainder cycle ? Easy

A TCS-style screen case uses the quantities in this question. Determine the requested result using the remainder cycle model: ■270 marked price has 35% discount. Find sale price.

Options: 175.5 | 193.5 | 157.5 | 211.5

Answer: 175.5

Solution: Context: TCS-style screen. $270 \times 65 / 100 = \blacksquare 175.5$. Shortcut: Multiply by $(100 - \text{discount}) / 100$.

Q1978 ? Divisibility test ? Medium

A banking case case uses the quantities in this question. Determine the requested result using the divisibility test model: ■280 marked price has 10% discount. Find sale price.

Options: 252 | 277 | 227 | 302

Answer: 252

Solution: Context: banking case. $280 \times 90 / 100 = \blacksquare 252$. Shortcut: Multiply by $(100 - \text{discount}) / 100$.

Q1979 ? Factorisation ? Hard

A consulting estimate case uses the quantities in this question. Determine the requested result using the factorisation model: ■290 marked price has 15% discount. Find sale price.

Options: 296.5 | 221.5 | 271.5 | 246.5

Answer: 246.5

Solution: Context: consulting estimate. $290 \times 85 / 100 = \blacksquare 246.5$. Shortcut: Multiply by $(100 - \text{discount}) / 100$.

Q1980 ? Approximation ? Placement

A retail audit case uses the quantities in this question. Determine the requested result using the approximation model: ■300 marked price has 20% discount. Find sale price.

Options: 240 | 264 | 216 | 288

Answer: 240

Solution: Context: retail audit. $300 \times 80 / 100 = \blacksquare 240$. Shortcut: Multiply by $(100 - \text{discount}) / 100$.

PRACTICE ? solutions hidden below**Q1981 ? Options elimination ? Beginner**

A campus drive case uses the quantities in this question. Determine the requested result using the options elimination model: ■310 marked price has 25%

discount. Find sale price.
Options: 232.5 | 255.5 | 209.5 | 278.5

Q1982 ? Boundary case ? Easy

A operations report case uses the quantities in this question. Determine the requested result using the boundary case model: ■320 marked price has 30% discount. Find sale price.
Options: 268 | 202 | 246 | 224

Q1983 ? Case split ? Medium

A finance question case uses the quantities in this question. Determine the requested result using the case split model: ■330 marked price has 35% discount. Find sale price.
Options: 256.5 | 235.5 | 214.5 | 193.5

Q1984 ? Contradiction check ? Hard

A mixed placement case case uses the quantities in this question. Determine the requested result using the contradiction check model: ■340 marked price has 10% discount. Find sale price.
Options: 306 | 337 | 275 | 368

Q1985 ? Statement I test ? Placement

A TCS-style screen case uses the quantities in this question. Determine the requested result using the statement i test model: ■350 marked price has 15% discount. Find sale price.
Options: 297.5 | 327.5 | 267.5 | 357.5

Q1986 ? Statement II test ? Beginner

A banking case case uses the quantities in this question. Determine the requested result using the statement ii test model: ■360 marked price has 20% discount. Find sale price.
Options: 346 | 259 | 317 | 288

Q1987 ? Combined statements ? Easy

A consulting estimate case uses the quantities in this question. Determine the requested result using the combined statements model: ■370 marked price has 25% discount. Find sale price.
Options: 277.5 | 305.5 | 249.5 | 333.5

Q1988 ? Counting restriction ? Medium

A retail audit case uses the quantities in this question. Determine the requested result using the counting restriction model: ■380 marked price has 30% discount. Find sale price.
Options: 266 | 293 | 239 | 320

Q1989 ? Circular arrangement ? Hard

A campus drive case uses the quantities in this question. Determine the requested result using the circular arrangement model: ■390 marked price has 35% discount. Find sale price.
Options: 303.5 | 228.5 | 278.5 | 253.5

Q1990 ? Complement method ? Placement

A operations report case uses the quantities in this question. Determine the requested result using the complement method model: ■400 marked price has 10% discount. Find sale price.
Options: 432 | 324 | 396 | 360

Q1991 ? Pattern: differences ? Beginner

A finance question case uses the quantities in this question. Determine the requested result using the pattern: differences model: ■410 marked price has 15% discount. Find sale price.
Options: 348.5 | 383.5 | 313.5 | 418.5

Q1992 ? Pattern: ratios ? Easy

A mixed placement case case uses the quantities in this question. Determine the requested result using the pattern: ratios model: ■420 marked price has 20% discount. Find sale price.
Options: 302 | 370 | 336 | 404

Q1993 ? Pattern: alternating ? Medium

A TCS-style screen case uses the quantities in this question. Determine the requested result using the pattern: alternating model: ■430 marked price has 25% discount. Find sale price.
Options: 386.5 | 290.5 | 354.5 | 322.5

Q1994 ? Pattern: squares / cubes ? Hard

A banking case case uses the quantities in this question. Determine the requested result using the pattern: squares / cubes model: ■440 marked price has 30% discount. Find sale price.
Options: 308 | 339 | 277 | 370

Q1995 ? Inference from facts ? Placement

A consulting estimate case uses the quantities in this question. Determine the requested result using the inference from facts model: ■450 marked price has 35% discount. Find sale price.
Options: 292.5 | 263.5 | 321.5 | 350.5

Q1996 ? Assumption check ? Beginner

A retail audit case uses the quantities in this question. Determine the requested result using the assumption check model: ■460 marked price has 10% discount. Find sale price.
Options: 414 | 455 | 373 | 496

Q1997 ? Cause vs correlation ? Easy

A campus drive case uses the quantities in this question. Determine the requested result using the cause vs correlation model: ■470 marked price has 15% discount. Find sale price.

Options: 479.5 | 359.5 | 439.5 | 399.5

Q1998 ? Grammar agreement ? Medium

A operations report case uses the quantities in this question. Determine the requested result using the grammar agreement model: ■480 marked price has 20% discount. Find sale price.

Options: 384 | 422 | 346 | 460

Q1999 ? Vocabulary in context ? Hard

A finance question case uses the quantities in this question. Determine the requested result using the vocabulary in context model: ■490 marked price has 25% discount. Find sale price.

Options: 330.5 | 367.5 | 441.5 | 404.5

Q2000 ? Mixed placement challenge ? Placement

A mixed placement case case uses the quantities in this question. Determine the requested result using the mixed placement challenge model: ■500 marked price has 30% discount. Find sale price.

Options: 420 | 315 | 385 | 350

Practice Answer Key & Solutions

Percentages

- Q21 ? Options elimination ? **189**. $35/100 \times 540 = 189$.
- Q22 ? Boundary case ? **224**. $40/100 \times 560 = 224$.
- Q23 ? Case split ? **261**. $45/100 \times 580 = 261$.
- Q24 ? Contradiction check ? **60**. $10/100 \times 600 = 60$.
- Q25 ? Statement I test ? **93**. $15/100 \times 620 = 93$.
- Q26 ? Statement II test ? **128**. $20/100 \times 640 = 128$.
- Q27 ? Combined statements ? **165**. $25/100 \times 660 = 165$.
- Q28 ? Counting restriction ? **204**. $30/100 \times 680 = 204$.
- Q29 ? Circular arrangement ? **245**. $35/100 \times 700 = 245$.
- Q30 ? Complement method ? **288**. $40/100 \times 720 = 288$.
- Q31 ? Pattern: differences ? **333**. $45/100 \times 740 = 333$.
- Q32 ? Pattern: ratios ? **76**. $10/100 \times 760 = 76$.
- Q33 ? Pattern: alternating ? **117**. $15/100 \times 780 = 117$.
- Q34 ? Pattern: squares / cubes ? **160**. $20/100 \times 800 = 160$.
- Q35 ? Inference from facts ? **205**. $25/100 \times 820 = 205$.
- Q36 ? Assumption check ? **252**. $30/100 \times 840 = 252$.
- Q37 ? Cause vs correlation ? **301**. $35/100 \times 860 = 301$.
- Q38 ? Grammar agreement ? **352**. $40/100 \times 880 = 352$.
- Q39 ? Vocabulary in context ? **405**. $45/100 \times 900 = 405$.
- Q40 ? Mixed placement challenge ? **92**. $10/100 \times 920 = 92$.

Profit & Loss

- Q61 ? Options elimination ? **870**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 870.
- Q62 ? Boundary case ? **937.5**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 937.5.
- Q63 ? Case split ? **1007.5**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1007.5.
- Q64 ? Contradiction check ? **840**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 840.
- Q65 ? Statement I test ? **907.5**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 907.5.
- Q66 ? Statement II test ? **977.5**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 977.5.
- Q67 ? Combined statements ? **1050**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1050.
- Q68 ? Counting restriction ? **1125**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1125.
- Q69 ? Circular arrangement ? **1202.5**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1202.5.
- Q70 ? Complement method ? **997.5**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 997.5.
- Q71 ? Pattern: differences ? **1072.5**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1072.5.
- Q72 ? Pattern: ratios ? **1150**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1150.
- Q73 ? Pattern: alternating ? **1230**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1230.
- Q74 ? Pattern: squares / cubes ? **1312.5**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1312.5.
- Q75 ? Inference from facts ? **1397.5**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1397.5.
- Q76 ? Assumption check ? **1155**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1155.
- Q77 ? Cause vs correlation ? **1237.5**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1237.5.
- Q78 ? Grammar agreement ? **1322.5**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1322.5.
- Q79 ? Vocabulary in context ? **1410**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1410.
- Q80 ? Mixed placement challenge ? **1500**. Apply the profit/loss relation to the stated cost-price scenario; the verified answer is 1500.

Simple Interest

- Q101 ? Options elimination ? **875**. Context: savings account. $SI = PRT/100 = 6250 \times 7 \times 2/100 = \blacksquare 875$.
- Q102 ? Boundary case ? **1560**. Context: equipment finance. $SI = PRT/100 = 6500 \times 8 \times 3/100 = \blacksquare 1560$.
- Q103 ? Case split ? **2430**. Context: invoice delay. $SI = PRT/100 = 6750 \times 9 \times 4/100 = \blacksquare 2430$.
- Q104 ? Contradiction check ? **3500**. Context: personal loan. $SI = PRT/100 = 7000 \times 10 \times 5/100 = \blacksquare 3500$.

Q105 ? Statement I test ? **797.5**. Context: loan repayment. $SI = PRT/100 = 7250 \times 11 \times 1/100 = \blacksquare 797.5$.

Q106 ? Statement II test ? **1800**. Context: fixed deposit. $SI = PRT/100 = 7500 \times 12 \times 2/100 = \blacksquare 1800$.

Q107 ? Combined statements ? **930**. Context: education loan. $SI = PRT/100 = 7750 \times 4 \times 3/100 = \blacksquare 930$.

Q108 ? Counting restriction ? **1600**. Context: working-capital advance. $SI = PRT/100 = 8000 \times 5 \times 4/100 = \blacksquare 1600$.

Q109 ? Circular arrangement ? **2475**. Context: savings account. $SI = PRT/100 = 8250 \times 6 \times 5/100 = \blacksquare 2475$.

Q110 ? Complement method ? **595**. Context: equipment finance. $SI = PRT/100 = 8500 \times 7 \times 1/100 = \blacksquare 595$.

Q111 ? Pattern: differences ? **1400**. Context: invoice delay. $SI = PRT/100 = 8750 \times 8 \times 2/100 = \blacksquare 1400$.

Q112 ? Pattern: ratios ? **2430**. Context: personal loan. $SI = PRT/100 = 9000 \times 9 \times 3/100 = \blacksquare 2430$.

Q113 ? Pattern: alternating ? **3700**. Context: loan repayment. $SI = PRT/100 = 9250 \times 10 \times 4/100 = \blacksquare 3700$.

Q114 ? Pattern: squares / cubes ? **5225**. Context: fixed deposit. $SI = PRT/100 = 9500 \times 11 \times 5/100 = \blacksquare 5225$.

Q115 ? Inference from facts ? **1170**. Context: education loan. $SI = PRT/100 = 9750 \times 12 \times 1/100 = \blacksquare 1170$.

Q116 ? Assumption check ? **800**. Context: working-capital advance. $SI = PRT/100 = 10000 \times 4 \times 2/100 = \blacksquare 800$.

Q117 ? Cause vs correlation ? **1537.5**. Context: savings account. $SI = PRT/100 = 10250 \times 5 \times 3/100 = \blacksquare 1537.5$.

Q118 ? Grammar agreement ? **2520**. Context: equipment finance. $SI = PRT/100 = 10500 \times 6 \times 4/100 = \blacksquare 2520$.

Q119 ? Vocabulary in context ? **3762.5**. Context: invoice delay. $SI = PRT/100 = 10750 \times 7 \times 5/100 = \blacksquare 3762.5$.

Q120 ? Mixed placement challenge ? **880**. Context: personal loan. $SI = PRT/100 = 11000 \times 8 \times 1/100 = \blacksquare 880$.

Compound Interest

Q141 ? Options elimination ? **1178.7500000000005**. Context: fund NAV. $CI = 11500[(1+5/100)^2-1] = \blacksquare 1178.7500000000005$.

Q142 ? Boundary case ? **2520.0000000000023**. Context: equipment lease. $CI = 12000[(1+10/100)^2-1] = \blacksquare 2520.0000000000023$.

Q143 ? Case split ? **5499.999999999999**. Context: savings target. $CI = 12500[(1+20/100)^2-1] = \blacksquare 5499.999999999999$.

Q144 ? Contradiction check ? **1332.5000000000005**. Context: rate comparison. $CI = 13000[(1+5/100)^2-1] = \blacksquare 1332.5000000000005$.

Q145 ? Statement I test ? **2835.0000000000027**. Context: investment ladder. $CI = 13500[(1+10/100)^2-1] = \blacksquare 2835.0000000000027$.

Q146 ? Statement II test ? **6159.999999999999**. Context: inflation forecast. $CI = 14000[(1+20/100)^2-1] = \blacksquare 6159.999999999999$.

Q147 ? Combined statements ? **1486.2500000000005**. Context: loan balance. $CI = 14500[(1+5/100)^2-1] = \blacksquare 1486.2500000000005$.

Q148 ? Counting restriction ? **3150.0000000000027**. Context: annual deposit. $CI = 15000[(1+10/100)^2-1] = \blacksquare 3150.0000000000027$.

Q149 ? Circular arrangement ? **6819.999999999999**. Context: fund NAV. $CI = 15500[(1+20/100)^2-1] = \blacksquare 6819.999999999999$.

Q150 ? Complement method ? **1640.0000000000005**. Context: equipment lease. $CI = 16000[(1+5/100)^2-1] = \blacksquare 1640.0000000000005$.

Q151 ? Pattern: differences ? **3465.0000000000003**. Context: savings target. $CI = 16500[(1+10/100)^2-1] = \blacksquare 3465.0000000000003$.

Q152 ? Pattern: ratios ? **7479.999999999999**. Context: rate comparison. $CI = 17000[(1+20/100)^2-1] = \blacksquare 7479.999999999999$.

Q153 ? Pattern: alternating ? **1793.7500000000007**. Context: investment ladder. $CI = 17500[(1+5/100)^2-1] = \blacksquare 1793.7500000000007$.

Q154 ? Pattern: squares / cubes ? **3780.0000000000003**. Context: inflation forecast. $CI = 18000[(1+10/100)^2-1] = \blacksquare 3780.0000000000003$.

Q155 ? Inference from facts ? **8139.999999999999**. Context: loan balance. $CI = 18500[(1+20/100)^2-1] = \blacksquare 8139.999999999999$.

Q156 ? Assumption check ? **1947.5000000000007**. Context: annual deposit. $CI = 19000[(1+5/100)^2-1] = \blacksquare 1947.5000000000007$.

Q157 ? Cause vs correlation ? **4095.0000000000036**. Context: fund NAV. $CI = 19500[(1+10/100)^2-1] = \blacksquare 4095.0000000000036$.

Q158 ? Grammar agreement ? **8799.999999999998**. Context: equipment lease. $CI = 20000[(1+20/100)^2-1] = \blacksquare 8799.999999999998$.

Q159 ? Vocabulary in context ? **2101.2500000000001**. Context: savings target. $CI = 20500[(1+5/100)^2-1] = \blacksquare 2101.2500000000001$.

Q160 ? Mixed placement challenge ? **4410.0000000000004**. Context: rate comparison. $CI = 21000[(1+10/100)^2-1] = \blacksquare 4410.0000000000004$.

Ratio & Proportion

Q181 ? Options elimination ? **123**. Context: map scale. Total parts = 6; first share = $246 \times 3/6 = \blacksquare 123$.

Q182 ? Boundary case ? **168**. Context: fleet split. Total parts = 9; first share = $378 \times 4/9 = \blacksquare 168$.

Q183 ? Case split ? **215**. Context: budget share. Total parts = 12; first share = $516 \times 5/12 = \blacksquare 215$.

Q184 ? Contradiction check ? **264**. Context: mixture ratio. Total parts = 9; first share = $396 \times 6/9 = \blacksquare 264$.

Q185 ? Statement I test ? **90**. Context: team allocation. Total parts = 7; first share = $315 \times 2/7 = \blacksquare 90$.

Q186 ? Statement II test ? **138**. Context: recipe scaling. Total parts = 10; first share = $460 \times 3/10 = \blacksquare 138$.

Q187 ? Combined statements ? **188**. Context: project staffing. Total parts = 7; first share = $329 \times 4/7 = \blacksquare 188$.

Q188 ? Counting restriction ? **240**. Context: screen resolution. Total parts = 10; first share = $480 \times 5/10 = \blacksquare 240$.

Q189 ? Circular arrangement ? **294**. Context: map scale. Total parts = 13; first share = $637 \times 6/13 = \blacksquare 294$.

Q190 ? Complement method ? **100**. Context: fleet split. Total parts = 5; first share = $250 \times 2/5 = \blacksquare 100$.

Q191 ? Pattern: differences ? **153**. Context: budget share. Total parts = 8; first share = $408 \times 3/8 = \blacksquare 153$.

Q192 ? Pattern: ratios ? **208**. Context: mixture ratio. Total parts = 11; first share = $572 \times 4/11 = \blacksquare 208$.

Q193 ? Pattern: alternating ? **265**. Context: team allocation. Total parts = 8; first share = $424 \times 5/8 = \blacksquare 265$.

Q194 ? Pattern: squares / cubes ? **324**. Context: recipe scaling. Total parts = 11; first share = $594 \times 6/11 = \blacksquare 324$.

Q195 ? Inference from facts ? **110**. Context: project staffing. Total parts = 9; first share = $495 \times 2/9 = \blacksquare 110$.

Q196 ? Assumption check ? **168**. Context: screen resolution. Total parts = 6; first share = $336 \times 3/6 = \blacksquare 168$.

Q197 ? Cause vs correlation ? **228**. Context: map scale. Total parts = 9; first share = $513 \times 4/9 = \blacksquare 228$.

Q198 ? Grammar agreement ? **290**. Context: fleet split. Total parts = 12; first share = $696 \times 5/12 = \blacksquare 290$.

Q199 ? Vocabulary in context ? **354**. Context: budget share. Total parts = 9; first share = $531 \times 6/9 = \blacksquare 354$.

Q200 ? Mixed placement challenge ? **120**. Context: mixture ratio. Total parts = 7; first share = $420 \times 2/7 = \blacksquare 120$.

Averages

Q221 ? Options elimination ? **39**. Context: match scores. The sum is 195. Divide by 5 to get 39.

Q222 ? Boundary case ? **40**. Context: delivery time. The sum is 200. Divide by 5 to get 40.

Q223 ? Case split ? **41**. Context: salary bands. The sum is 205. Divide by 5 to get 41.

Q224 ? Contradiction check ? **42**. Context: quality samples. The sum is 210. Divide by 5 to get 42.

Q225 ? Statement I test ? **43**. Context: class performance. The sum is 215. Divide by 5 to get 43.

Q226 ? Statement II test ? **44**. Context: weekly sales. The sum is 220. Divide by 5 to get 44.

Q227 ? Combined statements ? **45**. Context: sensor readings. The sum is 225. Divide by 5 to get 45.

Q228 ? Counting restriction ? **46**. Context: team output. The sum is 230. Divide by 5 to get 46.

Q229 ? Circular arrangement ? **47**. Context: match scores. The sum is 235. Divide by 5 to get 47.

Q230 ? Complement method ? **48**. Context: delivery time. The sum is 240. Divide by 5 to get 48.

Q231 ? Pattern: differences ? **49**. Context: salary bands. The sum is 245. Divide by 5 to get 49.

Q232 ? Pattern: ratios ? **50**. Context: quality samples. The sum is 250. Divide by 5 to get 50.

Q233 ? Pattern: alternating ? **51**. Context: class performance. The sum is 255. Divide by 5 to get 51.

Q234 ? Pattern: squares / cubes ? **52**. Context: weekly sales. The sum is 260. Divide by 5 to get 52.

Q235 ? Inference from facts ? **53**. Context: sensor readings. The sum is 265. Divide by 5 to get 53.

Q236 ? Assumption check ? **54**. Context: team output. The sum is 270. Divide by 5 to get 54.

Q237 ? Cause vs correlation ? **55**. Context: match scores. The sum is 275. Divide by 5 to get 55.

Q238 ? Grammar agreement ? **56**. Context: delivery time. The sum is 280. Divide by 5 to get 56.

Q239 ? Vocabulary in context ? **57**. Context: salary bands. The sum is 285. Divide by 5 to get 57.

Q240 ? Mixed placement challenge ? **58**. Context: quality samples. The sum is 290. Divide by 5 to get 58.

Time & Work

Q261 ? Options elimination ? **7.22**. Context: factory line. Combined rate = $1/11+1/21$. Time = $231/32 = 7.22$ days.

Q262 ? Boundary case ? **6**. Context: event setup. Combined rate = $1/12+1/12$. Time = $144/24 = 6$ days.

Q263 ? Case split ? **6.74**. Context: content pipeline. Combined rate = $1/13+1/14$. Time = $182/27 = 6.74$ days.

Q264 ? Contradiction check ? **7.47**. Context: audit team. Combined rate = $1/14+1/16$. Time = $224/30 = 7.47$ days.

Q265 ? Statement I test ? **8.18**. Context: software sprint. Combined rate = $1/15+1/18$. Time = $270/33 = 8.18$ days.

Q266 ? Statement II test ? **8.89**. Context: construction crew. Combined rate = $1/16+1/20$. Time = $320/36 = 8.89$ days.

Q267 ? Combined statements ? **5.87**. Context: data migration. Combined rate = $1/8+1/22$. Time = $176/30 = 5.87$ days.

Q268 ? Counting restriction ? **5.32**. Context: exam paper review. Combined rate = $1/9+1/13$. Time = $117/22 = 5.32$ days.

Q269 ? Circular arrangement ? **6**. Context: factory line. Combined rate = $1/10+1/15$. Time = $150/25 = 6$ days.

Q270 ? Complement method ? **6.68**. Context: event setup. Combined rate = $1/11+1/17$. Time = $187/28 = 6.68$ days.

Q271 ? Pattern: differences ? **7.35**. Context: content pipeline. Combined rate = $1/12+1/19$. Time = $228/31 = 7.35$ days.

Q272 ? Pattern: ratios ? **8.03**. Context: audit team. Combined rate = $1/13+1/21$. Time = $273/34 = 8.03$ days.

Q273 ? Pattern: alternating ? **6.46**. Context: software sprint. Combined rate = $1/14+1/12$. Time = $168/26 = 6.46$ days.

Q274 ? Pattern: squares / cubes ? **7.24**. Context: construction crew. Combined rate = $1/15+1/14$. Time = $210/29 = 7.24$ days.

Q275 ? Inference from facts ? **8**. Context: data migration. Combined rate = $1/16+1/16$. Time = $256/32 = 8$ days.

Q276 ? Assumption check ? **5.54**. Context: exam paper review. Combined rate = $1/8+1/18$. Time = $144/26 = 5.54$ days.

Q277 ? Cause vs correlation ? **6.21**. Context: factory line. Combined rate = $1/9+1/20$. Time = $180/29 = 6.21$ days.

Q278 ? Grammar agreement ? **6.88**. Context: event setup. Combined rate = $1/10+1/22$. Time = $220/32 = 6.88$ days.

Q279 ? Vocabulary in context ? **5.96**. Context: content pipeline. Combined rate = $1/11+1/13$. Time = $143/24 = 5.96$ days.

Q280 ? Mixed placement challenge ? **6.67**. Context: audit team. Combined rate = $1/12+1/15$. Time = $180/27 = 6.67$ days.

Time, Speed & Distance

Q301 ? Options elimination ? **165**. Context: cycling stage. Distance = speed $\times$ time = $55 \times 3 = 165$ km.

Q302 ? Boundary case ? **240**. Context: race track. Distance = speed $\times$ time = $60 \times 4 = 240$ km.

Q303 ? Case split ? **325**. Context: courier service. Distance = speed $\times$ time = $65 \times 5 = 325$ km.

Q304 ? Contradiction check ? **180**. Context: network latency. Distance = speed $\times$ time = $30 \times 6 = 180$ km.

Q305 ? Statement I test ? **70**. Context: delivery route. Distance = speed $\times$ time = $35 \times 2 = 70$ km.

Q306 ? Statement II test ? **120**. Context: commute. Distance = speed $\times$ time = $40 \times 3 = 120$ km.

Q307 ? Combined statements ? **180**. Context: train journey. Distance = speed $\times$ time = $45 \times 4 = 180$ km.

Q308 ? Counting restriction ? **250**. Context: drone flight. Distance = speed $\times$ time = $50 \times 5 = 250$ km.

Q309 ? Circular arrangement ? **330**. Context: cycling stage. Distance = speed $\times$ time = $55 \times 6 = 330$ km.

Q310 ? Complement method ? **120**. Context: race track. Distance = speed $\times$ time = $60 \times 2 = 120$ km.

Q311 ? Pattern: differences ? **195**. Context: courier service. Distance = speed $\times$ time = $65 \times 3 = 195$ km.

Q312 ? Pattern: ratios ? **120**. Context: network latency. Distance = speed $\times$ time = $30 \times 4 = 120$ km.

Q313 ? Pattern: alternating ? **175**. Context: delivery route. Distance = speed $\times$ time = $35 \times 5 = 175$ km.

Q314 ? Pattern: squares / cubes ? **240**. Context: commute. Distance = speed $\times$ time = $40 \times 6 = 240$ km.

Q315 ? Inference from facts ? **90**. Context: train journey. Distance = speed $\times$ time = $45 \times 2 = 90$ km.

Q316 ? Assumption check ? **150**. Context: drone flight. Distance = speed $\times$ time = $50 \times 3 = 150$ km.

Q317 ? Cause vs correlation ? **220**. Context: cycling stage. Distance = speed $\times$ time = $55 \times 4 = 220$ km.

Q318 ? Grammar agreement ? **300**. Context: race track. Distance = speed $\times$ time = $60 \times 5 = 300$ km.

Q319 ? Vocabulary in context ? **390**. Context: courier service. Distance = speed $\times$ time = $65 \times 6 = 390$ km.

Q320 ? Mixed placement challenge ? **60**. Context: network latency. Distance = speed $\times$ time = $30 \times 2 = 60$ km.

Trains

Q341 ? Options elimination ? **85.85**. Context: freight corridor. Speed = $310/13 = 23.85$ m/s = 85.85 km/h.

Q342 ? Boundary case ? **82.29**. Context: metro overtaking. Speed = $320/14 = 22.86$ m/s = 82.29 km/h.

Q343 ? Case split ? **79.2**. Context: track maintenance. Speed = $330/15 = 22$ m/s = 79.2 km/h.

Q344 ? Contradiction check ? **153**. Context: station approach. Speed = $340/8 = 42.5$ m/s = 153 km/h.

Q345 ? Statement I test ? **140**. Context: rail crossing. Speed = $350/9 = 38.89$ m/s = 140 km/h.

Q346 ? Statement II test ? **129.6**. Context: platform clearance. Speed = $360/10 = 36$ m/s = 129.6 km/h.

Q347 ? Combined statements ? **121.09**. Context: two-train dispatch. Speed = $370/11 = 33.64$ m/s = 121.09 km/h.

Q348 ? Counting restriction ? **114**. Context: signal timing. Speed = $380/12 = 31.67$ m/s = 114 km/h.

Q349 ? Circular arrangement ? **108**. Context: freight corridor. Speed = $390/13 = 30$ m/s = 108 km/h.

Q350 ? Complement method ? **102.86**. Context: metro overtaking. Speed = $400/14 = 28.57$ m/s = 102.86 km/h.

Q351 ? Pattern: differences ? **98.4**. Context: track maintenance. Speed = $410/15 = 27.33$ m/s = 98.4 km/h.

Q352 ? Pattern: ratios ? **189**. Context: station approach. Speed = $420/8 = 52.5$ m/s = 189 km/h.

Q353 ? Pattern: alternating ? **172**. Context: rail crossing. Speed = $430/9 = 47.78$ m/s = 172 km/h.

Q354 ? Pattern: squares / cubes ? **158.4**. Context: platform clearance. Speed = $440/10 = 44$ m/s = 158.4 km/h.

Q355 ? Inference from facts ? **147.27**. Context: two-train dispatch. Speed = $450/11 = 40.91$ m/s = 147.27 km/h.

Q356 ? Assumption check ? **138**. Context: signal timing. Speed = $460/12 = 38.33$ m/s = 138 km/h.

Q357 ? Cause vs correlation ? **130.15**. Context: freight corridor. Speed = $470/13 = 36.15$ m/s = 130.15 km/h.

Q358 ? Grammar agreement ? **123.43**. Context: metro overtaking. Speed = $480/14 = 34.29$ m/s = 123.43 km/h.

Q359 ? Vocabulary in context ? **117.6**. Context: track maintenance. Speed = $490/15 = 32.67$ m/s = 117.6 km/h.

Q360 ? Mixed placement challenge ? **225**. Context: station approach. Speed = $500/8 = 62.5$ m/s = 225 km/h.

Problems on Ages

Q381 ? Options elimination ? **87**. Context: generation puzzle. Father's age = $3 \times 29 = 87$ years.

Q382 ? Boundary case ? **120**. Context: team seniority. Father's age = $4 \times 30 = 120$ years.

Q383 ? Case split ? **155**. Context: historic record. Father's age = $5 \times 31 = 155$ years.

Q384 ? Contradiction check ? **64**. Context: age-ratio case. Father's age = $2 \times 32 = 64$ years.

Q385 ? Statement I test ? **99**. Context: family tree. Father's age = $3 \times 33 = 99$ years.

Q386 ? Statement II test ? **136**. Context: career timeline. Father's age = $4 \times 34 = 136$ years.

Q387 ? Combined statements ? **175**. Context: parent-child interview. Father's age = $5 \times 35 = 175$ years.

Q388 ? Counting restriction ? **72**. Context: birth-year audit. Father's age = $2 \times 36 = 72$ years.

Q389 ? Circular arrangement ? **111**. Context: generation puzzle. Father's age = $3 \times 37 = 111$ years.

Q390 ? Complement method ? **152**. Context: team seniority. Father's age = $4 \times 38 = 152$ years.

Q391 ? Pattern: differences ? **195**. Context: historic record. Father's age = $5 \times 39 = 195$ years.

Q392 ? Pattern: ratios ? **80**. Context: age-ratio case. Father's age = $2 \times 40 = 80$ years.

Q393 ? Pattern: alternating ? **123**. Context: family tree. Father's age = $3 \times 41 = 123$ years.

Q394 ? Pattern: squares / cubes ? **168**. Context: career timeline. Father's age = $4 \times 42 = 168$ years.

Q395 ? Inference from facts ? **215**. Context: parent-child interview. Father's age = $5 \times 43 = 215$ years.

Q396 ? Assumption check ? **88**. Context: birth-year audit. Father's age = $2 \times 44 = 88$ years.

Q397 ? Cause vs correlation ? **135**. Context: generation puzzle. Father's age = $3 \times 45 = 135$ years.

Q398 ? Grammar agreement ? **184**. Context: team seniority. Father's age = $4 \times 46 = 184$ years.

Q399 ? Vocabulary in context ? **235**. Context: historic record. Father's age = $5 \times 47 = 235$ years.

Q400 ? Mixed placement challenge ? **96**. Context: age-ratio case. Father's age = $2 \times 48 = 96$ years.

Probability

Q421 ? Options elimination ? **5/8**. Context: candidate choice. Favourable = 5; total = 8. $P = 5/8 = 5/8$.

Q422 ? Boundary case ? **6/11**. Context: medical screening. Favourable = 6; total = 11. $P = 6/11 = 6/11$.

Q423 ? Case split ? **1/2**. Context: urn experiment. Favourable = 7; total = 14. $P = 7/14 = 1/2$.

Q424 ? Contradiction check ? **2/11**. Context: risk estimate. Favourable = 2; total = 11. $P = 2/11 = 2/11$.

Q425 ? Statement I test ? **3/7**. Context: quality inspection. Favourable = 3; total = 7. $P = 3/7 = 3/7$.

Q426 ? Statement II test ? **2/5**. Context: card draw. Favourable = 4; total = 10. $P = 4/10 = 2/5$.

Q427 ? Combined statements ? **5/13**. Context: lottery selection. Favourable = 5; total = 13. $P = 5/13 = 5/13$.

Q428 ? Counting restriction ? **2/3**. Context: server failure. Favourable = 6; total = 9. $P = 6/9 = 2/3$.

Q429 ? Circular arrangement ? **7/12**. Context: candidate choice. Favourable = 7; total = 12. $P = 7/12 = 7/12$.

Q430 ? Complement method ? **2/9**. Context: medical screening. Favourable = 2; total = 9. $P = 2/9 = 2/9$.

Q431 ? Pattern: differences ? **1/4**. Context: urn experiment. Favourable = 3; total = 12. $P = 3/12 = 1/4$.

Q432 ? Pattern: ratios ? **1/2**. Context: risk estimate. Favourable = 4; total = 8. $P = 4/8 = 1/2$.

Q433 ? Pattern: alternating ? **5/11**. Context: quality inspection. Favourable = 5; total = 11. $P = 5/11 = 5/11$.

Q434 ? Pattern: squares / cubes ? **3/7**. Context: card draw. Favourable = 6; total = 14. $P = 6/14 = 3/7$.

Q435 ? Inference from facts ? **7/10**. Context: lottery selection. Favourable = 7; total = 10. $P = 7/10 = 7/10$.

Q436 ? Assumption check ? **2/7**. Context: server failure. Favourable = 2; total = 7. $P = 2/7 = 2/7$.

Q437 ? Cause vs correlation ? **3/10**. Context: candidate choice. Favourable = 3; total = 10. $P = 3/10 = 3/10$.

Q438 ? Grammar agreement ? **4/13**. Context: medical screening. Favourable = 4; total = 13. $P = 4/13 = 4/13$.

Q439 ? Vocabulary in context ? **5/9**. Context: urn experiment. Favourable = 5; total = 9. $P = 5/9 = 5/9$.

Q440 ? Mixed placement challenge ? **1/2**. Context: risk estimate. Favourable = 6; total = 12. $P = 6/12 = 1/2$.

HCF & LCM

Q461 ? Options elimination ? **27**. Context: batch size. By prime factors or Euclid's method, $HCF(54, 81) = 27$.

Q462 ? Boundary case ? **28**. Context: traffic signals. By prime factors or Euclid's method, $HCF(56, 84) = 28$.

Q463 ? Case split ? **29**. Context: medicine dosage. By prime factors or Euclid's method, $HCF(58, 87) = 29$.

Q464 ? Contradiction check ? **30**. Context: event recurrence. By prime factors or Euclid's method, $HCF(60, 90) = 30$.

Q465 ? Statement I test ? **31**. Context: maintenance cycle. By prime factors or Euclid's method, $HCF(62, 93) = 31$.

Q466 ? Statement II test ? **32**. Context: bell schedule. By prime factors or Euclid's method, $HCF(64, 96) = 32$.

Q467 ? Combined statements ? **33**. Context: tile packing. By prime factors or Euclid's method, $HCF(66, 99) = 33$.

Q468 ? Counting restriction ? **34**. Context: gear rotation. By prime factors or Euclid's method, $HCF(68, 102) = 34$.

Q469 ? Circular arrangement ? **35**. Context: batch size. By prime factors or Euclid's method, $HCF(70, 105) = 35$.

Q470 ? Complement method ? **36**. Context: traffic signals. By prime factors or Euclid's method, $HCF(72, 108) = 36$.

Q471 ? Pattern: differences ? **37**. Context: medicine dosage. By prime factors or Euclid's method, $HCF(74, 111) = 37$.

Q472 ? Pattern: ratios ? **38**. Context: event recurrence. By prime factors or Euclid's method, $HCF(76, 114) = 38$.

Q473 ? Pattern: alternating ? **39**. Context: maintenance cycle. By prime factors or Euclid's method, $HCF(78, 117) = 39$.

Q474 ? Pattern: squares / cubes ? **40**. Context: bell schedule. By prime factors or Euclid's method, $HCF(80, 120) = 40$.

Q475 ? Inference from facts ? **41**. Context: tile packing. By prime factors or Euclid's method, $HCF(82, 123) = 41$.

Q476 ? Assumption check ? **42**. Context: gear rotation. By prime factors or Euclid's method, $HCF(84, 126) = 42$.

Q477 ? Cause vs correlation ? **43**. Context: batch size. By prime factors or Euclid's method, $HCF(86, 129) = 43$.

Q478 ? Grammar agreement ? **44**. Context: traffic signals. By prime factors or Euclid's method, $HCF(88, 132) = 44$.

Q479 ? Vocabulary in context ? **45**. Context: medicine dosage. By prime factors or Euclid's method, $HCF(90, 135) = 45$.

Q480 ? Mixed placement challenge ? **46**. Context: event recurrence. By prime factors or Euclid's method, $HCF(92, 138) = 46$.

Quadratic Equations

Q501 ? Options elimination ? **5**. Context: rectangular field. Substitute $x=5$: $25+-2\times5-15=0$, so 5 is a root.

Q502 ? Boundary case ? **6**. Context: number pair. Substitute $x=6$: $36+-2\times6-24=0$, so 6 is a root.

Q503 ? Case split ? **7**. Context: motion model. Substitute $x=7$: $49+-2\times7-35=0$, so 7 is a root.

Q504 ? Contradiction check ? **8**. Context: equation roots. Substitute $x=8$: $64+-2\times8-48=0$, so 8 is a root.

Q505 ? Statement I test ? **9**. Context: projectile path. Substitute $x=9$: $81+-2\times9-63=0$, so 9 is a root.

Q506 ? Statement II test ? **10**. Context: area constraint. Substitute $x=10$: $100+-2\times10-80=0$, so 10 is a root.

Q507 ? Combined statements ? **2**. Context: profit maximum. Substitute $x=2$: $4+7\times2-18=0$, so 2 is a root.

Q508 ? Counting restriction ? **3**. Context: root condition. Substitute $x=3$: $9+0\times3-9=0$, so 3 is a root.

Q509 ? Circular arrangement ? **4**. Context: rectangular field. Substitute $x=4$: $16+0\times4-16=0$, so 4 is a root.

Q510 ? Complement method ? **5**. Context: number pair. Substitute $x=5$: $25+0\times5-25=0$, so 5 is a root.

Q511 ? Pattern: differences ? **6**. Context: motion model. Substitute $x=6$: $36+0\times6-36=0$, so 6 is a root.

Q512 ? Pattern: ratios ? **7**. Context: equation roots. Substitute $x=7$: $49+0\times7-49=0$, so 7 is a root.

Q513 ? Pattern: alternating ? **8**. Context: projectile path. Substitute $x=8$: $64+0\times8-64=0$, so 8 is a root.

Q514 ? Pattern: squares / cubes ? **9**. Context: area constraint. Substitute $x=9$: $81+0\times9-81=0$, so 9 is a root.

Q515 ? Inference from facts ? **10**. Context: profit maximum. Substitute $x=10$: $100+-7\times10-30=0$, so 10 is a root.

Q516 ? Assumption check ? **2**. Context: root condition. Substitute $x=2$: $4+2\times2-8=0$, so 2 is a root.

Q517 ? Cause vs correlation ? **3**. Context: rectangular field. Substitute $x=3$: $9+2\times3-15=0$, so 3 is a root.

Q518 ? Grammar agreement ? **4**. Context: number pair. Substitute $x=4$: $16+2\times4-24=0$, so 4 is a root.

Q519 ? Vocabulary in context ? **5**. Context: motion model. Substitute $x=5$: $25+2\times5-35=0$, so 5 is a root.

Q520 ? Mixed placement challenge ? **6**. Context: equation roots. Substitute $x=6$: $36+2\times6-48=0$, so 6 is a root.

Mensuration

Q541 ? Options elimination ? **116**. Context: paint coverage. Area = $l \times b = 29 \times 4 = 116 \text{ cm}^2$.

Q542 ? Boundary case ? **150**. Context: warehouse volume. Area = $l \times b = 30 \times 5 = 150 \text{ cm}^2$.

Q543 ? Case split ? **186**. Context: cone container. Area = $l \times b = 31 \times 6 = 186 \text{ cm}^2$.

Q544 ? Contradiction check ? **224**. Context: sphere packing. Area = $l \times b = 32 \times 7 = 224 \text{ cm}^2$.

Q545 ? Statement I test ? **264**. Context: tank capacity. Area = $l \times b = 33 \times 8 = 264 \text{ cm}^2$.

Q546 ? Statement II test ? **306**. Context: floor tiling. Area = $l \times b = 34 \times 9 = 306 \text{ cm}^2$.

Q547 ? Combined statements ? **350**. Context: cylindrical drum. Area = $l \times b = 35 \times 10 = 350 \text{ cm}^2$.

Q548 ? Counting restriction ? **144**. Context: circular track. Area = $l \times b = 36 \times 4 = 144 \text{ cm}^2$.

Q549 ? Circular arrangement ? **185**. Context: paint coverage. Area = $l \times b = 37 \times 5 = 185 \text{ cm}^2$.

Q550 ? Complement method ? **228**. Context: warehouse volume. Area = $l \times b = 38 \times 6 = 228 \text{ cm}^2$.

Q551 ? Pattern: differences ? **273**. Context: cone container. Area = $l \times b = 39 \times 7 = 273 \text{ cm}^2$.

Q552 ? Pattern: ratios ? **320**. Context: sphere packing. Area = $l \times b = 40 \times 8 = 320 \text{ cm}^2$.

Q553 ? Pattern: alternating ? **369**. Context: tank capacity. Area = $l \times b = 41 \times 9 = 369 \text{ cm}^2$.

Q554 ? Pattern: squares / cubes ? **420**. Context: floor tiling. Area = $l \times b = 42 \times 10 = 420 \text{ cm}^2$.

Q555 ? Inference from facts ? **172**. Context: cylindrical drum. Area = $l \times b = 43 \times 4 = 172 \text{ cm}^2$.

Q556 ? Assumption check ? **220**. Context: circular track. Area = $l \times b = 44 \times 5 = 220 \text{ cm}^2$.

Q557 ? Cause vs correlation ? **270**. Context: paint coverage. Area = $l \times b = 45 \times 6 = 270 \text{ cm}^2$.

Q558 ? Grammar agreement ? **322**. Context: warehouse volume. Area = $l \times b = 46 \times 7 = 322 \text{ cm}^2$.

Q559 ? Vocabulary in context ? **376**. Context: cone container. Area = $l \times b = 47 \times 8 = 376 \text{ cm}^2$.

Q560 ? Mixed placement challenge ? **432**. Context: sphere packing. Area = $l \times b = 48 \times 9 = 432 \text{ cm}^2$.

Series

Q581 ? Options elimination ? **42**. Context: stock movement. The common difference is 5; next = $37 + 5 = 42$.

Q582 ? Boundary case ? **47**. Context: machine output. The common difference is 6; next = $41 + 6 = 47$.

Q583 ? Case split ? **52**. Context: calendar code. The common difference is 7; next = $45 + 7 = 52$.

Q584 ? Contradiction check ? **33**. Context: exam puzzle. The common difference is 2; next = $31 + 2 = 33$.

Q585 ? Statement I test ? **38**. Context: code sequence. The common difference is 3; next = $35 + 3 = 38$.

Q586 ? Statement II test ? **43**. Context: production cycle. The common difference is 4; next = $39 + 4 = 43$.

Q587 ? Combined statements ? **48**. Context: number pattern. The common difference is 5; next = $43 + 5 = 48$.

Q588 ? Counting restriction ? **53**. Context: daily visitors. The common difference is 6; next = $47 + 6 = 53$.

Q589 ? Circular arrangement ? **58**. Context: stock movement. The common difference is 7; next = $51 + 7 = 58$.

Q590 ? Complement method ? **39**. Context: machine output. The common difference is 2; next = $37 + 2 = 39$.

Q591 ? Pattern: differences ? **44**. Context: calendar code. The common difference is 3; next = $41 + 3 = 44$.

Q592 ? Pattern: ratios ? **49**. Context: exam puzzle. The common difference is 4; next = $45 + 4 = 49$.

Q593 ? Pattern: alternating ? **54**. Context: code sequence. The common difference is 5; next = $49 + 5 = 54$.

Q594 ? Pattern: squares / cubes ? **59**. Context: production cycle. The common difference is 6; next = $53 + 6 = 59$.

Q595 ? Inference from facts ? **64**. Context: number pattern. The common difference is 7; next = $57 + 7 = 64$.

Q596 ? Assumption check ? **45**. Context: daily visitors. The common difference is 2; next = $43 + 2 = 45$.

Q597 ? Cause vs correlation ? **50**. Context: stock movement. The common difference is 3; next = $47 + 3 = 50$.

Q598 ? Grammar agreement ? **55**. Context: machine output. The common difference is 4; next = $51 + 4 = 55$.

Q599 ? Vocabulary in context ? **60**. Context: calendar code. The common difference is 5; next = $55 + 5 = 60$.

Q600 ? Mixed placement challenge ? **65**. Context: exam puzzle. The common difference is 6; next = $59 + 6 = 65$.

Direction Sense

Q621 ? Options elimination ? **24.52**. Context: treasure hunt. Distance = $\sqrt{(24^2+5^2)} = 24.52 \text{ km}$.

Q622 ? Boundary case ? **25.71**. Context: robot movement. Distance = $\sqrt{(25^2+6^2)} = 25.71 \text{ km}$.

Q623 ? Case split ? **26.93**. Context: warehouse aisle. Distance = $\sqrt{(26^2+7^2)} = 26.93 \text{ km}$.

Q624 ? Contradiction check ? **28.16**. Context: emergency response. Distance = $\sqrt{(27^2+8^2)} = 28.16 \text{ km}$.

Q625 ? Statement I test ? **28.28**. Context: campus navigation. Distance = $\sqrt{(28^2+4^2)} = 28.28 \text{ km}$.

Q626 ? Statement II test ? **29.43**. Context: drone route. Distance = $\sqrt{(29^2+5^2)} = 29.43 \text{ km}$.

Q627 ? Combined statements ? **30.59**. Context: delivery map. Distance = $\sqrt{(30^2+6^2)} = 30.59 \text{ km}$.

Q628 ? Counting restriction ? **31.78**. Context: city grid. Distance = $\sqrt{(31^2+7^2)} = 31.78 \text{ km}$.

Q629 ? Circular arrangement ? **32.98**. Context: treasure hunt. Distance = $\sqrt{(32^2+8^2)} = 32.98 \text{ km}$.

Q630 ? Complement method ? **33.24**. Context: robot movement. Distance = $\sqrt{(33^2+4^2)} = 33.24 \text{ km}$.

Q631 ? Pattern: differences ? **34.37**. Context: warehouse aisle. Distance = $\sqrt{(34^2+5^2)} = 34.37 \text{ km}$.

Q632 ? Pattern: ratios ? **35.51**. Context: emergency response. Distance = $\sqrt{(35^2+6^2)} = 35.51 \text{ km}$.

Q633 ? Pattern: alternating ? **36.67**. Context: campus navigation. Distance = $\sqrt{(36^2+7^2)} = 36.67 \text{ km}$.

Q634 ? Pattern: squares / cubes ? **37.85**. Context: drone route. Distance = $\sqrt{(37^2+8^2)} = 37.85 \text{ km}$.

Q635 ? Inference from facts ? **38.21**. Context: delivery map. Distance = $\sqrt{(38^2+4^2)} = 38.21 \text{ km}$.

Q636 ? Assumption check ? **39.32**. Context: city grid. Distance = $\sqrt{(39^2+5^2)} = 39.32 \text{ km}$.

Q637 ? Cause vs correlation ? **40.45**. Context: treasure hunt. Distance = $\sqrt{(40^2+6^2)} = 40.45 \text{ km}$.

Q638 ? Grammar agreement ? **41.59**. Context: robot movement. Distance = $\sqrt{(41^2+7^2)} = 41.59 \text{ km}$.

Q639 ? Vocabulary in context ? **42.76**. Context: warehouse aisle. Distance = $\sqrt{(42^2+8^2)} = 42.76 \text{ km}$.

Q640 ? Mixed placement challenge ? **43.19**. Context: emergency response. Distance = $\sqrt{(43^2+4^2)} = 43.19 \text{ km}$.

Coding–Decoding

Q661 ? Options elimination ? **ECV**. Context: pattern code. $C \rightarrow E$, $A \rightarrow C$, $T \rightarrow V$; answer ECV.

Q662 ? Boundary case ? **FDW**. Context: symbol language. $C \rightarrow F$, $A \rightarrow D$, $T \rightarrow W$; answer FDW.

Q663 ? Case split ? **GEX**. Context: ID transformation. $C \rightarrow G$, $A \rightarrow E$, $T \rightarrow X$; answer GEX.

Q664 ? Contradiction check ? **HFY**. Context: message protocol. $C \rightarrow H$, $A \rightarrow F$, $T \rightarrow Y$; answer HFY.

Q665 ? Statement I test ? **DBU**. Context: secure token. $C \rightarrow D, A \rightarrow B, T \rightarrow U$; answer DBU.

Q666 ? Statement II test ? **ECV**. Context: letter shift. $C \rightarrow E, A \rightarrow C, T \rightarrow V$; answer ECV.

Q667 ? Combined statements ? **FDW**. Context: phone keypad. $C \rightarrow F, A \rightarrow D, T \rightarrow W$; answer FDW.

Q668 ? Counting restriction ? **GEX**. Context: encrypted label. $C \rightarrow G, A \rightarrow E, T \rightarrow X$; answer GEX.

Q669 ? Circular arrangement ? **HFY**. Context: pattern code. $C \rightarrow H, A \rightarrow F, T \rightarrow Y$; answer HFY.

Q670 ? Complement method ? **DBU**. Context: symbol language. $C \rightarrow D, A \rightarrow B, T \rightarrow U$; answer DBU.

Q671 ? Pattern: differences ? **ECV**. Context: ID transformation. $C \rightarrow E, A \rightarrow C, T \rightarrow V$; answer ECV.

Q672 ? Pattern: ratios ? **FDW**. Context: message protocol. $C \rightarrow F, A \rightarrow D, T \rightarrow W$; answer FDW.

Q673 ? Pattern: alternating ? **GEX**. Context: secure token. $C \rightarrow G, A \rightarrow E, T \rightarrow X$; answer GEX.

Q674 ? Pattern: squares / cubes ? **HFY**. Context: letter shift. $C \rightarrow H, A \rightarrow F, T \rightarrow Y$; answer HFY.

Q675 ? Inference from facts ? **DBU**. Context: phone keypad. $C \rightarrow D, A \rightarrow B, T \rightarrow U$; answer DBU.

Q676 ? Assumption check ? **ECV**. Context: encrypted label. $C \rightarrow E, A \rightarrow C, T \rightarrow V$; answer ECV.

Q677 ? Cause vs correlation ? **FDW**. Context: pattern code. $C \rightarrow F, A \rightarrow D, T \rightarrow W$; answer FDW.

Q678 ? Grammar agreement ? **GEX**. Context: symbol language. $C \rightarrow G, A \rightarrow E, T \rightarrow X$; answer GEX.

Q679 ? Vocabulary in context ? **HFY**. Context: ID transformation. $C \rightarrow H, A \rightarrow F, T \rightarrow Y$; answer HFY.

Q680 ? Mixed placement challenge ? **DBU**. Context: message protocol. $C \rightarrow D, A \rightarrow B, T \rightarrow U$; answer DBU.

Data Interpretation

Q701 ? Options elimination ? **40**. Context: budget sheet. Increase = 90. Increase% = $90/225 \times 100 = 40\%$.

Q702 ? Boundary case ? **39.13**. Context: product funnel. Increase = 90. Increase% = $90/230 \times 100 = 39.13\%$.

Q703 ? Case split ? **38.3**. Context: survey graph. Increase = 90. Increase% = $90/235 \times 100 = 38.3\%$.

Q704 ? Contradiction check ? **37.5**. Context: regional KPI. Increase = 90. Increase% = $90/240 \times 100 = 37.5\%$.

Q705 ? Statement I test ? **36.73**. Context: quarterly dashboard. Increase = 90. Increase% = $90/245 \times 100 = 36.73\%$.

Q706 ? Statement II test ? **36**. Context: placement report. Increase = 90. Increase% = $90/250 \times 100 = 36\%$.

Q707 ? Combined statements ? **35.29**. Context: sales chart. Increase = 90. Increase% = $90/255 \times 100 = 35.29\%$.

Q708 ? Counting restriction ? **34.62**. Context: attendance table. Increase = 90. Increase% = $90/260 \times 100 = 34.62\%$.

Q709 ? Circular arrangement ? **33.96**. Context: budget sheet. Increase = 90. Increase% = $90/265 \times 100 = 33.96\%$.

Q710 ? Complement method ? **33.33**. Context: product funnel. Increase = 90. Increase% = $90/270 \times 100 = 33.33\%$.

Q711 ? Pattern: differences ? **32.73**. Context: survey graph. Increase = 90. Increase% = $90/275 \times 100 = 32.73\%$.

Q712 ? Pattern: ratios ? **32.14**. Context: regional KPI. Increase = 90. Increase% = $90/280 \times 100 = 32.14\%$.

Q713 ? Pattern: alternating ? **31.58**. Context: quarterly dashboard. Increase = 90. Increase% = $90/285 \times 100 = 31.58\%$.

Q714 ? Pattern: squares / cubes ? **31.03**. Context: placement report. Increase = 90. Increase% = $90/290 \times 100 = 31.03\%$.

Q715 ? Inference from facts ? **30.51**. Context: sales chart. Increase = 90. Increase% = $90/295 \times 100 = 30.51\%$.

Q716 ? Assumption check ? **30**. Context: attendance table. Increase = 90. Increase% = $90/300 \times 100 = 30\%$.

Q717 ? Cause vs correlation ? **29.51**. Context: budget sheet. Increase = 90. Increase% = $90/305 \times 100 = 29.51\%$.

Q718 ? Grammar agreement ? **29.03**. Context: product funnel. Increase = 90. Increase% = $90/310 \times 100 = 29.03\%$.

Q719 ? Vocabulary in context ? **28.57**. Context: survey graph. Increase = 90. Increase% = $90/315 \times 100 = 28.57\%$.

Q720 ? Mixed placement challenge ? **28.13**. Context: regional KPI. Increase = 90. Increase% = $90/320 \times 100 = 28.13\%$.

Clocks

Q741 ? Options elimination ? **165**. Context: time-zone display. Angle = $|30H - 5.5M| = 165^\circ$.

Q742 ? Boundary case ? **170**. Context: race timer. Angle = $|30H - 5.5M| = 170^\circ$.

Q743 ? Case split ? **145**. Context: office clock. Angle = $|30H - 5.5M| = 145^\circ$.

Q744 ? Contradiction check ? **90**. Context: gain-loss audit. Angle = $|30H - 5.5M| = 90^\circ$.

Q745 ? Statement I test ? **65**. Context: analog dial. Angle = $|30H - 5.5M| = 65^\circ$.

Q746 ? Statement II test ? **40**. Context: watch calibration. Angle = $|30H - 5.5M| = 40^\circ$.

Q747 ? Combined statements ? **15**. Context: factory shift. Angle = $|30H - 5.5M| = 15^\circ$.

Q748 ? Counting restriction ? **10**. Context: clock-angle puzzle. Angle = $|30H - 5.5M| = 10^\circ$.

Q749 ? Circular arrangement ? **35**. Context: time-zone display. Angle = $|30H - 5.5M| = 35^\circ$.

Q750 ? Complement method ? **90**. Context: race timer. Angle = $|30H - 5.5M| = 90^\circ$.

Q751 ? Pattern: differences ? **115**. Context: office clock. Angle = $|30H - 5.5M| = 115^\circ$.

Q752 ? Pattern: ratios ? **140**. Context: gain-loss audit. Angle = $|30H - 5.5M| = 140^\circ$.

Q753 ? Pattern: alternating ? **135**. Context: analog dial. Angle = $|30H - 5.5M| = 135^\circ$.

Q754 ? Pattern: squares / cubes ? **160**. Context: watch calibration. Angle = $|30H - 5.5M| = 160^\circ$.

Q755 ? Inference from facts ? **175**. Context: factory shift. Angle = $|30H - 5.5M| = 175^\circ$.

Q756 ? Assumption check ? **120**. Context: clock-angle puzzle. Angle = $|30H - 5.5M| = 120^\circ$.

Q757 ? Cause vs correlation ? **95**. Context: time-zone display. Angle = $|30H - 5.5M| = 95^\circ$.

Q758 ? Grammar agreement ? **70**. Context: race timer. Angle = $|30H - 5.5M| = 70^\circ$.

Q759 ? Vocabulary in context ? **45**. Context: office clock. Angle = $|30H - 5.5M| = 45^\circ$.

Q760 ? Mixed placement challenge ? **20**. Context: gain-loss audit. Angle = $|30H - 5.5M| = 20^\circ$.

Verbal Reasoning

Q781 ? Options elimination ? **Neither answer is correct.**. Context: cloze test. This option follows standard agreement, pronoun, and preposition rules.

Q782 ? Boundary case ? **She is senior to me.**. Context: sentence completion. This option follows standard agreement, pronoun, and preposition rules.

Q783 ? Case split ? **The committee has submitted its report.**. Context: parallelism. This option follows standard agreement, pronoun, and preposition rules.

Q784 ? Contradiction check ? **Each of the players has a locker.**. Context: pronoun reference. This option follows standard agreement, pronoun, and preposition rules.

Q785 ? Statement I test ? **Neither answer is correct.**. Context: grammar audit. This option follows standard agreement, pronoun, and preposition rules.

Q786 ? Statement II test ? **She is senior to me.**. Context: sentence correction. This option follows standard agreement, pronoun, and preposition rules.

Q787 ? Combined statements ? **The committee has submitted its report.**. Context: usage choice. This option follows standard agreement, pronoun, and preposition rules.

Q788 ? Counting restriction ? **Each of the players has a locker.**. Context: error spotting. This option follows standard agreement, pronoun, and preposition rules.

Q789 ? Circular arrangement ? **Neither answer is correct.**. Context: cloze test. This option follows standard agreement, pronoun, and preposition rules.

Q790 ? Complement method ? **She is senior to me.**. Context: sentence completion. This option follows standard agreement, pronoun, and preposition rules.

Q791 ? Pattern: differences ? **The committee has submitted its report.**. Context: parallelism. This option follows standard agreement, pronoun, and preposition rules.

Q792 ? Pattern: ratios ? **Each of the players has a locker.**. Context: pronoun reference. This option follows standard agreement, pronoun, and preposition rules.

Q793 ? Pattern: alternating ? **Neither answer is correct.**. Context: grammar audit. This option follows standard agreement, pronoun, and preposition rules.

Q794 ? Pattern: squares / cubes ? **She is senior to me.**. Context: sentence correction. This option follows standard agreement, pronoun, and preposition rules.

Q795 ? Inference from facts ? **The committee has submitted its report.**. Context: usage choice. This option follows standard agreement, pronoun, and preposition rules.

Q796 ? Assumption check ? **Each of the players has a locker.**. Context: error spotting. This option follows standard agreement, pronoun, and preposition rules.

Q797 ? Cause vs correlation ? **Neither answer is correct.**. Context: cloze test. This option follows standard agreement, pronoun, and preposition rules.

Q798 ? Grammar agreement ? **She is senior to me.**. Context: sentence completion. This option follows standard agreement, pronoun, and preposition rules.

Q799 ? Vocabulary in context ? **The committee has submitted its report.**. Context: parallelism. This option follows standard agreement, pronoun, and preposition rules.

Q800 ? Mixed placement challenge ? **Each of the players has a locker.**. Context: pronoun reference. This option follows standard agreement, pronoun, and preposition rules.

Partnership

Q821 ? Options elimination ? **8493.15**. Context: agency ownership. Money-month ratio=372:66. A's share= $10000 \times 372/438 = \blacksquare 8493.15$.

Q822 ? Boundary case ? **8205.13**. Context: franchise investment. Money-month ratio=384:84. A's share= $10000 \times 384/468 = \blacksquare 8205.13$.

Q823 ? Case split ? **7920**. Context: joint research. Money-month ratio=396:104. A's share= $10000 \times 396/500 = \blacksquare 7920$.

Q824 ? Contradiction check ? **7640.45**. Context: co-founder exit. Money-month ratio=408:126. A's share= $10000 \times 408/534 = \blacksquare 7640.45$.

Q825 ? Statement I test ? **7368.42**. Context: startup equity split. Money-month ratio=420:150. A's share= $10000 \times 420/570 = \blacksquare 7368.42$.

Q826 ? Statement II test ? **7105.26**. Context: consulting venture. Money-month ratio=432:176. A's share= $10000 \times 432/608 = \blacksquare 7105.26$.

Q827 ? Combined statements ? **8222.22**. Context: restaurant partners. Money-month ratio=444:96. A's share= $10000 \times 444/540 = \blacksquare 8222.22$.

Q828 ? Counting restriction ? **8941.18**. Context: campus project. Money-month ratio=456:54. A's share= $10000 \times 456/510 = \blacksquare 8941.18$.

Q829 ? Circular arrangement ? **8698.88**. Context: agency ownership. Money-month ratio=468:70. A's share= $10000 \times 468/538 = \blacksquare 8698.88$.

Q830 ? Complement method ? **8450.7**. Context: franchise investment. Money-month ratio=480:88. A's share= $10000 \times 480/568 = \blacksquare 8450.7$.

Q831 ? Pattern: differences ? **8200**. Context: joint research. Money-month ratio=492:108. A's share= $10000 \times 492/600 = \blacksquare 8200$.

Q832 ? Pattern: ratios ? **7949.53**. Context: co-founder exit. Money-month ratio=504:130. A's share= $10000 \times 504/634 = \blacksquare 7949.53$.

Q833 ? Pattern: alternating ? **7701.49**. Context: startup equity split. Money-month ratio=516:154. A's share= $10000 \times 516/670 = \blacksquare 7701.49$.

Q834 ? Pattern: squares / cubes ? **7457.63**. Context: consulting venture. Money-month ratio=528:180. A's share= $10000 \times 528/708 = \blacksquare 7457.63$.

Q835 ? Inference from facts ? **8490.57**. Context: restaurant partners. Money-month ratio=540:96. A's share= $10000 \times 540/636 = \blacksquare 8490.57$.

Q836 ? Assumption check ? **9078.95**. Context: campus project. Money-month ratio=552:56. A's share= $10000 \times 552 / 608 = \blacksquare 9078.95$.

Q837 ? Cause vs correlation ? **8867.92**. Context: agency ownership. Money-month ratio=564:72. A's share= $10000 \times 564 / 636 = \blacksquare 8867.92$.

Q838 ? Grammar agreement ? **8648.65**. Context: franchise investment. Money-month ratio=576:90. A's share= $10000 \times 576 / 666 = \blacksquare 8648.65$.

Q839 ? Vocabulary in context ? **8424.07**. Context: joint research. Money-month ratio=588:110. A's share= $10000 \times 588 / 698 = \blacksquare 8424.07$.

Q840 ? Mixed placement challenge ? **8196.72**. Context: co-founder exit. Money-month ratio=600:132. A's share= $10000 \times 600 / 732 = \blacksquare 8196.72$.

Mixture & Alligation

Q861 ? Options elimination ? **50.29**. Context: food-grade mix. Total cost= $\blacksquare 352$; quantity=7 kg; price= $\blacksquare 50.29/\text{kg}$.

Q862 ? Boundary case ? **50.89**. Context: paint formulation. Total cost= $\blacksquare 458$; quantity=9 kg; price= $\blacksquare 50.89/\text{kg}$.

Q863 ? Case split ? **51.64**. Context: feed composition. Total cost= $\blacksquare 568$; quantity=11 kg; price= $\blacksquare 51.64/\text{kg}$.

Q864 ? Contradiction check ? **58.44**. Context: water treatment. Total cost= $\blacksquare 526$; quantity=9 kg; price= $\blacksquare 58.44/\text{kg}$.

Q865 ? Statement I test ? **52.5**. Context: fuel blend. Total cost= $\blacksquare 315$; quantity=6 kg; price= $\blacksquare 52.5/\text{kg}$.

Q866 ? Statement II test ? **53.5**. Context: coffee recipe. Total cost= $\blacksquare 428$; quantity=8 kg; price= $\blacksquare 53.5/\text{kg}$.

Q867 ? Combined statements ? **54.5**. Context: metal alloy. Total cost= $\blacksquare 545$; quantity=10 kg; price= $\blacksquare 54.5/\text{kg}$.

Q868 ? Counting restriction ? **61.75**. Context: chemical concentration. Total cost= $\blacksquare 494$; quantity=8 kg; price= $\blacksquare 61.75/\text{kg}$.

Q869 ? Circular arrangement ? **61.5**. Context: food-grade mix. Total cost= $\blacksquare 615$; quantity=10 kg; price= $\blacksquare 61.5/\text{kg}$.

Q870 ? Complement method ? **47.14**. Context: paint formulation. Total cost= $\blacksquare 330$; quantity=7 kg; price= $\blacksquare 47.14/\text{kg}$.

Q871 ? Pattern: differences ? **48.78**. Context: feed composition. Total cost= $\blacksquare 439$; quantity=9 kg; price= $\blacksquare 48.78/\text{kg}$.

Q872 ? Pattern: ratios ? **60.57**. Context: water treatment. Total cost= $\blacksquare 424$; quantity=7 kg; price= $\blacksquare 60.57/\text{kg}$.

Q873 ? Pattern: alternating ? **59.67**. Context: fuel blend. Total cost= $\blacksquare 537$; quantity=9 kg; price= $\blacksquare 59.67/\text{kg}$.

Q874 ? Pattern: squares / cubes ? **59.45**. Context: coffee recipe. Total cost= $\blacksquare 654$; quantity=11 kg; price= $\blacksquare 59.45/\text{kg}$.

Q875 ? Inference from facts ? **50**. Context: metal alloy. Total cost= $\blacksquare 400$; quantity=8 kg; price= $\blacksquare 50/\text{kg}$.

Q876 ? Assumption check ? **62.67**. Context: chemical concentration. Total cost= $\blacksquare 376$; quantity=6 kg; price= $\blacksquare 62.67/\text{kg}$.

Q877 ? Cause vs correlation ? **62**. Context: food-grade mix. Total cost= $\blacksquare 496$; quantity=8 kg; price= $\blacksquare 62/\text{kg}$.

Q878 ? Grammar agreement ? **62**. Context: paint formulation. Total cost= $\blacksquare 620$; quantity=10 kg; price= $\blacksquare 62/\text{kg}$.

Q879 ? Vocabulary in context ? **62.33**. Context: feed composition. Total cost= $\blacksquare 748$; quantity=12 kg; price= $\blacksquare 62.33/\text{kg}$.

Q880 ? Mixed placement challenge ? **52**. Context: water treatment. Total cost= $\blacksquare 260$; quantity=5 kg; price= $\blacksquare 52/\text{kg}$.

Pipes & Cisterns

Q901 ? Options elimination ? **7.54**. Context: production vessel. Rate= $1/11 + 1/24$; time= $264/35 = 7.54$ hours.

Q902 ? Boundary case ? **8.11**. Context: water treatment plant. Rate= $1/12 + 1/25$; time= $300/37 = 8.11$ hours.

Q903 ? Case split ? **8.67**. Context: battery charge cycle. Rate= $1/13 + 1/26$; time= $338/39 = 8.67$ hours.

Q904 ? Contradiction check ? **7.24**. Context: pipeline network. Rate= $1/14 + 1/15$; time= $210/29 = 7.24$ hours.

Q905 ? Statement I test ? **7.74**. Context: reservoir controls. Rate= $1/15 + 1/16$; time= $240/31 = 7.74$ hours.

Q906 ? Statement II test ? **8.24**. Context: swimming pool. Rate= $1/16 + 1/17$; time= $272/33 = 8.24$ hours.

Q907 ? Combined statements ? **8.74**. Context: server upload queue. Rate= $1/17 + 1/18$; time= $306/35 = 8.74$ hours.

Q908 ? Counting restriction ? **9.24**. Context: irrigation tank. Rate= $1/18 + 1/19$; time= $342/37 = 9.24$ hours.

Q909 ? Circular arrangement ? **9.74**. Context: production vessel. Rate= $1/19 + 1/20$; time= $380/39 = 9.74$ hours.

Q910 ? Complement method ? **6.77**. Context: water treatment plant. Rate= $1/10 + 1/21$; time= $210/31 = 6.77$ hours.

Q911 ? Pattern: differences ? **7.33**. Context: battery charge cycle. Rate= $1/11 + 1/22$; time= $242/33 = 7.33$ hours.

Q912 ? Pattern: ratios ? **7.89**. Context: pipeline network. Rate= $1/12 + 1/23$; time= $276/35 = 7.89$ hours.

Q913 ? Pattern: alternating ? **8.43**. Context: reservoir controls. Rate= $1/13 + 1/24$; time= $312/37 = 8.43$ hours.

Q914 ? Pattern: squares / cubes ? **8.97**. Context: swimming pool. Rate= $1/14 + 1/25$; time= $350/39 = 8.97$ hours.

Q915 ? Inference from facts ? **9.51**. Context: server upload queue. Rate= $1/15 + 1/26$; time= $390/41 = 9.51$ hours.

Q916 ? Assumption check ? **7.74**. Context: irrigation tank. Rate= $1/16 + 1/15$; time= $240/31 = 7.74$ hours.

Q917 ? Cause vs correlation ? **8.24**. Context: production vessel. Rate= $1/17 + 1/16$; time= $272/33 = 8.24$ hours.

Q918 ? Grammar agreement ? **8.74**. Context: water treatment plant. Rate= $1/18 + 1/17$; time= $306/35 = 8.74$ hours.

Q919 ? Vocabulary in context ? **9.24**. Context: battery charge cycle. Rate= $1/19 + 1/18$; time= $342/37 = 9.24$ hours.

Q920 ? Mixed placement challenge ? **6.55**. Context: pipeline network. Rate= $1/10 + 1/19$; time= $190/29 = 6.55$ hours.

Boats & Streams

Q941 ? Options elimination ? **19**. Context: lake patrol. Boat speed= $(21+17)/2 = 19$ km/h.

Q942 ? Boundary case ? **20**. Context: rowing trial. Boat speed= $(23+17)/2=20$ km/h.

Q943 ? Case split ? **21**. Context: stream survey. Boat speed= $(25+17)/2=21$ km/h.

Q944 ? Contradiction check ? **10**. Context: harbour transfer. Boat speed= $(15+5)/2=10$ km/h.

Q945 ? Statement I test ? **11**. Context: river ferry. Boat speed= $(12+10)/2=11$ km/h.

Q946 ? Statement II test ? **12**. Context: rescue boat. Boat speed= $(14+10)/2=12$ km/h.

Q947 ? Combined statements ? **13**. Context: canal logistics. Boat speed= $(16+10)/2=13$ km/h.

Q948 ? Counting restriction ? **14**. Context: upstream cargo. Boat speed= $(18+10)/2=14$ km/h.

Q949 ? Circular arrangement ? **15**. Context: lake patrol. Boat speed= $(20+10)/2=15$ km/h.

Q950 ? Complement method ? **16**. Context: rowing trial. Boat speed= $(17+15)/2=16$ km/h.

Q951 ? Pattern: differences ? **17**. Context: stream survey. Boat speed= $(19+15)/2=17$ km/h.

Q952 ? Pattern: ratios ? **18**. Context: harbour transfer. Boat speed= $(21+15)/2=18$ km/h.

Q953 ? Pattern: alternating ? **19**. Context: river ferry. Boat speed= $(23+15)/2=19$ km/h.

Q954 ? Pattern: squares / cubes ? **20**. Context: rescue boat. Boat speed= $(25+15)/2=20$ km/h.

Q955 ? Inference from facts ? **21**. Context: canal logistics. Boat speed= $(22+20)/2=21$ km/h.

Q956 ? Assumption check ? **10**. Context: upstream cargo. Boat speed= $(12+8)/2=10$ km/h.

Q957 ? Cause vs correlation ? **11**. Context: lake patrol. Boat speed= $(14+8)/2=11$ km/h.

Q958 ? Grammar agreement ? **12**. Context: rowing trial. Boat speed= $(16+8)/2=12$ km/h.

Q959 ? Vocabulary in context ? **13**. Context: stream survey. Boat speed= $(18+8)/2=13$ km/h.

Q960 ? Mixed placement challenge ? **14**. Context: harbour transfer. Boat speed= $(15+13)/2=14$ km/h.

Calendar

Q981 ? Options elimination ? **Wednesday**. Context: leap-year check. $51 \bmod 7=2$; moving 2 days gives Wednesday.

Q982 ? Boundary case ? **Friday**. Context: delivery calendar. $52 \bmod 7=3$; moving 3 days gives Friday.

Q983 ? Case split ? **Sunday**. Context: meeting rotation. $53 \bmod 7=4$; moving 4 days gives Sunday.

Q984 ? Contradiction check ? **Tuesday**. Context: roster cycle. $54 \bmod 7=5$; moving 5 days gives Tuesday.

Q985 ? Statement I test ? **Thursday**. Context: release schedule. $55 \bmod 7=6$; moving 6 days gives Thursday.

Q986 ? Statement II test ? **Saturday**. Context: weekday audit. $56 \bmod 7=0$; moving 0 days gives Saturday.

Q987 ? Combined statements ? **Monday**. Context: festival planning. $57 \bmod 7=1$; moving 1 days gives Monday.

Q988 ? Counting restriction ? **Wednesday**. Context: exam timetable. $58 \bmod 7=2$; moving 2 days gives Wednesday.

Q989 ? Circular arrangement ? **Friday**. Context: leap-year check. $59 \bmod 7=3$; moving 3 days gives Friday.

Q990 ? Complement method ? **Sunday**. Context: delivery calendar. $60 \bmod 7=4$; moving 4 days gives Sunday.

Q991 ? Pattern: differences ? **Tuesday**. Context: meeting rotation. $61 \bmod 7=5$; moving 5 days gives Tuesday.

Q992 ? Pattern: ratios ? **Thursday**. Context: roster cycle. $62 \bmod 7=6$; moving 6 days gives Thursday.

Q993 ? Pattern: alternating ? **Saturday**. Context: release schedule. $63 \bmod 7=0$; moving 0 days gives Saturday.

Q994 ? Pattern: squares / cubes ? **Monday**. Context: weekday audit. $64 \bmod 7=1$; moving 1 days gives Monday.

Q995 ? Inference from facts ? **Wednesday**. Context: festival planning. $65 \bmod 7=2$; moving 2 days gives Wednesday.

Q996 ? Assumption check ? **Friday**. Context: exam timetable. $66 \bmod 7=3$; moving 3 days gives Friday.

Q997 ? Cause vs correlation ? **Sunday**. Context: leap-year check. $67 \bmod 7=4$; moving 4 days gives Sunday.

Q998 ? Grammar agreement ? **Tuesday**. Context: delivery calendar. $68 \bmod 7=5$; moving 5 days gives Tuesday.

Q999 ? Vocabulary in context ? **Thursday**. Context: meeting rotation. $69 \bmod 7=6$; moving 6 days gives Thursday.

Q1000 ? Mixed placement challenge ? **Saturday**. Context: roster cycle. $70 \bmod 7=0$; moving 0 days gives Saturday.

Permutation

Q1021 ? Options elimination ? **20**. Context: team roles. $5P2=5!/(3)!=20$.

Q1022 ? Boundary case ? **120**. Context: route ordering. $6P3=6!/(3)!=120$.

Q1023 ? Case split ? **840**. Context: task scheduling. $7P4=7!/(3)!=840$.

Q1024 ? Contradiction check ? **56**. Context: ID generation. $8P2=8!/(6)!=56$.

Q1025 ? Statement I test ? **24**. Context: password design. $4P3=4!/(1)!=24$.

Q1026 ? Statement II test ? **120**. Context: podium finish. $5P4=5!/(1)!=120$.

Q1027 ? Combined statements ? **30**. Context: seat allocation. $6P2=6!/(4)!=30$.

Q1028 ? Counting restriction ? **210**. Context: letter arrangement. $7P3=7!/(4)!=210$.

Q1029 ? Circular arrangement ? **1680**. Context: team roles. $8P4=8!/(4)!=1680$.
 Q1030 ? Complement method ? **12**. Context: route ordering. $4P2=4!/(2)!=12$.
 Q1031 ? Pattern: differences ? **60**. Context: task scheduling. $5P3=5!/(2)!=60$.
 Q1032 ? Pattern: ratios ? **360**. Context: ID generation. $6P4=6!/(2)!=360$.
 Q1033 ? Pattern: alternating ? **42**. Context: password design. $7P2=7!/(5)!=42$.
 Q1034 ? Pattern: squares / cubes ? **336**. Context: podium finish. $8P3=8!/(5)!=336$.
 Q1035 ? Inference from facts ? **24**. Context: seat allocation. $4P4=4!/(0)!=24$.
 Q1036 ? Assumption check ? **20**. Context: letter arrangement. $5P2=5!/(3)!=20$.
 Q1037 ? Cause vs correlation ? **120**. Context: team roles. $6P3=6!/(3)!=120$.
 Q1038 ? Grammar agreement ? **840**. Context: route ordering. $7P4=7!/(3)!=840$.
 Q1039 ? Vocabulary in context ? **56**. Context: task scheduling. $8P2=8!/(6)!=56$.
 Q1040 ? Mixed placement challenge ? **24**. Context: ID generation. $4P3=4!/(1)!=24$.

Combination

Q1061 ? Options elimination ? **28**. Context: scholarship group. $8C2=8!/[2!6!] = 28$.
 Q1062 ? Boundary case ? **84**. Context: menu bundle. $9C3=9!/[3!6!] = 84$.
 Q1063 ? Case split ? **210**. Context: hiring shortlist. $10C4=10!/[4!6!] = 210$.
 Q1064 ? Contradiction check ? **10**. Context: lottery sample. $5C2=5!/[2!3!] = 10$.
 Q1065 ? Statement I test ? **20**. Context: committee selection. $6C3=6!/[3!3!] = 20$.
 Q1066 ? Statement II test ? **35**. Context: team shortlist. $7C4=7!/[4!3!] = 35$.
 Q1067 ? Combined statements ? **28**. Context: feature selection. $8C2=8!/[2!6!] = 28$.
 Q1068 ? Counting restriction ? **84**. Context: panel formation. $9C3=9!/[3!6!] = 84$.
 Q1069 ? Circular arrangement ? **210**. Context: scholarship group. $10C4=10!/[4!6!] = 210$.
 Q1070 ? Complement method ? **10**. Context: menu bundle. $5C2=5!/[2!3!] = 10$.
 Q1071 ? Pattern: differences ? **20**. Context: hiring shortlist. $6C3=6!/[3!3!] = 20$.
 Q1072 ? Pattern: ratios ? **35**. Context: lottery sample. $7C4=7!/[4!3!] = 35$.
 Q1073 ? Pattern: alternating ? **28**. Context: committee selection. $8C2=8!/[2!6!] = 28$.
 Q1074 ? Pattern: squares / cubes ? **84**. Context: team shortlist. $9C3=9!/[3!6!] = 84$.
 Q1075 ? Inference from facts ? **210**. Context: feature selection. $10C4=10!/[4!6!] = 210$.
 Q1076 ? Assumption check ? **10**. Context: panel formation. $5C2=5!/[2!3!] = 10$.
 Q1077 ? Cause vs correlation ? **20**. Context: scholarship group. $6C3=6!/[3!3!] = 20$.
 Q1078 ? Grammar agreement ? **35**. Context: menu bundle. $7C4=7!/[4!3!] = 35$.
 Q1079 ? Vocabulary in context ? **28**. Context: hiring shortlist. $8C2=8!/[2!6!] = 28$.
 Q1080 ? Mixed placement challenge ? **84**. Context: lottery sample. $9C3=9!/[3!6!] = 84$.

Number System

Q1101 ? Options elimination ? **1**. Context: billing sequence. $457=6 \times 76+1$; remainder=1.
 Q1102 ? Boundary case ? **5**. Context: integer game. $474=7 \times 67+5$; remainder=5.
 Q1103 ? Case split ? **3**. Context: factor table. $491=8 \times 61+3$; remainder=3.
 Q1104 ? Contradiction check ? **4**. Context: base conversion. $508=9 \times 56+4$; remainder=4.
 Q1105 ? Statement I test ? **5**. Context: remainder audit. $525=10 \times 52+5$; remainder=5.
 Q1106 ? Statement II test ? **3**. Context: digit puzzle. $542=11 \times 49+3$; remainder=3.
 Q1107 ? Combined statements ? **1**. Context: number machine. $559=3 \times 186+1$; remainder=1.
 Q1108 ? Counting restriction ? **0**. Context: modular code. $576=4 \times 144+0$; remainder=0.
 Q1109 ? Circular arrangement ? **3**. Context: billing sequence. $593=5 \times 118+3$; remainder=3.
 Q1110 ? Complement method ? **4**. Context: integer game. $610=6 \times 101+4$; remainder=4.
 Q1111 ? Pattern: differences ? **4**. Context: factor table. $627=7 \times 89+4$; remainder=4.
 Q1112 ? Pattern: ratios ? **4**. Context: base conversion. $644=8 \times 80+4$; remainder=4.
 Q1113 ? Pattern: alternating ? **4**. Context: remainder audit. $661=9 \times 73+4$; remainder=4.
 Q1114 ? Pattern: squares / cubes ? **8**. Context: digit puzzle. $678=10 \times 67+8$; remainder=8.
 Q1115 ? Inference from facts ? **2**. Context: number machine. $695=11 \times 63+2$; remainder=2.

Q1116 ? Assumption check ? **1**. Context: modular code. $712=3\times 237+1$; remainder=1.
 Q1117 ? Cause vs correlation ? **1**. Context: billing sequence. $729=4\times 182+1$; remainder=1.
 Q1118 ? Grammar agreement ? **1**. Context: integer game. $746=5\times 149+1$; remainder=1.
 Q1119 ? Vocabulary in context ? **1**. Context: factor table. $763=6\times 127+1$; remainder=1.
 Q1120 ? Mixed placement challenge ? **3**. Context: base conversion. $780=7\times 111+3$; remainder=3.

Divisibility

Q1141 ? Options elimination ? **204**. Context: serial code. $204/4=51$, an integer.
 Q1142 ? Boundary case ? **205**. Context: digit audit. $205/5=41$, an integer.
 Q1143 ? Case split ? **207**. Context: packet routing. $207/9=23$, an integer.
 Q1144 ? Contradiction check ? **216**. Context: number lock. $216/3=72$, an integer.
 Q1145 ? Statement I test ? **220**. Context: barcode validation. $220/4=55$, an integer.
 Q1146 ? Statement II test ? **220**. Context: invoice check. $220/5=44$, an integer.
 Q1147 ? Combined statements ? **225**. Context: seat-number rule. $225/9=25$, an integer.
 Q1148 ? Counting restriction ? **231**. Context: data checksum. $231/3=77$, an integer.
 Q1149 ? Circular arrangement ? **236**. Context: serial code. $236/4=59$, an integer.
 Q1150 ? Complement method ? **240**. Context: digit audit. $240/5=48$, an integer.
 Q1151 ? Pattern: differences ? **243**. Context: packet routing. $243/9=27$, an integer.
 Q1152 ? Pattern: ratios ? **246**. Context: number lock. $246/3=82$, an integer.
 Q1153 ? Pattern: alternating ? **252**. Context: barcode validation. $252/4=63$, an integer.
 Q1154 ? Pattern: squares / cubes ? **255**. Context: invoice check. $255/5=51$, an integer.
 Q1155 ? Inference from facts ? **252**. Context: seat-number rule. $252/9=28$, an integer.
 Q1156 ? Assumption check ? **264**. Context: data checksum. $264/3=88$, an integer.
 Q1157 ? Cause vs correlation ? **268**. Context: serial code. $268/4=67$, an integer.
 Q1158 ? Grammar agreement ? **270**. Context: digit audit. $270/5=54$, an integer.
 Q1159 ? Vocabulary in context ? **270**. Context: packet routing. $270/9=30$, an integer.
 Q1160 ? Mixed placement challenge ? **279**. Context: number lock. $279/3=93$, an integer.

Simplification

Q1181 ? Options elimination ? **40**. Context: engineering estimate. BODMAS: $3\times 3=9$; then $31+9=40$.
 Q1182 ? Boundary case ? **48**. Context: invoice arithmetic. BODMAS: $4\times 4=16$; then $32+16=48$.
 Q1183 ? Case split ? **58**. Context: nested operation. BODMAS: $5\times 5=25$; then $33+25=58$.
 Q1184 ? Contradiction check ? **70**. Context: mental-math drill. BODMAS: $6\times 6=36$; then $34+36=70$.
 Q1185 ? Statement I test ? **49**. Context: spreadsheet formula. BODMAS: $7\times 2=14$; then $35+14=49$.
 Q1186 ? Statement II test ? **60**. Context: billing expression. BODMAS: $8\times 3=24$; then $36+24=60$.
 Q1187 ? Combined statements ? **73**. Context: unit-cost calculation. BODMAS: $9\times 4=36$; then $37+36=73$.
 Q1188 ? Counting restriction ? **53**. Context: exam simplification. BODMAS: $3\times 5=15$; then $38+15=53$.
 Q1189 ? Circular arrangement ? **63**. Context: engineering estimate. BODMAS: $4\times 6=24$; then $39+24=63$.
 Q1190 ? Complement method ? **50**. Context: invoice arithmetic. BODMAS: $5\times 2=10$; then $40+10=50$.
 Q1191 ? Pattern: differences ? **59**. Context: nested operation. BODMAS: $6\times 3=18$; then $41+18=59$.
 Q1192 ? Pattern: ratios ? **70**. Context: mental-math drill. BODMAS: $7\times 4=28$; then $42+28=70$.
 Q1193 ? Pattern: alternating ? **83**. Context: spreadsheet formula. BODMAS: $8\times 5=40$; then $43+40=83$.
 Q1194 ? Pattern: squares / cubes ? **98**. Context: billing expression. BODMAS: $9\times 6=54$; then $44+54=98$.
 Q1195 ? Inference from facts ? **51**. Context: unit-cost calculation. BODMAS: $3\times 2=6$; then $45+6=51$.
 Q1196 ? Assumption check ? **58**. Context: exam simplification. BODMAS: $4\times 3=12$; then $46+12=58$.
 Q1197 ? Cause vs correlation ? **67**. Context: engineering estimate. BODMAS: $5\times 4=20$; then $47+20=67$.
 Q1198 ? Grammar agreement ? **78**. Context: invoice arithmetic. BODMAS: $6\times 5=30$; then $48+30=78$.
 Q1199 ? Vocabulary in context ? **91**. Context: nested operation. BODMAS: $7\times 6=42$; then $49+42=91$.
 Q1200 ? Mixed placement challenge ? **66**. Context: mental-math drill. BODMAS: $8\times 2=16$; then $50+16=66$.

Surds & Indices

Q1221 ? Options elimination ? **9**. Context: signal amplitude. $\sqrt{81}=9$, the principal square root.

Q1222 ? Boundary case ? **10**. Context: area side-length. $\sqrt{100}=10$, the principal square root.

Q1223 ? Case split ? **11**. Context: index comparison. $\sqrt{121}=11$, the principal square root.

Q1224 ? Contradiction check ? **12**. Context: radical simplification. $\sqrt{144}=12$, the principal square root.

Q1225 ? Statement I test ? **13**. Context: square-root measurement. $\sqrt{169}=13$, the principal square root.

Q1226 ? Statement II test ? **14**. Context: scaling law. $\sqrt{196}=14$, the principal square root.

Q1227 ? Combined statements ? **15**. Context: scientific notation. $\sqrt{225}=15$, the principal square root.

Q1228 ? Counting restriction ? **2**. Context: geometric growth. $\sqrt{4}=2$, the principal square root.

Q1229 ? Circular arrangement ? **3**. Context: signal amplitude. $\sqrt{9}=3$, the principal square root.

Q1230 ? Complement method ? **4**. Context: area side-length. $\sqrt{16}=4$, the principal square root.

Q1231 ? Pattern: differences ? **5**. Context: index comparison. $\sqrt{25}=5$, the principal square root.

Q1232 ? Pattern: ratios ? **6**. Context: radical simplification. $\sqrt{36}=6$, the principal square root.

Q1233 ? Pattern: alternating ? **7**. Context: square-root measurement. $\sqrt{49}=7$, the principal square root.

Q1234 ? Pattern: squares / cubes ? **8**. Context: scaling law. $\sqrt{64}=8$, the principal square root.

Q1235 ? Inference from facts ? **9**. Context: scientific notation. $\sqrt{81}=9$, the principal square root.

Q1236 ? Assumption check ? **10**. Context: geometric growth. $\sqrt{100}=10$, the principal square root.

Q1237 ? Cause vs correlation ? **11**. Context: signal amplitude. $\sqrt{121}=11$, the principal square root.

Q1238 ? Grammar agreement ? **12**. Context: area side-length. $\sqrt{144}=12$, the principal square root.

Q1239 ? Vocabulary in context ? **13**. Context: index comparison. $\sqrt{169}=13$, the principal square root.

Q1240 ? Mixed placement challenge ? **14**. Context: radical simplification. $\sqrt{196}=14$, the principal square root.

Logarithms

Q1261 ? Options elimination ? **3**. Context: earthquake magnitude. $3^3=27$, hence $\log_3(27)=3$.

Q1262 ? Boundary case ? **4**. Context: financial log-return. $4^4=256$, hence $\log_4(256)=4$.

Q1263 ? Case split ? **5**. Context: population model. $5^5=3125$, hence $\log_5(3125)=5$.

Q1264 ? Contradiction check ? **2**. Context: base conversion. $2^2=4$, hence $\log_2(4)=2$.

Q1265 ? Statement I test ? **3**. Context: decibel scale. $3^3=27$, hence $\log_3(27)=3$.

Q1266 ? Statement II test ? **4**. Context: pH reading. $4^4=256$, hence $\log_4(256)=4$.

Q1267 ? Combined statements ? **5**. Context: compound growth. $5^5=3125$, hence $\log_5(3125)=5$.

Q1268 ? Counting restriction ? **2**. Context: algorithm complexity. $2^2=4$, hence $\log_2(4)=2$.

Q1269 ? Circular arrangement ? **3**. Context: earthquake magnitude. $3^3=27$, hence $\log_3(27)=3$.

Q1270 ? Complement method ? **4**. Context: financial log-return. $4^4=256$, hence $\log_4(256)=4$.

Q1271 ? Pattern: differences ? **5**. Context: population model. $5^5=3125$, hence $\log_5(3125)=5$.

Q1272 ? Pattern: ratios ? **2**. Context: base conversion. $2^2=4$, hence $\log_2(4)=2$.

Q1273 ? Pattern: alternating ? **3**. Context: decibel scale. $3^3=27$, hence $\log_3(27)=3$.

Q1274 ? Pattern: squares / cubes ? **4**. Context: pH reading. $4^4=256$, hence $\log_4(256)=4$.

Q1275 ? Inference from facts ? **5**. Context: compound growth. $5^5=3125$, hence $\log_5(3125)=5$.

Q1276 ? Assumption check ? **2**. Context: algorithm complexity. $2^2=4$, hence $\log_2(4)=2$.

Q1277 ? Cause vs correlation ? **3**. Context: earthquake magnitude. $3^3=27$, hence $\log_3(27)=3$.

Q1278 ? Grammar agreement ? **4**. Context: financial log-return. $4^4=256$, hence $\log_4(256)=4$.

Q1279 ? Vocabulary in context ? **5**. Context: population model. $5^5=3125$, hence $\log_5(3125)=5$.

Q1280 ? Mixed placement challenge ? **2**. Context: base conversion. $2^2=4$, hence $\log_2(4)=2$.

Algebra

Q1301 ? Options elimination ? **23**. Context: inventory balance. $3x=69$; $x=69/3=23$.

Q1302 ? Boundary case ? **24**. Context: coordinate relation. $4x=96$; $x=96/4=24$.

Q1303 ? Case split ? **25**. Context: production target. $5x=125$; $x=125/5=25$.

Q1304 ? Contradiction check ? **26**. Context: linear forecast. $6x=156$; $x=156/6=26$.

Q1305 ? Statement I test ? **27**. Context: unknown price. $2x=54$; $x=54/2=27$.

Q1306 ? Statement II test ? **28**. Context: work-rate equation. $3x=84$; $x=84/3=28$.

Q1307 ? Combined statements ? **29**. Context: age variable. $4x=116$; $x=116/4=29$.

Q1308 ? Counting restriction ? **30**. Context: ticket equation. $5x=150$; $x=150/5=30$.

Q1309 ? Circular arrangement ? **31**. Context: inventory balance. $6x=186$; $x=186/6=31$.

Q1310 ? Complement method ? **32**. Context: coordinate relation. $2x=64$; $x=64/2=32$.

Q1311 ? Pattern: differences ? **33**. Context: production target. $3x=99$; $x=99/3=33$.

Q1312 ? Pattern: ratios ? **34**. Context: linear forecast. $4x=136$; $x=136/4=34$.

Q1313 ? Pattern: alternating ? **35**. Context: unknown price. $5x=175$; $x=175/5=35$.

Q1314 ? Pattern: squares / cubes ? **36**. Context: work-rate equation. $6x=216$; $x=216/6=36$.

Q1315 ? Inference from facts ? **37**. Context: age variable. $2x=74$; $x=74/2=37$.

Q1316 ? Assumption check ? **38**. Context: ticket equation. $3x=114$; $x=114/3=38$.

Q1317 ? Cause vs correlation ? **39**. Context: inventory balance. $4x=156$; $x=156/4=39$.

Q1318 ? Grammar agreement ? **40**. Context: coordinate relation. $5x=200$; $x=200/5=40$.

Q1319 ? Vocabulary in context ? **41**. Context: production target. $6x=246$; $x=246/6=41$.

Q1320 ? Mixed placement challenge ? **42**. Context: linear forecast. $2x=84$; $x=84/2=42$.

Geometry

Q1341 ? Options elimination ? **48**. Context: polygon design. Third angle= $180-51-81=48^\circ$.

Q1342 ? Boundary case ? **46**. Context: coordinate map. Third angle= $180-52-82=46^\circ$.

Q1343 ? Case split ? **44**. Context: architectural plan. Third angle= $180-53-83=44^\circ$.

Q1344 ? Contradiction check ? **42**. Context: angle chase. Third angle= $180-54-84=42^\circ$.

Q1345 ? Statement I test ? **40**. Context: land survey. Third angle= $180-55-85=40^\circ$.

Q1346 ? Statement II test ? **38**. Context: triangular park. Third angle= $180-56-86=38^\circ$.

Q1347 ? Combined statements ? **36**. Context: angle of elevation. Third angle= $180-57-87=36^\circ$.

Q1348 ? Counting restriction ? **34**. Context: road junction. Third angle= $180-58-88=34^\circ$.

Q1349 ? Circular arrangement ? **32**. Context: polygon design. Third angle= $180-59-89=32^\circ$.

Q1350 ? Complement method ? **60**. Context: coordinate map. Third angle= $180-60-60=60^\circ$.

Q1351 ? Pattern: differences ? **58**. Context: architectural plan. Third angle= $180-61-61=58^\circ$.

Q1352 ? Pattern: ratios ? **56**. Context: angle chase. Third angle= $180-62-62=56^\circ$.

Q1353 ? Pattern: alternating ? **54**. Context: land survey. Third angle= $180-63-63=54^\circ$.

Q1354 ? Pattern: squares / cubes ? **52**. Context: triangular park. Third angle= $180-64-64=52^\circ$.

Q1355 ? Inference from facts ? **50**. Context: angle of elevation. Third angle= $180-65-65=50^\circ$.

Q1356 ? Assumption check ? **48**. Context: road junction. Third angle= $180-66-66=48^\circ$.

Q1357 ? Cause vs correlation ? **46**. Context: polygon design. Third angle= $180-67-67=46^\circ$.

Q1358 ? Grammar agreement ? **44**. Context: coordinate map. Third angle= $180-68-68=44^\circ$.

Q1359 ? Vocabulary in context ? **42**. Context: architectural plan. Third angle= $180-69-69=42^\circ$.

Q1360 ? Mixed placement challenge ? **40**. Context: angle chase. Third angle= $180-70-70=40^\circ$.

Data Sufficiency

Q1381 ? Options elimination ? **Statement I alone is sufficient**. Context: budget authorization. I fixes one value, while II permits many; I alone is sufficient.

Q1382 ? Boundary case ? **Statement I alone is sufficient**. Context: quality release. I fixes one value, while II permits many; I alone is sufficient.

Q1383 ? Case split ? **Statement I alone is sufficient**. Context: project estimate. I fixes one value, while II permits many; I alone is sufficient.

Q1384 ? Contradiction check ? **Statement I alone is sufficient**. Context: schedule decision. I fixes one value, while II permits many; I alone is sufficient.

Q1385 ? Statement I test ? **Statement I alone is sufficient**. Context: loan approval. I fixes one value, while II permits many; I alone is sufficient.

Q1386 ? Statement II test ? **Statement I alone is sufficient**. Context: inventory decision. I fixes one value, while II permits many; I alone is sufficient.

Q1387 ? Combined statements ? **Statement I alone is sufficient**. Context: candidate eligibility. I fixes one value, while II permits many; I alone is sufficient.

Q1388 ? Counting restriction ? **Statement I alone is sufficient**. Context: route selection. I fixes one value, while II permits many; I alone is sufficient.

Q1389 ? Circular arrangement ? **Statement I alone is sufficient**. Context: budget authorization. I fixes one value, while II permits many; I alone is sufficient.

Q1390 ? Complement method ? **Statement I alone is sufficient**. Context: quality release. I fixes one value, while II permits many; I alone is sufficient.

Q1391 ? Pattern: differences ? **Statement I alone is sufficient**. Context: project estimate. I fixes one value, while II permits many; I alone is sufficient.

Q1392 ? Pattern: ratios ? **Statement I alone is sufficient**. Context: schedule decision. I fixes one value, while II permits many; I alone is sufficient.

Q1393 ? Pattern: alternating ? **Statement I alone is sufficient**. Context: loan approval. I fixes one value, while II permits many; I alone is sufficient.

Q1394 ? Pattern: squares / cubes ? **Statement I alone is sufficient**. Context: inventory decision. I fixes one value, while II permits many; I alone is sufficient.

Q1395 ? Inference from facts ? **Statement I alone is sufficient**. Context: candidate eligibility. I fixes one value, while II permits many; I alone is sufficient.

Q1396 ? Assumption check ? **Statement I alone is sufficient.** Context: route selection. I fixes one value, while II permits many; I alone is sufficient.

Q1397 ? Cause vs correlation ? **Statement I alone is sufficient.** Context: budget authorization. I fixes one value, while II permits many; I alone is sufficient.

Q1398 ? Grammar agreement ? **Statement I alone is sufficient.** Context: quality release. I fixes one value, while II permits many; I alone is sufficient.

Q1399 ? Vocabulary in context ? **Statement I alone is sufficient.** Context: project estimate. I fixes one value, while II permits many; I alone is sufficient.

Q1400 ? Mixed placement challenge ? **Statement I alone is sufficient.** Context: schedule decision. I fixes one value, while II permits many; I alone is sufficient.

Venn Diagrams

Q1421 ? Options elimination ? **72.** Context: product overlap. Inclusion–exclusion: $61+26-15=72$.

Q1422 ? Boundary case ? **73.** Context: language survey. Inclusion–exclusion: $62+27-16=73$.

Q1423 ? Case split ? **74.** Context: candidate preferences. Inclusion–exclusion: $63+28-17=74$.

Q1424 ? Contradiction check ? **83.** Context: market research. Inclusion–exclusion: $64+29-10=83$.

Q1425 ? Statement I test ? **84.** Context: course enrollment. Inclusion–exclusion: $65+30-11=84$.

Q1426 ? Statement II test ? **85.** Context: customer segments. Inclusion–exclusion: $66+31-12=85$.

Q1427 ? Combined statements ? **86.** Context: skill survey. Inclusion–exclusion: $67+32-13=86$.

Q1428 ? Counting restriction ? **87.** Context: club membership. Inclusion–exclusion: $68+33-14=87$.

Q1429 ? Circular arrangement ? **88.** Context: product overlap. Inclusion–exclusion: $69+34-15=88$.

Q1430 ? Complement method ? **79.** Context: language survey. Inclusion–exclusion: $70+25-16=79$.

Q1431 ? Pattern: differences ? **80.** Context: candidate preferences. Inclusion–exclusion: $71+26-17=80$.

Q1432 ? Pattern: ratios ? **89.** Context: market research. Inclusion–exclusion: $72+27-10=89$.

Q1433 ? Pattern: alternating ? **90.** Context: course enrollment. Inclusion–exclusion: $73+28-11=90$.

Q1434 ? Pattern: squares / cubes ? **91.** Context: customer segments. Inclusion–exclusion: $74+29-12=91$.

Q1435 ? Inference from facts ? **92.** Context: skill survey. Inclusion–exclusion: $75+30-13=92$.

Q1436 ? Assumption check ? **93.** Context: club membership. Inclusion–exclusion: $76+31-14=93$.

Q1437 ? Cause vs correlation ? **94.** Context: product overlap. Inclusion–exclusion: $77+32-15=94$.

Q1438 ? Grammar agreement ? **95.** Context: language survey. Inclusion–exclusion: $78+33-16=95$.

Q1439 ? Vocabulary in context ? **96.** Context: candidate preferences. Inclusion–exclusion: $79+34-17=96$.

Q1440 ? Mixed placement challenge ? **95.** Context: market research. Inclusion–exclusion: $80+25-10=95$.

Blood Relations

Q1461 ? Options elimination ? **mother.** Context: team relation. The statement directly says K is M's mother.

Q1462 ? Boundary case ? **brother.** Context: hospital form. The statement directly says K is M's brother.

Q1463 ? Case split ? **uncle.** Context: family invitation. The statement directly says K is M's uncle.

Q1464 ? Contradiction check ? **father.** Context: tree diagram. The statement directly says K is M's father.

Q1465 ? Statement I test ? **mother.** Context: family photograph. The statement directly says K is M's mother.

Q1466 ? Statement II test ? **brother.** Context: genealogy chart. The statement directly says K is M's brother.

Q1467 ? Combined statements ? **uncle.** Context: seating clue. The statement directly says K is M's uncle.

Q1468 ? Counting restriction ? **father.** Context: inheritance record. The statement directly says K is M's father.

Q1469 ? Circular arrangement ? **mother.** Context: team relation. The statement directly says K is M's mother.

Q1470 ? Complement method ? **brother.** Context: hospital form. The statement directly says K is M's brother.

Q1471 ? Pattern: differences ? **uncle.** Context: family invitation. The statement directly says K is M's uncle.

Q1472 ? Pattern: ratios ? **father.** Context: tree diagram. The statement directly says K is M's father.

Q1473 ? Pattern: alternating ? **mother.** Context: family photograph. The statement directly says K is M's mother.

Q1474 ? Pattern: squares / cubes ? **brother.** Context: genealogy chart. The statement directly says K is M's brother.

Q1475 ? Inference from facts ? **uncle.** Context: seating clue. The statement directly says K is M's uncle.

Q1476 ? Assumption check ? **father.** Context: inheritance record. The statement directly says K is M's father.

Q1477 ? Cause vs correlation ? **mother.** Context: team relation. The statement directly says K is M's mother.

Q1478 ? Grammar agreement ? **brother.** Context: hospital form. The statement directly says K is M's brother.

Q1479 ? Vocabulary in context ? **uncle.** Context: family invitation. The statement directly says K is M's uncle.

Q1480 ? Mixed placement challenge ? **father.** Context: tree diagram. The statement directly says K is M's father.

Analogy

Q1501 ? Options elimination ? **School.** Context: cause-effect. Doctor relates to Hospital in the same way Teacher relates to School.

Q1502 ? Boundary case ? **Cat**. Context: part-whole. Puppy relates to Dog in the same way Kitten relates to Cat.

Q1503 ? Case split ? **Listen**. Context: synonym relation. Book relates to Read in the same way Song relates to Listen.

Q1504 ? Contradiction check ? **Hive**. Context: process-output. Bird relates to Nest in the same way Bee relates to Hive.

Q1505 ? Statement I test ? **School**. Context: tool-function pair. Doctor relates to Hospital in the same way Teacher relates to School.

Q1506 ? Statement II test ? **Cat**. Context: profession-workplace. Puppy relates to Dog in the same way Kitten relates to Cat.

Q1507 ? Combined statements ? **Listen**. Context: animal-home. Book relates to Read in the same way Song relates to Listen.

Q1508 ? Counting restriction ? **Hive**. Context: object-material. Bird relates to Nest in the same way Bee relates to Hive.

Q1509 ? Circular arrangement ? **School**. Context: cause-effect. Doctor relates to Hospital in the same way Teacher relates to School.

Q1510 ? Complement method ? **Cat**. Context: part-whole. Puppy relates to Dog in the same way Kitten relates to Cat.

Q1511 ? Pattern: differences ? **Listen**. Context: synonym relation. Book relates to Read in the same way Song relates to Listen.

Q1512 ? Pattern: ratios ? **Hive**. Context: process-output. Bird relates to Nest in the same way Bee relates to Hive.

Q1513 ? Pattern: alternating ? **School**. Context: tool-function pair. Doctor relates to Hospital in the same way Teacher relates to School.

Q1514 ? Pattern: squares / cubes ? **Cat**. Context: profession-workplace. Puppy relates to Dog in the same way Kitten relates to Cat.

Q1515 ? Inference from facts ? **Listen**. Context: animal-home. Book relates to Read in the same way Song relates to Listen.

Q1516 ? Assumption check ? **Hive**. Context: object-material. Bird relates to Nest in the same way Bee relates to Hive.

Q1517 ? Cause vs correlation ? **School**. Context: cause-effect. Doctor relates to Hospital in the same way Teacher relates to School.

Q1518 ? Grammar agreement ? **Cat**. Context: part-whole. Puppy relates to Dog in the same way Kitten relates to Cat.

Q1519 ? Vocabulary in context ? **Listen**. Context: synonym relation. Book relates to Read in the same way Song relates to Listen.

Q1520 ? Mixed placement challenge ? **Hive**. Context: process-output. Bird relates to Nest in the same way Bee relates to Hive.

Classification

Q1541 ? Options elimination ? **Cube**. Context: odd process. Cube does not share the defining property of the other three.

Q1542 ? Boundary case ? **Mango**. Context: odd relation. Mango does not share the defining property of the other three.

Q1543 ? Case split ? **Monday**. Context: odd symbol. Monday does not share the defining property of the other three.

Q1544 ? Contradiction check ? **9**. Context: odd object. 9 does not share the defining property of the other three.

Q1545 ? Statement I test ? **Cube**. Context: odd number. Cube does not share the defining property of the other three.

Q1546 ? Statement II test ? **Mango**. Context: odd word. Mango does not share the defining property of the other three.

Q1547 ? Combined statements ? **Monday**. Context: odd shape. Monday does not share the defining property of the other three.

Q1548 ? Counting restriction ? **9**. Context: odd category. 9 does not share the defining property of the other three.

Q1549 ? Circular arrangement ? **Cube**. Context: odd process. Cube does not share the defining property of the other three.

Q1550 ? Complement method ? **Mango**. Context: odd relation. Mango does not share the defining property of the other three.

Q1551 ? Pattern: differences ? **Monday**. Context: odd symbol. Monday does not share the defining property of the other three.

Q1552 ? Pattern: ratios ? **9**. Context: odd object. 9 does not share the defining property of the other three.

Q1553 ? Pattern: alternating ? **Cube**. Context: odd number. Cube does not share the defining property of the other three.

Q1554 ? Pattern: squares / cubes ? **Mango**. Context: odd word. Mango does not share the defining property of the other three.

Q1555 ? Inference from facts ? **Monday**. Context: odd shape. Monday does not share the defining property of the other three.

Q1556 ? Assumption check ? **9**. Context: odd category. 9 does not share the defining property of the other three.

Q1557 ? Cause vs correlation ? **Cube**. Context: odd process. Cube does not share the defining property of the other three.

Q1558 ? Grammar agreement ? **Mango**. Context: odd relation. Mango does not share the defining property of the other three.

Q1559 ? Vocabulary in context ? **Monday**. Context: odd symbol. Monday does not share the defining property of the other three.

Q1560 ? Mixed placement challenge ? **9**. Context: odd object. 9 does not share the defining property of the other three.

Ranking

Q1581 ? Options elimination ? **41**. Context: queue position. $51-11+1=41$.

Q1582 ? Boundary case ? **41**. Context: height order. $52-12+1=41$.

Q1583 ? Case split ? **41**. Context: performance list. $53-13+1=41$.

Q1584 ? Contradiction check ? **41**. Context: selection order. $54-14+1=41$.

Q1585 ? Statement I test ? **41**. Context: leaderboard. $55-15+1=41$.

Q1586 ? Statement II test ? **41**. Context: exam rank. $56-16+1=41$.

Q1587 ? Combined statements ? **41**. Context: race finish. $57-17+1=41$.

Q1588 ? Counting restriction ? **41**. Context: salary order. $58-18+1=41$.

Q1589 ? Circular arrangement ? **41**. Context: queue position. $59-19+1=41$.
 Q1590 ? Complement method ? **56**. Context: height order. $60-5+1=56$.
 Q1591 ? Pattern: differences ? **56**. Context: performance list. $61-6+1=56$.
 Q1592 ? Pattern: ratios ? **56**. Context: selection order. $62-7+1=56$.
 Q1593 ? Pattern: alternating ? **56**. Context: leaderboard. $63-8+1=56$.
 Q1594 ? Pattern: squares / cubes ? **56**. Context: exam rank. $64-9+1=56$.
 Q1595 ? Inference from facts ? **56**. Context: race finish. $65-10+1=56$.
 Q1596 ? Assumption check ? **56**. Context: salary order. $66-11+1=56$.
 Q1597 ? Cause vs correlation ? **56**. Context: queue position. $67-12+1=56$.
 Q1598 ? Grammar agreement ? **56**. Context: height order. $68-13+1=56$.
 Q1599 ? Vocabulary in context ? **56**. Context: performance list. $69-14+1=56$.
 Q1600 ? Mixed placement challenge ? **56**. Context: selection order. $70-15+1=56$.

Seating Arrangement

Q1621 ? Options elimination ? **24**. Context: circular committee. Circular arrangements= $(5-1)!=24$.
 Q1622 ? Boundary case ? **120**. Context: bench arrangement. Circular arrangements= $(6-1)!=120$.
 Q1623 ? Case split ? **720**. Context: classroom seats. Circular arrangements= $(7-1)!=720$.
 Q1624 ? Contradiction check ? **5040**. Context: dinner seating. Circular arrangements= $(8-1)!=5040$.
 Q1625 ? Statement I test ? **6**. Context: round table. Circular arrangements= $(4-1)!=6$.
 Q1626 ? Statement II test ? **24**. Context: theatre row. Circular arrangements= $(5-1)!=24$.
 Q1627 ? Combined statements ? **120**. Context: office desks. Circular arrangements= $(6-1)!=120$.
 Q1628 ? Counting restriction ? **720**. Context: conference table. Circular arrangements= $(7-1)!=720$.
 Q1629 ? Circular arrangement ? **5040**. Context: circular committee. Circular arrangements= $(8-1)!=5040$.
 Q1630 ? Complement method ? **6**. Context: bench arrangement. Circular arrangements= $(4-1)!=6$.
 Q1631 ? Pattern: differences ? **24**. Context: classroom seats. Circular arrangements= $(5-1)!=24$.
 Q1632 ? Pattern: ratios ? **120**. Context: dinner seating. Circular arrangements= $(6-1)!=120$.
 Q1633 ? Pattern: alternating ? **720**. Context: round table. Circular arrangements= $(7-1)!=720$.
 Q1634 ? Pattern: squares / cubes ? **5040**. Context: theatre row. Circular arrangements= $(8-1)!=5040$.
 Q1635 ? Inference from facts ? **6**. Context: office desks. Circular arrangements= $(4-1)!=6$.
 Q1636 ? Assumption check ? **24**. Context: conference table. Circular arrangements= $(5-1)!=24$.
 Q1637 ? Cause vs correlation ? **120**. Context: circular committee. Circular arrangements= $(6-1)!=120$.
 Q1638 ? Grammar agreement ? **720**. Context: bench arrangement. Circular arrangements= $(7-1)!=720$.
 Q1639 ? Vocabulary in context ? **5040**. Context: classroom seats. Circular arrangements= $(8-1)!=5040$.
 Q1640 ? Mixed placement challenge ? **6**. Context: dinner seating. Circular arrangements= $(4-1)!=6$.

Puzzle Solving

Q1661 ? Options elimination ? **39**. Context: room distribution. Common increase=4; next=39.
 Q1662 ? Boundary case ? **44**. Context: logic grid. Common increase=5; next=44.
 Q1663 ? Case split ? **49**. Context: interview order. Common increase=6; next=49.
 Q1664 ? Contradiction check ? **54**. Context: constraint puzzle. Common increase=7; next=54.
 Q1665 ? Statement I test ? **39**. Context: floor puzzle. Common increase=3; next=39.
 Q1666 ? Statement II test ? **44**. Context: box allocation. Common increase=4; next=44.
 Q1667 ? Combined statements ? **49**. Context: day scheduling. Common increase=5; next=49.
 Q1668 ? Counting restriction ? **54**. Context: team assignment. Common increase=6; next=54.
 Q1669 ? Circular arrangement ? **59**. Context: room distribution. Common increase=7; next=59.
 Q1670 ? Complement method ? **44**. Context: logic grid. Common increase=3; next=44.
 Q1671 ? Pattern: differences ? **49**. Context: interview order. Common increase=4; next=49.
 Q1672 ? Pattern: ratios ? **54**. Context: constraint puzzle. Common increase=5; next=54.
 Q1673 ? Pattern: alternating ? **59**. Context: floor puzzle. Common increase=6; next=59.
 Q1674 ? Pattern: squares / cubes ? **64**. Context: box allocation. Common increase=7; next=64.
 Q1675 ? Inference from facts ? **49**. Context: day scheduling. Common increase=3; next=49.

Q1676 ? Assumption check ? **54**. Context: team assignment. Common increase=4; next=54.

Q1677 ? Cause vs correlation ? **59**. Context: room distribution. Common increase=5; next=59.

Q1678 ? Grammar agreement ? **64**. Context: logic grid. Common increase=6; next=64.

Q1679 ? Vocabulary in context ? **69**. Context: interview order. Common increase=7; next=69.

Q1680 ? Mixed placement challenge ? **54**. Context: constraint puzzle. Common increase=3; next=54.

Logical Reasoning

Q1701 ? Options elimination ? **Some analysts are graduates**. Context: decision rule. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1702 ? Boundary case ? **Some analysts are graduates**. Context: course of action. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1703 ? Case split ? **Some analysts are graduates**. Context: inference chain. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1704 ? Contradiction check ? **Some analysts are graduates**. Context: propositional logic. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1705 ? Statement I test ? **Some analysts are graduates**. Context: syllogism. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1706 ? Statement II test ? **Some analysts are graduates**. Context: statement conclusion. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1707 ? Combined statements ? **Some analysts are graduates**. Context: assumption test. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1708 ? Counting restriction ? **Some analysts are graduates**. Context: cause-effect. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1709 ? Circular arrangement ? **Some analysts are graduates**. Context: decision rule. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1710 ? Complement method ? **Some analysts are graduates**. Context: course of action. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1711 ? Pattern: differences ? **Some analysts are graduates**. Context: inference chain. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1712 ? Pattern: ratios ? **Some analysts are graduates**. Context: propositional logic. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1713 ? Pattern: alternating ? **Some analysts are graduates**. Context: syllogism. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1714 ? Pattern: squares / cubes ? **Some analysts are graduates**. Context: statement conclusion. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1715 ? Inference from facts ? **Some analysts are graduates**. Context: assumption test. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1716 ? Assumption check ? **Some analysts are graduates**. Context: cause-effect. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1717 ? Cause vs correlation ? **Some analysts are graduates**. Context: decision rule. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1718 ? Grammar agreement ? **Some analysts are graduates**. Context: course of action. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1719 ? Vocabulary in context ? **Some analysts are graduates**. Context: inference chain. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Q1720 ? Mixed placement challenge ? **Some analysts are graduates**. Context: propositional logic. Analysts who are engineers belong to the graduate set, so some analysts are graduates.

Non-Verbal Reasoning

Q1741 ? Options elimination ? **5**. Context: visual series. The visual rule adds one side: $4+1=5$.

Q1742 ? Boundary case ? **6**. Context: pattern matrix. The visual rule adds one side: $5+1=6$.

Q1743 ? Case split ? **7**. Context: shape counting. The visual rule adds one side: $6+1=7$.

Q1744 ? Contradiction check ? **8**. Context: spatial transformation. The visual rule adds one side: $7+1=8$.

Q1745 ? Statement I test ? **4**. Context: figure rotation. The visual rule adds one side: $3+1=4$.

Q1746 ? Statement II test ? **5**. Context: mirror image. The visual rule adds one side: $4+1=5$.

Q1747 ? Combined statements ? **6**. Context: paper fold. The visual rule adds one side: $5+1=6$.

Q1748 ? Counting restriction ? **7**. Context: cube net. The visual rule adds one side: $6+1=7$.

Q1749 ? Circular arrangement ? **8**. Context: visual series. The visual rule adds one side: $7+1=8$.

Q1750 ? Complement method ? **4**. Context: pattern matrix. The visual rule adds one side: $3+1=4$.

Q1751 ? Pattern: differences ? **5**. Context: shape counting. The visual rule adds one side: $4+1=5$.

Q1752 ? Pattern: ratios ? **6**. Context: spatial transformation. The visual rule adds one side: $5+1=6$.

Q1753 ? Pattern: alternating ? **7**. Context: figure rotation. The visual rule adds one side: $6+1=7$.

Q1754 ? Pattern: squares / cubes ? **8**. Context: mirror image. The visual rule adds one side: $7+1=8$.

Q1755 ? Inference from facts ? **4**. Context: paper fold. The visual rule adds one side: $3+1=4$.

Q1756 ? Assumption check ? **5**. Context: cube net. The visual rule adds one side: $4+1=5$.

Q1757 ? Cause vs correlation ? **6**. Context: visual series. The visual rule adds one side: $5+1=6$.

Q1758 ? Grammar agreement ? **7**. Context: pattern matrix. The visual rule adds one side: $6+1=7$.

Q1759 ? Vocabulary in context ? **8**. Context: shape counting. The visual rule adds one side: $7+1=8$.

Q1760 ? Mixed placement challenge ? **4**. Context: spatial transformation. The visual rule adds one side: $3+1=4$.

Critical Reasoning

Q1781 ? Options elimination ? **The campaign correlates with the rise but causation is unproven**. Context: survey inference. Timing shows correlation, but other causes remain possible.

Q1782 ? Boundary case ? **The campaign correlates with the rise but causation is unproven**. Context: argument flaw. Timing shows correlation, but other causes remain possible.

Q1783 ? Case split ? **The campaign correlates with the rise but causation is unproven**. Context: evidence test. Timing shows correlation, but other causes remain possible.

Q1784 ? Contradiction check ? **The campaign correlates with the rise but causation is unproven**. Context: assumption audit. Timing shows correlation, but other causes remain possible.

Q1785 ? Statement I test ? **The campaign correlates with the rise but causation is unproven**. Context: business claim. Timing shows correlation, but other causes remain possible.

Q1786 ? Statement II test ? **The campaign correlates with the rise but causation is unproven**. Context: research conclusion. Timing shows correlation, but other causes remain possible.

Q1787 ? Combined statements ? **The campaign correlates with the rise but causation is unproven**. Context: advertising study. Timing shows correlation, but other causes remain possible.

Q1788 ? Counting restriction ? **The campaign correlates with the rise but causation is unproven**. Context: policy proposal. Timing shows correlation, but other causes remain possible.

Q1789 ? Circular arrangement ? **The campaign correlates with the rise but causation is unproven**. Context: survey inference. Timing shows correlation, but other causes remain possible.

Q1790 ? Complement method ? **The campaign correlates with the rise but causation is unproven**. Context: argument flaw. Timing shows correlation, but other causes remain possible.

Q1791 ? Pattern: differences ? **The campaign correlates with the rise but causation is unproven**. Context: evidence test. Timing shows correlation, but other causes remain possible.

Q1792 ? Pattern: ratios ? **The campaign correlates with the rise but causation is unproven**. Context: assumption audit. Timing shows correlation, but other causes remain possible.

Q1793 ? Pattern: alternating ? **The campaign correlates with the rise but causation is unproven**. Context: business claim. Timing shows correlation, but other causes remain possible.

Q1794 ? Pattern: squares / cubes ? **The campaign correlates with the rise but causation is unproven**. Context: research conclusion. Timing shows correlation, but other causes remain possible.

Q1795 ? Inference from facts ? **The campaign correlates with the rise but causation is unproven**. Context: advertising study. Timing shows correlation, but other causes remain possible.

Q1796 ? Assumption check ? **The campaign correlates with the rise but causation is unproven**. Context: policy proposal. Timing shows correlation, but other causes remain possible.

Q1797 ? Cause vs correlation ? **The campaign correlates with the rise but causation is unproven**. Context: survey inference. Timing shows correlation, but other causes remain possible.

Q1798 ? Grammar agreement ? **The campaign correlates with the rise but causation is unproven**. Context: argument flaw. Timing shows correlation, but other causes remain possible.

Q1799 ? Vocabulary in context ? **The campaign correlates with the rise but causation is unproven**. Context: evidence test. Timing shows correlation, but other causes remain possible.

Q1800 ? Mixed placement challenge ? **The campaign correlates with the rise but causation is unproven**. Context: assumption audit. Timing shows correlation, but other causes remain possible.

Quantitative Aptitude

Q1821 ? Options elimination ? **51**. Context: finance case. Total= $41+10=51$.

Q1822 ? Boundary case ? **53**. Context: operations metric. Total= $42+11=53$.

Q1823 ? Case split ? **55**. Context: exam word problem. Total= $43+12=55$.

Q1824 ? Contradiction check ? **49**. Context: workplace calculation. Total= $44+5=49$.

Q1825 ? Statement I test ? **51**. Context: placement screening. Total= $45+6=51$.

Q1826 ? Statement II test ? **53**. Context: business arithmetic. Total= $46+7=53$.
 Q1827 ? Combined statements ? **55**. Context: campus survey. Total= $47+8=55$.
 Q1828 ? Counting restriction ? **57**. Context: engineering estimate. Total= $48+9=57$.
 Q1829 ? Circular arrangement ? **59**. Context: finance case. Total= $49+10=59$.
 Q1830 ? Complement method ? **61**. Context: operations metric. Total= $50+11=61$.
 Q1831 ? Pattern: differences ? **63**. Context: exam word problem. Total= $51+12=63$.
 Q1832 ? Pattern: ratios ? **57**. Context: workplace calculation. Total= $52+5=57$.
 Q1833 ? Pattern: alternating ? **59**. Context: placement screening. Total= $53+6=59$.
 Q1834 ? Pattern: squares / cubes ? **61**. Context: business arithmetic. Total= $54+7=61$.
 Q1835 ? Inference from facts ? **63**. Context: campus survey. Total= $55+8=63$.
 Q1836 ? Assumption check ? **65**. Context: engineering estimate. Total= $56+9=65$.
 Q1837 ? Cause vs correlation ? **67**. Context: finance case. Total= $57+10=67$.
 Q1838 ? Grammar agreement ? **69**. Context: operations metric. Total= $58+11=69$.
 Q1839 ? Vocabulary in context ? **71**. Context: exam word problem. Total= $59+12=71$.
 Q1840 ? Mixed placement challenge ? **65**. Context: workplace calculation. Total= $60+5=65$.

Arithmetic

Q1861 ? Options elimination ? **496**. Context: unit economics. $n(n+1)/2=31 \times 32/2=496$.
 Q1862 ? Boundary case ? **528**. Context: household expense. $n(n+1)/2=32 \times 33/2=528$.
 Q1863 ? Case split ? **561**. Context: schedule arithmetic. $n(n+1)/2=33 \times 34/2=561$.
 Q1864 ? Contradiction check ? **595**. Context: exam calculation. $n(n+1)/2=34 \times 35/2=595$.
 Q1865 ? Statement I test ? **630**. Context: shopping bill. $n(n+1)/2=35 \times 36/2=630$.
 Q1866 ? Statement II test ? **666**. Context: travel budget. $n(n+1)/2=36 \times 37/2=666$.
 Q1867 ? Combined statements ? **703**. Context: production count. $n(n+1)/2=37 \times 38/2=703$.
 Q1868 ? Counting restriction ? **741**. Context: rate calculation. $n(n+1)/2=38 \times 39/2=741$.
 Q1869 ? Circular arrangement ? **780**. Context: unit economics. $n(n+1)/2=39 \times 40/2=780$.
 Q1870 ? Complement method ? **820**. Context: household expense. $n(n+1)/2=40 \times 41/2=820$.
 Q1871 ? Pattern: differences ? **861**. Context: schedule arithmetic. $n(n+1)/2=41 \times 42/2=861$.
 Q1872 ? Pattern: ratios ? **903**. Context: exam calculation. $n(n+1)/2=42 \times 43/2=903$.
 Q1873 ? Pattern: alternating ? **946**. Context: shopping bill. $n(n+1)/2=43 \times 44/2=946$.
 Q1874 ? Pattern: squares / cubes ? **990**. Context: travel budget. $n(n+1)/2=44 \times 45/2=990$.
 Q1875 ? Inference from facts ? **1035**. Context: production count. $n(n+1)/2=45 \times 46/2=1035$.
 Q1876 ? Assumption check ? **1081**. Context: rate calculation. $n(n+1)/2=46 \times 47/2=1081$.
 Q1877 ? Cause vs correlation ? **1128**. Context: unit economics. $n(n+1)/2=47 \times 48/2=1128$.
 Q1878 ? Grammar agreement ? **1176**. Context: household expense. $n(n+1)/2=48 \times 49/2=1176$.
 Q1879 ? Vocabulary in context ? **1225**. Context: schedule arithmetic. $n(n+1)/2=49 \times 50/2=1225$.
 Q1880 ? Mixed placement challenge ? **1275**. Context: exam calculation. $n(n+1)/2=50 \times 51/2=1275$.

Modern Math

Q1901 ? Options elimination ? **52**. Context: function mapping. $41+16-5=52$.
 Q1902 ? Boundary case ? **53**. Context: distribution problem. $42+17-6=53$.
 Q1903 ? Case split ? **54**. Context: series application. $43+18-7=54$.
 Q1904 ? Contradiction check ? **55**. Context: set intersection. $44+19-8=55$.
 Q1905 ? Statement I test ? **56**. Context: set selection. $45+20-9=56$.
 Q1906 ? Statement II test ? **57**. Context: progression model. $46+21-10=57$.
 Q1907 ? Combined statements ? **58**. Context: counting rule. $47+22-11=58$.
 Q1908 ? Counting restriction ? **66**. Context: sequence design. $48+23-5=66$.
 Q1909 ? Circular arrangement ? **67**. Context: function mapping. $49+24-6=67$.
 Q1910 ? Complement method ? **58**. Context: distribution problem. $50+15-7=58$.
 Q1911 ? Pattern: differences ? **59**. Context: series application. $51+16-8=59$.
 Q1912 ? Pattern: ratios ? **60**. Context: set intersection. $52+17-9=60$.

Q1913 ? Pattern: alternating ? **61**. Context: set selection. $53+18-10=61$.

Q1914 ? Pattern: squares / cubes ? **62**. Context: progression model. $54+19-11=62$.

Q1915 ? Inference from facts ? **70**. Context: counting rule. $55+20-5=70$.

Q1916 ? Assumption check ? **71**. Context: sequence design. $56+21-6=71$.

Q1917 ? Cause vs correlation ? **72**. Context: function mapping. $57+22-7=72$.

Q1918 ? Grammar agreement ? **73**. Context: distribution problem. $58+23-8=73$.

Q1919 ? Vocabulary in context ? **74**. Context: series application. $59+24-9=74$.

Q1920 ? Mixed placement challenge ? **65**. Context: set intersection. $60+15-10=65$.

Basic Statistics

Q1941 ? Options elimination ? **25**. Context: salary data. Sum=125; divide by 5 to get 25.

Q1942 ? Boundary case ? **26**. Context: delivery times. Sum=130; divide by 5 to get 26.

Q1943 ? Case split ? **27**. Context: survey summary. Sum=135; divide by 5 to get 27.

Q1944 ? Contradiction check ? **28**. Context: performance report. Sum=140; divide by 5 to get 28.

Q1945 ? Statement I test ? **29**. Context: quality sample. Sum=145; divide by 5 to get 29.

Q1946 ? Statement II test ? **30**. Context: student marks. Sum=150; divide by 5 to get 30.

Q1947 ? Combined statements ? **31**. Context: sales spread. Sum=155; divide by 5 to get 31.

Q1948 ? Counting restriction ? **32**. Context: sensor readings. Sum=160; divide by 5 to get 32.

Q1949 ? Circular arrangement ? **33**. Context: salary data. Sum=165; divide by 5 to get 33.

Q1950 ? Complement method ? **34**. Context: delivery times. Sum=170; divide by 5 to get 34.

Q1951 ? Pattern: differences ? **35**. Context: survey summary. Sum=175; divide by 5 to get 35.

Q1952 ? Pattern: ratios ? **36**. Context: performance report. Sum=180; divide by 5 to get 36.

Q1953 ? Pattern: alternating ? **37**. Context: quality sample. Sum=185; divide by 5 to get 37.

Q1954 ? Pattern: squares / cubes ? **38**. Context: student marks. Sum=190; divide by 5 to get 38.

Q1955 ? Inference from facts ? **39**. Context: sales spread. Sum=195; divide by 5 to get 39.

Q1956 ? Assumption check ? **40**. Context: sensor readings. Sum=200; divide by 5 to get 40.

Q1957 ? Cause vs correlation ? **41**. Context: salary data. Sum=205; divide by 5 to get 41.

Q1958 ? Grammar agreement ? **42**. Context: delivery times. Sum=210; divide by 5 to get 42.

Q1959 ? Vocabulary in context ? **43**. Context: survey summary. Sum=215; divide by 5 to get 43.

Q1960 ? Mixed placement challenge ? **44**. Context: performance report. Sum=220; divide by 5 to get 44.

Frequently Asked

Q1981 ? Options elimination ? **232.5**. Context: campus drive. $310 \times 75 / 100 = \blacksquare 232.5$.

Q1982 ? Boundary case ? **224**. Context: operations report. $320 \times 70 / 100 = \blacksquare 224$.

Q1983 ? Case split ? **214.5**. Context: finance question. $330 \times 65 / 100 = \blacksquare 214.5$.

Q1984 ? Contradiction check ? **306**. Context: mixed placement case. $340 \times 90 / 100 = \blacksquare 306$.

Q1985 ? Statement I test ? **297.5**. Context: TCS-style screen. $350 \times 85 / 100 = \blacksquare 297.5$.

Q1986 ? Statement II test ? **288**. Context: banking case. $360 \times 80 / 100 = \blacksquare 288$.

Q1987 ? Combined statements ? **277.5**. Context: consulting estimate. $370 \times 75 / 100 = \blacksquare 277.5$.

Q1988 ? Counting restriction ? **266**. Context: retail audit. $380 \times 70 / 100 = \blacksquare 266$.

Q1989 ? Circular arrangement ? **253.5**. Context: campus drive. $390 \times 65 / 100 = \blacksquare 253.5$.

Q1990 ? Complement method ? **360**. Context: operations report. $400 \times 90 / 100 = \blacksquare 360$.

Q1991 ? Pattern: differences ? **348.5**. Context: finance question. $410 \times 85 / 100 = \blacksquare 348.5$.

Q1992 ? Pattern: ratios ? **336**. Context: mixed placement case. $420 \times 80 / 100 = \blacksquare 336$.

Q1993 ? Pattern: alternating ? **322.5**. Context: TCS-style screen. $430 \times 75 / 100 = \blacksquare 322.5$.

Q1994 ? Pattern: squares / cubes ? **308**. Context: banking case. $440 \times 70 / 100 = \blacksquare 308$.

Q1995 ? Inference from facts ? **292.5**. Context: consulting estimate. $450 \times 65 / 100 = \blacksquare 292.5$.

Q1996 ? Assumption check ? **414**. Context: retail audit. $460 \times 90 / 100 = \blacksquare 414$.

Q1997 ? Cause vs correlation ? **399.5**. Context: campus drive. $470 \times 85 / 100 = \blacksquare 399.5$.

Q1998 ? Grammar agreement ? **384**. Context: operations report. $480 \times 80 / 100 = \blacksquare 384$.

Q1999 ? Vocabulary in context ? **367.5**. Context: finance question. $490 \times 75 / 100 = \blacksquare 367.5$.

Q2000 ? Mixed placement challenge ? **350**. Context: mixed placement case. $500 \times 70 / 100 = \blacksquare 350$.